

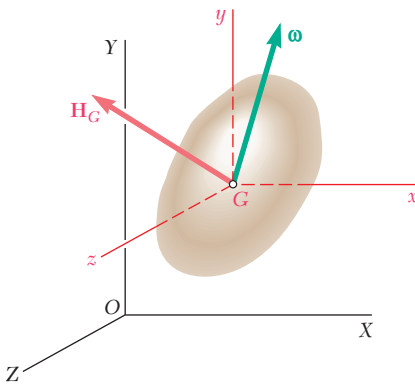


Fig. 18.3 In general, the angular momentum and the angular velocity are not in the same direction.

The relations in Eq. (18.7) show that the operation transforming the vector $\boldsymbol{\omega}$ into the vector $\mathbf{H}_G$ (Fig. 18.3) is characterized by the array of moments and products of inertia as

Inertia tensor

$$\begin{pmatrix} \bar{I}_x & -\bar{I}_{xy} & -\bar{I}_{xz} \\ -\bar{I}_{yx} & \bar{I}_y & -\bar{I}_{yz} \\ -\bar{I}_{zx} & -\bar{I}_{zy} & \bar{I}_z \end{pmatrix} \quad (18.8)$$

The array in Eq. (18.8) defines the **inertia tensor** of the body at its mass center G .[†] We obtain a new array of moments and products of inertia if we use a different system of axes. The angular momentum $\mathbf{H}_G$ corresponding to a given angular velocity $\boldsymbol{\omega}$ is independent of the choice of the coordinate axes.

As we showed in *Statics*, Sec. 9.6, or in Appendix B, it is always possible to select a system of axes $Gx'y'z'$, called *principal axes of inertia*, with respect to which all the products of inertia of a given body are zero. The array of Eq. (18.8) then takes the diagonalized form as

$$\begin{pmatrix} \bar{I}_{x'} & 0 & 0 \\ 0 & \bar{I}_{y'} & 0 \\ 0 & 0 & \bar{I}_{z'} \end{pmatrix} \quad (18.9)$$

where $\bar{I}_{x'}, \bar{I}_{y'}, \bar{I}_{z'}$ represent the *principal centroidal moments of inertia* of the body, and the relations in Eq. (18.7) reduce to

$$H_{x'} = \bar{I}_{x'}\omega_{x'} \quad H_{y'} = \bar{I}_{y'}\omega_{y'} \quad H_{z'} = \bar{I}_{z'}\omega_{z'} \quad (18.10)$$

Note that if the three principal centroidal moments of inertia $\bar{I}_{x'}, \bar{I}_{y'}, \bar{I}_{z'}$ are equal, the components $H_{x'}, H_{y'}, H_{z'}$ of the angular momentum about G are proportional to the components $\omega_{x'}, \omega_{y'}, \omega_{z'}$ of the angular velocity, and the vectors $\mathbf{H}_G$ and $\boldsymbol{\omega}$ are collinear. In general, however, the principal moments of inertia are different, and the vectors $\mathbf{H}_G$ and $\boldsymbol{\omega}$ have different directions except when two of the three components of $\boldsymbol{\omega}$ happen to be zero, i.e., when $\boldsymbol{\omega}$ is directed along one of the coordinate axes. Thus,

The angular momentum $\mathbf{H}_G$ of a rigid body and its angular velocity $\boldsymbol{\omega}$ have the same direction if, and only if, $\boldsymbol{\omega}$ is directed along a principal axis of inertia.[‡]

This condition is satisfied in the case of the plane motion of a rigid body that is symmetrical with respect to the reference plane, so in Secs. 16.1 and 17.2, we were able to represent the angular momentum $\mathbf{H}_G$ of such a body by the vector $\bar{I}\boldsymbol{\omega}$. We must realize, however, that this result cannot be extended to the case of the plane motion of a nonsymmetrical body or to

[†]Setting $\bar{I}_x = I_{11}$, $\bar{I}_y = I_{22}$, $\bar{I}_z = I_{33}$, and $-\bar{I}_{xy} = I_{12}$, $-\bar{I}_{xz} = I_{13}$, etc., we can write the inertia tensor of Eq. (18.8) in the standard form

$$\begin{pmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{pmatrix}$$

[‡]In the particular case when $\bar{I}_{x'} = \bar{I}_{y'} = \bar{I}_{z'}$, any line through G can be considered to be a principal axis of inertia, and the vectors $\mathbf{H}_G$ and $\boldsymbol{\omega}$ are always collinear.

the case of the three-dimensional motion of a rigid body. Except when $\boldsymbol{\omega}$ happens to be directed along a principal axis of inertia, the angular momentum and angular velocity of a rigid body have different directions, and you must use the relation in Eq. (18.7) or (18.10) to determine $\mathbf{H}_G$ from $\boldsymbol{\omega}$.

Reduction of the Momenta of the Particles of a Rigid Body to a Momentum Vector and a Couple at G .

We saw in Sec. 17.2A that we can reduce the system formed by the momenta of the various particles of a rigid body to a vector $\mathbf{L}$ that is attached at the mass center G of the body, representing the linear momentum of the body, and to a couple $\mathbf{H}_G$, representing the angular momentum of the body about G (Fig. 18.4). We are now in a position to determine the vector $\mathbf{L}$ and the couple $\mathbf{H}_G$ in the most general case of three-dimensional motion of a rigid body. As in the case of the two-dimensional motion considered earlier, the linear momentum $\mathbf{L}$ of the body is equal to the product $m\bar{\mathbf{v}}$ of its mass m and velocity $\bar{\mathbf{v}}$ of its mass center G . However, we can no longer obtain the angular momentum $\mathbf{H}_G$ by simply multiplying the angular velocity $\boldsymbol{\omega}$ of the body by the scalar $\bar{I}$. Instead, we obtain it from the components of $\boldsymbol{\omega}$ and from the centroidal moments and products of inertia of the body through the use of Eq. (18.7) or (18.10).

We should also note that once we have determined the linear momentum $m\bar{\mathbf{v}}$ and the angular momentum $\mathbf{H}_G$ of a rigid body, we can obtain its angular momentum $\mathbf{H}_O$ about any given point O by adding the moments about O of vector $m\bar{\mathbf{v}}$ and of couple $\mathbf{H}_G$. We have

$$\mathbf{H}_O = \bar{\mathbf{r}} \times m\bar{\mathbf{v}} + \mathbf{H}_G \quad (18.11)$$

Angular Momentum of a Rigid Body Constrained to Rotate about a Fixed Point.

In the particular case of a rigid body constrained to rotate in three-dimensional space about a fixed point O (Fig. 18.5a), it is sometimes convenient to determine the angular momentum $\mathbf{H}_O$ of the body about O . Although we could obtain $\mathbf{H}_O$ by first computing $\mathbf{H}_G$ as indicated previously and then using Eq. (18.11), it is often advantageous to determine $\mathbf{H}_O$ directly from the angular velocity $\boldsymbol{\omega}$ of the body and its moments and products of inertia with respect to a frame $Oxyz$ centered at O . From Eq. (14.7), we have

$$\mathbf{H}_O = \sum_{i=1}^n (\mathbf{r}_i \times \mathbf{v}_i \Delta m_i) \quad (18.12)$$

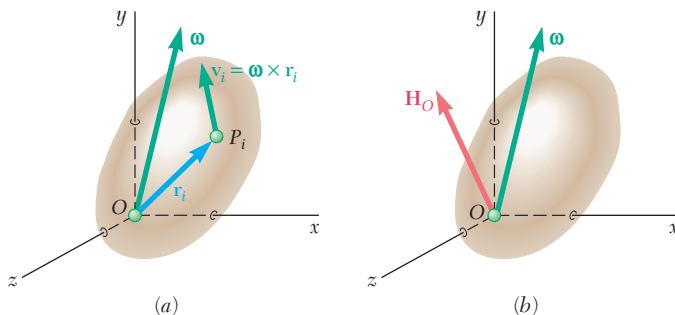


Fig. 18.5 (a) The velocity of particle P_i of a rigid body rotating with angular velocity $\boldsymbol{\omega}$; (b) angular velocity and angular momentum of a rigid body.



Photo 18.1 The design of a robotic welder for an automobile assembly line requires a three-dimensional study of both kinematics and kinetics.

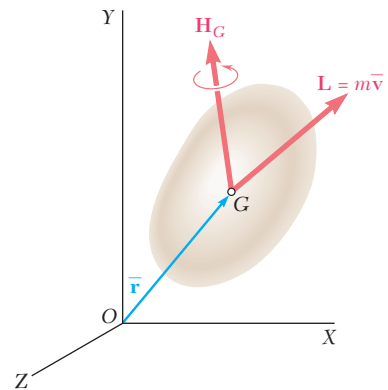


Fig. 18.4 A momentum vector attached to the mass center of a rigid body and the angular momentum of the body about its mass center.

where $\mathbf{r}_i$ and $\mathbf{v}_i$ denote, respectively, the position vector and the velocity of particle P_i with respect to the fixed frame $Oxyz$. Substituting $\mathbf{v}_i = \boldsymbol{\omega} \times \mathbf{r}_i$ and after making manipulations similar to those used in the earlier part of this section, we find that the components of the angular momentum $\mathbf{H}_O$ (Fig. 18.5b) are given by the relations

**Angular momentum about
a fixed point O**

$$\begin{aligned} H_x &= +I_x \omega_x - I_{xy} \omega_y - I_{xz} \omega_z \\ H_y &= -I_{yx} \omega_x + I_y \omega_y - I_{yz} \omega_z \\ H_z &= -I_{zx} \omega_x - I_{zy} \omega_y + I_z \omega_z \end{aligned} \quad (18.13)$$

where we compute the moments of inertia I_x, I_y, I_z and the products of inertia I_{xy}, I_{yz}, I_{zx} with respect to the frame $Oxyz$ centered at the fixed point O .

*18.1B Applying the Principle of Impulse and Momentum to the Three-Dimensional Motion of a Rigid Body

Before we can apply the fundamental equation (18.2) to the solution of problems involving the three-dimensional motion of a rigid body, we must be able to compute the derivative of the vector $\mathbf{H}_G$. We show how to do this in Sec. 18.2A. However, we can use the results obtained already to solve problems using the impulse–momentum method.

Recall that the system formed by the momenta of the particles of a rigid body reduces to a linear momentum vector $m\bar{\mathbf{v}}$ attached at the mass center G of the body and an angular momentum couple $\mathbf{H}_G$. We can represent the fundamental relation

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2 \quad (17.14)$$

graphically by means of the impulse–momentum diagram shown in Fig. 18.6. To solve a given problem, we can use this diagram to write appropriate component and moment equations, keeping in mind that the components of the angular momentum $\mathbf{H}_G$ are related to the components of the angular velocity $\boldsymbol{\omega}$ by Eqs. (18.7).



Photo 18.2 As a result of the impulsive force applied by the bowling ball, a pin acquires both linear momentum and angular momentum.

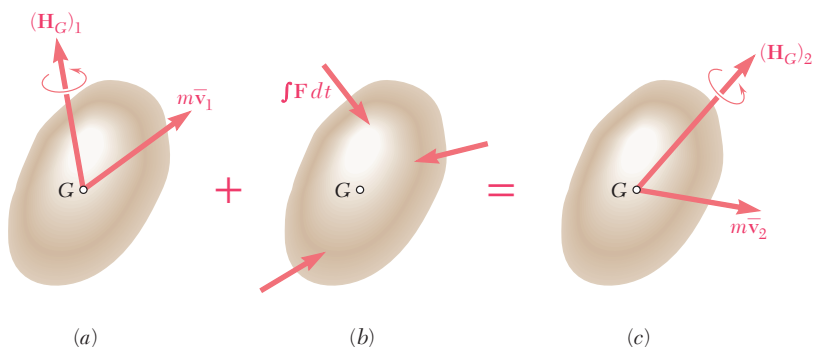


Fig. 18.6 Impulse–momentum diagram for applying the principle of impulse and momentum to the motion of a rigid body in space.

In solving problems dealing with the motion of a body rotating about a fixed point O , it will be convenient to eliminate the impulse of the reaction at O by writing an equation involving the moments of the momenta and impulses about O . Recall that you can obtain the angular momentum $\mathbf{H}_O$ of the body about the fixed point O either directly from Eqs. (18.13) or by first computing its linear momentum $m\bar{\mathbf{v}}$ and its angular momentum $\mathbf{H}_G$ and then using Eq. (18.11).

*18.1C Kinetic Energy of a Rigid Body in Three Dimensions

Consider a rigid body with a mass m in three-dimensional motion. Recall from Sec. 14.2A that, if we express the absolute velocity $\mathbf{v}_i$ of each particle P_i of the body as the sum of velocity $\bar{\mathbf{v}}$ of the mass center G of the body and velocity $\mathbf{v}'_i$ of the particle relative to a frame $Gxyz$ attached to G and of fixed orientation (Fig. 18.7), we can write the kinetic energy of the system of particles forming the rigid body as

$$T = \frac{1}{2}m\bar{\mathbf{v}}^2 + \frac{1}{2} \sum_{i=1}^n \Delta m_i v_i'^2 \quad (18.14)$$

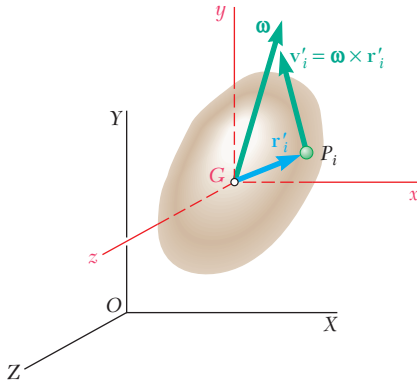


Fig. 18.7 The relative velocity of a particle P_i with respect to the mass center is $\boldsymbol{\omega} \times \mathbf{r}'_i$.

Here the last term represents the kinetic energy T' of the body relative to the centroidal frame $Gxyz$. Since $v'_i = |\mathbf{v}'_i| = |\boldsymbol{\omega} \times \mathbf{r}'_i|$, we have

$$T' = \frac{1}{2} \sum_{i=1}^n \Delta m_i v_i'^2 = \frac{1}{2} \sum_{i=1}^n |\boldsymbol{\omega} \times \mathbf{r}'_i|^2 \Delta m_i$$

Expressing the square in terms of the rectangular components of the vector product and replacing the sums by integrals, we have

$$\begin{aligned} T' &= \frac{1}{2} \int [(\omega_x y - \omega_y x)^2 + (\omega_y z - \omega_z y)^2 + (\omega_z x - \omega_x z)^2] dm \\ &= \frac{1}{2} [\omega_x^2 \int (y^2 + z^2) dm + \omega_y^2 \int (z^2 + x^2) dm + \omega_z^2 \int (x^2 + y^2) dm \\ &\quad - 2\omega_x \omega_y \int xy dm - 2\omega_y \omega_z \int yz dm - 2\omega_z \omega_x \int zx dm] \end{aligned}$$

or recalling the relations of Eqs. (18.5) and (18.6), we have

$$T' = \frac{1}{2} (\bar{I}_x \omega_x^2 + \bar{I}_y \omega_y^2 + \bar{I}_z \omega_z^2 - 2\bar{I}_{xy} \omega_x \omega_y - 2\bar{I}_{yz} \omega_y \omega_z - 2\bar{I}_{zx} \omega_z \omega_x) \quad (18.15)$$

Substituting Eq. (18.15) for the kinetic energy of the body relative to centroidal axes into Eq. (18.14), we obtain

Kinetic energy of a rigid body

$$T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}(\bar{I}_x\omega_x^2 + \bar{I}_y\omega_y^2 + \bar{I}_z\omega_z^2 - 2\bar{I}_{xy}\omega_x\omega_y - 2\bar{I}_{yz}\omega_y\omega_z - 2\bar{I}_{zx}\omega_z\omega_x) \quad (18.16)$$

If we choose the axes of coordinates so that they coincide with the principal axes x', y', z' of the body at the instant considered, this relation reduces to

$$T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}(\bar{I}_{x'}\omega_{x'}^2 + \bar{I}_{y'}\omega_{y'}^2 + \bar{I}_{z'}\omega_{z'}^2) \quad (18.17)$$

where $\bar{v}$ = velocity of mass center

ω = angular velocity

m = mass of rigid body

$\bar{I}_{x'}, \bar{I}_{y'}, \bar{I}_{z'}$ = principal centroidal moments of inertia

These results enable us to apply the principles of work and energy (Sec. 17.1A) and the conservation of energy (Sec. 17.1E) to the three-dimensional motion of a rigid body.

Kinetic Energy of a Rigid Body with a Fixed Point. In the particular case of a rigid body rotating in three-dimensional space about a fixed point O , we can express the kinetic energy of the body in terms of its moments and products of inertia with respect to axes attached at O (Fig. 18.8). Recalling the definition of the kinetic energy of a system of particles and substituting $v_i = |\mathbf{v}_i| = |\omega \times \mathbf{r}_i|$, we have

$$T = \frac{1}{2} \sum_{i=1}^n \Delta m_i v_i^2 = \frac{1}{2} \sum_{i=1}^n |\omega \times \mathbf{r}_i|^2 \Delta m_i \quad (18.18)$$

Manipulations similar to those used to derive Eq. (18.15) yield

$$T = \frac{1}{2}(I_x\omega_x^2 + I_y\omega_y^2 + I_z\omega_z^2 - 2I_{xy}\omega_x\omega_y - 2I_{yz}\omega_y\omega_z - 2I_{zx}\omega_z\omega_x) \quad (18.19)$$

or if we choose the principal axes x', y', z' of the body at the origin O as coordinate axes, we have

$$T = \frac{1}{2}(I_{x'}\omega_{x'}^2 + I_{y'}\omega_{y'}^2 + I_{z'}\omega_{z'}^2) \quad (18.20)$$

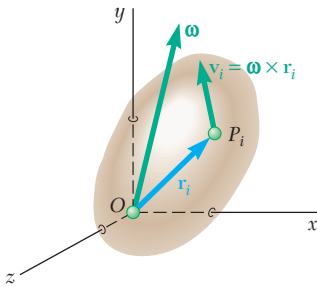
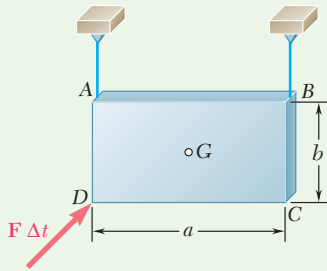


Fig. 18.8 The velocity of every particle P_i of a rigid body undergoing fixed axis rotation is $\omega \times \mathbf{r}_i$.



Sample Problem 18.1

A rectangular plate with a mass m is suspended from two wires at A and B and is hit at D in a direction perpendicular to the plate. Denoting the impulse applied at D by $\mathbf{F} \Delta t$, determine immediately after the impact (a) the velocity of the mass center G , (b) the angular velocity of the plate.

STRATEGY: Since you have an impulse applied to the plate, use the principle of impulse and momentum.

MODELING: Choose the plate to be your system and model it as a rigid body undergoing three-dimensional motion.

ANALYSIS: Assume that the wires remain taut. Therefore, the components $\bar{v}_y$ of $\bar{\mathbf{v}}$ and ω_z of $\boldsymbol{\omega}$ are zero after the impact. Then you have

$$\bar{\mathbf{v}} = \bar{v}_x \mathbf{i} + \bar{v}_z \mathbf{k} \quad \boldsymbol{\omega} = \omega_x \mathbf{i} + \omega_y \mathbf{j}$$

The x, y, z axes are principal axes of inertia, so you have

$$\mathbf{H}_G = \bar{I}_x \omega_x \mathbf{i} + \bar{I}_y \omega_y \mathbf{j} \quad \mathbf{H}_G = \frac{1}{12} m b^2 \omega_x \mathbf{i} + \frac{1}{12} m a^2 \omega_y \mathbf{j} \quad (1)$$

Principle of Impulse and Momentum. Since the initial momenta are zero, the system of the impulses must be equivalent to the system of the final momenta (Fig. 1).

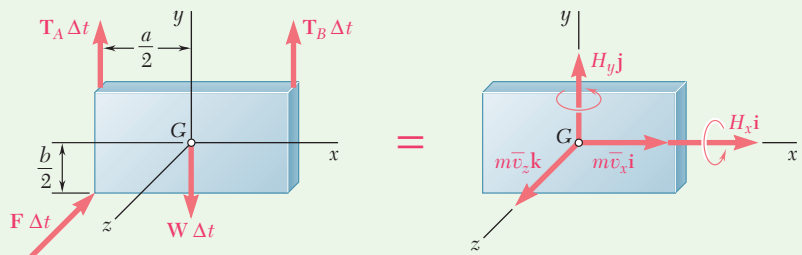


Fig. 1 Impulse-momentum diagram for the plate.

a. Velocity of Mass Center. Equate the components of the impulses and momenta in the x and z directions as

$$\begin{aligned} x \text{ components:} \quad & 0 = m \bar{v}_x & \bar{v}_x &= 0 \\ z \text{ components:} \quad & -F \Delta t = m \bar{v}_z & \bar{v}_z &= -F \Delta t / m \\ & & \bar{\mathbf{v}} &= \bar{v}_x \mathbf{i} + \bar{v}_z \mathbf{k} & \bar{\mathbf{v}} &= -(F \Delta t / m) \mathbf{k} \quad \blacktriangleleft \end{aligned}$$

b. Angular Velocity. Equate the moments of the impulses and momenta about the x and y axes as

$$\begin{aligned} \text{About } x \text{ axis:} \quad & \frac{1}{2} b F \Delta t = H_x \\ \text{About } y \text{ axis:} \quad & -\frac{1}{2} a F \Delta t = H_y \\ & \mathbf{H}_G = H_x \mathbf{i} + H_y \mathbf{j} & \mathbf{H}_G &= \frac{1}{2} b F \Delta t \mathbf{i} - \frac{1}{2} a F \Delta t \mathbf{j} \quad (2) \end{aligned}$$

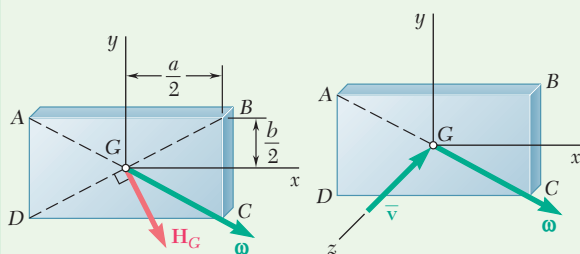


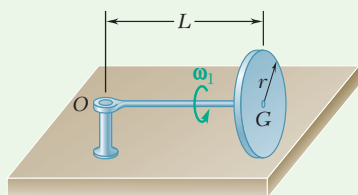
Fig. 2 Directions of the angular velocity, angular momentum, and velocity of G immediately after the impulse.

Comparing Eqs. (1) and (2), you can conclude that

$$\begin{aligned}\omega_x &= 6F \Delta t / mb & \omega_y &= -6F \Delta t / ma \\ \boldsymbol{\omega} &= \omega_x \mathbf{i} + \omega_y \mathbf{j} & \boldsymbol{\omega} &= (6F \Delta t / mab)(a\mathbf{i} - b\mathbf{j})\end{aligned}$$

Note that $\boldsymbol{\omega}$ is directed along the diagonal AC (Fig. 2).

REFLECT and THINK: Equating the y components of the impulses and momenta and their moments about the z axis, you can obtain two additional equations that yield $T_A = T_B = \frac{1}{2}W$. This verifies that the wires remain taut and that the initial assumption was correct. If the impulse was at G , this would reduce to a two-dimensional problem.



Sample Problem 18.2

A homogeneous disk of radius r and mass m is mounted on an axle OG of length L and negligible mass. The axle is pivoted at the fixed point O , and the disk is constrained to roll on a horizontal floor. The disk rotates counterclockwise at the rate ω_1 about the axle OG . Determine (a) the angular velocity of the disk, (b) its angular momentum about O , (c) its kinetic energy, (d) the linear momentum and angular momentum about G of the disk.

STRATEGY: Recognizing that the wheel rolls without slip, you can use kinematics to calculate the angular velocity of the bar around O . Then you can determine the kinetic energy and momenta of the system.

MODELING and ANALYSIS:

a. Angular Velocity. As the disk rotates about the axle OG , it also rotates with the axle about the y axis at a rate of ω_2 clockwise (Fig. 1). The total angular velocity of the disk is therefore

$$\boldsymbol{\omega} = \omega_1 \mathbf{i} - \omega_2 \mathbf{j} \quad (1)$$

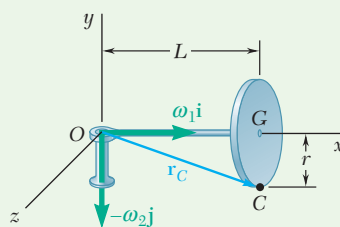


Fig. 1 Angular velocity of the system.

(continued)

The disk is rolling, so set the velocity of C to zero to determine ω_2 as

$$\begin{aligned}\mathbf{v}_C &= \boldsymbol{\omega} \times \mathbf{r}_C = 0 \\ (\omega_1 \mathbf{i} - \omega_2 \mathbf{j}) \times (L \mathbf{i} - r \mathbf{j}) &= 0 \\ (L\omega_2 - r\omega_1) \mathbf{k} &= 0 \quad \omega_2 = r\omega_1/L\end{aligned}$$

Substituting into Eq. (1) for ω_2 gives

$$\boldsymbol{\omega} = \omega_1 \mathbf{i} - (r\omega_1/L) \mathbf{j} \quad \blacktriangleleft$$

b. Angular Momentum about O . Assuming the axle to be part of the disk, you can consider the disk to have a fixed point at O . Since the x , y , and z axes are principal axes of inertia for the disk, you have

$$\begin{aligned}H_x &= I_x \omega_x = (\tfrac{1}{2}mr^2)\omega_1 \\ H_y &= I_y \omega_y = (mL^2 + \tfrac{1}{4}mr^2)(-r\omega_1/L) \\ H_z &= I_z \omega_z = (mL^2 + \tfrac{1}{4}mr^2)0 = 0 \\ \mathbf{H}_O &= \tfrac{1}{2}mr^2\omega_1 \mathbf{i} - m(L^2 + \tfrac{1}{4}r^2)(r\omega_1/L) \mathbf{j} \quad \blacktriangleleft\end{aligned}$$

c. Kinetic Energy. Using the values obtained for the moments of inertia and the components of $\boldsymbol{\omega}$, you have

$$\begin{aligned}T &= \tfrac{1}{2}(I_x \omega_x^2 + I_y \omega_y^2 + I_z \omega_z^2) = \tfrac{1}{2}[\tfrac{1}{2}mr^2\omega_1^2 + m(L^2 + \tfrac{1}{4}r^2)(-r\omega_1/L)^2] \\ T &= \tfrac{1}{8}mr^2 \left(6 + \frac{r^2}{L^2}\right) \omega_1^2 \quad \blacktriangleleft\end{aligned}$$

d. Linear Momentum and Angular Momentum about G . The linear momentum vector $m\bar{\mathbf{v}}$ and the angular momentum couple $\mathbf{H}_G$ are (Fig.2)

$$m\bar{\mathbf{v}} = mr\omega_1 \mathbf{k} \quad \blacktriangleleft$$

and

$$\begin{aligned}\mathbf{H}_G &= \bar{I}_x \omega_x \mathbf{i} + \bar{I}_y \omega_y \mathbf{j} + \bar{I}_z \omega_z \mathbf{k} = \tfrac{1}{2}mr^2\omega_1 \mathbf{i} + \tfrac{1}{4}mr^2(-r\omega_1/L) \mathbf{j} \\ \mathbf{H}_G &= \tfrac{1}{2}mr^2\omega_1 \left(\mathbf{i} - \frac{r}{2L} \mathbf{j} \right) \quad \blacktriangleleft\end{aligned}$$

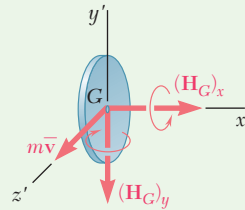


Fig. 2 Linear and angular momenta for the system.

REFLECT and THINK: If the mass of the axle was not negligible and it was instead modeled as a slender rod with a mass M_{axle} , it would also contribute to the kinetic energy $T_{\text{axle}} = \tfrac{1}{2}(\tfrac{1}{3}M_{\text{axle}}L^2)\omega_2^2$ and to the angular momentum $\mathbf{H}_{\text{axle}} = -(\tfrac{1}{3}M_{\text{axle}}L^2)\omega_2 \mathbf{j}$ of the system.

SOLVING PROBLEMS ON YOUR OWN

In this section, you saw how to compute the **angular momentum of a rigid body in three dimensions** and to apply the principle of impulse and momentum to the three-dimensional motion of a rigid body. You also learned how to compute the **kinetic energy of a rigid body in three dimensions**. It is important for you to keep in mind that, except for very special situations, the angular momentum of a rigid body in three dimensions *cannot* be expressed as the product $\bar{I}\boldsymbol{\omega}$ and, therefore, does not have the same direction as the angular velocity $\boldsymbol{\omega}$ (Fig. 18.3).

1. To compute the angular momentum $\mathbf{H}_G$ of a rigid body about its mass center G , you must first determine the angular velocity $\boldsymbol{\omega}$ of the body with respect to a system of axes centered at G and of fixed orientation. Since you will be asked in the problems to determine the angular momentum of the body *at a given instant only*, select the system of axes that will be most convenient for your computations.

a. If the principal axes of inertia of the body at G are known, use these axes as coordinate axes x' , y' , and z' , since the corresponding products of inertia of the body are equal to zero. Resolve $\boldsymbol{\omega}$ into components $\omega_{x'}$, $\omega_{y'}$, and $\omega_{z'}$ along these axes and compute the principal moments of inertia as $\bar{I}_{x'}$, $\bar{I}_{y'}$, $\bar{I}_{z'}$. The corresponding components of the angular momentum $\mathbf{H}_G$ are

$$H_{x'} = \bar{I}_{x'}\omega_{x'} \quad H_{y'} = \bar{I}_{y'}\omega_{y'} \quad H_{z'} = \bar{I}_{z'}\omega_{z'} \quad (18.10)$$

b. If the principal axes of inertia of the body at G are not known, you must use Eqs. (18.7) to determine the components of the angular momentum $\mathbf{H}_G$. These equations require prior computation of the products of inertia of the body as well as prior computation of its moments of inertia with respect to the selected axes.

c. The magnitude and direction cosines of $\mathbf{H}_G$ are obtained from formulas similar to those used in *Statics* [Sec. 2.4A]. We have

$$H_G = \sqrt{H_x^2 + H_y^2 + H_z^2}$$

$$\cos \theta_x = \frac{H_x}{H_G} \quad \cos \theta_y = \frac{H_y}{H_G} \quad \cos \theta_z = \frac{H_z}{H_G}$$

d. Once you have determined $\mathbf{H}_G$, you can obtain the angular momentum of the body about any given point O by observing from Fig. (18.4) that

$$\mathbf{H}_O = \bar{\mathbf{r}} \times m\bar{\mathbf{v}} + \mathbf{H}_G \quad (18.11)$$

where $\bar{\mathbf{r}}$ is the position vector of G relative to O and $m\bar{\mathbf{v}}$ is the linear momentum of the body.

2. To compute the angular momentum $\mathbf{H}_O$ of a rigid body with a fixed point O , follow the procedure described in paragraph 1, except that you should now use axes centered at the fixed point O . Alternatively, you can use Eq. 18.11.

a. If you know the principal axes of inertia of the body at O , resolve $\boldsymbol{\omega}$ into components along these axes [Sample Prob. 18.2]. Obtain the corresponding components of the angular momentum $\mathbf{H}_G$ from equations similar to Eqs. (18.10).

b. If you do not know the principal axes of inertia of the body at O , you must compute the products as well as the moments of inertia of the body with respect to the axes that you have selected. Then use Eqs. (18.13) to determine the components of the angular momentum $\mathbf{H}_O$.

3. To apply the principle of impulse and momentum to the solution of a problem involving the three-dimensional motion of a rigid body, use the same vector equation that you used for plane motion in Chap. 17:

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2 \quad (17.14)$$

where the initial and final systems of momenta are each represented by a *linear-momentum vector* $m\bar{\mathbf{v}}$ and an *angular-momentum couple* $\mathbf{H}_G$. Now, however, these vector-and-couple systems should be represented in three dimensions, as shown in Fig. 18.6, and $\mathbf{H}_G$ should be determined as explained in paragraph 1.

a. In problems involving the application of a known impulse to a rigid body, draw the impulse–momentum diagram corresponding to Eq. (17.14). Equating the components of the vectors involved, you can determine the final linear momentum $m\bar{\mathbf{v}}$ of the body and, thus, the corresponding velocity $\bar{\mathbf{v}}$ of its mass center. Equating moments about G , you can determine the final angular momentum $\mathbf{H}_G$ of the body. Then substitute the values obtained for the components of $\mathbf{H}_G$ into Eq. (18.10) or (18.7) and solve for the corresponding values of the components of the angular velocity $\boldsymbol{\omega}$ of the body [Sample Prob. 18.1].

b. In problems involving unknown impulses, draw the impulse–momentum diagram corresponding to Eq. (17.14) and write equations that do not involve the unknown impulses. You can obtain such equations by equating moments about the point or line of impact.

4. To compute the kinetic energy of a rigid body with a fixed point O , resolve the angular velocity $\boldsymbol{\omega}$ into components along axes of your choice and compute the moments and products of inertia of the body with respect to these axes. As was the case for the computation of the angular momentum, use the principal axes of inertia x' , y' , and z' if you can easily determine them. The products of inertia are then zero [Sample Prob. 18.2], and the expression for the kinetic energy reduces to

$$T = \frac{1}{2}(I_{x'}\omega_{x'}^2 + I_{y'}\omega_{y'}^2 + I_{z'}\omega_{z'}^2) \quad (18.20)$$

If you must use axes other than the principal axes of inertia, express the kinetic energy of the body as shown in Eq. (18.19).

5. To compute the kinetic energy of a rigid body in general motion, consider the motion as the sum of a *translation with the mass center G* and a *rotation about G* . The kinetic energy associated with the translation is $\frac{1}{2}m\bar{\mathbf{v}}^2$. If you can use principal axes of inertia, express the kinetic energy associated with the rotation about G in the form used in Eq. (18.20). The total kinetic energy of the rigid body is then

$$T = \frac{1}{2}m\bar{\mathbf{v}}^2 + \frac{1}{2}(\bar{I}_{x'}\omega_{x'}^2 + \bar{I}_{y'}\omega_{y'}^2 + \bar{I}_{z'}\omega_{z'}^2) \quad (18.17)$$

If you must use axes other than the principal axes of inertia to determine the kinetic energy associated with the rotation about G , express the total kinetic energy of the body as shown in Eq. (18.16).

Problems

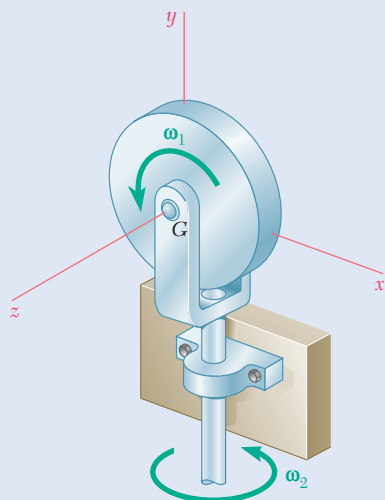


Fig. P18.1

18.1 A thin, homogeneous disk of mass m and radius r spins at the constant rate ω_1 about an axle held by a fork-ended vertical rod that rotates at the constant rate ω_2 . Determine the angular momentum $\mathbf{H}_G$ of the disk about its mass center G .

18.2 A thin rectangular plate of weight 15 lb rotates about its vertical diagonal AB with an angular velocity ω . Knowing that the z axis is perpendicular to the plate and that ω is constant and equal to 5 rad/s, determine the angular momentum of the plate about its mass center G .

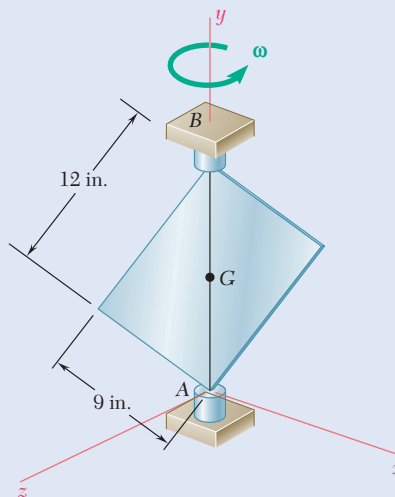


Fig. P18.2

18.3 Two uniform rods AB and CE , each of weight 3 lb and length 2 ft, are welded to each other at their midpoints. Knowing that this assembly has an angular velocity of constant magnitude $\omega = 12$ rad/s, determine the magnitude and direction of the angular momentum $\mathbf{H}_D$ of the assembly about D .

18.4 A homogeneous disk of weight $W = 6$ lb rotates at the constant rate $\omega_1 = 16$ rad/s with respect to arm ABC , which is welded to a shaft DCE rotating at the constant rate $\omega_2 = 8$ rad/s. Determine the angular momentum $\mathbf{H}_A$ of the disk about its center A .

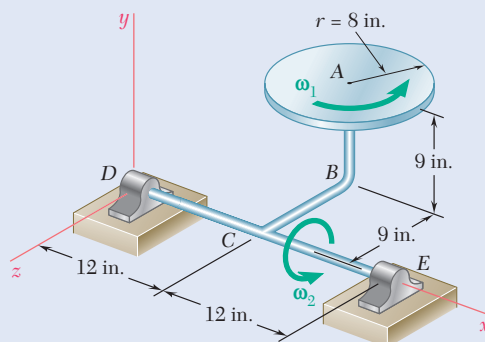


Fig. P18.4

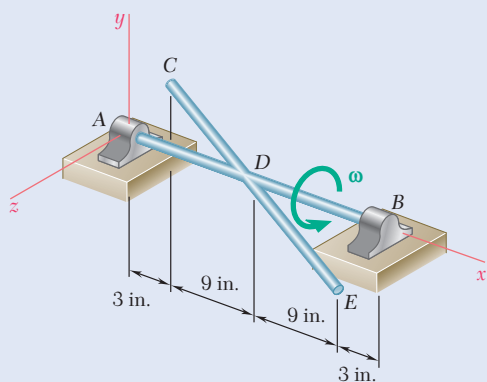


Fig. P18.3

18.5 A thin disk of mass $m = 4$ kg rotates at the constant rate $\omega_2 = 15$ rad/s with respect to arm ABC , which itself rotates at the constant rate $\omega_1 = 5$ rad/s about the y axis. Determine the angular momentum of the disk about its center C .

18.6 A solid rectangular parallelepiped of mass m has a square base of side a and a length $2a$. Knowing that it rotates at the constant rate ω about its diagonal AC' and that its rotation is observed from A as counterclockwise, determine (a) the magnitude of the angular momentum $\mathbf{H}_G$ of the parallelepiped about its mass center G , (b) the angle that $\mathbf{H}_G$ forms with the diagonal AC' .

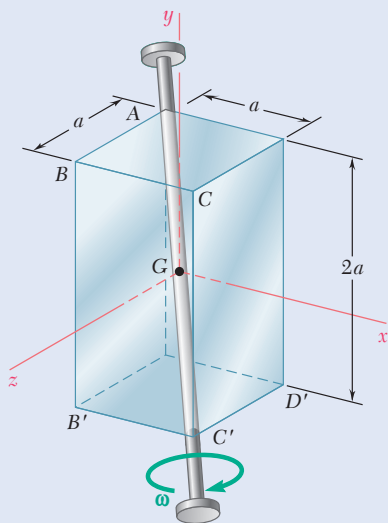


Fig. P18.6

18.7 Solve Prob. 18.6, assuming that the solid rectangular parallelepiped has been replaced by a hollow one consisting of six thin metal plates welded together.

18.8 A thin homogeneous disk with a mass m and radius r is mounted on the horizontal axle AB . The plane of the disk forms an angle of $\beta = 20^\circ$ with the vertical. Knowing that the axle rotates with an angular velocity ω , determine the angle θ formed by the axle and the angular momentum of the disk about G .

18.9 Determine the angular momentum $\mathbf{H}_D$ of the disk of Prob. 18.4 about point D .

18.10 Determine the angular momentum of the disk of Prob. 18.5 about point A .

18.11 Determine the angular momentum $\mathbf{H}_O$ of the disk of Sample Prob. 18.2 from the expressions obtained for its linear momentum $m\bar{\mathbf{v}}$ and its angular momentum $\mathbf{H}_G$, using Eqs. (18.11). Verify that the result obtained is the same as that obtained by direct computation.

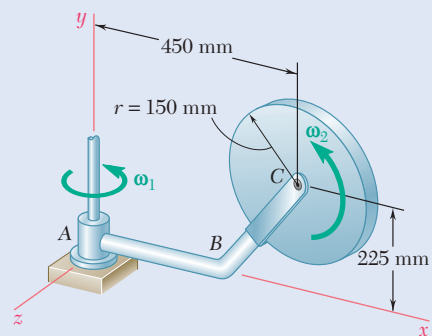


Fig. P18.5

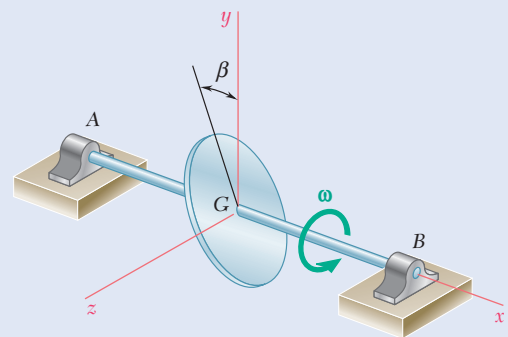


Fig. P18.8

- 18.12** The 100-kg projectile shown has a radius of gyration of 100 mm about its axis of symmetry Gx and a radius of gyration of 250 mm about the transverse axis Gy . Its angular velocity ω can be resolved into two components; one component, directed along Gx , measures the *rate of spin* of the projectile, while the other component, directed along GD , measures its *rate of precession*. Knowing that $\theta = 6^\circ$ and that the angular momentum of the projectile about its mass center G is $\mathbf{H}_G = (500 \text{ g}\cdot\text{m}^2/\text{s})\mathbf{i} - (10 \text{ g}\cdot\text{m}^2/\text{s})\mathbf{j}$, determine (a) the rate of spin, (b) the rate of precession.

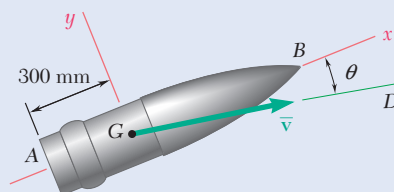


Fig. P18.12

- 18.13** Determine the angular momentum $\mathbf{H}_A$ of the projectile of Prob. 18.12 about the center A of its base, knowing that its mass center G has a velocity $\bar{\mathbf{v}}$ of 750 m/s. Give your answer in terms of components respectively parallel to the x and y axes shown and to a third axis z pointing toward you.

- 18.14** (a) Show that the angular momentum $\mathbf{H}_B$ of a rigid body about point B can be obtained by adding to the angular momentum $\mathbf{H}_A$ of that body about point A the vector product of the vector $\mathbf{r}_{A/B}$ drawn from B to A and the linear momentum $m\bar{\mathbf{v}}$ of the body:

$$\mathbf{H}_B = \mathbf{H}_A + \mathbf{r}_{A/B} \times m\bar{\mathbf{v}}$$

(b) Further show that when a rigid body rotates about a fixed axis, its angular momentum is the same about any two points A and B located on the fixed axis ($\mathbf{H}_A = \mathbf{H}_B$) if, and only if, the mass center G of the body is located on the fixed axis.

- 18.15** Two L-shaped arms each have a mass of 5 kg and are welded at the one-third points of the 600-mm shaft AB to form the assembly shown. Knowing that the assembly rotates at the constant rate of 360 rpm, determine (a) the angular momentum $\mathbf{H}_A$ of the assembly about point A , (b) the angle formed by $\mathbf{H}_A$ and AB .

- 18.16** For the assembly of Prob. 18.15, determine (a) the angular momentum $\mathbf{H}_B$ of the assembly about point B , (b) the angle formed by $\mathbf{H}_B$ and BA .

- 18.17** A 10-lb rod of uniform cross section is used to form the shaft shown. Knowing that the shaft rotates with a constant angular velocity ω of magnitude 12 rad/s, determine (a) the angular momentum $\mathbf{H}_G$ of the shaft about its mass center G , (b) the angle formed by $\mathbf{H}_G$ and the axis AB .

- 18.18** Determine the angular momentum of the shaft of Prob. 18.17 about (a) point A , (b) point B .

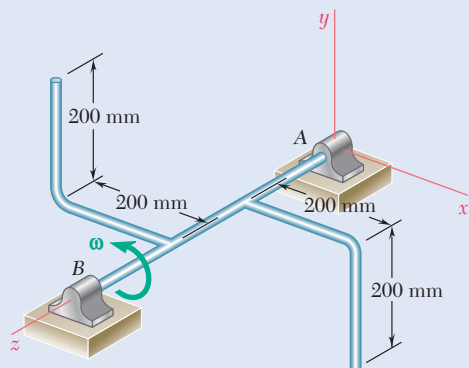


Fig. P18.15

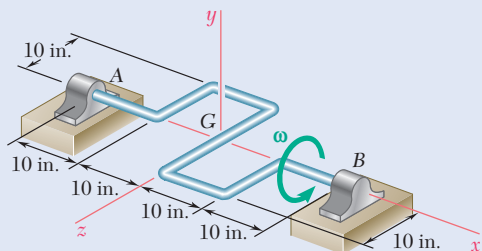


Fig. P18.17

18.19 Two triangular plates each have a mass of 8 kg and are welded to a vertical shaft AB . Knowing that the system rotates at the constant rate of $\omega = 6 \text{ rad/s}$, determine its angular momentum about G .

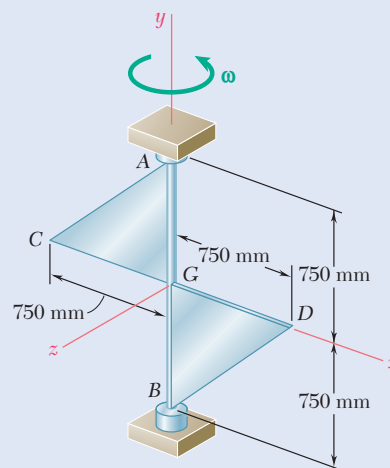


Fig. P18.19

18.20 The assembly shown consists of two pieces of sheet aluminum with a uniform thickness and total mass of 1.6 kg welded to a light axle supported by bearings A and B . Knowing that the assembly rotates with an angular velocity of constant magnitude $\omega = 20 \text{ rad/s}$, determine the angular momentum $\mathbf{H}_G$ of the assembly about point G .

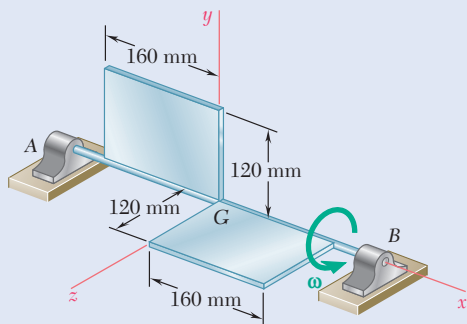


Fig. P18.20

18.21 One of the sculptures displayed on a university campus consists of a hollow cube made of six aluminum sheets, each $1.5 \times 1.5 \text{ m}$, welded together and reinforced with internal braces of negligible weight. The cube is mounted on a fixed base at A and can rotate freely about its vertical diagonal AB . As she passes by this display on the way to a class in mechanics, an engineering student grabs corner C of the cube and pushes it for 1.2 s in a direction perpendicular to the plane ABC with an average force of 50 N. Having observed that it takes 5 s for the cube to complete one full revolution, she flips out her calculator and proceeds to determine the mass of the cube. What is the result of her calculation? (Hint: The perpendicular distance from the diagonal joining two vertices of a cube to any of its other six vertices can be obtained by multiplying the side of the cube by $\sqrt{2/3}$.)

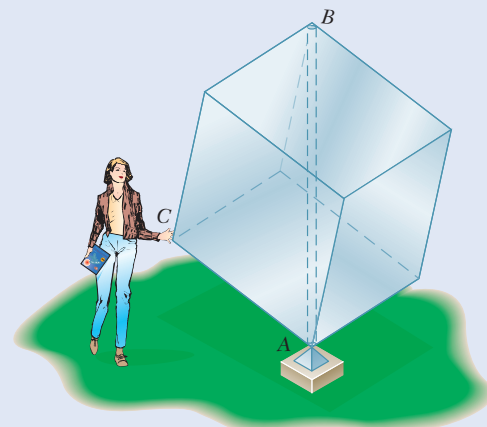


Fig. P18.21

18.22 If the aluminum cube of Prob. 18.21 were replaced by a cube of the same size, made of six plywood sheets with mass 8 kg each, how long would it take for that cube to complete one full revolution if the student pushed its corner C in the same way that she pushed the corner of the aluminum cube?

18.23 A uniform rod of total mass m is bent into the shape shown and is suspended by a wire attached at B . The bent rod is hit at D in a direction perpendicular to the plane containing the rod (in the negative z direction). Denoting the corresponding impulse by $\mathbf{F} \Delta t$, determine (a) the velocity of the mass center of the rod, (b) the angular velocity of the rod.

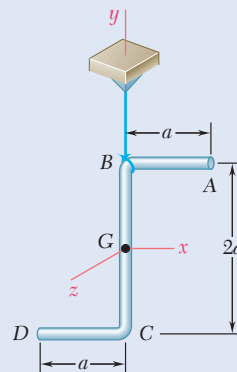


Fig. P18.23

18.24 Solve Prob. 18.23, assuming that the bent rod is hit at C .

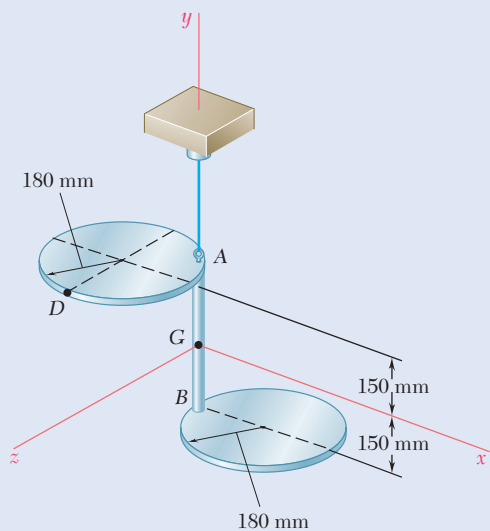


Fig. P18.27 and P18.28

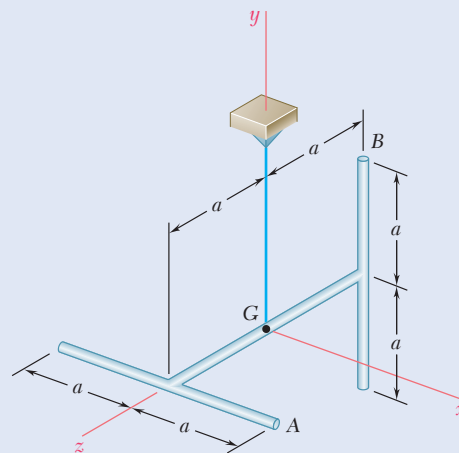


Fig. P18.25

- 18.25** Three slender rods, each of mass m and length $2a$, are welded together to form the assembly shown. The assembly is hit at A in a vertical downward direction. Denoting the corresponding impulse by $\mathbf{F} \Delta t$, determine immediately after the impact (a) the velocity of the mass center G , (b) the angular velocity of the rod.

- 18.26** Solve Prob. 18.25, assuming that the assembly is hit at B in the negative x direction.

- 18.27** Two circular plates, each of mass 4 kg, are rigidly connected by a rod AB of negligible mass and are suspended from point A as shown. Knowing that an impulse $\mathbf{F} \Delta t = -(2.4 \text{ N}\cdot\text{s})\mathbf{k}$ is applied at point D , determine (a) the velocity of the mass center G of the assembly, (b) the angular velocity of the assembly.

- 18.28** Two circular plates, each of mass 4 kg, are rigidly connected by a rod AB of negligible mass and are suspended from point A as shown. Knowing that an impulse $\mathbf{F} \Delta t = (2.4 \text{ N}\cdot\text{s})\mathbf{j}$ is applied at point D , determine (a) the velocity of the mass center G of the assembly, (b) the angular velocity of the assembly.

- 18.29** A circular plate of mass m is falling with a velocity $\bar{v}_0$ and no angular velocity when its edge C strikes an obstruction. A line passing through the origin and parallel to the line CG makes a 45° angle with the x -axis. Assuming the impact to be perfectly plastic ($e = 0$), determine the angular velocity of the plate immediately after the impact.

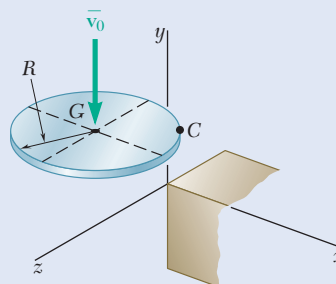


Fig. P18.29

- 18.30** For the plate of Prob. 18.29, determine (a) the velocity of its mass center G immediately after the impact, (b) the impulse exerted on the plate by the obstruction during the impact.

- 18.31** A square plate of side a and mass m supported by a ball-and-socket joint at A is rotating about the y axis with a constant angular velocity $\boldsymbol{\omega} = \omega_0 \mathbf{j}$ when an obstruction is suddenly introduced at B in the xy plane. Assuming the impact at B to be perfectly plastic ($e = 0$), determine immediately after the impact (a) the angular velocity of the plate, (b) the velocity of its mass center G .

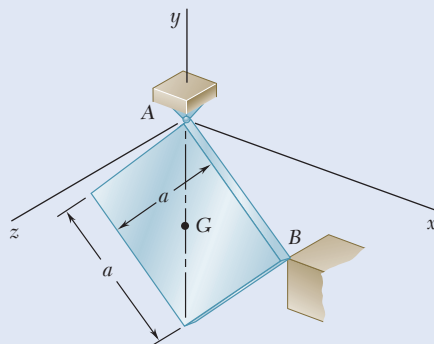


Fig. P18.31

- 18.32** Determine the impulse exerted on the plate of Prob. 18.31 during the impact by (a) the obstruction at B , (b) the support at A .
- 18.33** The coordinate axes shown represent the principal centroidal axes of inertia of a 3000-lb space probe whose radii of gyration are $k_x = 1.375$ ft, $k_y = 1.425$ ft, and $k_z = 1.250$ ft. The probe has no angular velocity when a 5-oz meteorite strikes one of its solar panels at point A with a velocity $\mathbf{v}_0 = (2400 \text{ ft/s})\mathbf{i} - (3000 \text{ ft/s})\mathbf{j} + (3200 \text{ ft/s})\mathbf{k}$ relative to the probe. Knowing that the meteorite emerges on the other side of the panel with no change in the direction of its velocity, but with a speed reduced by 20 percent, determine the final angular velocity of the probe.

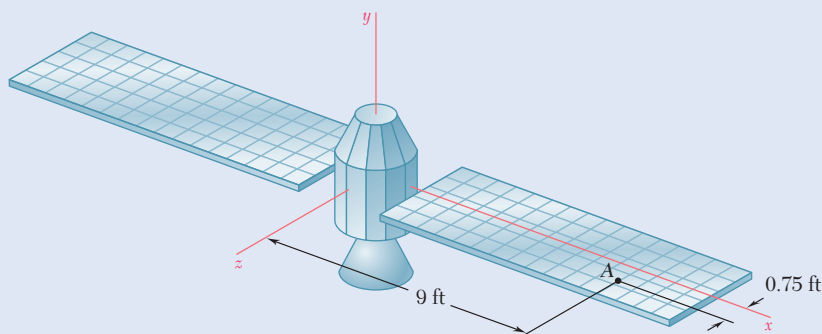


Fig. P18.33 and P18.34

- 18.34** The coordinate axes shown represent the principal centroidal axes of inertia of a 3000-lb space probe whose radii of gyration are $k_x = 1.375$ ft, $k_y = 1.425$ ft, and $k_z = 1.250$ ft. The probe has no angular velocity when a 5-oz meteorite strikes one of its solar panels at point A and emerges on the other side of the panel with no change in the direction of its velocity, but with a speed reduced by 25 percent. Knowing that the final angular velocity of the probe is $\boldsymbol{\omega} = (0.05 \text{ rad/s})\mathbf{i} - (0.12 \text{ rad/s})\mathbf{j} + \omega_z \mathbf{k}$ and that the x -component of the resulting change in the velocity of the mass center of the probe is -0.675 in./s, determine (a) the component ω_z of the final angular velocity of the probe, (b) the relative velocity $\mathbf{v}_0$ with which the meteorite strikes the panel.

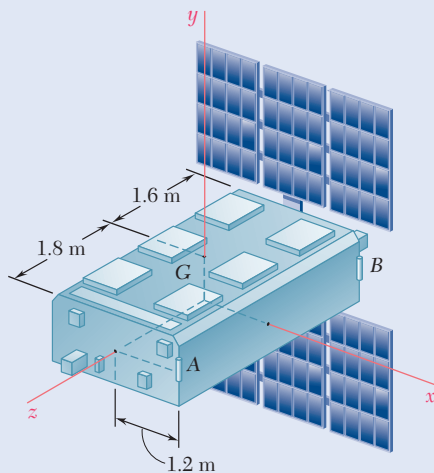


Fig. P18.35

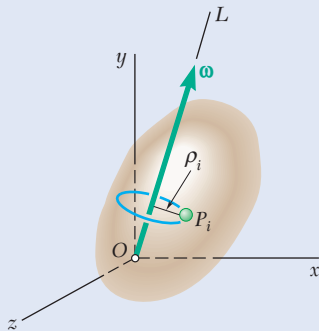


Fig. P18.38

18.35 A 1200-kg satellite designed to study the sun has an angular velocity of $\omega_0 = (0.050 \text{ rad/s})\mathbf{i} + (0.075 \text{ rad/s})\mathbf{k}$ when two small jets are activated at A and B in a direction parallel to the y axis. Knowing that the coordinate axes are principal centroidal axes, that the radii of gyration of the satellite are $\bar{k}_x = 1.120 \text{ m}$, $\bar{k}_y = 1.200 \text{ m}$, and $\bar{k}_z = 0.900 \text{ m}$, and that each jet produces a 50-N thrust, determine (a) the required operating time of each jet if the angular velocity of the satellite is to be reduced to zero, (b) the resulting change in the velocity of the mass center G .

18.36 If jet A in Prob. 18.35 is inoperative, determine (a) the required operating time of jet B to reduce the x -component of the angular velocity of the satellite to zero, (b) the resulting final angular velocity, (c) the resulting change in the velocity of the mass center G .

18.37 Denoting, respectively, by ω , $\mathbf{H}_O$, and T the angular velocity, the angular momentum, and the kinetic energy of a rigid body with a fixed point O , (a) prove that $\mathbf{H}_O \cdot \omega = 2T$; (b) show that the angle θ between ω and $\mathbf{H}_O$ will always be acute.

18.38 Show that the kinetic energy of a rigid body with a fixed point O can be expressed as $T = \frac{1}{2}I_{OL}\omega^2$, where ω is the instantaneous angular velocity of the body and I_{OL} is its moment of inertia about the line of action OL of ω . Derive this expression (a) from Eqs. (9.46) (or Eq. B.19 in the Appendix) and (18.19), (b) by considering T as the sum of the kinetic energies of particles P_i describing circles of radius ρ_i about line OL .

18.39 Determine the kinetic energy of the disk of Prob. 18.1.

18.40 Determine the kinetic energy of the plate of Prob. 18.2.

18.41 Determine the kinetic energy of the assembly of Prob. 18.3.

18.42 Determine the kinetic energy of the disk of Prob. 18.4.

18.43 Determine the kinetic energy of the disk of Prob. 18.5.

18.44 Determine the kinetic energy of the solid parallelepiped of Prob. 18.6.

18.45 Determine the kinetic energy of the hollow parallelepiped of Prob. 18.7.

18.46 Determine the kinetic energy of the disk of Prob. 18.8.

18.47 Determine the kinetic energy of the assembly of Prob. 18.15.

18.48 Determine the kinetic energy of the shaft of Prob. 18.17.

18.49 Determine the kinetic energy of the assembly of Prob. 18.19.

18.50 Determine the kinetic energy imparted to the cube of Prob. 18.21.

18.51 Determine the kinetic energy lost when edge C of the plate of Prob. 18.29 hits the obstruction.

18.52 Determine the kinetic energy lost when the plate of Prob. 18.31 hits the obstruction at B .

18.53 Determine the kinetic energy of the space probe of Prob. 18.33 in its motion about its mass center after its collision with the meteorite.

18.54 Determine the kinetic energy of the space probe of Prob. 18.34 in its motion about its mass center after its collision with the meteorite.

*18.2 MOTION OF A RIGID BODY IN THREE DIMENSIONS

As indicated in Sec. 18.1A, the fundamental equations

$$\Sigma \mathbf{F} = m\bar{\mathbf{a}} \quad (18.1)$$

$$\Sigma \mathbf{M}_G = \dot{\mathbf{H}}_G \quad (18.2)$$

remain valid in the most general case of the motion of a rigid body. Before we could apply Eq. (18.2) to the three-dimensional motion of a rigid body, however, it was necessary to derive Eqs. (18.7), which relate the components of the angular momentum $\mathbf{H}_G$ and those of the angular velocity $\boldsymbol{\omega}$. It still remains for us to find an effective and convenient way to compute the components of the derivative $\dot{\mathbf{H}}_G$ of the angular momentum. In this section, we do that first and then show how we can use the results to analyze motion of a rigid body in space.

18.2A Rate of Change of Angular Momentum

The notation $\mathbf{H}_G$ represents the angular momentum of a rigid body in its motion relative to centroidal axes $GX'Y'Z'$ with a fixed orientation (Fig. 18.9). Since $\dot{\mathbf{H}}_G$ represents the rate of change of $\mathbf{H}_G$ with respect to the same axes, it would seem natural to use components of $\boldsymbol{\omega}$ and $\mathbf{H}_G$ along the axes X', Y', Z' in writing the relations of Eq. (18.7). However, since the body rotates, its moments and products of inertia change continually, and it would be necessary to determine their values as functions of time. It is therefore more convenient to use axes x, y, z attached to the body, ensuring that its moments and products of inertia maintain the same values during the motion. The angular velocity $\boldsymbol{\omega}$, however, still should be *defined* with respect to the frame $GX'Y'Z'$ with a fixed orientation. We can then *resolve* the vector $\boldsymbol{\omega}$ into components along the rotating x, y , and z axes. Applying the relations of Eq. (18.7), we obtain the *components* of vector $\mathbf{H}_G$ along the rotating axes. Vector $\mathbf{H}_G$, however, represents the angular momentum about G of the body in its motion relative to the frame $GX'Y'Z'$.

Differentiating the components of the angular momentum in Eq. (18.7) with respect to t , we define the rate of change of vector $\mathbf{H}_G$ with respect to the rotating frame $Gxyz$ as

$$(\dot{\mathbf{H}}_G)_{Gxyz} = \dot{H}_x \mathbf{i} + \dot{H}_y \mathbf{j} + \dot{H}_z \mathbf{k} \quad (18.21)$$

where $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are the unit vectors along the rotating axes. Recall from Sec. 15.5A that the rate of change $\dot{\mathbf{H}}_G$ of vector $\mathbf{H}_G$ with respect to the frame $GX'Y'Z'$ is found by adding $(\dot{\mathbf{H}}_G)_{Gxyz}$ to the vector product $\boldsymbol{\Omega} \times \mathbf{H}_G$, where $\boldsymbol{\Omega}$ denotes the angular velocity of the rotating frame. That is,

$$\dot{\mathbf{H}}_G = (\dot{\mathbf{H}}_G)_{Gxyz} + \boldsymbol{\Omega} \times \mathbf{H}_G \quad (18.22)$$

where $\mathbf{H}_G$ = angular momentum of the body with respect to frame $GX'Y'Z'$ with a fixed orientation

$(\dot{\mathbf{H}}_G)_{Gxyz}$ = rate of change of $\mathbf{H}_G$ with respect to rotating frame $Gxyz$ to be computed from the relations in Eqs. (18.7) and (18.21)

$\boldsymbol{\Omega}$ = angular velocity of rotating frame $Gxyz$

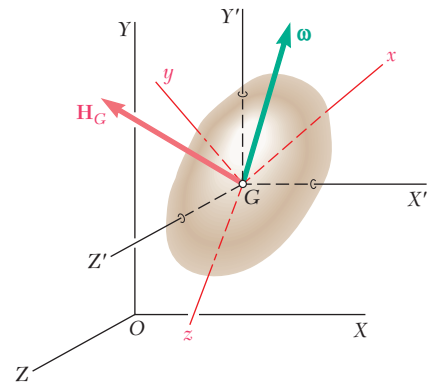


Fig. 18.9 Angular velocity and angular momentum of a rigid body with centroidal axes $X'Y'Z'$ of fixed orientation and centroidal axes xyz attached to the body.

Substituting for $\dot{\mathbf{H}}_G$ from Eq. (18.22) into Eq. (18.2), we have

$$\Sigma \mathbf{M}_G = (\dot{\mathbf{H}}_G)_{Gxyz} + \boldsymbol{\Omega} \times \mathbf{H}_G \quad (18.23)$$

If the rotating frame is attached to the body as we have assumed in this discussion, its angular velocity $\boldsymbol{\Omega}$ is identically equal to the angular velocity $\boldsymbol{\omega}$ of the body. In many applications, however, it is advantageous to use a frame of reference that is not actually attached to the body but rotates in an independent manner. For example, if the body considered is axisymmetric, as in Sample Prob. 18.5 or Sec. 18.3, it is possible to select a frame of reference where the moments and products of inertia of the body remain constant, but which rotate less than the body itself. As a result, it is possible to obtain simpler expressions for the angular velocity $\boldsymbol{\omega}$ and the angular momentum $\mathbf{H}_G$ of the body than we could have obtained if the frame of reference had actually been attached to the body. It is clear that in such cases the angular velocity $\boldsymbol{\Omega}$ of the rotating frame and the angular velocity $\boldsymbol{\omega}$ of the body are different.

*18.2B Euler's Equations of Motion

If we choose the x , y , and z axes to coincide with the principal axes of inertia of the body, we can use the simplified relations in Eq. (18.10) to determine the components of the angular momentum $\mathbf{H}_G$. Omitting the primes from the subscripts, we have

$$\mathbf{H}_G = \bar{I}_x \omega_x \mathbf{i} + \bar{I}_y \omega_y \mathbf{j} + \bar{I}_z \omega_z \mathbf{k} \quad (18.24)$$

where $\bar{I}_x$, $\bar{I}_y$, and $\bar{I}_z$ denote the principal centroidal moments of inertia of the body. Substituting for $\mathbf{H}_G$ from Eq. (18.24) into Eq. (18.23) and setting $\boldsymbol{\Omega} = \boldsymbol{\omega}$, we obtain the three scalar equations:

**Euler's equations
of motion**

$$\begin{aligned} \Sigma M_x &= \bar{I}_x \dot{\omega}_x - (\bar{I}_y - \bar{I}_z) \omega_y \omega_z \\ \Sigma M_y &= \bar{I}_y \dot{\omega}_y - (\bar{I}_z - \bar{I}_x) \omega_z \omega_x \\ \Sigma M_z &= \bar{I}_z \dot{\omega}_z - (\bar{I}_x - \bar{I}_y) \omega_x \omega_y \end{aligned} \quad (18.25)$$

We can use these equations, called **Euler's equations of motion** after the Swiss mathematician Leonhard Euler (1707–1783), to analyze the motion of a rigid body about its mass center. In the following sections, however, we will use Eq. (18.23) in preference to Eqs. (18.25), since Eq. (18.23) is more general, and the compact vectorial form in which it is expressed is easier to remember.

Writing Eq. (18.1) in scalar form, we obtain the three additional equations of

$$\Sigma F_x = m\bar{a}_x \quad \Sigma F_y = m\bar{a}_y \quad \Sigma F_z = m\bar{a}_z \quad (18.26)$$

Together with Euler's equations, these form a system of six differential equations. Given appropriate initial conditions, these differential equations have a unique solution. Thus, the motion of a rigid body in three dimensions is completely defined by the resultant and the moment resultant of the

external forces acting on it. This result is a generalization of a similar result obtained in Sec. 16.1C in the case of the plane motion of a rigid body. It follows that in three as well as in two dimensions, two systems of forces that are equipollent are also equivalent; that is, they have the same effect on a given rigid body.

Considering in particular the system of the external forces acting on a rigid body (Fig. 18.10a) and the system of the inertial terms associated with the particles forming the rigid body (Fig. 18.10b), we can state that the two systems—which were shown in Sec. 14.1A to be equipollent—are also equivalent. Replacing the inertia terms in Fig. 18.10b by $m\bar{\mathbf{a}}$ and $\dot{\mathbf{H}}_G$, we can verify that the system of the external forces acting on a rigid body in three-dimensional motion is equivalent to the system consisting of the vector $m\bar{\mathbf{a}}$ attached at the mass center G of the body and the couple of moment $\dot{\mathbf{H}}_G$ (Fig. 18.11), where we obtain $\dot{\mathbf{H}}_G$ from the relations in Eqs. (18.7) and (18.22). Note that the equivalence of the systems of vectors shown in Figs. 18.10 and 18.11 has been indicated by *red* equals signs. You can solve problems involving the three-dimensional motion of a rigid body by considering the free-body diagram and kinetic diagram represented in Fig. 18.11 and by writing appropriate scalar equations relating the components or moments of the external forces and the inertial terms (see Sample Prob. 18.3).

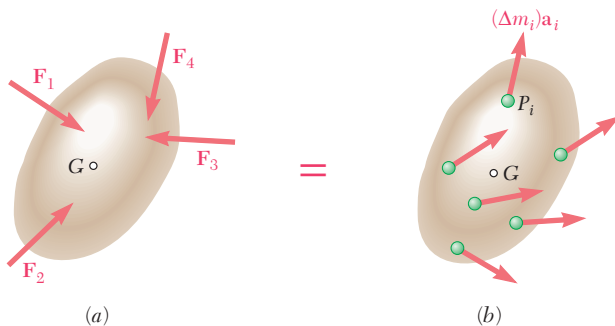


Fig. 18.10 (a) The system of external forces acting on a rigid body is equivalent to (b) the system of inertia terms associated with the particles of the rigid body.

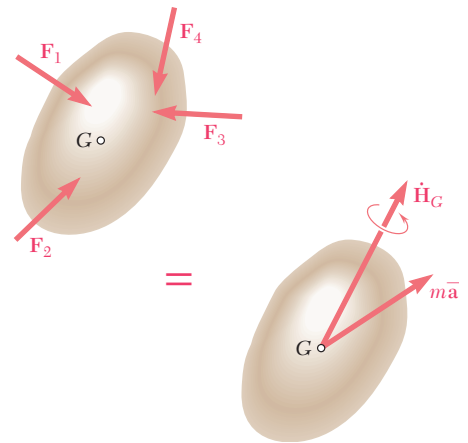


Fig. 18.11 The free-body diagram and kinetic diagram show that the system of external forces is equivalent to the system consisting of the vectors $m\bar{\mathbf{a}}$ attached at the mass center G and $\dot{\mathbf{H}}_G$.

*18.2C Motion of a Rigid Body About a Fixed Point

If we want to analyze the motion of a rigid body constrained to rotate about a fixed point O , it is useful to write an equation involving the moments about O of the external forces and of the inertial terms, since this equation contains the unknown reaction at O . Although we can obtain such an equation from Fig. 18.11, it may be more convenient to write it by considering the rate of change of the angular momentum $\mathbf{H}_O$ of the body about the fixed point O (Fig. 18.12). Recalling Eq. (14.11), we have

$$\Sigma \mathbf{M}_O = \dot{\mathbf{H}}_O \quad (18.27)$$

where $\dot{\mathbf{H}}_O$ denotes the rate of change of the vector $\mathbf{H}_O$ with respect to the fixed frame $OXYZ$. A derivation similar to that used in Sec. 18.2A enables

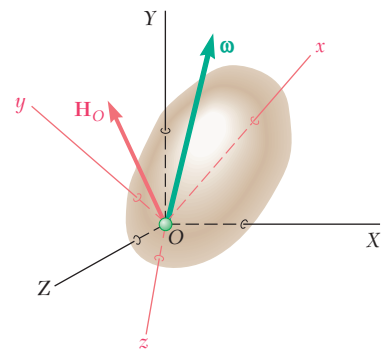


Fig. 18.12 Angular velocity and angular momentum of a rigid body rotating about a fixed point.



Photo 18.3 The revolving radio telescope is an example of a structure constrained to rotate about a fixed point.

us to relate $\dot{\mathbf{H}}_O$ to the rate of change $(\dot{\mathbf{H}}_O)_{Oxyz}$ of $\mathbf{H}_O$ with respect to the rotating frame $Oxyz$. Substitution into Eq. (18.27) leads to

$$\Sigma \mathbf{M}_O = (\dot{\mathbf{H}}_O)_{Oxyz} + \boldsymbol{\Omega} \times \mathbf{H}_O \quad (18.28)$$

where $\Sigma \mathbf{M}_O$ = sum of moments about O of forces applied to the rigid body

$\mathbf{H}_O$ = angular momentum of the body with respect to fixed frame $OXYZ$

$(\dot{\mathbf{H}}_O)_{Oxyz}$ = rate of change of $\mathbf{H}_O$ with respect to rotating frame $Oxyz$ to be computed from relations in Eq. (18.13)

$\boldsymbol{\Omega}$ = angular velocity of rotating frame $Oxyz$

If the rotating frame is attached to the body, its angular velocity $\boldsymbol{\Omega}$ is identically equal to the angular velocity $\boldsymbol{\omega}$ of the body. However, as indicated in the last paragraph of Sec. 18.2A, in many applications it is advantageous to use a frame of reference that is not actually attached to the body but rotates in an independent manner.

*18.2D Rotation of a Rigid Body About a Fixed Axis

We can use Eq. (18.28) to analyze the motion of a rigid body constrained to rotate about a fixed axis AB (Fig. 18.13). First, we note that the angular velocity of the body with respect to the fixed frame $OXYZ$ is represented by the vector $\boldsymbol{\omega}$ directed along the axis of rotation. Attaching the moving frame of reference $Oxyz$ to the body, with the z axis along AB , we have $\boldsymbol{\omega} = \omega \mathbf{k}$. Substituting $\omega_x = 0$, $\omega_y = 0$, and $\omega_z = \omega$ into the relations of Eq. (18.13), we obtain the components along the rotating axes of the angular momentum $\mathbf{H}_O$ of the body about O as

$$H_x = -I_{xz}\omega \quad H_y = -I_{yz}\omega \quad H_z = I_z\omega$$

Since the frame $Oxyz$ is attached to the body, we have $\boldsymbol{\Omega} = \boldsymbol{\omega}$, and Eq. (18.28) yields

$$\begin{aligned} \Sigma \mathbf{M}_O &= (\dot{\mathbf{H}}_O)_{Oxyz} + \boldsymbol{\omega} \times \mathbf{H}_O \\ &= (-I_{xz}\mathbf{i} - I_{yz}\mathbf{j} + I_z\mathbf{k})\dot{\omega} + \omega \mathbf{k} \times (-I_{xz}\mathbf{i} - I_{yz}\mathbf{j} + I_z\mathbf{k})\omega \\ &= (-I_{xz}\mathbf{i} - I_{yz}\mathbf{j} + I_z\mathbf{k})\alpha + (-I_{xz}\mathbf{j} + I_{yz}\mathbf{i})\omega^2 \end{aligned}$$

We can express this result by the three scalar equations

$$\begin{aligned} \Sigma M_x &= -I_{xz}\alpha + I_{yz}\omega^2 \\ \Sigma M_y &= -I_{yz}\alpha - I_{xz}\omega^2 \\ \Sigma M_z &= I_z\alpha \end{aligned} \quad (18.29)$$

When the forces and moments applied to the body are known, you can obtain the angular acceleration α from the last of Eqs. (18.29). You can then determine the angular velocity ω by integration and substitute the values obtained for α and ω into the first two of Eqs. (18.29). You can then use these equations plus the three equations (18.26) that define the motion of the mass center of the body to determine the reactions at the bearings A and B .

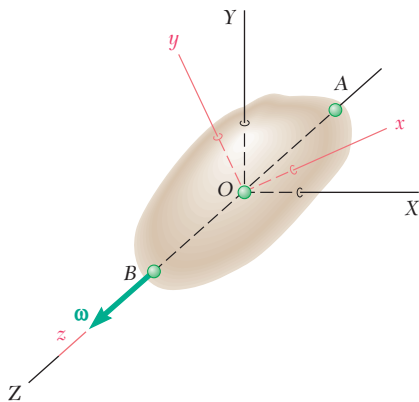


Fig. 18.13 Angular velocity of a rigid body rotating about a fixed axis AB .

It is possible to select axes other than those shown in Fig. 18.13 to analyze the rotation of a rigid body about a fixed axis. In many cases, the principal axes of inertia of the body will be more advantageous. It is therefore a good idea to revert to Eq. (18.28) and select the system of axes that best fits the problem under consideration.

If the rotating body is symmetrical with respect to the xy plane, the products of inertia I_{xz} and I_{yz} are equal to zero. Then Eqs. (18.29) reduce to

$$\Sigma M_x = 0 \quad \Sigma M_y = 0 \quad \Sigma M_z = I_z \alpha \quad (18.30)$$

which is in agreement with the results obtained in Chap. 16. If, on the other hand, the products of inertia I_{xz} and I_{yz} are different from zero, the sum of the moments of the external forces about the x and y axes are also different from zero, even when the body rotates at a constant rate ω . Indeed, in this case, Eqs. (18.29) yield

$$\Sigma M_x = I_{yz} \omega^2 \quad \Sigma M_y = -I_{xz} \omega^2 \quad \Sigma M_z = 0 \quad (18.31)$$

This last observation leads us to discuss the **balancing of rotating shafts**. Consider, for instance, the crankshaft shown in Fig. 18.14a that is symmetrical about its mass center G . We first observe that, when the crankshaft is at rest, it exerts no lateral thrust on its supports, since its center of gravity G is located directly above A . The shaft is said to be *statically balanced*. The reaction at A is often referred to as a *static reaction* and is vertical, and its magnitude is equal to the weight W of the shaft. Let us now assume that the shaft rotates with a constant angular velocity ω . Attaching our frame of reference to the shaft with its origin at G , the z axis along AB , and the y axis in the plane of symmetry of the shaft (Fig. 18.14b), we note that I_{xz} is zero and that I_{yz} is positive. According to Eqs. (18.31), there is an inertial term $I_{yz} \omega^2 \mathbf{i}$. Summing the moments about G in the x direction and applying Eq. (18.31), we have

$$\mathbf{A}_y = \frac{I_{yz} \omega^2}{l} \mathbf{j} \quad \mathbf{B} = -\frac{I_{yz} \omega^2}{l} \mathbf{j} \quad (18.32)$$

Since the bearing reactions are proportional to ω^2 , the shaft has a tendency to tear away from its bearings when rotating at high speeds. Moreover, since the bearing reactions $\mathbf{A}_y$ and $\mathbf{B}$, which are called *dynamic reactions*, are contained in the yz plane, they rotate with the shaft and cause the structure supporting it to vibrate. These undesirable effects can be avoided if, by rearranging the distribution of mass around the shaft or by adding corrective masses, we let I_{yz} become equal to zero. Then the dynamic reactions $\mathbf{A}_y$ and $\mathbf{B}$ vanish and the reactions at the bearings reduce to the static reaction $\mathbf{A}_z$ —the direction of which is fixed. The shaft is then **dynamically as well as statically balanced**.



Photo 18.4 The rotating automobile crankshaft causes static and dynamic reactions on its bearings. The crankshaft can be designed to minimize dynamic imbalances and reduce these reaction forces.

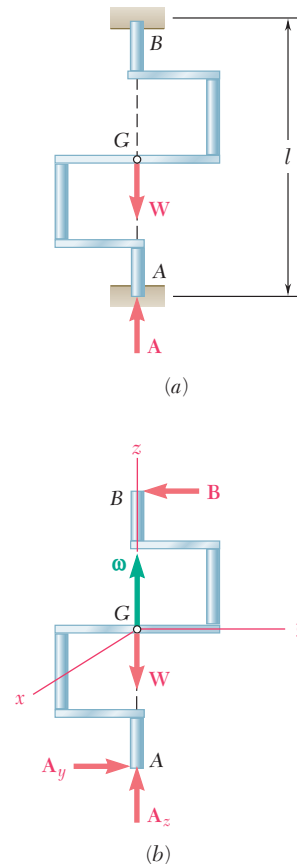


Fig. 18.14 (a) The crankshaft at rest is statically balanced; (b) the crankshaft rotating with constant angular velocity may or may not be dynamically balanced.

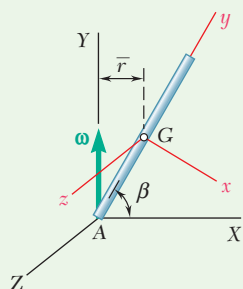
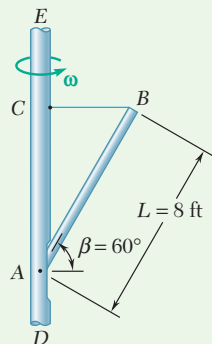


Fig. 1 Angular velocity of the rod.

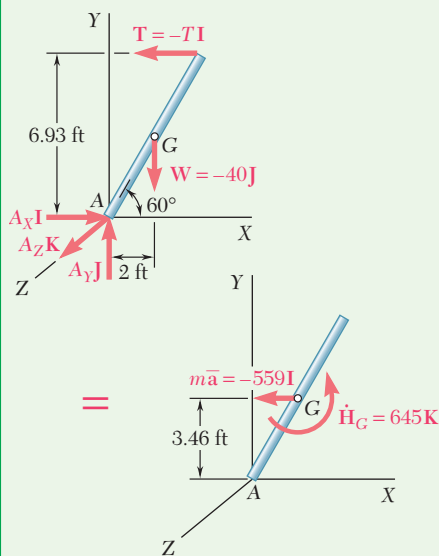


Fig. 2 Free-body diagram and kinetic diagram of the rod.

Sample Problem 18.3

A slender rod AB with a length of $L = 8$ ft and a weight of $W = 40$ lb is pinned at A to a vertical axle DE that rotates with a constant angular velocity ω of 15 rad/s. The rod is maintained in position by means of a horizontal wire BC attached to the axle and to end B of the rod. Determine the tension in the wire and the reaction at A .

STRATEGY: Since you have a rigid body that is not symmetrical with respect to the plane of motion, you need to use the three-dimensional form of Newton's second law.

MODELING: Choose the rod AB as your system. The angular velocity is shown in Fig. 1, and the free-body and kinetic diagrams consisting of the vector $m\bar{\mathbf{a}}$ attached at G and the couple $\dot{\mathbf{H}}_G$ are shown in Fig. 2.

ANALYSIS: Since G describes a horizontal circle with a radius of $\bar{r} = \frac{1}{2}L \cos \beta$ and BG rotates at the constant rate ω (Fig. 1), you have

$$\begin{aligned}\bar{\mathbf{a}} &= \mathbf{a}_n = -\bar{r}\omega^2 \mathbf{I} = -\left(\frac{1}{2}L \cos \beta\right)\omega^2 \mathbf{I} = -(450 \text{ ft/s}^2) \mathbf{I} \\ m\bar{\mathbf{a}} &= \frac{40}{g}(-450 \mathbf{I}) = -(559 \text{ lb}) \mathbf{I}\end{aligned}$$

Determination of $\dot{\mathbf{H}}_G$. First compute the angular momentum $\mathbf{H}_G$. Using the principal centroidal axes of inertia x , y , and z , you have

$$\begin{aligned}\bar{I}_x &= \frac{1}{12}mL^2 & \bar{I}_y &= 0 & \bar{I}_z &= \frac{1}{12}mL^2 \\ \omega_x &= -\omega \cos \beta & \omega_y &= \omega \sin \beta & \omega_z &= 0 \\ \mathbf{H}_G &= \bar{I}_x \omega_x \mathbf{i} + \bar{I}_y \omega_y \mathbf{j} + \bar{I}_z \omega_z \mathbf{k} \\ \mathbf{H}_G &= -\frac{1}{12}mL^2 \omega \cos \beta \mathbf{i}\end{aligned}$$

Obtain the rate of change $\dot{\mathbf{H}}_G$ of $\mathbf{H}_G$ with respect to axes of fixed orientation from Eq. (18.22). Observe that the rate of change $(\dot{\mathbf{H}}_G)_{Gxyz}$ of $\mathbf{H}_G$ with respect to the rotating frame $Gxyz$ is zero and that the angular velocity $\boldsymbol{\Omega}$ of that frame is equal to the angular velocity $\boldsymbol{\omega}$ of the rod. Thus, you have

$$\begin{aligned}\dot{\mathbf{H}}_G &= (\dot{\mathbf{H}}_G)_{Gxyz} + \boldsymbol{\omega} \times \mathbf{H}_G \\ \dot{\mathbf{H}}_G &= 0 + (-\omega \cos \beta \mathbf{i} + \omega \sin \beta \mathbf{j}) \times \left(-\frac{1}{12}mL^2 \omega \cos \beta \mathbf{i}\right) \\ \dot{\mathbf{H}}_G &= \frac{1}{12}mL^2 \omega^2 \sin \beta \cos \beta \mathbf{k} = (645 \text{ lb}\cdot\text{ft}) \mathbf{k}\end{aligned}$$

Equations of Motion. The system of the external forces is equivalent to the inertia terms (Fig. 2). This gives

$$\begin{aligned}\Sigma \mathbf{M}_A &= \dot{\mathbf{H}}_A = \mathbf{r} \times m\bar{\mathbf{a}} + \dot{\mathbf{H}}_G: \\ 6.93 \mathbf{j} \times (-T \mathbf{i}) + 2 \mathbf{i} \times (-40 \mathbf{j}) &= 3.46 \mathbf{j} \times (-559 \mathbf{i}) + 645 \mathbf{k} \\ (6.93T - 80) \mathbf{k} &= (1934 + 645) \mathbf{k} & T &= 384 \text{ lb} \\ \Sigma \mathbf{F} &= m\bar{\mathbf{a}}: & A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k} - 384 \mathbf{i} - 40 \mathbf{j} &= -559 \mathbf{i} \\ & & \mathbf{A} &= -(175 \text{ lb}) \mathbf{i} + (40 \text{ lb}) \mathbf{j}\end{aligned}$$

REFLECT and THINK: You could have obtained the value of T from $\mathbf{H}_A$ and Eq. (18.28). Even though the rod rotates with a constant angular velocity, the asymmetry of the rod causes a moment about the z axis. Note that we calculated the inertial term $\dot{\mathbf{H}}_A$ by adding $\mathbf{r} \times m\bar{\mathbf{a}}$ and the couple $\dot{\mathbf{H}}_G$.

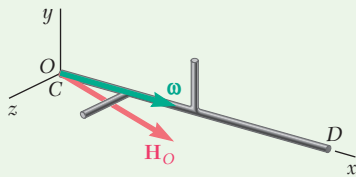
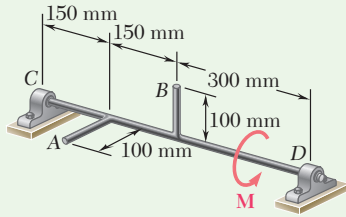


Fig. 1 Angular momentum and angular velocity of the system.

Sample Problem 18.4

Two 100-mm rods *A* and *B* each have a mass of 300 g and are welded to shaft *CD* that is supported by bearings at *C* and *D*. If a couple **M** with a magnitude of 6 N·m is applied to the shaft, determine the components of the dynamic reactions at *C* and *D* at the instant when the shaft has reached an angular velocity of 1200 rpm. Neglect the moment of inertia of the shaft itself.

STRATEGY: Use the three-dimensional form of Newton's second law in the form of Eq. (18.28) for the case of rotation about a fixed axis, where $\mathbf{\Omega} = \boldsymbol{\omega}$.

MODELING: Choose the shaft and the two rods as your system. The angular momentum and angular velocity are shown in Fig. 1, and a free-body diagram is shown in Fig. 2.

ANALYSIS:

Angular Momentum About O. Attach the frame of reference *Oxyz* to the body and note that the axes chosen are not principal axes of inertia for the body. Since the body rotates about the *x* axis, you know that $\omega_x = \omega$ and $\omega_y = \omega_z = 0$ (Fig. 1). Substituting into Eqs. (18.13), you have

$$\begin{aligned} H_x &= I_x \omega & H_y &= -I_{xy} \omega & H_z &= -I_{xz} \omega \\ \mathbf{H}_O &= (I_x \mathbf{i} - I_{xy} \mathbf{j} - I_{xz} \mathbf{k}) \omega \end{aligned}$$

Moments of the External Forces About O. Since the frame of reference rotates with the angular velocity $\boldsymbol{\omega}$ and the only angular acceleration term is $\alpha_x = \alpha$, Eq. (18.28) gives

$$\begin{aligned} \Sigma \mathbf{M}_O &= (\dot{\mathbf{H}}_O)_{Oxyz} + \boldsymbol{\omega} \times \mathbf{H}_O \\ &= (I_x \mathbf{i} - I_{xy} \mathbf{j} - I_{xz} \mathbf{k}) \alpha + \boldsymbol{\omega} \times (I_x \mathbf{i} - I_{xy} \mathbf{j} - I_{xz} \mathbf{k}) \omega \\ &= I_x \alpha \mathbf{i} - (I_{xy} \alpha - I_{xz} \omega^2) \mathbf{j} - (I_{xz} \alpha + I_{xy} \omega^2) \mathbf{k} \end{aligned} \quad (1)$$

Dynamic Reaction at D. The external forces consist of the weights of the shaft and rods, the couple **M**, the static reactions at *C* and *D*, and the dynamic reactions at *C* and *D*. Since the weights and static reactions are balanced, the external forces reduce to the couple **M** and the dynamic reactions **C** and **D**, as shown in Fig. 2. Taking moments about *O*, you have

$$\Sigma \mathbf{M}_O = L \mathbf{i} \times (D_y \mathbf{j} + D_z \mathbf{k}) + M \mathbf{i} = M \mathbf{i} - D_z L \mathbf{j} + D_y L \mathbf{k} \quad (2)$$

Equating the coefficients of the unit vector **i** in Eqs. (1) and (2) gives

$$M = I_x \alpha \quad M = 2\left(\frac{1}{3} m c^2\right) \alpha \quad \alpha = 3M/2mc^2$$

Equating the coefficients of **k** and **j** in Eqs. (1) and (2) provides

$$D_y = -(I_{xz} \alpha + I_{xy} \omega^2)/L \quad D_z = (I_{xy} \alpha - I_{xz} \omega^2)/L \quad (3)$$

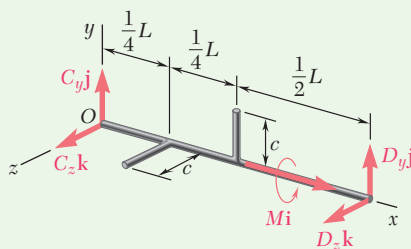


Fig. 2 Free-body diagram for the system.

Using the parallel-axis theorem and noting that the product of inertia of each rod is zero with respect to their own centroidal axes, you obtain

$$I_{xy} = \sum m \bar{x} \bar{y} = m(\tfrac{1}{2}L)(\tfrac{1}{2}c) = \tfrac{1}{4}mLc$$

$$I_{xz} = \sum m \bar{x} \bar{z} = m(\tfrac{1}{4}L)(\tfrac{1}{2}c) = \tfrac{1}{8}mLc$$

Substituting into Eq. (3) the values found for I_{xy} , I_{xz} , and α gives

$$D_y = -\tfrac{3}{16}(M/c) - \tfrac{1}{4}mc\omega^2 \quad D_z = \tfrac{3}{8}(M/c) - \tfrac{1}{8}mc\omega^2$$

Substituting $\omega = 1200 \text{ rpm} = 125.7 \text{ rad/s}$, $c = 0.100 \text{ m}$, $M = 6 \text{ N}\cdot\text{m}$, and $m = 0.300 \text{ kg}$, you have

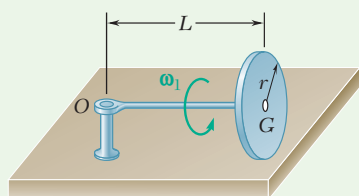
$$D_y = -129.8 \text{ N} \quad D_z = -36.8 \text{ N} \quad \blacktriangleleft$$

Dynamic Reaction at C. Using a frame of reference attached at D , you obtain equations similar to Eqs. (3) that yield

$$C_y = -152.2 \text{ N} \quad C_z = -155.2 \text{ N} \quad \blacktriangle$$

REFLECT and THINK: The dynamic forces are larger at C than at D . Rod A is closer to this end of the bar, so you would expect it to affect this end more than the other end. Note that two small 300-g rods end up causing forces of over 150 N. You often have to account for these large forces when designing mechanical systems involving rotary equipment (e.g., automobiles, turbines, mills).

Sample Problem 18.5



A homogeneous disk with radius r and mass m is mounted on an axle OG with length L and a negligible mass. The axle is pivoted at the fixed point O , and the disk is constrained to roll on a horizontal surface. The disk rotates counterclockwise at the constant rate ω_1 about the axle. Determine (a) the force (assumed vertical) exerted by the floor on the disk, (b) the reaction at the pivot O .

STRATEGY: Use the three-dimensional form of Newton's second law; that is, Eqs (18.1) and (18.2).

MODELING: Choose the disk as your system and model it as a rigid body. The angular momentum and angular velocity are shown in Fig. 1, and free-body and kinetic diagrams consisting of the vector $m\bar{\mathbf{a}}$ attached at G and the couple $\mathbf{H}_G$ are shown in Fig. 2.

ANALYSIS: From Sample Prob. 18.2, the axle rotates about the y axis at the rate $\omega_2 = r\omega_1/L$, so you have

$$m\bar{\mathbf{a}} = -mL\omega_2^2\mathbf{i} = -mL(r\omega_1/L)^2\mathbf{i} = -(mr^2\omega_1^2/L)\mathbf{i} \quad (1)$$

(continued)

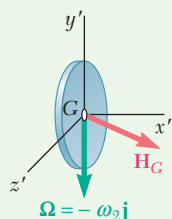


Fig. 1 The angular momentum and angular velocity of the disk.

Determination of $\dot{\mathbf{H}}_G$. Allow the x, y, z axes to rotate with the bar OG but not with the disk; the x', y', z' axes rotate with both the bar and the disk. Recall from Sample Prob. 18.2 that the angular momentum of the disk about G is

$$\mathbf{H}_G = \frac{1}{2}mr^2\omega_1\left(\mathbf{i} - \frac{r}{2L}\mathbf{j}\right)$$

where $\mathbf{H}_G$ is resolved into components along the rotating axes x', y', z' ; x' is along OG ; and y' is vertical at the instant shown (Fig. 1). Obtain the rate of change $\dot{\mathbf{H}}_G$ of $\mathbf{H}_G$ with respect to axes of fixed orientation from Eq. (18.22). Note that the rate of change $(\dot{\mathbf{H}}_G)_{Gx'y'z'}$ of $\mathbf{H}_G$ with respect to the rotating frame is zero and that the angular velocity $\boldsymbol{\Omega}$ of that frame is

$$\boldsymbol{\Omega} = -\omega_2\mathbf{j} = -\frac{r\omega_1}{L}\mathbf{j}$$

Then you have

$$\begin{aligned}\dot{\mathbf{H}}_G &= (\dot{\mathbf{H}}_G)_{Gx'y'z'} + \boldsymbol{\Omega} \times \mathbf{H}_G \\ &= 0 - \frac{r\omega_1}{L}\mathbf{j} \times \frac{1}{2}mr^2\omega_1\left(\mathbf{i} - \frac{r}{2L}\mathbf{j}\right) \\ &= \frac{1}{2}mr^2(r/L)\omega_1^2\mathbf{k}\end{aligned}\quad (2)$$

Equations of Motion. The system of the external forces is equivalent to the system of the inertial terms (Fig. 2), so you have

$$\begin{aligned}\Sigma \mathbf{M}_O &= \dot{\mathbf{H}}_G: & Li \times (N\mathbf{j} - W\mathbf{j}) &= \dot{\mathbf{H}}_G \\ & & (N - W)L\mathbf{k} &= \frac{1}{2}mr^2(r/L)\omega_1^2\mathbf{k} \\ N &= W + \frac{1}{2}mr(r/L)^2\omega_1^2 & \mathbf{N} &= [W + \frac{1}{2}mr(r/L)^2\omega_1^2]\mathbf{j} \quad (3) \quad \blacktriangleleft\end{aligned}$$

$$\Sigma \mathbf{F} = m\bar{\mathbf{a}}: \quad \mathbf{R} + N\mathbf{j} - W\mathbf{j} = m\bar{\mathbf{a}}$$

Substituting for N from Eq. (3), for $m\bar{\mathbf{a}}$ from Eq. (1), and solving for $\mathbf{R}$, you have

$$\begin{aligned}\mathbf{R} &= -(mr^2\omega_1^2/L)\mathbf{i} - \frac{1}{2}mr(r/L)^2\omega_1^2\mathbf{j} \\ \mathbf{R} &= -\frac{mr^2\omega_1^2}{L}\left(\mathbf{i} + \frac{r}{2L}\mathbf{j}\right) \quad \blacktriangleleft\end{aligned}$$

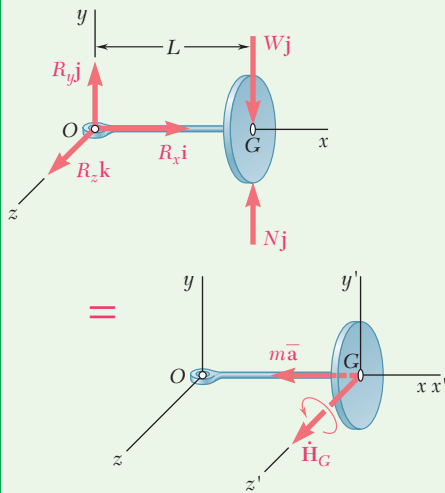


Fig. 2 Free-body diagram and kinetic diagram for the system.

REFLECT and THINK: This is a case where the coordinate system attached to the rotating object has its own angular velocity. The change in direction of the angular momentum of the disk ends up increasing the normal force.

SOLVING PROBLEMS ON YOUR OWN

In this section, you were asked to solve problems involving the three-dimensional motion of rigid bodies. The method you used is basically the same one you used in Chap. 16 in your study of the plane motion of rigid bodies. You made free-body and kinetic diagrams showing that the system of the external forces is equivalent to the system of the inertia terms. You equated sums of components and sums of moments on both sides of this equation. Now, however, the system of the inertia terms is represented by the vector $m\mathbf{\bar{a}}$ and a couple vector $\dot{\mathbf{H}}_G$, which are explained next in paragraphs 1 and 2.

To solve a problem involving the three-dimensional motion of a rigid body, you should take the following steps.

1. Determine the angular momentum $\mathbf{H}_G$ of the body about its mass center G from its angular velocity $\boldsymbol{\omega}$ with respect to a frame of reference $GX'Y'Z'$ of fixed orientation. This is an operation you learned to perform in Sect. 18.1. However, since the configuration of the body is changing with time, it is now necessary for you to use an auxiliary system of axes $Gx'y'z'$ (Fig. 18.9) to compute the components of $\boldsymbol{\omega}$ and the moments and products of inertia of the body. These axes may be rigidly attached to the body, in which case their angular velocity is equal to $\boldsymbol{\omega}$ [Sample Probs. 18.3 and 18.4], or they may have an angular velocity $\boldsymbol{\Omega}$ of their own [Sample Prob. 18.5].

Recall the following ideas from the preceding section.

a. If you know the principal axes of inertia of the body at G , use these axes as coordinate axes x' , y' , and z' , since the corresponding products of inertia of the body are equal to zero. (Note that if the body is axisymmetric, these axes do not need to be rigidly attached to the body.) Resolve $\boldsymbol{\omega}$ into components $\omega_{x'}$, $\omega_{y'}$, and $\omega_{z'}$ along these axes and compute the principal moments of inertia $\bar{I}_{x'}$, $\bar{I}_{y'}$, and $\bar{I}_{z'}$. The corresponding components of the angular momentum $\mathbf{H}_G$ are

$$H_{x'} = \bar{I}_{x'}\omega_{x'} \quad H_{y'} = \bar{I}_{y'}\omega_{y'} \quad H_{z'} = \bar{I}_{z'}\omega_{z'} \quad (18.10)$$

b. If you do not know the principal axes of inertia of the body at G , you must use Eqs. (18.7) to determine the components of the angular momentum $\mathbf{H}_G$. These equations require your prior computation of the products of inertia of the body—as well as of its moments of inertia—with respect to the selected axes.

2. Compute the rate of change $\dot{\mathbf{H}}_G$ of the angular momentum $\mathbf{H}_G$ with respect to the frame $Gx'y'z'$. Note that this frame has a *fixed orientation*, whereas the frame $Gx'y'z'$ you used when calculating the components of the vector $\boldsymbol{\omega}$ was a *rotating frame*. (Review the discussion in Sec. 15.5A of the rate of change of a vector with respect to a rotating frame.) Recalling Eq. (15.31), you can express the rate of change $\dot{\mathbf{H}}_G$ as

$$\dot{\mathbf{H}}_G = (\dot{\mathbf{H}}_G)_{Gx'y'z'} + \boldsymbol{\Omega} \times \mathbf{H}_G \quad (18.22)$$

The first term in the right-hand side of Eq. (18.22) represents the rate of change of $\mathbf{H}_G$ with respect to the rotating frame $Gx'y'z'$. This term drops out if $\boldsymbol{\omega}$ —and thus $\mathbf{H}_G$ —remains constant in both magnitude and direction when viewed from that frame. On the other hand, if any of the time derivatives $\dot{\omega}_{x'}$, $\dot{\omega}_{y'}$, or $\dot{\omega}_{z'}$ is different from zero, $(\dot{\mathbf{H}}_G)_{Gx'y'z'}$ is also different from zero, and you should determine its components by differentiating Eqs. (18.10) with respect to t . Finally, we remind you that if the rotating frame is rigidly attached to the body, its angular velocity is the same as that of the body, and $\boldsymbol{\Omega}$ can be replaced by $\boldsymbol{\omega}$.

3. Draw the free-body and kinetic diagrams for the rigid body showing that the system of the external forces exerted on the body is equivalent to the vector $m\bar{\mathbf{a}}$ applied at G and the couple vector $\dot{\mathbf{H}}_G$ (Fig. 18.11). By equating components in any direction and moments about any point, you can write as many as six independent scalar equations of motion [Sample Probs. 18.3 and 18.5].

4. When solving problems involving the motion of a rigid body about a fixed point O , you may find it convenient to use the following equation that was derived in Sec. 18.2C and eliminates the components of the reaction at the support O . So

$$\Sigma \mathbf{M}_O = (\dot{\mathbf{H}}_O)_{Oxyz} + \boldsymbol{\Omega} \times \mathbf{H}_O \quad (18.28)$$

Here the first term on the right-hand side represents the rate of change of $\mathbf{H}_O$ with respect to the rotating frame $Oxyz$, and $\boldsymbol{\Omega}$ is the angular velocity of that frame.

5. When determining the reactions at the bearings of a rotating shaft, use Eq. (18.28) and take the following steps.

a. Place the fixed point O at one of the two bearings supporting the shaft and attach the rotating frame $Oxyz$ to the shaft with one of the axes directed along it. Assuming, for instance, that the x axis has been aligned with the shaft, you will have $\boldsymbol{\Omega} = \boldsymbol{\omega} = \omega \mathbf{i}$ [Sample Prob. 18.4].

b. Since the selected axes are usually not the principal axes of inertia at O , you must compute the products of inertia of the shaft—as well as its moments of inertia—with respect to these axes and use Eqs. (18.13) to determine $\mathbf{H}_O$. Assuming again that the x axis has been aligned with the shaft, Eqs. (18.13) reduce to

$$H_x = I_x \omega \quad H_y = -I_{yx} \omega \quad H_z = -I_{zx} \omega \quad (18.13')$$

These equations show that $\mathbf{H}_O$ is not directed along the shaft.

(continued)

c. To obtain $\dot{\mathbf{H}}_O$, substitute these expressions into Eq. (18.28), and let $\mathbf{\Omega} = \mathbf{\omega} = \omega \mathbf{i}$. If the angular velocity of the shaft is constant, the first term in the right-hand side of the equation drops out. However, if the shaft has an angular acceleration $\mathbf{\alpha} = \alpha \mathbf{i}$, the first term is not zero and must be determined by differentiating the expressions in Eq. (18.13') with respect to t . The result will be equations similar to Eqs. (18.13') with ω replaced by α . The result also can be expressed by the three scalar equations of Eq. (18.29).

d. Since point O coincides with one of the bearings, you can solve the three scalar equations corresponding to Eq. (18.28) for the components of the dynamic reaction at the other bearing. If the mass center G of the shaft is located on the line joining the two bearings, the inertial term $m\bar{\mathbf{a}}$ is zero. Drawing the free-body diagram and kinetic diagram of the shaft, you then observe that the components of the dynamic reaction at the first bearing must be equal and opposite to those you have just determined. If G is not located on the line joining the two bearings, you can determine the reaction at the first bearing by placing the fixed point O at the second bearing and repeating the earlier procedure [Sample Prob. 18.4]; or you can obtain additional equations of motion from the free-body and kinetic diagrams of the shaft, making sure to first determine and include the inertial term $m\bar{\mathbf{a}}$ applied at G .

e. Most problems call for the determination of the “dynamic reactions” at the bearings, that is, for the additional forces exerted by the bearings on the shaft when the shaft is rotating. When determining dynamic reactions, ignore the effect of static loads, such as the weight of the shaft.

Problems

- 18.55** Determine the rate of change $\dot{\mathbf{H}}_G$ of the angular momentum $\mathbf{H}_G$ of the disk of Prob. 18.1.
- 18.56** Determine the rate of change $\dot{\mathbf{H}}_G$ of the angular momentum $\mathbf{H}_G$ of the plate of Prob. 18.2.
- 18.57** Determine the rate of change $\dot{\mathbf{H}}_D$ of the angular momentum $\mathbf{H}_D$ of the assembly of Prob. 18.3.
- 18.58** Determine the rate of change $\dot{\mathbf{H}}_A$ of the angular momentum $\mathbf{H}_A$ of the disk of Prob. 18.4.
- 18.59** Determine the rate of change $\dot{\mathbf{H}}_C$ of the angular momentum $\mathbf{H}_C$ of the disk of Prob. 18.5.
- 18.60** Determine the rate of change $\dot{\mathbf{H}}_G$ of the angular momentum $\mathbf{H}_G$ of the disk of Prob. 18.8 for an arbitrary value of β , knowing that its angular velocity $\boldsymbol{\omega}$ remains constant.
- 18.61** Determine the rate of change $\dot{\mathbf{H}}_D$ of the angular momentum $\mathbf{H}_D$ of the assembly of Prob. 18.3, assuming that at the instant considered the assembly has an angular velocity $\boldsymbol{\omega} = (12 \text{ rad/s})\mathbf{i}$ and an angular acceleration $\boldsymbol{\alpha} = -(96 \text{ rad/s}^2)\mathbf{i}$.
- 18.62** Determine the rate of change $\dot{\mathbf{H}}_D$ of the angular momentum $\mathbf{H}_D$ of the assembly of Prob. 18.3, assuming that at the instant considered the assembly has an angular velocity $\boldsymbol{\omega} = (12 \text{ rad/s})\mathbf{i}$ and an angular acceleration $\boldsymbol{\alpha} = (96 \text{ rad/s}^2)\mathbf{i}$.
- 18.63** A thin, homogeneous square of mass m and side a is welded to a vertical shaft AB with which it forms an angle of 45° . Knowing that the shaft rotates with an angular velocity $\boldsymbol{\omega} = \omega\mathbf{j}$ and an angular acceleration $\boldsymbol{\alpha} = \alpha\mathbf{j}$, determine the rate of change $\dot{\mathbf{H}}_A$ of the angular momentum $\mathbf{H}_A$ of the plate assembly.
- 18.64** Determine the rate of change $\dot{\mathbf{H}}_G$ of the angular momentum $\mathbf{H}_G$ of the disk of Prob. 18.8 for an arbitrary value of β , knowing that the disk has an angular velocity $\boldsymbol{\omega} = \omega\mathbf{i}$ and an angular acceleration $\boldsymbol{\alpha} = \alpha\mathbf{i}$.
- 18.65** A slender, uniform rod AB of mass m and a vertical shaft CD , each of length $2b$, are welded together at their midpoints G . Knowing that the shaft rotates at the constant rate ω , determine the dynamic reactions at C and D .

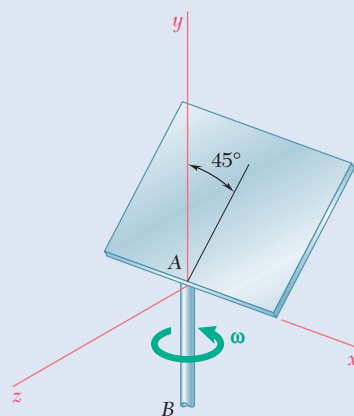


Fig. P18.63

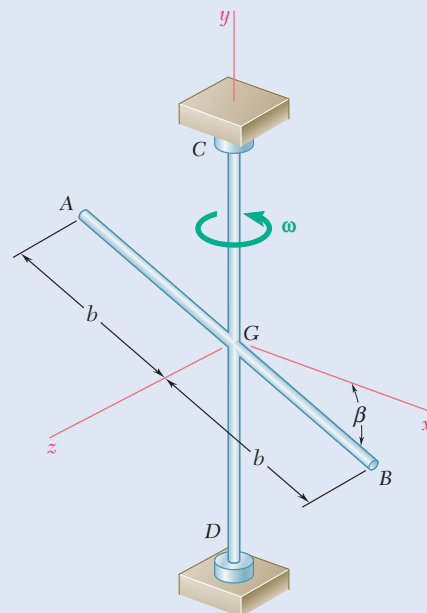


Fig. P18.65

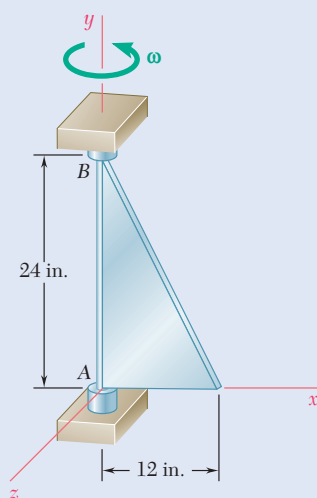


Fig. P18.66

18.66 A thin, homogeneous triangular plate of weight 10 lb is welded to a light, vertical axle supported by bearings at A and B. Knowing that the plate rotates at the constant rate $\omega = 8$ rad/s, determine the dynamic reactions at A and B.

18.67 The assembly shown consists of pieces of sheet aluminum of uniform thickness and of total weight 2.7 lb welded to a light axle supported by bearings at A and B. Knowing that the assembly rotates at the constant rate $\omega = 240$ rpm, determine the dynamic reactions at A and B.

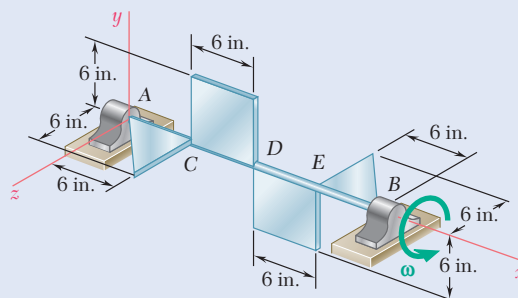


Fig. P18.67

18.68 The 8-kg shaft shown has a uniform cross section. Knowing that the shaft rotates at the constant rate $\omega = 12$ rad/s, determine the dynamic reactions at A and B.

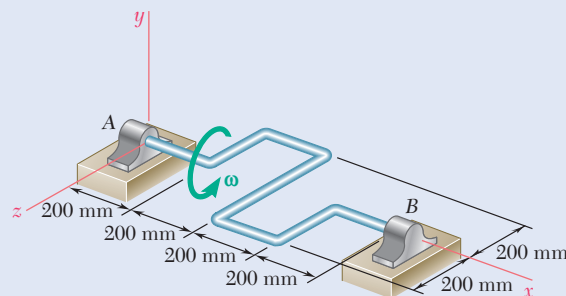


Fig. P18.68

18.69 After attaching the 18-kg wheel shown to a balancing machine and making it spin at the rate of 15 rev/s, a mechanic has found that to balance the wheel both statically and dynamically, he should use two corrective masses, a 170-g mass placed at B and a 56-g mass placed at D. Using a right-handed frame of reference rotating with the wheel (with the z axis perpendicular to the plane of the figure), determine before the corrective masses have been attached (a) the distance from the axis of rotation to the mass center of the wheel and the products of inertia I_{xy} and I_{zx} , (b) the force-couple system at C equivalent to the forces exerted by the wheel on the machine.

18.70 When the 18-kg wheel shown is attached to a balancing machine and made to spin at a rate of 12.5 rev/s, it is found that the forces exerted by the wheel on the machine are equivalent to a force-couple system consisting of a force $\mathbf{F} = (160 \text{ N})\mathbf{j}$ applied at C and a couple $\mathbf{M}_C = (14.7 \text{ N}\cdot\text{m})\mathbf{k}$, where the unit vectors form a triad that rotates with the wheel. (a) Determine the distance from the axis of rotation to the mass center of the wheel and the products of inertia I_{xy} and I_{zx} . (b) If only two corrective masses are to be used to balance the wheel statically and dynamically, what should these masses be and at which of the points A, B, D, or E should they be placed?

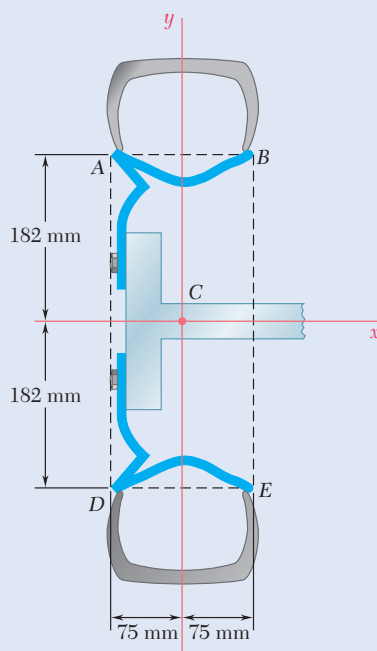


Fig. P18.69 and P18.70

- 18.71** Knowing that the assembly of Prob. 18.65 is initially at rest ($\omega = 0$) when a couple of moment $\mathbf{M}_0 = M_0\mathbf{j}$ is applied to shaft CD , determine (a) the resulting angular acceleration of the assembly, (b) the dynamic reactions at C and D immediately after the couple is applied.
- 18.72** Knowing that the plate of Prob. 18.66 is initially at rest ($\omega = 0$) when a couple of moment $\mathbf{M}_0 = (0.75 \text{ ft}\cdot\text{lb})\mathbf{j}$ is applied to it, determine (a) the resulting angular acceleration of the plate, (b) the dynamic reactions at A and B immediately after the couple has been applied.
- 18.73** The assembly of Prob. 18.67 is initially at rest ($\omega = 0$) when a couple $\mathbf{M}_0$ is applied to axle AB . Knowing that the resulting angular acceleration of the assembly is $\boldsymbol{\alpha} = (150 \text{ rad/s}^2)\mathbf{i}$, determine (a) the couple $\mathbf{M}_0$, (b) the dynamic reactions at A and B immediately after the couple is applied.
- 18.74** The shaft of Prob. 18.68 is initially at rest ($\omega = 0$) when a couple $\mathbf{M}_0$ is applied to it. Knowing that the resulting angular acceleration of the shaft is $\boldsymbol{\alpha} = (20 \text{ rad/s}^2)\mathbf{i}$, determine (a) the couple $\mathbf{M}_0$, (b) the dynamic reactions at A and B immediately after the couple is applied.
- 18.75** The assembly shown weighs 12 lb and consists of 4 thin 16-in.-diameter semicircular aluminum plates welded to a light 40-in.-long shaft AB . The assembly is at rest ($\omega = 0$) at time $t = 0$ when a couple $\mathbf{M}_0$ is applied to it as shown, causing the assembly to complete one full revolution in 2 s. Determine (a) the couple $\mathbf{M}_0$, (b) the dynamic reactions at A and B at $t = 0$.
- 18.76** For the assembly of Prob. 18.75, determine the dynamic reactions at A and B at $t = 2$ s.
- 18.77** The sheet-metal component shown is of uniform thickness and has a mass of 600 g. It is attached to a light axle supported by bearings at A and B located 150 mm apart. The component is at rest when it is subjected to a couple $\mathbf{M}_0$ as shown. If the resulting angular acceleration is $\boldsymbol{\alpha} = (12 \text{ rad/s}^2)\mathbf{k}$, determine (a) the couple $\mathbf{M}_0$, (b) the dynamic reactions at A and B immediately after the couple has been applied.

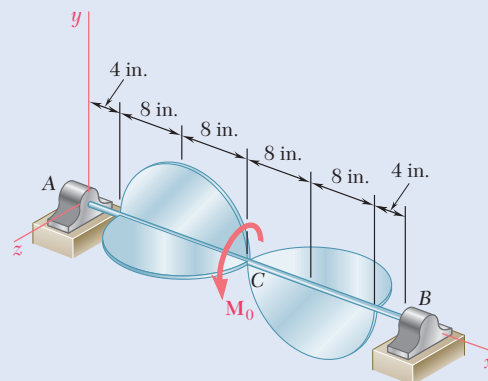


Fig. P18.75

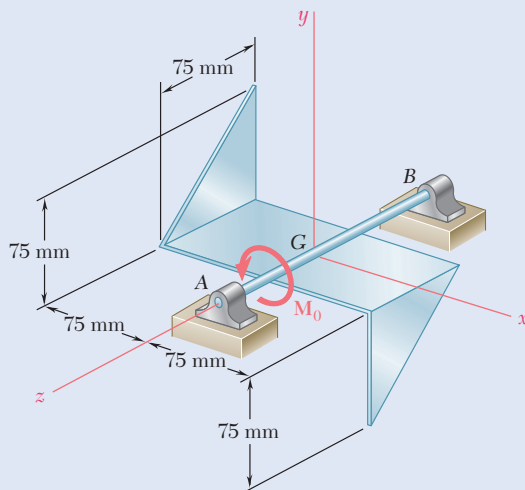


Fig. P18.77

- 18.78** For the sheet-metal component of Prob. 18.77, determine (a) the angular velocity of the component 0.6 s after the couple $\mathbf{M}_0$ has been applied to it, (b) the magnitude of the dynamic reactions at A and B at that time.

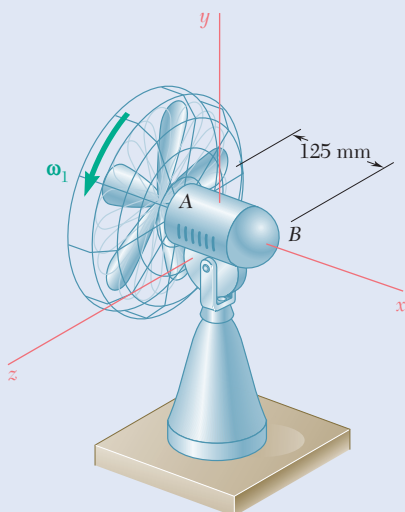


Fig. P18.79

18.79 The blade of an oscillating fan and the rotor of its motor have a total mass of 300 g and a combined radius of gyration of 75 mm. They are supported by bearings at A and B, 125 mm apart, and rotate at the rate $\omega_1 = 1800$ rpm. Determine the dynamic reactions at A and B when the motor casing has an angular velocity $\omega_2 = (0.6 \text{ rad/s})\mathbf{j}$.

18.80 The blade of a portable saw and the rotor of its motor have a total weight of 2.5 lb and a combined radius of gyration of 1.5 in. Knowing that the blade rotates as shown at the rate $\omega_1 = 1500$ rpm, determine the magnitude and direction of the couple $\mathbf{M}$ that a worker must exert on the handle of the saw to rotate it with a constant angular velocity $\omega_2 = -(2.4 \text{ rad/s})\mathbf{j}$.

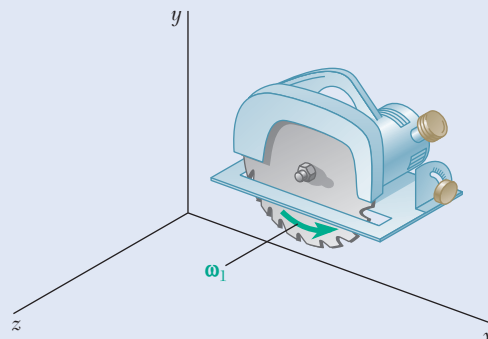


Fig. P18.80

18.81 The flywheel of an automobile engine, which is rigidly attached to the crankshaft, is equivalent to a 400-mm-diameter, 15-mm-thick steel plate. Determine the magnitude of the couple exerted by the flywheel on the horizontal crankshaft as the automobile travels around an unbanked curve of 200-m radius at a speed of 90 km/h, with the flywheel rotating at 2700 rpm. Assume the automobile to have (a) a rear-wheel drive with the engine mounted longitudinally, (b) a front-wheel drive with the engine mounted transversely. (Density of steel = 7860 kg/m³.)

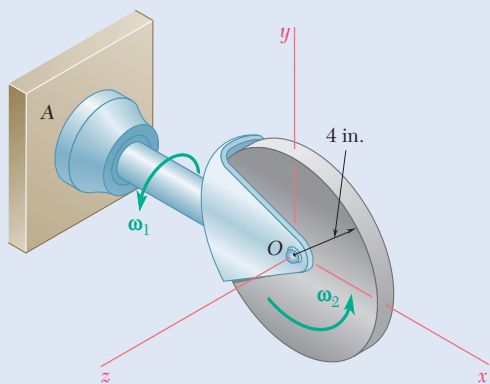


Fig. P18.83

18.82 Each wheel of an automobile has a mass of 22 kg, a diameter of 575 mm, and a radius of gyration of 225 mm. The automobile travels around an unbanked curve of radius 150 m at a speed of 95 km/h. Knowing that the transverse distance between the wheels is 1.5 m, determine the additional normal force exerted by the ground on each outside wheel due to the motion of the car.

18.83 The uniform, thin 5-lb disk spins at a constant rate $\omega_2 = 6$ rad/s about an axis held by a housing attached to a horizontal rod that rotates at the constant rate $\omega_1 = 3$ rad/s. Determine the couple that represents the dynamic reaction at the support A.

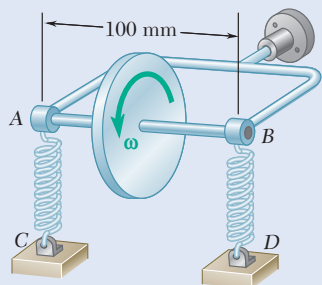


Fig. P18.84

18.84 The essential structure of a certain type of aircraft turn indicator is shown. Each spring has a constant of 500 N/m, and the 200-g uniform disk of 40-mm radius spins at the rate of 10 000 rpm. The springs are stretched and exert equal vertical forces on yoke AB when the airplane is traveling in a straight path. Determine the angle through which the yoke will rotate when the pilot executes a horizontal turn of 750-m radius to the right at a speed of 800 km/h. Indicate whether point A will move up or down.

- 18.85** A slender rod is bent to form a square frame of side 6 in. The frame is attached by a collar at A to a vertical shaft that rotates with a constant angular velocity ω . Determine the value of ω for which line AB forms an angle $\beta = 48^\circ$ with the horizontal x axis.

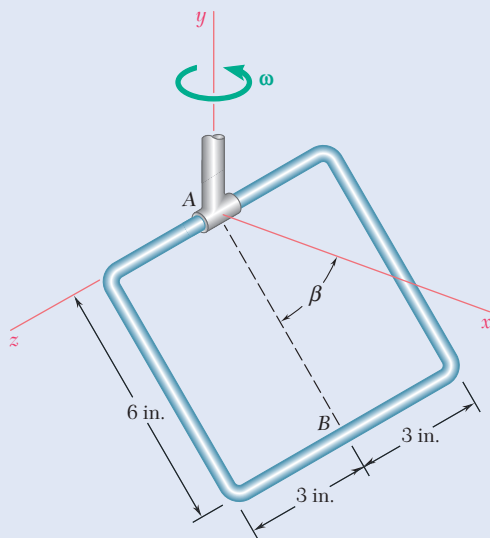


Fig. P18.85

- 18.86** A uniform square plate with side $a = 225$ mm is hinged at points A and B to a clevis that rotates with a constant angular velocity ω about a vertical axis. Determine (a) the constant angle β that the plate forms with the horizontal x axis when $\omega = 12$ rad/s, (b) the largest value of ω for which the plate remains vertical ($\beta = 90^\circ$).

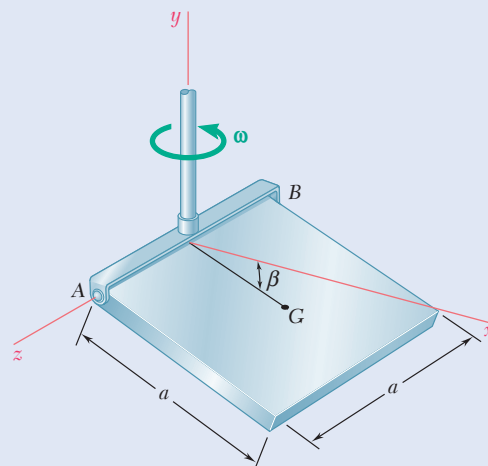


Fig. P18.86 and P18.87

- 18.87** A uniform square plate with side $a = 300$ mm is hinged at points A and B to a clevis that rotates with a constant angular velocity ω about a vertical axis. Determine (a) the value of ω for which the plate forms a constant angle $\beta = 60^\circ$ with the horizontal x axis, (b) the largest value of ω for which the plate remains vertical ($\beta = 90^\circ$).

- 18.88** The 2-lb gear A is constrained to roll on the fixed gear B but is free to rotate about axle AD . Axle AD has a length of 20 in., a negligible weight, and is connected by a clevis to the vertical shaft DE that rotates as shown with a constant angular velocity ω_1 . Assuming that gear A can be approximated by a thin disk with a radius of 4 in., determine the largest allowable value of ω_1 if gear A is not to lose contact with gear B .

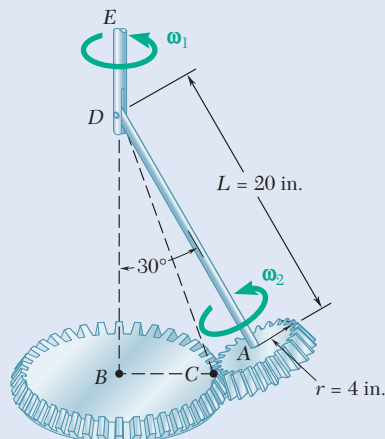


Fig. P18.88

- 18.89** Determine the force $\mathbf{F}$ exerted by gear B on gear A of Prob. 18.88 when shaft DE rotates with the constant angular speed of $\omega_1 = 4$ rad/s. (Hint: The force $\mathbf{F}$ must be perpendicular to the line drawn from D to C .)

18.90 and 18.91 The slender rod AB is attached by a clevis to arm BCD that rotates with a constant angular velocity ω about the centerline of its vertical portion CD . Determine the magnitude of the angular velocity ω .

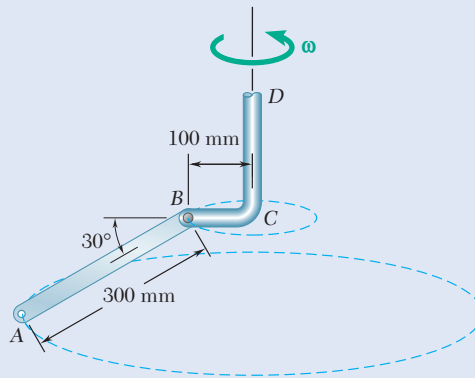


Fig. P18.90

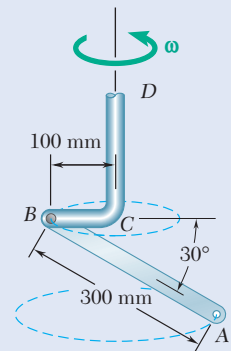


Fig. P18.91

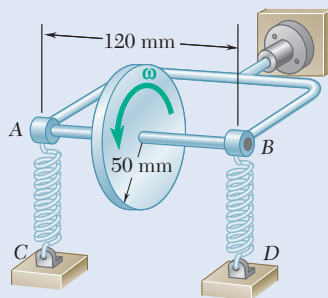


Fig. P18.92

18.92 The essential structure of a certain type of aircraft turn indicator is shown. Springs AC and BD are initially stretched and exert equal vertical forces at A and B when the airplane is traveling in a straight path. Each spring has a constant of 600 N/m and the uniform disk has a mass of 250 g and spins at the rate of $12\,000 \text{ rpm}$. Determine the angle through which the yoke will rotate when the pilot executes a horizontal turn of 800-m radius to the right at a speed of 720 km/h . Indicate whether point A will move up or down.

18.93 The 10-oz disk shown spins at the rate $\omega_1 = 750 \text{ rpm}$, while axle AB rotates as shown with an angular velocity ω_2 of magnitude 6 rad/s . Determine the dynamic reactions at A and B .

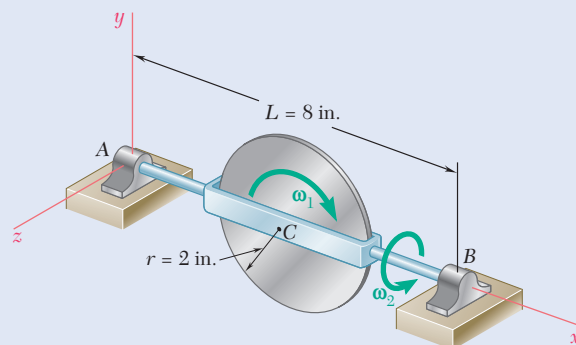


Fig. P18.93 and P18.94

18.94 The 10-oz disk shown spins at the rate $\omega_1 = 750 \text{ rpm}$, while axle AB rotates as shown with an angular velocity ω_2 . Determine the maximum allowable magnitude of ω_2 if the dynamic reactions at A and B are not to exceed 0.25 lb each.

18.95 Two disks each have a mass of 5 kg and a radius 300 mm. They spin as shown at the rate of $\omega_1 = 1200$ rpm about a rod AB of negligible mass that rotates about the horizontal z axis at the rate of $\omega_2 = 60$ rpm. (a) Determine the dynamic reactions at points C and D . (b) Solve part (a) assuming that the direction of spin of disk A is reversed.

18.96 Two disks each have a mass of 5 kg and a radius of 300 mm. They spin as shown at the rate of $\omega_1 = 1200$ rpm about a rod AB of negligible mass that rotates about the horizontal z axis at the rate ω_2 . Determine the maximum allowable value of ω_2 if the magnitudes of the dynamic reactions at points C and D are not to exceed 350 N each.

18.97 A stationary horizontal plate is attached to the ceiling by means of a fixed vertical tube. A wheel of radius a and mass m is mounted on a light axle AC that is attached by means of a clevis at A to a rod AB fitted inside the vertical tube. The rod AB is made to rotate with a constant angular velocity Ω causing the wheel to roll on the lower face of the stationary plate. Determine the minimum angular velocity Ω for which contact is maintained between the wheel and the plate. Consider the particular cases (a) when the mass of the wheel is concentrated in the rim, (b) when the wheel is equivalent to a thin disk of radius a .

18.98 Assuming that the wheel of Prob. 18.97 weighs 8 lb, has a radius $a = 4$ in., and a radius of gyration of 3 in., and that $R = 20$ in., determine the force exerted by the plate on the wheel when $\Omega = 25$ rad/s.

18.99 A thin disk of mass $m = 4$ kg rotates with an angular velocity ω_2 with respect to arm ABC , which itself rotates with an angular velocity ω_1 about the y axis. Knowing that $\omega_1 = 5$ rad/s and $\omega_2 = 15$ rad/s and that both are constant, determine the force-couple system representing the dynamic reaction at the support at A .

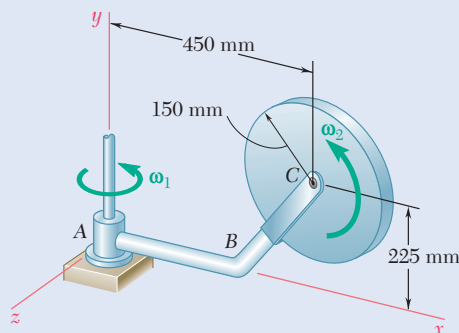


Fig. P18.99

18.100 An experimental Fresnel-lens solar-energy concentrator can rotate about the horizontal axis AB that passes through its mass center G . It is supported at A and B by a steel framework that can rotate about the vertical y axis. The concentrator has a mass of 30 Mg, a radius of gyration of 12 m about its axis of symmetry CD , and a radius of gyration of 10 m about any transverse axis through G . Knowing that the angular velocities ω_1 and ω_2 have constant magnitudes equal to 0.20 rad/s and 0.25 rad/s, respectively, determine for the position $\theta = 60^\circ$ (a) the forces exerted on the concentrator at A and B , (b) the couple $M_2 \mathbf{k}$ applied to the concentrator at that instant.

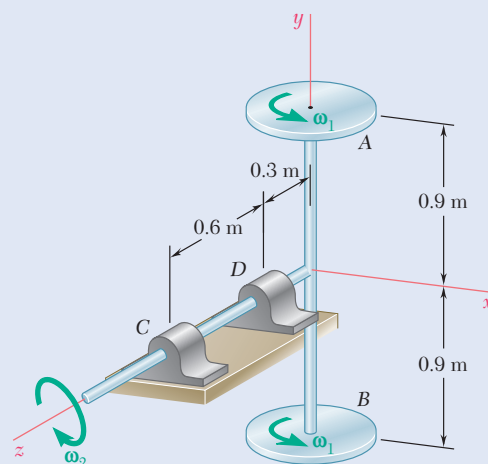


Fig. P18.95 and P18.96

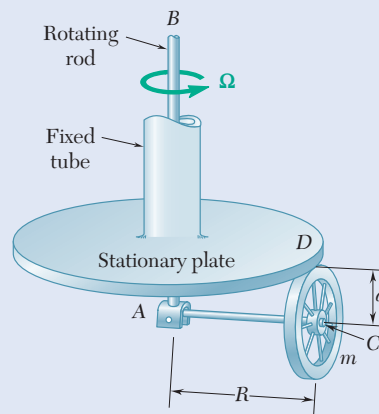


Fig. P18.97

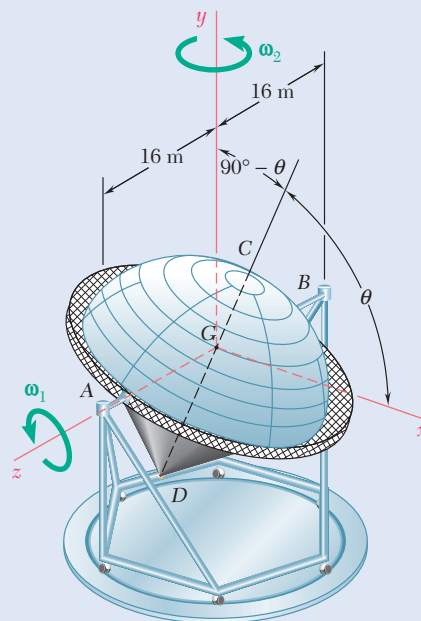


Fig. P18.100

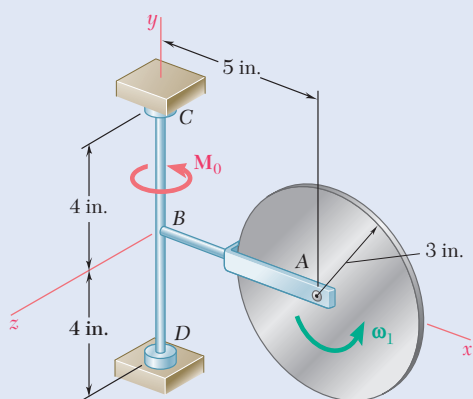


Fig. P18.101 and P18.102

18.101 A 6-lb homogeneous disk of radius 3 in. spins as shown at the constant rate $\omega_1 = 60$ rad/s. The disk is supported by the fork-ended rod AB , which is welded to the vertical shaft CBD . The system is at rest when a couple $\mathbf{M}_0 = (0.25 \text{ ft}\cdot\text{lb})\mathbf{j}$ is applied to the shaft for 2 s and then removed. Determine the dynamic reactions at C and D after the couple has been removed.

18.102 A 6-lb homogeneous disk of radius 3 in. spins as shown at the constant rate $\omega_1 = 60$ rad/s. The disk is supported by the fork-ended rod AB , which is welded to the vertical shaft CBD . The system is at rest when a couple $\mathbf{M}_0$ is applied as shown to the shaft for 3 s and then removed. Knowing that the maximum angular velocity reached by the shaft is 18 rad/s, determine (a) the couple $\mathbf{M}_0$, (b) the dynamic reactions at C and D after the couple has been removed.

18.103 A 2.5-kg homogeneous disk of radius 80 mm rotates with an angular velocity ω_1 with respect to arm ABC , which is welded to a shaft DCE rotating as shown at the constant rate $\omega_2 = 12$ rad/s. Friction in the bearing at A causes ω_1 to decrease at the rate of 15 rad/s^2 . Determine the dynamic reactions at D and E at a time when ω_1 has decreased to 50 rad/s.

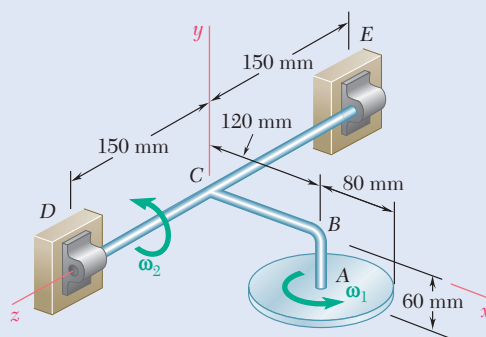


Fig. P18.103 and P18.104

18.104 A 2.5-kg homogeneous disk of radius 80 mm rotates at the constant rate $\omega_1 = 50$ rad/s with respect to arm ABC , which is welded to a shaft DCE . Knowing that at the instant shown, shaft DCE has an angular velocity $\omega_2 = (12 \text{ rad/s})\mathbf{k}$ and an angular acceleration $\alpha_2 = (8 \text{ rad/s}^2)\mathbf{k}$, determine (a) the couple that must be applied to shaft DCE to produce that acceleration, (b) the corresponding dynamic reactions at D and E .

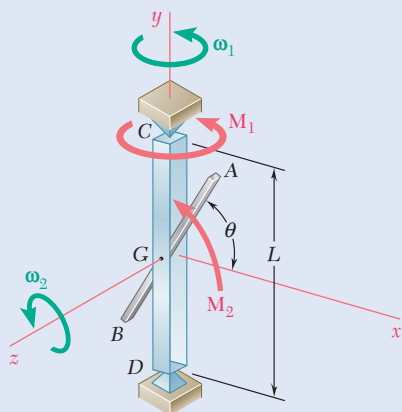


Fig. P18.106

18.105 For the disk of Prob. 18.99, determine (a) the couple $M_1\mathbf{j}$ that should be applied to arm ABC to give it an angular acceleration $\alpha_1 = -(7.5 \text{ rad/s}^2)\mathbf{j}$ when $\omega_1 = 5$ rad/s, knowing that the disk rotates at the constant rate $\omega_2 = 15$ rad/s, (b) the force-couple system representing the dynamic reaction at A at that instant. Assume that ABC has a negligible mass.

***18.106** A slender homogeneous rod AB of mass m and length L is made to rotate at a constant rate ω_2 about the horizontal z axis, while frame CD is made to rotate at the constant rate ω_1 about the y axis. Express as a function of the angle θ (a) the couple $\mathbf{M}_1$ required to maintain the rotation of the frame, (b) the couple $\mathbf{M}_2$ required to maintain the rotation of the rod, (c) the dynamic reactions at the supports C and D .

*18.3 MOTION OF A GYROSCOPE

A **gyroscope** consists essentially of a rotor that can spin freely about its geometric axis. When mounted in a Cardan's suspension (Fig. 18.15), a gyroscope can assume any orientation, but its mass center must remain fixed in space. Because a gyroscope can measure its orientation in space and maintain that orientation, it has become an indispensable part of modern navigational equipment. In this section, we examine the motion of a gyroscope as a practical example of analyzing the motion of a rigid body in three dimensions.

18.3A Eulerian Angles

In order to define the position of a gyroscope at a given instant, let us select a fixed frame of reference $OXYZ$ with the origin O located at the mass center of the gyroscope and the Z axis directed along the line defined by the bearings A and A' of the outer gimbal. We consider a reference position of the gyroscope in which the two gimbals and a given diameter DD' of the rotor are located in the fixed YZ plane (Fig. 18.15a). The gyroscope can be brought from this reference position into any arbitrary position (Fig. 18.15b) by means of the following steps.

1. A rotation of the outer gimbal through an angle ϕ about the axis AA' .
2. A rotation of the inner gimbal through θ about BB' .
3. A rotation of the rotor through ψ about CC' .

The angles ϕ , θ , and ψ are called the **Eulerian angles**; they completely characterize the position of the gyroscope at any given instant. Their derivatives $\dot{\phi}$, $\dot{\theta}$, and $\dot{\psi}$ define, respectively, the rate of **precession**, the rate of **nutation**, and the rate of **spin** of the gyroscope at the instant considered. Precession is the revolution of the axis BB' about the Z -axis, and nutation is the back-and-forth motion of CC' as the object precesses.

In order to compute the components of the angular velocity and of the angular momentum of the gyroscope, we will use a rotating system of axes $Oxyz$ attached to the inner gimbal, with the y axis along BB' and the z axis along CC' (Fig. 18.16). These axes are principal axes of inertia for the gyroscope. Although they follow it in its precession and nutation, they do not spin with $\dot{\psi}$; for that reason, they are more convenient to use than axes actually attached to the gyroscope. The angular velocity $\boldsymbol{\omega}$ of the gyroscope with respect to the fixed frame of reference $OXYZ$ now can be expressed as the sum of three partial angular velocities that correspond to the precession, the nutation, and the spin of the gyroscope, respectively. Denoting the unit vectors along the rotating axes by $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ and the unit vector along the fixed Z axis by $\mathbf{K}$, we have

$$\boldsymbol{\omega} = \dot{\phi}\mathbf{K} + \dot{\theta}\mathbf{j} + \dot{\psi}\mathbf{k} \quad (18.33)$$

Since the vector components obtained for $\boldsymbol{\omega}$ in Eq. (18.33) are not orthogonal (Fig. 18.16), we resolve the unit vector $\mathbf{K}$ into components along the x and z axes; we obtain

$$\mathbf{K} = -\sin\theta\mathbf{i} + \cos\theta\mathbf{k} \quad (18.34)$$

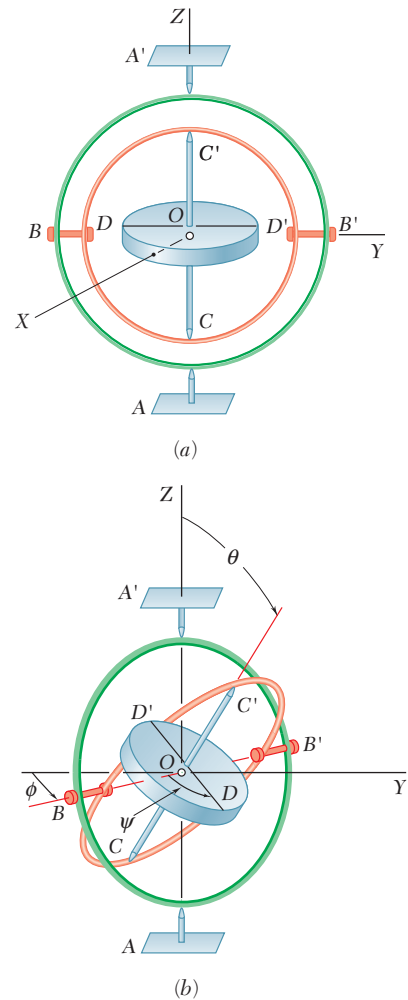


Fig. 18.15 (a) Reference position of a gyroscope; (b) arbitrary position of the gyroscope by rotation through the three Eulerian angles.

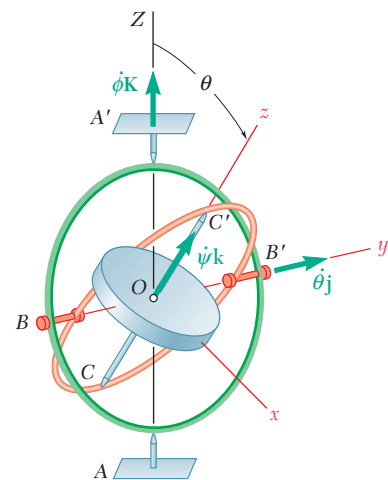


Fig. 18.16 Precession $\dot{\phi}$, nutation $\dot{\theta}$, and spin $\dot{\psi}$ of a gyroscope.

Then, substituting for $\mathbf{K}$ into Eq. (18.33), we have

$$\boldsymbol{\omega} = -\dot{\phi} \sin \theta \mathbf{i} + \dot{\theta} \mathbf{j} + (\dot{\psi} + \dot{\phi} \cos \theta) \mathbf{k} \quad (18.35)$$

The coordinate axes are principal axes of inertia, so we can obtain the components of the angular momentum $\mathbf{H}_O$ by multiplying the components of $\boldsymbol{\omega}$ by the moments of inertia of the rotor about the x , y , and z axes, respectively. Denoting the moment of inertia of the rotor about its spin axis by I , its moment of inertia about a transverse axis through O by I' , and neglecting the mass of the gimbals, we have

$$\mathbf{H}_O = -I' \dot{\phi} \sin \theta \mathbf{i} + I' \dot{\theta} \mathbf{j} + I(\dot{\psi} + \dot{\phi} \cos \theta) \mathbf{k} \quad (18.36)$$

Recalling that the rotating axes are attached to the inner gimbal and thus do not spin with $\dot{\psi}$, we express their angular velocity as the sum

$$\boldsymbol{\Omega} = \dot{\phi} \mathbf{K} + \dot{\theta} \mathbf{j} \quad (18.37)$$

or substituting for $\mathbf{K}$ from Eq. (18.34), we have

$$\boldsymbol{\Omega} = -\dot{\phi} \sin \theta \mathbf{i} + \dot{\theta} \mathbf{j} + \dot{\phi} \cos \theta \mathbf{k} \quad (18.38)$$

Substituting for $\mathbf{H}_O$ and $\boldsymbol{\Omega}$ from Eqs. (18.36) and (18.38) into the equation gives

$$\Sigma \mathbf{M}_O = (\dot{\mathbf{H}}_O)_{Oxyz} + \boldsymbol{\Omega} \times \mathbf{H}_O \quad (18.28)$$

We now obtain the three differential equations

$$\begin{aligned} \Sigma M_x &= -I'(\ddot{\phi} \sin \theta + 2\dot{\theta}\dot{\phi} \cos \theta) + I\dot{\theta}(\dot{\psi} + \dot{\phi} \cos \theta) \\ \Sigma M_y &= I'(\ddot{\theta} - \dot{\phi}^2 \sin \theta \cos \theta) + I\dot{\phi} \sin \theta (\dot{\psi} + \dot{\phi} \cos \theta) \\ \Sigma M_z &= I \frac{d}{dt} (\dot{\psi} + \dot{\phi} \cos \theta) \end{aligned} \quad (18.39)$$

Equations (18.39) define the motion of a gyroscope subjected to a given system of forces when the mass of its gimbals is neglected. We can also use them to define the motion of an **axisymmetric body** (or body of revolution) attached at a point on its axis of symmetry and to define the motion of an axisymmetric body about its mass center. The gimbals of the gyroscope helped us visualize the Eulerian angles, but it is clear that we can use these angles to define the position of any rigid body with respect to axes centered at a point of the body—regardless of the way in which the body is actually supported.

Because Eqs. (18.39) are nonlinear, it is not possible to express the Eulerian angles ϕ , θ , and ψ as analytical functions of time t in general, and you may need to use numerical methods of solution. However, as you will see in the rest of this section, several particular cases of interest can be analyzed easily.

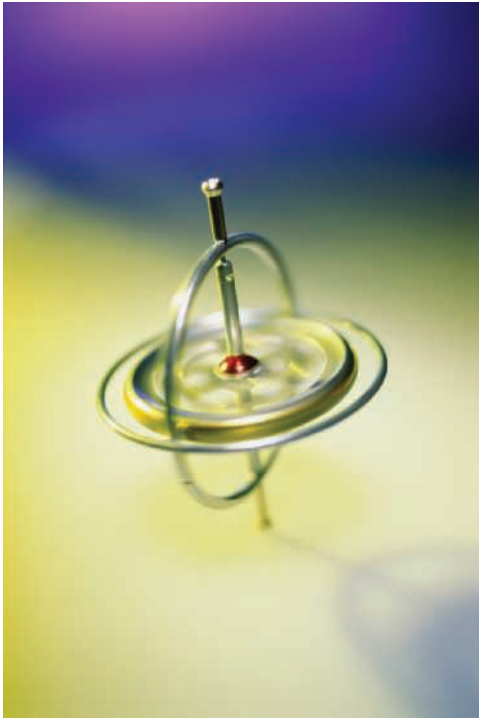


Photo 18.5 A gyroscope can be used for measuring orientation and is capable of maintaining the same absolute direction in space.

*18.3B Steady Precession of a Gyroscope

Let us now investigate the particular case of gyroscopic motion in which the angle θ , the rate of precession $\dot{\phi}$, and the rate of spin $\dot{\psi}$ remain constant. We propose to determine the forces that must be applied to the gyroscope to maintain this motion, which is known as the **steady precession** of a gyroscope.

Instead of applying the general equations (18.39), we determine the sum of the moments of the required forces by computing the rate of change of the angular momentum of the gyroscope in the particular case considered. We first note that the angular velocity $\boldsymbol{\omega}$ of the gyroscope, its angular momentum $\mathbf{H}_O$, and the angular velocity $\boldsymbol{\Omega}$ of the rotating frame of reference (Fig. 18.17) reduce, respectively, to

$$\boldsymbol{\omega} = -\dot{\phi} \sin \theta \mathbf{i} + \omega_z \mathbf{k} \quad (18.40)$$

$$\mathbf{H}_O = -I' \dot{\phi} \sin \theta \mathbf{i} + I \omega_z \mathbf{k} \quad (18.41)$$

$$\boldsymbol{\Omega} = -\dot{\phi} \sin \theta \mathbf{i} + \dot{\phi} \cos \theta \mathbf{k} \quad (18.42)$$

where $\omega_z = \dot{\psi} + \dot{\phi} \cos \theta$ is the rectangular component along the spin axis of the total angular velocity of the gyroscope.

Since θ , $\dot{\phi}$, and $\dot{\psi}$ are constant, the vector $\mathbf{H}_O$ is constant in magnitude and direction with respect to the rotating frame of reference. Therefore its rate of change $(\dot{\mathbf{H}}_O)_{Oxyz}$ with respect to that frame is zero. Thus, Eq. (18.28) reduces to

$$\Sigma \mathbf{M}_O = \boldsymbol{\Omega} \times \mathbf{H}_O \quad (18.43)$$

which yields, after substitutions from Eqs. (18.41) and (18.42),

$$\Sigma \mathbf{M}_O = (I \omega_z - I' \dot{\phi} \cos \theta) \dot{\phi} \sin \theta \mathbf{j} \quad (18.44)$$

The mass center of the gyroscope is fixed in space, so using Eq. (18.1), we have $\Sigma \mathbf{F} = 0$. Thus, the forces that must be applied to the gyroscope to maintain its steady precession reduce to a couple of moment equal to the right-hand side of Eq. (18.44). Note that *this couple should be applied about an axis perpendicular to the precession axis and to the spin axis of the gyroscope* (Fig. 18.18).

In the particular case when the precession axis and the spin axis are at a right angle to each other, we have $\theta = 90^\circ$, and Eq. (18.44) reduces to

$$\Sigma \mathbf{M}_O = I \dot{\psi} \dot{\phi} \mathbf{j} \quad (18.45)$$

Thus, if we apply a couple $\mathbf{M}_O$ to the gyroscope about an axis perpendicular to its axis of spin, the gyroscope precesses about an axis perpendicular to both the spin axis and the couple axis. The sense of the precession is such that the vectors representing the spin, the couple, and the precession, respectively, form a right-handed triad (Fig. 18.19). The relationship of this triad also can be represented by writing Eq. (18.45) as the vector equation of

$$\Sigma \mathbf{M}_O = \dot{\phi} \times I \dot{\psi} \quad (18.45')$$

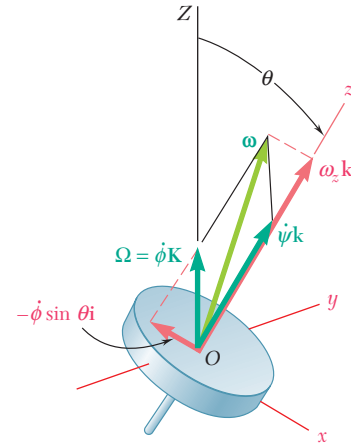


Fig. 18.17 Kinematic quantities used to determine the steady rate of precession of a gyroscope.

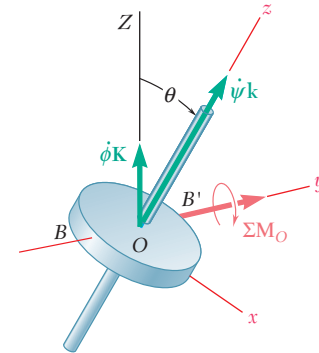


Fig. 18.18 To maintain a gyroscope in steady precession, a couple must be applied about an axis perpendicular to the precession and spin axis.

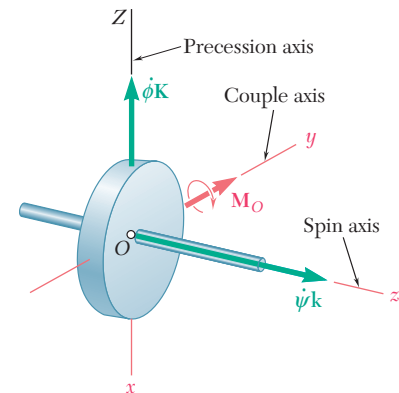


Fig. 18.19 A right-handed triad of the spin, couple, and precession axes.

Because of the relatively large couples required to change the orientation of their axes, gyroscopes are used as stabilizers in torpedoes and ships. Spinning bullets and shells remain tangent to their trajectory because of gyroscopic action. Also, a bicycle is easier to keep balanced at high speeds because of the stabilizing effect of its spinning wheels. However, gyroscopic action is not always welcome; it must be taken into account in the design of bearings supporting rotating shafts subjected to forced precession. The reactions exerted on an airplane by its propellers, which changes the direction of flight, also must be taken into consideration and compensated for whenever possible.

*18.3C Motion of an Axisymmetric Body Under no Force

We can now analyze the motion of an axisymmetric body about its mass center under no force except its own weight. Examples of such motion are furnished by projectiles (if air resistance is neglected) and by satellites and space vehicles after the burnout of their launching rockets.

The sum of the moments of the external forces about the mass center G of the body is zero, so Eq. (18.2) yields $\dot{\mathbf{H}}_G = 0$. It follows that the angular momentum $\mathbf{H}_G$ of the body about G is constant. Thus, the direction of $\mathbf{H}_G$ is fixed in space and can be used to define the Z axis, or axis of precession (Fig. 18.20). Let us select a rotating system of axes $Gxyz$ with the z axis along the axis of symmetry of the body, the x axis in the plane defined by the Z and z axes, and the y axis pointing away from you (Fig. 18.21). This gives us

$$H_x = -H_G \sin \theta \quad H_y = 0 \quad H_z = H_G \cos \theta \quad (18.46)$$

where θ represents the angle formed by the Z and z axes and H_G denotes the constant magnitude of the angular momentum of the body about G . Since the x , y , and z axes are principal axes of inertia for the body considered, we have

$$H_x = I' \omega_x \quad H_y = I' \omega_y \quad H_z = I \omega_z \quad (18.47)$$

where I denotes the moment of inertia of the body about its axis of symmetry and I' denotes its moment of inertia about a transverse axis through G . It follows from Eqs. (18.46) and (18.47) that

$$\omega_x = -\frac{H_G \sin \theta}{I'} \quad \omega_y = 0 \quad \omega_z = \frac{H_G \cos \theta}{I} \quad (18.48)$$

The second of these relations shows that the angular velocity $\boldsymbol{\omega}$ has no component along the y axis, i.e., along an axis perpendicular to the Z - z plane. Thus, the angle θ formed by the Z and z axes remains constant and *the body is in steady precession about the Z axis.*

Dividing the first of the relations in Eqs. (18.48) by the third, and observing from Fig. 18.21 that $-\omega_x/\omega_z = \tan \gamma$, we obtain the following relation between the angles γ and θ that the vectors $\boldsymbol{\omega}$ and $\mathbf{H}_G$, respectively, form with the axis of symmetry of the body.

$$\tan \gamma = \frac{I}{I'} \tan \theta \quad (18.49)$$

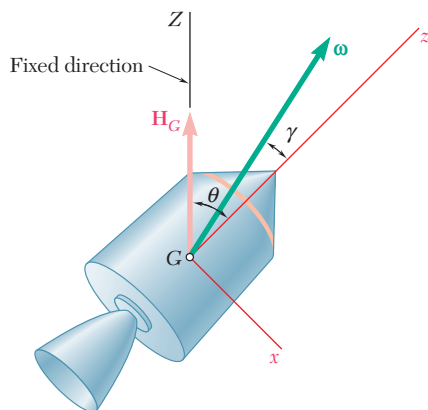


Fig. 18.20 For an axisymmetric body under no force other than its own weight, the angular momentum has a constant direction.

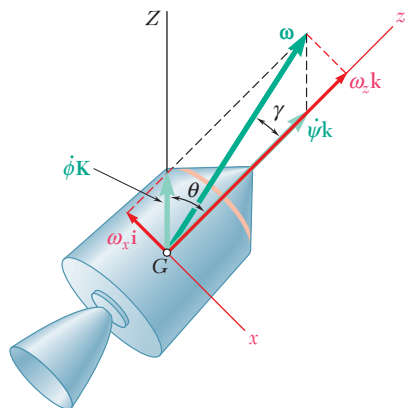


Fig. 18.21 The angular velocity of an axisymmetric body expressed in terms of body-fixed coordinates xyz .

Two particular cases of motion of an axisymmetric body under no force involve no precession.

1. If the body is set to spin about its axis of symmetry, we have $\omega_x = 0$ and, by Eq. (18.47), $H_x = 0$. Thus, the vectors $\boldsymbol{\omega}$ and $\mathbf{H}_G$ have the same orientation, and the body keeps spinning about its axis of symmetry (Fig. 18.22a).
2. If the body is set to spin about a transverse axis, we have $\omega_z = 0$ and, by Eq. (18.47), $H_z = 0$. Again $\boldsymbol{\omega}$ and $\mathbf{H}_G$ have the same orientation, and the body keeps spinning about the given transverse axis (Fig. 18.22b).

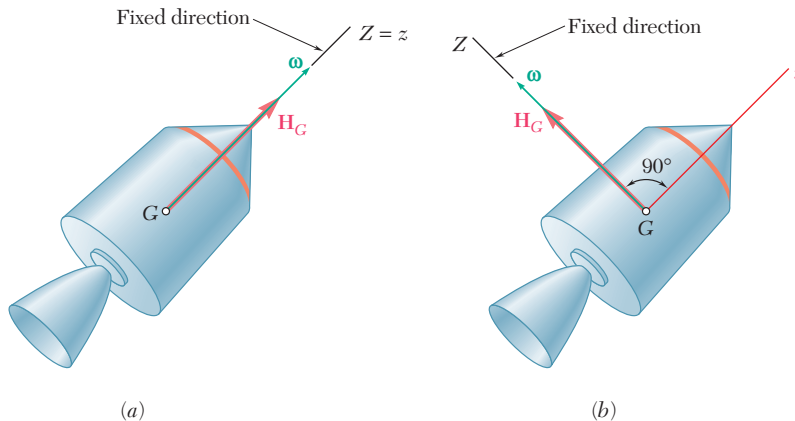


Fig. 18.22 (a) A body spinning about its axis of symmetry; (b) a body spinning about a transverse axis.

Considering now the general case represented in Fig. 18.21, recall from Sec. 15.6A that we can represent the motion of a body about a fixed point—or about its mass center—by the motion of a body cone rolling on a space cone. In the case of steady precession, the two cones are circular, since the angles γ and $\theta - \gamma$ that the angular velocity $\boldsymbol{\omega}$ forms, respectively, with the axis of symmetry of the body and with the precession axis are constant. Two cases should be distinguished.

1. $I < I'$. This is the case of an elongated body, such as the space vehicle of Fig. 18.23. From Eq. (18.49), we have $\gamma < \theta$. The vector $\boldsymbol{\omega}$ lies inside the angle ZGz ; the space cone and the body cone are tangent externally; and the spin and the precession are both observed as counterclockwise from the positive z axis. The precession is said to be *direct*.
2. $I > I'$. This is the case of a flattened body, such as the satellite of Fig. 18.24. From Eq. (18.49), we have $\gamma > \theta$. Since the vector $\boldsymbol{\omega}$ must lie outside the angle ZGz , the vector $\dot{\psi}\mathbf{k}$ has a sense opposite to that of the z axis; the space cone is inside the body cone; and the precession and the spin have opposite senses. The precession is said to be *retrograde*.

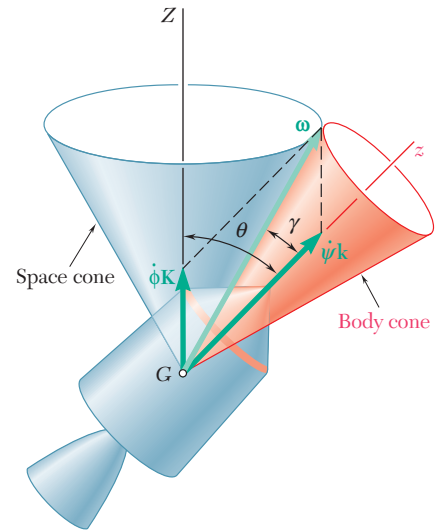


Fig. 18.23 Space cone and body cone for an elongated body ($I < I'$) in direct precession.

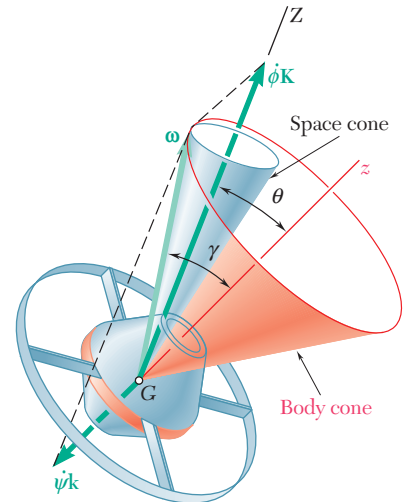
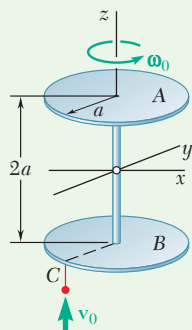


Fig. 18.24 Space cone and body cone for a flattened body ($I > I'$) in retrograde precession.



Sample Problem 18.6

A space satellite with mass m can be modeled as two thin disks of equal mass. The disks have a radius of $a = 800$ mm and are rigidly connected by a light rod with a length of $2a$. Initially, the satellite is spinning freely about its axis of symmetry at the rate $\omega_0 = 60$ rpm. A meteorite with a mass of $m_0 = m/1000$ is traveling with a velocity $\mathbf{v}_0$ of 2000 m/s relative to the satellite, strikes the satellite, and becomes embedded at C . Determine (a) the angular velocity of the satellite immediately after impact, (b) the precession axis of the ensuing motion, (c) the rates of precession and spin of the ensuing motion.

STRATEGY: Since an impact occurs, use the principle of impulse and momentum. Then you can use the relations in this section to determine the gyroscopic motion of the satellite.

MODELING: Choose the meteorite and the satellite as your system. The linear and angular momenta of the system before and after the impact are shown in Fig. 1.

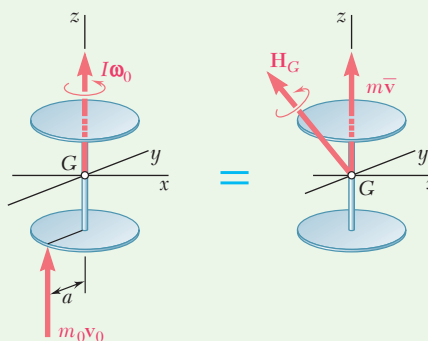


Fig. 1 Momenta before and after the impact.

ANALYSIS:

Moments of Inertia. Note that the axes shown are principal axes of inertia for the satellite. Thus, you have

$$I = I_z = \frac{1}{2}ma^2 \quad I' = I_x = I_y = 2\left[\frac{1}{4}\left(\frac{1}{2}m\right)a^2 + \left(\frac{1}{2}m\right)a^2\right] = \frac{5}{4}ma^2$$

Principle of Impulse and Momentum. Since no external force acts on the system, the momenta before and after impact are equal (Fig. 1). Taking moments about G , you have

$$\begin{aligned} -a\mathbf{j} \times m_0v_0\mathbf{k} + I\omega_0\mathbf{k} &= \mathbf{H}_G \\ \mathbf{H}_G &= -m_0v_0a\mathbf{i} + I\omega_0\mathbf{k} \end{aligned} \quad (1)$$

Angular Velocity After Impact. Substitute the values obtained for the components of $\mathbf{H}_G$ in Eq. (1) and for the moments of inertia into

$$H_x = I_x \omega_x \quad H_y = I_y \omega_y \quad H_z = I_z \omega_z$$

The result is

$$\begin{aligned} -m_0 v_0 a &= I' \omega_x = \frac{5}{4} m a^2 \omega_x & 0 &= I' \omega_y & I \omega_0 &= I \omega_z \\ \omega_x &= -\frac{4}{5} \frac{m_0 v_0}{m a} & \omega_y &= 0 & \omega_z &= \omega_0 \end{aligned} \quad (2)$$

For the satellite considered, you have $\omega_0 = 60 \text{ rpm} = 6.283 \text{ rad/s}$, $m_0/m = 1/1000$, $a = 0.800 \text{ m}$, and $v_0 = 2000 \text{ m/s}$. You obtain

$$\begin{aligned} \omega_x &= -2 \text{ rad/s} & \omega_y &= 0 & \omega_z &= 6.283 \text{ rad/s} \\ \omega &= \sqrt{\omega_x^2 + \omega_z^2} = 6.594 \text{ rad/s} & \tan \gamma &= \frac{-\omega_x}{\omega_z} = +0.3183 \\ & & \omega &= 63.0 \text{ rpm} & \gamma &= 17.7^\circ \end{aligned}$$

Precession Axis. In free motion, the direction of the angular momentum $\mathbf{H}_G$ is fixed in space, so the satellite precesses about this direction. The angle θ formed by the precession axis and the z axis is (Fig. 2)

$$\tan \theta = \frac{-H_x}{H_z} = \frac{m_0 v_0 a}{I \omega_0} = \frac{2 m_0 v_0}{m a \omega_0} = 0.796 \quad \theta = 38.5^\circ$$

Rates of Precession and Spin. Sketch the space and body cones for the free motion of the satellite (Fig. 3). Using the law of sines, compute the rates of precession and spin.

$$\begin{aligned} \frac{\omega}{\sin \theta} &= \frac{\dot{\phi}}{\sin \gamma} = \frac{\dot{\psi}}{\sin (\theta - \gamma)} \\ \dot{\phi} &= 30.8 \text{ rpm} & \dot{\psi} &= 35.9 \text{ rpm} \end{aligned}$$

REFLECT and THINK: If you applied the principle of impulse and momentum in the z -direction, you would find that $P \Delta t = m \bar{v}$ where $P \Delta t$ is the impulse the meteorite applies to the satellite. In this problem, we were interested in the three-dimensional rotation of the satellite and modeled it as a rigid body. In Chap. 12, we were concerned with the orbits of satellites over the earth and modeled the satellite as a particle. As engineers, how we model a system depends on what type of problem we are trying to solve.

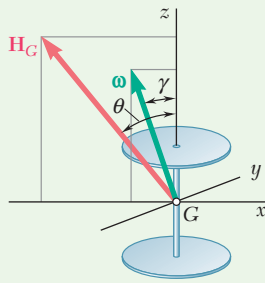


Fig. 2 Angles between the z -axis and the angular velocity and the angular momentum.

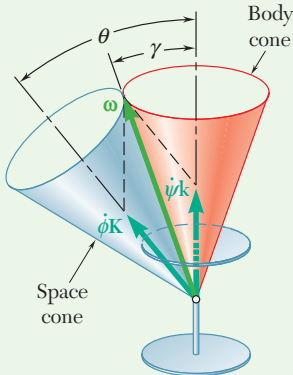


Fig. 3 Space and body cones for the satellite.

SOLVING PROBLEMS ON YOUR OWN

In this section, we analyzed the motion of **gyroscopes** and of other **axisymmetric bodies** with a fixed point O . In order to define the position of these bodies at any given instant, we introduced the three **Eulerian angles** ϕ , θ , and ψ (Fig. 18.15) and noted that their time derivatives define, respectively, the rate of **precession**, the rate of **nutation**, and the rate of **spin** (Fig. 18.16). The problems you encountered in this section fall into one of the following categories.

1. Steady precession. This is the motion of a gyroscope or other axisymmetric body with a fixed point located on its axis of symmetry in which the angle θ , the rate of precession $\dot{\phi}$, and the rate of spin $\dot{\psi}$ all remain constant.

a. Using the rotating frame of reference $Oxyz$ shown in Fig. 18.17, which *precesses* with the body, *but does not spin* with it, we obtained the expressions for the angular velocity $\boldsymbol{\omega}$ of the body, its angular momentum $\mathbf{H}_O$, and the angular velocity $\boldsymbol{\Omega}$ of the frame $Oxyz$ as

$$\boldsymbol{\omega} = -\dot{\phi} \sin \theta \mathbf{i} + \omega_z \mathbf{k} \quad (18.40)$$

$$\mathbf{H}_O = -I' \dot{\phi} \sin \theta \mathbf{i} + I \omega_z \mathbf{k} \quad (18.41)$$

$$\boldsymbol{\Omega} = -\dot{\phi} \sin \theta \mathbf{i} + \dot{\phi} \cos \theta \mathbf{k} \quad (18.42)$$

where I = moment of inertia of body about its axis of symmetry

I' = moment of inertia of body about a transverse axis through O

ω_z = rectangular component of $\boldsymbol{\omega}$ along z axis = $\dot{\psi} + \dot{\phi} \cos \theta$

b. The sum of the moments about O of the forces applied to the body is equal to the rate of change of its angular momentum, as expressed by Eq. (18.28). But, since θ and the rates of change $\dot{\phi}$ and $\dot{\psi}$ are constant, it follows from Eq. (18.41) that $\mathbf{H}_O$ remains constant in magnitude and direction when viewed from the frame $Oxyz$. Thus, its rate of change is zero with respect to that frame, and you have

$$\Sigma \mathbf{M}_O = \boldsymbol{\Omega} \times \mathbf{H}_O \quad (18.43)$$

where $\boldsymbol{\Omega}$ and $\mathbf{H}_O$ are defined by Eqs. (18.42) and (18.41), respectively. Equation (18.43) shows that the moment resultant at O of the forces applied to the body is perpendicular to both the axis of precession and the axis of spin (Fig. 18.18).

c. Keep in mind that the method described applies not only to gyroscopes, where the fixed point O coincides with the mass center G , but also to any axisymmetric body with a fixed point O located on its axis of symmetry. This method, therefore, can be used to analyze the *steady precession of a top* on a rough floor.

d. When an axisymmetric body has no fixed point but is in steady precession about its mass center G , you should draw a free-body diagram and a kinetic diagram showing that the system of the external forces exerted on the body (including the body's weight) is equivalent to the vector $m\bar{\mathbf{a}}$ applied at G and the couple vector $\dot{\mathbf{H}}_G$.

You can use Eqs. (18.40) through (18.42), replacing $\mathbf{H}_O$ with $\mathbf{H}_G$, and express the moment of the couple as

$$\dot{\mathbf{H}}_G = \boldsymbol{\Omega} \times \mathbf{H}_G$$

You can then use the free-body and kinetic diagrams to write as many as six independent scalar equations.

2. Motion of an axisymmetric body under no force, except its own weight. We have $\Sigma \mathbf{M}_G = 0$ and thus $\dot{\mathbf{H}}_G = 0$; it follows that the angular momentum $\mathbf{H}_G$ is constant in magnitude and direction (Sec. 18.3C). The body is in **steady precession** with the precession axis GZ directed along $\mathbf{H}_G$ (Fig. 18.20). Using the rotating frame $Gxyz$ and denoting by γ the angle that $\boldsymbol{\omega}$ forms with the spin axis Gz (Fig. 18.21), we obtained the relation between γ and the angle θ formed by the precession and spin axes as

$$\tan \gamma = \frac{I}{I'} \tan \theta \quad (18.49)$$

The precession is said to be *direct* if $I < I'$ (Fig. 18.23) and *retrograde* if $I > I'$ (Fig. 18.24).

a. In many problems dealing with the motion of an axisymmetric body under no force, you will be asked to determine the precession axis and the rates of precession and spin of the body when given the magnitude of its angular velocity $\boldsymbol{\omega}$ and the angle γ that it forms with the axis of symmetry Gz (Fig. 18.21). From Eq. (18.49), determine the angle θ that the precession axis GZ forms with Gz and resolve $\boldsymbol{\omega}$ into its two oblique components $\dot{\phi}\mathbf{K}$ and $\dot{\psi}\mathbf{k}$. Using the law of sines, you then can determine the rate of precession $\dot{\phi}$ and the rate of spin $\dot{\psi}$.

b. In other problems, the body is subjected to a given impulse and you will first determine the resulting angular momentum $\mathbf{H}_G$. Using Eqs. (18.10), you can calculate the rectangular components of the angular velocity $\boldsymbol{\omega}$, its magnitude ω , and the angle γ that it forms with the axis of symmetry. You then determine the precession axis and the rates of precession and spin as described previously [Sample Prob. 18.6].

3. General motion of an axisymmetric body with a fixed point O located on its axis of symmetry, and subjected only to its own weight. This is a motion in which the angle θ is allowed to vary. At any given instant you should take into account the rate of precession $\dot{\phi}$, the rate of spin $\dot{\psi}$, and the rate of nutation $\dot{\theta}$ —none of which will remain constant. An example of such a motion is the motion of a top, which is discussed in Probs. 18.137 and 18.138. The rotating frame of reference $Oxyz$ that you will use is still the one shown in Fig. 18.18, but this frame now rotates about the y axis at the rate $\dot{\theta}$. Equations (18.40), (18.41), and (18.42), therefore, should be replaced by

$$\boldsymbol{\omega} = -\dot{\phi} \sin \theta \mathbf{i} + \dot{\theta} \mathbf{j} + (\dot{\psi} + \dot{\phi} \cos \theta) \mathbf{k} \quad (18.40')$$

$$\mathbf{H}_O = -I' \dot{\phi} \sin \theta \mathbf{i} + I' \dot{\theta} \mathbf{j} + I(\dot{\psi} + \dot{\phi} \cos \theta) \mathbf{k} \quad (18.41')$$

$$\boldsymbol{\Omega} = -\dot{\phi} \sin \theta \mathbf{i} + \dot{\theta} \mathbf{j} + \dot{\phi} \cos \theta \mathbf{k} \quad (18.42')$$

Since substituting these expressions into Eq. (18.44) would lead to nonlinear differential equations, it is preferable, whenever feasible, to apply the following conservation principles.

a. Conservation of energy. Denoting the distance between the fixed point O and the mass center G of the body by c and the total energy by E , you have

$$T + V = E: \quad \frac{1}{2}(I'\omega_x^2 + I'\omega_y^2 + I\omega_z^2) + mgc \cos \theta = E$$

Then substitute the expressions obtained in Eq. (18.40') for the components of $\boldsymbol{\omega}$. Note that c is positive or negative depending upon the position of G relative to O . Also, $c = 0$ if G coincides with O ; the kinetic energy is then conserved.

b. Conservation of the angular momentum about the axis of precession.

Since the support at O is located on the Z axis and the weight of the body and the Z axis are both vertical and thus are parallel to each other, it follows that $\Sigma M_Z = 0$. Thus, H_Z remains constant. We can express this by writing that the scalar product $\mathbf{K} \cdot \mathbf{H}_O$ is constant, where $\mathbf{K}$ is the unit vector along the Z axis.

c. Conservation of the angular momentum about the axis of spin. Since the support at O and the center of gravity G are both located on the z axis, it follows that $\Sigma M_z = 0$ and, thus, that H_z remains constant. Thus, the coefficient of the unit vector $\mathbf{k}$ in Eq. (18.41') is constant. Note that this last conservation principle cannot be applied when the body is restrained from spinning about its axis of symmetry, but in that case, the only variables are θ and ϕ .

Problems

18.107 A uniform thin disk with a 6-in. diameter is attached to the end of a rod AB of negligible mass that is supported by a ball-and-socket joint at point A . Knowing that the disk is observed to precess about the vertical axis AC at the constant rate of 36 rpm in the sense indicated and that its axis of symmetry AB forms an angle $\beta = 60^\circ$ with AC , determine the rate at which the disk spins about rod AB .

18.108 A uniform thin disk with a 6-in. diameter is attached to the end of a rod AB of negligible mass that is supported by a ball-and-socket joint at point A . Knowing that the disk is spinning about its axis of symmetry AB at the rate of 2100 rpm in the sense indicated and that AB forms an angle $\beta = 45^\circ$ with the vertical axis AC , determine the two possible rates of steady precession of the disk about the axis AC .

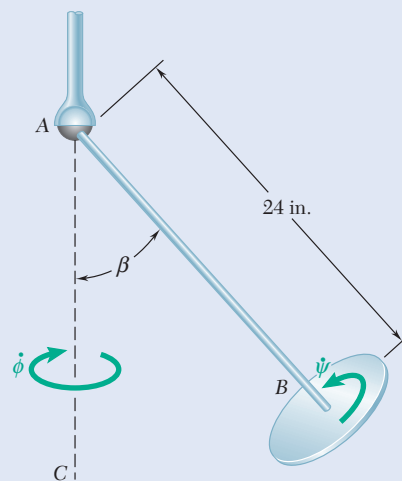


Fig. P18.107 and P18.108

18.109 The 85-g top shown is supported at the fixed point O . The radii of gyration of the top with respect to its axis of symmetry and with respect to a transverse axis through O are 21 mm and 45 mm, respectively. Knowing that $c = 37.5$ mm and that the rate of spin of the top about its axis of symmetry is 1800 rpm, determine the two possible rates of steady precession corresponding to $\theta = 30^\circ$.

18.110 The top shown is supported at the fixed point O and its moments of inertia about its axis of symmetry and about a transverse axis through O are denoted, respectively, by I and I' . (a) Show that the condition for steady precession of the top is

$$(I\omega_z - I'\dot{\phi} \cos \theta) \dot{\phi} = Wc$$

where $\dot{\phi}$ is the rate of precession and ω_z is the rectangular component of the angular velocity along the axis of symmetry of the top. (b) Show that if the rate of spin $\dot{\psi}$ of the top is very large compared with its rate of precession $\dot{\phi}$, the condition for steady precession is $I\dot{\psi}\dot{\phi} \approx Wc$. (c) Determine the percentage error introduced when this last relation is used to approximate the slower of the two rates of precession obtained for the top of Prob. 18.109.

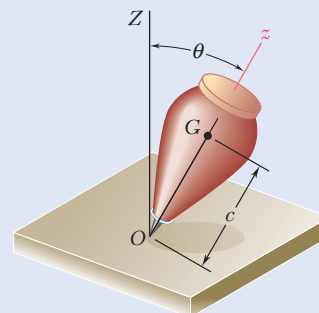


Fig. P18.109 and P18.110

18.111 A solid aluminum sphere of radius 4 in. is welded to the end of a 10-in.-long rod AB of negligible mass that is supported by a ball-and-socket joint at A . Knowing that the sphere is observed to precess about a vertical axis at the constant rate of 60 rpm in the sense indicated and that rod AB forms an angle $\beta = 20^\circ$ with the vertical, determine the rate of spin of the sphere about line AB .

18.112 A solid aluminum sphere of radius 4 in. is welded to the end of a 10-in.-long rod AB of negligible mass that is supported by a ball-and-socket joint at A . Knowing that the sphere spins as shown about line AB at the rate of 600 rpm, determine the angle β for which the sphere will precess about a vertical axis at the constant rate of 60 rpm in the sense indicated.

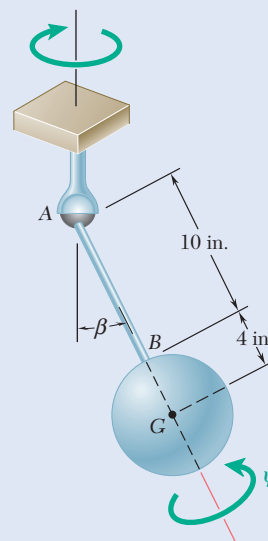


Fig. P18.111 and P18.112

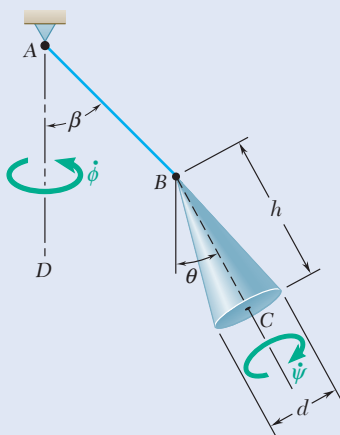


Fig. P18.113 and P18.114

18.113 A homogeneous cone with a height h and a base with a diameter $d < h$ is attached as shown to a cord AB . The cone spins about its axis BC at the constant rate $\dot{\psi}$ and precesses about the vertical through A at the constant rate $\dot{\phi}$. Determine the angle β for which the axis BC of the cone is aligned with cord AB ($\theta = \beta$).

18.114 A homogeneous cone with a height of $h = 12$ in. and a base with a diameter of $d = 6$ in. is attached as shown to a cord AB . Knowing that the angles that cord AB and the axis BC of the cone form with the vertical are, respectively, $\beta = 45^\circ$ and $\theta = 30^\circ$ and that the cone precesses at the constant rate $\dot{\phi} = 8$ rad/s in the sense indicated, determine (a) the rate of spin $\dot{\psi}$ of the cone about its axis BC , (b) the length of cord AB .

18.115 A solid cube of side $c = 80$ mm is attached as shown to cord AB . It is observed to spin at the rate $\dot{\psi} = 40$ rad/s about its diagonal BC and to precess at the constant rate $\dot{\phi} = 5$ rad/s about the vertical axis AD . Knowing that $\beta = 30^\circ$, determine the angle θ that the diagonal BC forms with the vertical. (Hint: The moment of inertia of a cube about an axis through its center is independent of the orientation of that axis.)

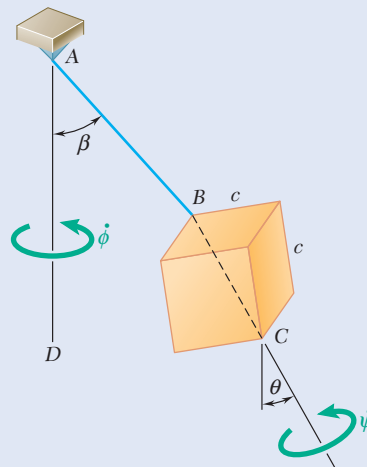


Fig. P18.115 and P18.116

18.116 A solid cube of side $c = 120$ mm is attached as shown to a cord AB of length 240 mm. The cube spins about its diagonal BC and precesses about the vertical axis AD . Knowing that $\theta = 25^\circ$ and $\beta = 40^\circ$, determine (a) the rate of spin of the cube, (b) its rate of precession. (See hint of Prob. 18.115.)

18.117 A high-speed photographic record shows that a certain projectile was fired with a horizontal velocity $\bar{v}$ of 2000 ft/s and with its axis of symmetry forming an angle $\beta = 3^\circ$ with the horizontal. The rate of spin $\dot{\psi}$ of the projectile was 6000 rpm, and the atmospheric drag was equivalent to a force $\mathbf{D}$ of 25 lb acting at the center of pressure C_P located at a distance $c = 6$ in. from G . (a) Knowing that the projectile has a weight of 45 lb and a radius of gyration of 2 in. with respect to its axis of symmetry, determine its approximate rate of steady precession. (b) If it is further known that the radius of gyration of the projectile with respect to a transverse axis through G is 8 in., determine the exact values of the two possible rates of precession.

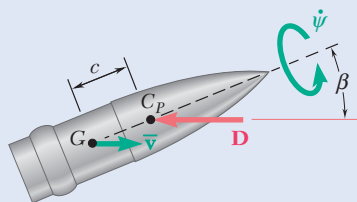


Fig. P18.117

18.118 If the earth were a sphere, the gravitational attraction of the sun, moon, and planets would at all times be equivalent to a single force $\mathbf{R}$ acting at the mass center of the earth. However, the earth is actually an oblate spheroid and the gravitational system acting on the earth is equivalent to a force $\mathbf{R}$ and a couple $\mathbf{M}$. Knowing that the effect of the couple $\mathbf{M}$ is to cause the axis of the earth to precess about the axis GA at the rate of one revolution in 25 800 years, determine the average magnitude of the couple $\mathbf{M}$ applied to the earth. Assume that the average density of the earth is 5.51 g/cm^3 , that the average radius of the earth is 6370 km, and that $\bar{I} = \frac{2}{5}mR^2$. (Note: This forced precession is known as the precession of the equinoxes and is not to be confused with the free precession discussed in Prob. 18.123.)

18.119 Show that for an axisymmetric body under no force, the rates of precession and spin can be expressed, respectively, as

$$\dot{\phi} = \frac{H_G}{I'}$$

and

$$\dot{\psi} = \frac{H_G \cos \theta (I' - I)}{II'}$$

where H_G is the constant value of the angular momentum of the body.

18.120 (a) Show that for an axisymmetric body under no force, the rate of precession can be expressed as

$$\dot{\phi} = \frac{I\omega_2}{I' \cos \theta}$$

where ω_2 is the rectangular component of $\boldsymbol{\omega}$ along the axis of symmetry of the body. (b) Use this result to check that the condition (18.44) for steady precession is satisfied by an axisymmetric body under no force.

18.121 Show that the angular velocity vector $\boldsymbol{\omega}$ of an axisymmetric body under no force is observed from the body itself to rotate about the axis of symmetry at the constant rate

$$n = \frac{I' - I}{I'} \omega_2$$

where ω_2 is the rectangular component of $\boldsymbol{\omega}$ along the axis of symmetry of the body.

18.122 For an axisymmetric body under no force, prove (a) that the rate of retrograde precession can never be less than twice the rate of spin of the body about its axis of symmetry, (b) that in Fig. 18.24 the axis of symmetry of the body can never lie within the space cone.

18.123 Using the relation given in Prob. 18.121, determine the period of precession of the north pole of the earth about the axis of symmetry of the earth. The earth may be approximated by an oblate spheroid of axial moment of inertia I and of transverse moment of inertia $I' = 0.9967I$. (Note: Actual observations show a period of precession of the north pole of about 432.5 mean solar days; the difference between the observed and computed periods is due to the fact that the earth is not a perfectly rigid body. The free precession considered here should not be confused with the much slower precession of the equinoxes, which is a forced precession. See Prob. 18.118.)

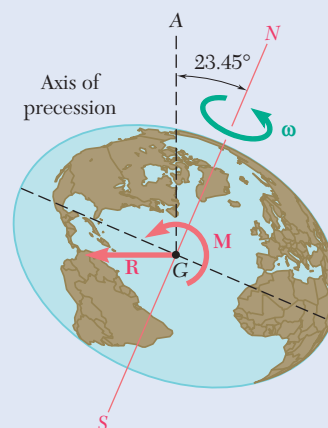


Fig. P18.118

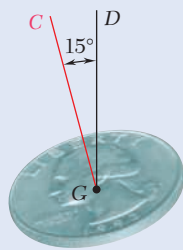


Fig. P18.124

18.124 A coin is tossed into the air. It is observed to spin at the rate of 600 rpm about an axis GC perpendicular to the coin and to precess about the vertical direction GD . Knowing that GC forms an angle of 15° with GD , determine (a) the angle that the angular velocity ω of the coin forms with GD , (b) the rate of precession of the coin about GD .

18.125 The angular velocity vector of a football that has just been kicked is horizontal, and its axis of symmetry OC is oriented as shown. Knowing that the magnitude of the angular velocity is 200 rpm and that the ratio of the axis and transverse moments of inertia is $I/I' = \frac{1}{3}$, determine (a) the orientation of the axis of precession OA , (b) the rates of precession and spin.

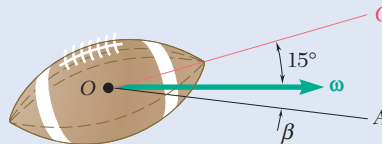


Fig. P18.125

18.126 A space station consists of two sections A and B of equal masses that are rigidly connected. Each section is dynamically equivalent to a homogeneous cylinder with a length of 15 m and a radius of 3 m. Knowing that the station is precessing about the fixed direction GD at the constant rate of 2 rev/h, determine the rate of spin of the station about its axis of symmetry CC' .

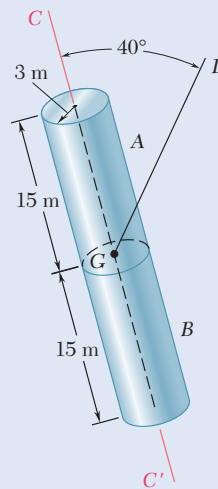


Fig. P18.126 and P18.127

18.127 If the connection between sections A and B of the space station of Prob. 18.126 is severed when the station is oriented as shown and if the two sections are gently pushed apart along their common axis of symmetry, determine (a) the angle between the spin axis and the new precession axis of section A , (b) the rate of precession of section A , (c) its rate of spin.

18.128 Solve Sample Prob. 18.6, assuming that the meteorite strikes the satellite at C with a velocity $\mathbf{v}_0 = (2000 \text{ m/s})\mathbf{i}$.

18.129 An 800-lb geostationary satellite is spinning with an angular velocity $\omega_0 = (1.5 \text{ rad/s})\mathbf{j}$ when it is hit at B by a 6-oz meteorite traveling with a velocity $\mathbf{v}_0 = -(1600 \text{ ft/s})\mathbf{i} + (1300 \text{ ft/s})\mathbf{j} + (4000 \text{ ft/s})\mathbf{k}$ relative to the satellite. Knowing that $b = 20 \text{ in.}$ and that the radii of gyration of the satellite are $\bar{k}_x = \bar{k}_z = 28.8 \text{ in.}$ and $\bar{k}_y = 32.4 \text{ in.}$, determine the precession axis and the rates of precession and spin of the satellite after the impact.

18.130 Solve Prob. 18.129, assuming that the meteorite hits the satellite at A instead of B .

18.131 A homogeneous rectangular plate of mass m and sides c and $2c$ is held at A and B by a fork-ended shaft of negligible mass that is supported by a bearing at C . The plate is free to rotate about AB , and the shaft is free to rotate about a horizontal axis through C . Knowing that, initially, $\theta_0 = 40^\circ$, $\dot{\theta}_0 = 0$, and $\dot{\phi}_0 = 10 \text{ rad/s}$, determine for the ensuing motion (a) the range of values of θ , (b) the minimum value of $\dot{\phi}$, (c) the maximum value of $\dot{\theta}$.

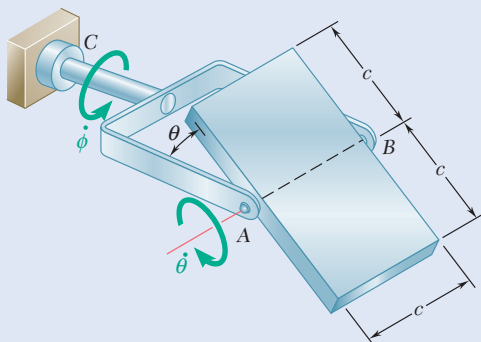


Fig. P18.131 and P18.132

18.132 A homogeneous rectangular plate of mass m and sides c and $2c$ is held at A and B by a fork-ended shaft of negligible mass that is supported by a bearing at C . The plate is free to rotate about AB , and the shaft is free to rotate about a horizontal axis through C . Initially the plate lies in the plane of the fork ($\theta_0 = 0$) and the shaft has an angular velocity $\dot{\phi}_0 = 10 \text{ rad/s}$. If the plate is slightly disturbed, determine for the ensuing motion (a) the minimum value of $\dot{\phi}$, (b) the maximum value of θ .

18.133 A homogeneous square plate with a mass m and side c is held at points A and B by a frame of negligible mass that is supported by bearings at points C and D . The plate is free to rotate about AB , and the frame is free to rotate about the vertical CD . Knowing that, initially, $\theta_0 = 45^\circ$, $\dot{\theta}_0 = 0$, and $\dot{\phi}_0 = 8 \text{ rad/s}$, determine for the ensuing motion (a) the range of values of θ , (b) the minimum value of $\dot{\phi}$, (c) the maximum value of $\dot{\theta}$.

18.134 A homogeneous square plate with a mass m and side c is held at points A and B by a frame of negligible mass that is supported by bearings at points C and D . The plate is free to rotate about AB , and the frame is free to rotate about the vertical CD . Initially, the plate lies in the plane of the frame ($\theta_0 = 90^\circ$), and the frame has an angular velocity of $\dot{\phi} = 8 \text{ rad/s}$. If the plate is slightly disturbed, determine for the ensuing motion (a) the minimum value of $\dot{\phi}$, (b) the maximum value of $\dot{\theta}$.

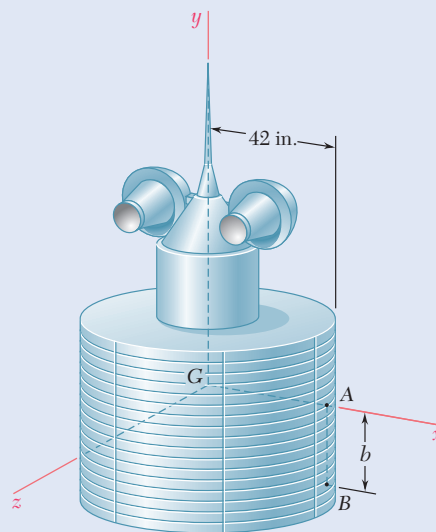


Fig. P18.129

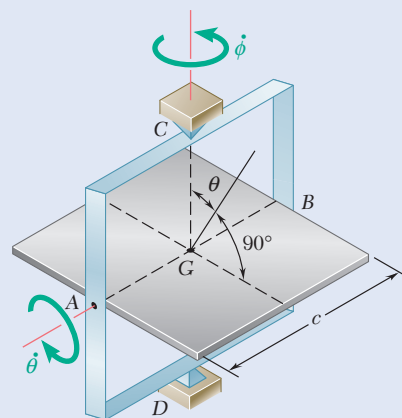


Fig. P18.133 and P18.134

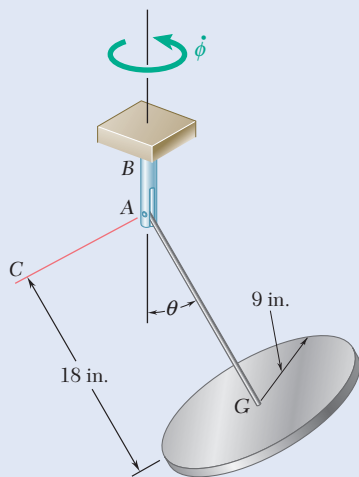


Fig. P18.135 and P18.136

18.135 A homogeneous disk with a radius of 9 in. is welded to a rod AG with a length of 18 in. and of negligible weight that is connected by a clevis to a vertical shaft AB . The rod and disk can rotate freely about a horizontal axis AC , and shaft AB can rotate freely about a vertical axis. Initially, rod AG is horizontal ($\theta_0 = 90^\circ$) and has no angular velocity about AC . Knowing that the maximum value $\dot{\phi}_m$ of the angular velocity of shaft AB in the ensuing motion is twice its initial value $\dot{\phi}_0$, determine (a) the minimum value of θ , (b) the initial angular velocity $\dot{\phi}_0$ of shaft AB .

18.136 A homogeneous disk with a radius of 9 in. is welded to a rod AG with a length of 18 in. and of negligible weight that is connected by a clevis to a vertical shaft AB . The rod and disk can rotate freely about a horizontal axis AC , and shaft AB can rotate freely about a vertical axis. Initially, rod AG is horizontal ($\theta_0 = 90^\circ$) and has no angular velocity about AC . Knowing that the smallest value of θ in the ensuing motion is 30° , determine (a) the initial angular velocity of shaft AB , (b) its maximum angular velocity.

***18.137** The top shown is supported at the fixed point O . Denoting by ϕ , θ , and ψ the Eulerian angles defining the position of the top with respect to a fixed frame of reference, consider the general motion of the top in which all Eulerian angles vary.

(a) Observing that $\Sigma M_Z = 0$ and $\Sigma M_z = 0$, and denoting by I and I' , respectively, the moments of inertia of the top about its axis of symmetry and about a transverse axis through O , derive the two first-order differential equations of motion

$$I' \dot{\phi} \sin^2 \theta + I(\dot{\psi} + \dot{\phi} \cos \theta) \cos \theta = \alpha \quad (1)$$

$$I(\dot{\psi} + \dot{\phi} \cos \theta) = \beta \quad (2)$$

where α and β are constants depending upon the initial conditions. These equations express that the angular momentum of the top is conserved about both the Z and z axes, i.e., that the rectangular component of $\mathbf{H}_O$ along each of these axes is constant.

(b) Use Eqs. (1) and (2) to show that the rectangular component ω_z of the angular velocity of the top is constant and that the rate of precession $\dot{\phi}$ depends upon the value of the angle of nutation θ .

***18.138** (a) Applying the principle of conservation of energy, derive a third differential equation for the general motion of the top of Prob. 18.137.

(b) Eliminating the derivatives $\dot{\phi}$ and $\dot{\psi}$ from the equation obtained and from the two equations of Prob. 18.137, show that the rate of nutation $\dot{\theta}$ is defined by the differential equation $\dot{\theta}^2 = f(\theta)$, where

$$f(\theta) = \frac{1}{I'} \left(2E - \frac{\beta^2}{I} - 2mgc \cos \theta \right) - \left(\frac{\alpha - \beta \cos \theta}{I' \sin \theta} \right)^2 \quad (1)$$

(c) Further show, by introducing the auxiliary variable $x = \cos \theta$, that the maximum and minimum values of θ can be obtained by solving for x the cubic equation

$$\left(2E - \frac{\beta^2}{I} - 2mgcx \right) (1 - x^2) - \frac{1}{I'} (\alpha - \beta x)^2 = 0 \quad (2)$$

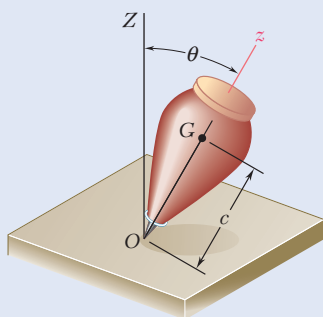


Fig. P18.137 and P18.138

- *18.139** A solid cone of height 180 mm with a circular base of radius 60 mm is supported by a ball and socket at A. The cone is released from the position $\theta_0 = 30^\circ$ with a rate of spin $\dot{\psi}_0 = 300$ rad/s, a rate of precession $\dot{\phi}_0 = 20$ rad/s, and a zero rate of nutation. Determine (a) the maximum value of θ in the ensuing motion, (b) the corresponding values of the rates of spin and precession. [Hint: Use Eq. (2) of Prob. 18.138; you can either solve this equation numerically or reduce it to a quadratic equation, since one of its roots is known.]

- *18.140** A solid cone of height 180 mm with a circular base of radius 60 mm is supported by a ball and socket at A. The cone is released from the position $\theta_0 = 30^\circ$ with a rate of spin $\dot{\psi}_0 = 300$ rad/s, a rate of precession $\dot{\phi}_0 = -4$ rad/s, and a zero rate of nutation. Determine (a) the maximum value of θ in the ensuing motion, (b) the corresponding values of the rates of spin and precession, (c) the value of θ for which the sense of the precession is reversed. (See hint of Prob. 18.139.)

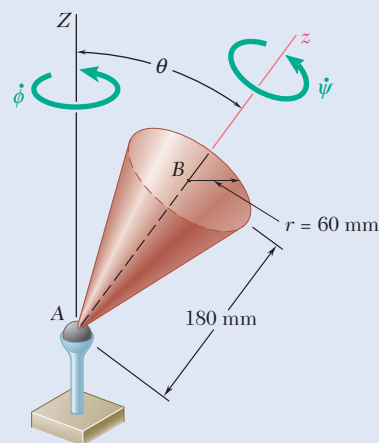


Fig. P18.139 and P18.140

- *18.141** A homogeneous sphere of mass m and radius a is welded to a rod AB of negligible mass, which is held by a ball-and-socket support at A. The sphere is released in the position $\beta = 0$ with a rate of precession $\dot{\phi} = \sqrt{17 g/11a}$ with no spin or nutation. Determine the largest value of β in the ensuing motion.

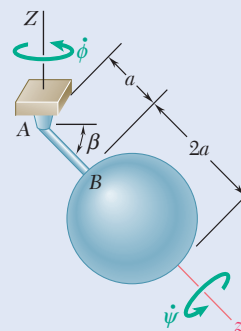


Fig. P18.141 and P18.142

- *18.142** A homogeneous sphere of mass m and radius a is welded to a rod AB of negligible mass, which is held by a ball-and-socket support at A. The sphere is released in the position $\beta = 0$ with a rate of precession $\dot{\phi} = \dot{\phi}_0$ with no spin or nutation. Knowing that the largest value of β in the ensuing motion is 30° , determine (a) the rate of precession $\dot{\phi}_0$ of the sphere in its initial position, (b) the rates of precession and spin when $\beta = 30^\circ$.

- *18.143** Consider a rigid body of arbitrary shape that is attached at its mass center O and subjected to no force other than its weight and the reaction of the support at O .

(a) Prove that the angular momentum $\mathbf{H}_O$ of the body about the fixed point O is constant in magnitude and direction, that the kinetic energy T of the body is constant, and that the projection along $\mathbf{H}_O$ of the angular velocity $\boldsymbol{\omega}$ of the body is constant.

(b) Show that the tip of the vector $\boldsymbol{\omega}$ describes a curve on a fixed plane in space (called the *invariable plane*), which is perpendicular to $\mathbf{H}_O$ and at a distance $2T/H_O$ from O .

(c) Show that with respect to a frame of reference attached to the body and coinciding with its principal axes of inertia, the tip of the vector $\boldsymbol{\omega}$ appears to describe a curve on an ellipsoid of equation

$$I_x \omega_x^2 + I_y \omega_y^2 + I_z \omega_z^2 = 2T = \text{constant}$$

The ellipsoid (called the *Poinsot ellipsoid*) is rigidly attached to the body and is of the same shape as the ellipsoid of inertia, but of a different size.

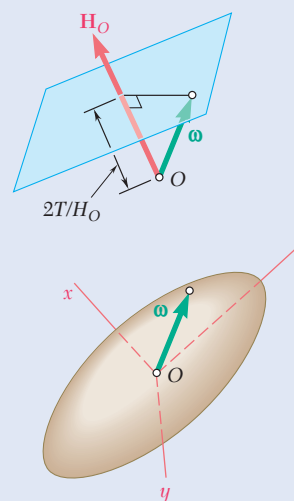


Fig. P18.143

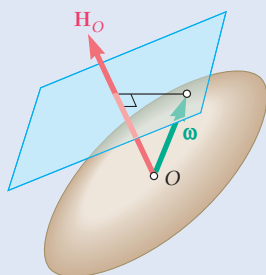


Fig. P18.144

***18.144** Referring to Prob. 18.143, (a) prove that the Poincaré ellipsoid is tangent to the invariable plane, (b) show that the motion of the rigid body must be such that the Poincaré ellipsoid appears to roll on the invariable plane. [Hint: In part a, show that the normal to the Poincaré ellipsoid at the tip of ω is parallel to $\mathbf{H}_O$. It is recalled that the direction of the normal to a surface of equation $F(x, y, z) = \text{constant}$ at a point P is the same as that of $\text{grad } F$ at point P .]

***18.145** Using the results obtained in Probs. 18.143 and 18.144, show that for an axisymmetric body attached at its mass center O and under no force other than its weight and the reaction at O , the Poincaré ellipsoid is an ellipsoid of revolution and the space and body cones are both circular and are tangent to each other. Further show that (a) the two cones are tangent externally, and the precession is direct, when $I < I'$, where I and I' denote, respectively, the axial and transverse moment of inertia of the body, (b) the space cone is inside the body cone, and the precession is retrograde, when $I > I'$.

***18.146** Refer to Probs. 18.143 and 18.144.

(a) Show that the curve (called *polhode*) described by the tip of the vector ω with respect to a frame of reference coinciding with the principal axes of inertia of the rigid body is defined by the equations

$$I_x \omega_x^2 + I_y \omega_y^2 + I_z \omega_z^2 = 2T = \text{constant} \quad (1)$$

$$I_x^2 \omega_x^2 + I_y^2 \omega_y^2 + I_z^2 \omega_z^2 = H_O^2 = \text{constant} \quad (2)$$

and that this curve can, therefore, be obtained by intersecting the Poincaré ellipsoid with the ellipsoid defined by Eq. (2).

(b) Further show, assuming $I_x > I_y > I_z$, that the polhodes obtained for various values of H_O have the shapes indicated in the figure.

(c) Using the result obtained in part b, show that a rigid body under no force can rotate about a fixed centroidal axis if, and only if, that axis coincides with one of the principal axes of inertia of the body, and that the motion will be stable if the axis of rotation coincides with the major or minor axis of the Poincaré ellipsoid (z or x axis in the figure) and unstable if it coincides with the intermediate axis (y axis).

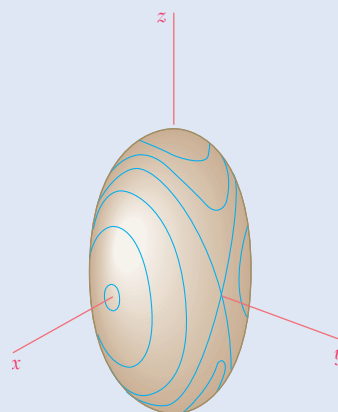


Fig. P18.146

Review and Summary

This chapter was devoted to the kinetic analysis of the motion of rigid bodies in three dimensions.

Fundamental Equations of Motion for a Rigid Body

We first noted that the two fundamental equations derived in Chap. 14 for the motion of a system of particles,

$$\Sigma \mathbf{F} = m\bar{\mathbf{a}} \quad (18.1)$$

$$\Sigma \mathbf{M}_G = \dot{\mathbf{H}}_G \quad (18.2)$$

provide the foundation of our analysis, just as they did in Chap. 16 in the case of the plane motion of rigid bodies. The computation of the angular momentum $\mathbf{H}_G$ of the body and of its derivative $\dot{\mathbf{H}}_G$, however, are now considerably more involved.

Angular Momentum of a Rigid Body in Three Dimensions

In Sec. 18.1A, we saw that we can express the rectangular components of the angular momentum $\mathbf{H}_G$ of a rigid body in terms of the components of its angular velocity $\boldsymbol{\omega}$ and of its centroidal moments and products of inertia:

$$\begin{aligned} H_x &= +\bar{I}_x \omega_x - \bar{I}_{xy} \omega_y - \bar{I}_{xz} \omega_z \\ H_y &= -\bar{I}_{yx} \omega_x + \bar{I}_y \omega_y - \bar{I}_{yz} \omega_z \\ H_z &= -\bar{I}_{zx} \omega_x - \bar{I}_{zy} \omega_y + \bar{I}_z \omega_z \end{aligned} \quad (18.7)$$

If we use **principal axes of inertia** $Gx'y'z'$, these relations reduce to

$$H_{x'} = \bar{I}_{x'} \omega_{x'} \quad H_{y'} = \bar{I}_{y'} \omega_{y'} \quad H_{z'} = \bar{I}_{z'} \omega_{z'} \quad (18.10)$$

We observed that, in general, *the angular momentum $\mathbf{H}_G$ and the angular velocity $\boldsymbol{\omega}$ do not have the same direction* (Fig. 18.25). They do, however, have the same direction if $\boldsymbol{\omega}$ is directed along one of the principal axes of inertia of the body.

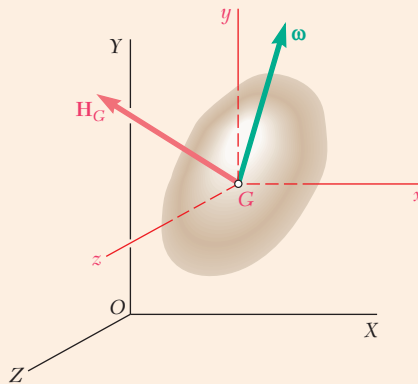


Fig. 18.25

Angular Momentum About a Given Point

Recalling that the system of the momenta of the particles forming a rigid body can be reduced to the vector $m\bar{\mathbf{v}}$ attached at G and the couple $\mathbf{H}_G$ (Fig. 18.26), we noted that, once we have determined the linear momentum $m\bar{\mathbf{v}}$ and the angular momentum $\mathbf{H}_G$ of a rigid body, we can obtain the angular momentum $\mathbf{H}_O$ of the body about any given point O from

$$\mathbf{H}_O = \bar{\mathbf{r}} \times m\bar{\mathbf{v}} + \mathbf{H}_G \quad (18.11)$$

Rigid Body with a Fixed Point

In the particular case of a rigid body *constrained to rotate about a fixed point* O , we can obtain the components of the angular momentum $\mathbf{H}_O$ of the body about O directly from the components of its angular velocity and from its moments and products of inertia with respect to axes through O .

$$\begin{aligned} H_x &= +I_x \omega_x - I_{xy} \omega_y - I_{xz} \omega_z \\ H_y &= -I_{yx} \omega_x + I_y \omega_y - I_{yz} \omega_z \\ H_z &= -I_{zx} \omega_x - I_{zy} \omega_y + I_z \omega_z \end{aligned} \quad (18.13)$$

Principle of Impulse and Momentum

The *principle of impulse and momentum* for a rigid body in three-dimensional motion [Sec. 18.1B] is expressed by the same fundamental formula that was used in Chap. 17 for a rigid body in plane motion as

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2 \quad (17.14)$$

However, the systems of the initial and final momenta should now be represented as shown in Fig. 18.26, and $\mathbf{H}_G$ should be computed from the relations in Eqs. (18.7) or (18.10) [Sample Probs. 18.1 and 18.2].

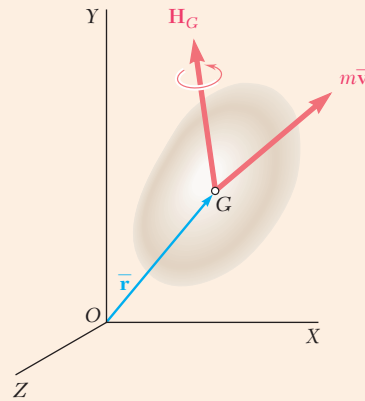


Fig. 18.26

Kinetic Energy of a Rigid Body in Three Dimensions

The kinetic energy of a rigid body in three-dimensional motion can be divided into two parts [Sec. 18.1C]: one associated with the motion of its mass center G and the other with its motion about G . Using principal centroidal axes x', y', z' , we wrote

$$T = \frac{1}{2} m \bar{v}^2 + \frac{1}{2} (\bar{I}_{x'} \omega_{x'}^2 + \bar{I}_{y'} \omega_{y'}^2 + \bar{I}_{z'} \omega_{z'}^2) \quad (18.17)$$

where $\bar{\mathbf{v}}$ = velocity of mass center

$\boldsymbol{\omega}$ = angular velocity

m = mass of rigid body

$\bar{I}_{x'}, \bar{I}_{y'}, \bar{I}_{z'}$ = principal centroidal moments of inertia

We also noted that, in the case of a rigid body *constrained to rotate about a fixed point* O , we can express the kinetic energy of the body as

$$T = \frac{1}{2}(I_{x'}\omega_{x'}^2 + I_{y'}\omega_{y'}^2 + I_{z'}\omega_{z'}^2) \quad (18.20)$$

where the x' , y' , and z' axes are the principal axes of inertia of the body about O . These results make it possible to extend the application of the principle of work and energy and the principle of conservation of energy to the three-dimensional motion of a rigid body.

Using a Rotating Frame to Write the Equations of Motion of a Rigid Body in Space

Section 18.2 was devoted to applying the fundamental equations

$$\Sigma \mathbf{F} = m\bar{\mathbf{a}} \quad (18.1)$$

$$\Sigma \mathbf{M}_G = \dot{\mathbf{H}}_G \quad (18.2)$$

to the motion of a rigid body in three dimensions. We first recalled [Sec. 18.2A] that $\mathbf{H}_G$ represents the angular momentum of the body relative to a centroidal frame $GX'Y'Z'$ of fixed orientation (Fig. 18.27) and that $\dot{\mathbf{H}}_G$ in Eq. (18.2)

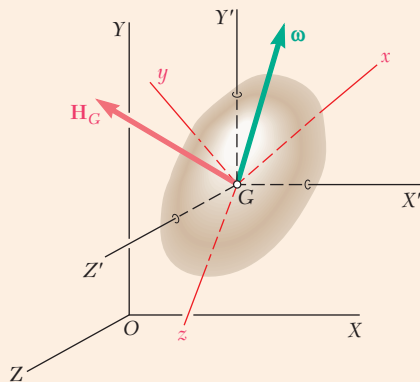


Fig. 18.27

represents the rate of change of $\mathbf{H}_G$ with respect to that frame. We noted that, as the body rotates, its moments and products of inertia with respect to the frame $GX'Y'Z'$ change continually. Therefore, it is more convenient to use a rotating frame $Gxyz$ when resolving $\boldsymbol{\omega}$ into components and computing the moments and products of inertia that are used to determine $\mathbf{H}_G$ from Eqs. (18.7) or (18.10). However, since $\dot{\mathbf{H}}_G$ in Eq. (18.2) represents the rate of change of $\mathbf{H}_G$ with respect to the frame $GX'Y'Z'$ of fixed orientation, we must use the method of Sec. 15.5A to determine its value. Recalling Eq. (15.31), we wrote

$$\dot{\mathbf{H}}_G = (\dot{\mathbf{H}}_G)_{Gxyz} + \boldsymbol{\Omega} \times \mathbf{H}_G \quad (18.22)$$

where $\mathbf{H}_G$ = angular momentum of body with respect to frame $GX'Y'Z'$ of fixed orientation

$(\dot{\mathbf{H}}_G)_{Gxyz}$ = rate of change of $\mathbf{H}_G$ with respect to rotating frame $Gxyz$ to be computed from relations in Eq. (18.7)

$\boldsymbol{\Omega}$ = angular velocity of the rotating frame $Gxyz$

Substituting for $\dot{\mathbf{H}}_G$ from Eq. (18.22) into Eq. (18.2), we obtained

$$\Sigma \mathbf{M}_G = (\dot{\mathbf{H}}_G)_{Gxyz} + \boldsymbol{\Omega} \times \mathbf{H}_G \quad (18.23)$$

If the rotating frame is actually attached to the body, its angular velocity Ω is identically equal to the angular velocity ω of the body. In many applications, however, it is advantageous to use a frame of reference that is not attached to the body but rotates in an independent manner [Sample Prob. 18.5].

Euler's Equations of Motion

Setting $\Omega = \omega$ in Eq. (18.23), using principal axes, and writing this equation in scalar form, we obtained **Euler's equations of motion** [Sec. 18.2B]. Then we extended Newton's second law to the three-dimensional motion of a rigid body, showing that the system of the external forces acting on the rigid body is not only equipollent but actually *equivalent* to the inertial terms of the body represented by the vector $m\bar{a}$ and the couple $\dot{\mathbf{H}}_G$ (Fig. 18.28). You can solve problems involving the three-dimensional motion of a rigid body by considering the free-body and kinetic diagrams represented in Fig. 18.28 and writing appropriate scalar equations relating the components or moments of the external forces and inertial terms [Sample Probs. 18.3 and 18.5].

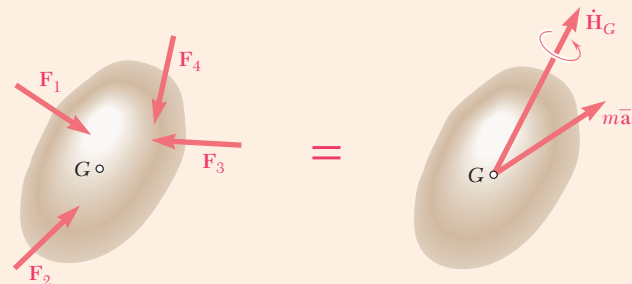


Fig. 18.28

Rigid Body with a Fixed Point

In the case of a rigid body *constrained to rotate about a fixed point* O , we can use an alternative method of solution involving the moments of the forces and the rate of change of the angular momentum about point O . We wrote [Sec. 18.2C]

$$\Sigma \mathbf{M}_O = (\dot{\mathbf{H}}_O)_{Oxyz} + \Omega \times \mathbf{H}_O \quad (18.28)$$

You can use this approach to solve some types of problems involving the rotation of a rigid body about a fixed axis [Sec. 18.2D]; for example, an unbalanced rotating shaft [Sample Prob. 18.4].

Motion of a Gyroscope

In Section 18.3, we considered the motion of **gyroscopes** and other **axisymmetric bodies**. We introduced the **Eulerian angles** ϕ , θ , and ψ to define the position of a gyroscope (Fig. 18.29), and we observed that their derivatives $\dot{\phi}$, $\dot{\theta}$, and $\dot{\psi}$ represent, respectively, the rates of **precession**, **nutation**, and **spin** of the gyroscope [Sec. 18.3A]. Expressing the angular velocity ω in terms of these derivatives, we wrote

$$\omega = -\dot{\phi} \sin \theta \mathbf{i} + \dot{\theta} \mathbf{j} + (\dot{\psi} + \dot{\phi} \cos \theta) \mathbf{k} \quad (18.35)$$

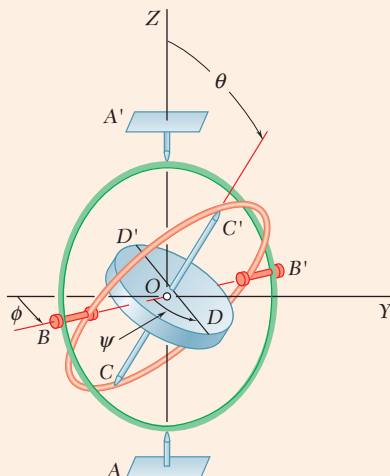


Fig. 18.29

where the unit vectors are associated with a frame $Oxyz$ attached to the inner gimbal of the gyroscope (Fig. 18.30). These vectors rotate, therefore, with the angular velocity

$$\mathbf{\Omega} = -\dot{\phi} \sin \theta \mathbf{i} + \dot{\theta} \mathbf{j} + \dot{\phi} \cos \theta \mathbf{k} \quad (18.38)$$

Denoting the moment of inertia of the gyroscope with respect to its spin axis z by I and its moment of inertia with respect to a transverse axis through O by I' , we wrote

$$\mathbf{H}_O = -I'\dot{\phi} \sin \theta \mathbf{i} + I'\dot{\theta} \mathbf{j} + I(\dot{\psi} + \dot{\phi} \cos \theta) \mathbf{k} \quad (18.36)$$

Substituting for $\mathbf{H}_O$ and $\mathbf{\Omega}$ into Eq. (18.28) led us to the differential equations defining the motion of a gyroscope.

Steady Precession

In the particular case of the **steady precession** of a gyroscope [Sec. 18.3B], the angle θ , the rate of precession $\dot{\phi}$, and the rate of spin $\dot{\psi}$ remain constant. We saw that such a motion is possible only if the moments of the external forces about O satisfy the relation

$$\Sigma \mathbf{M}_O = (I\omega_z - I'\dot{\phi} \cos \theta) \dot{\phi} \sin \theta \mathbf{j} \quad (18.44)$$

i.e., if the external forces reduce to a couple of moment equal to the right-hand side of Eq. (18.44) and applied about an axis perpendicular to the precession axis and to the spin axis (Fig. 18.31). This chapter ended with a discussion of the motion of an axisymmetric body spinning and precessing under no force [Sec. 18.3C; Sample Prob. 18.6].

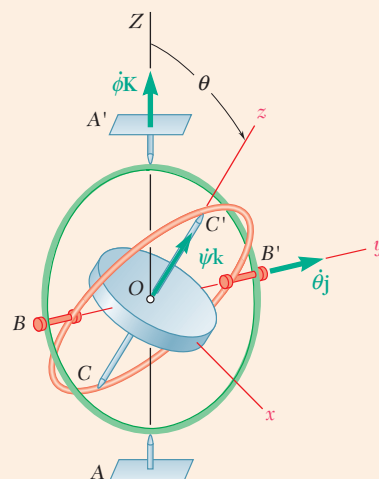


Fig. 18.30

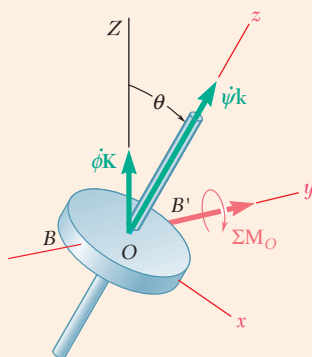


Fig. 18.31

Review Problems

- 18.147** Three 25-lb rotor disks are attached to a shaft that rotates at 720 rpm. Disk A is attached eccentrically so that its mass center is $\frac{1}{4}$ in. from the axis of rotation, while disks B and C are attached so that their mass centers coincide with the axis of rotation. Where should 2-lb weights be bolted to disks B and C to balance the system dynamically?

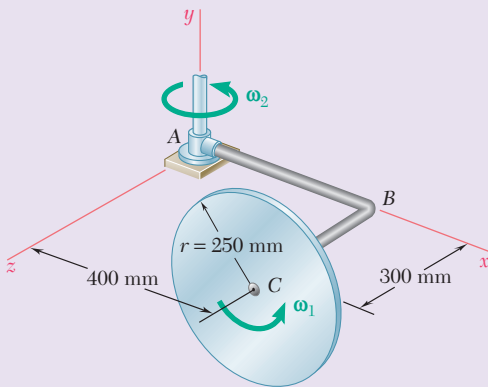


Fig. P18.148

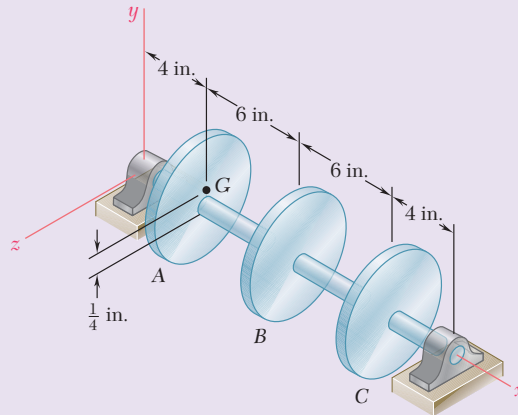


Fig. P18.147

- 18.148** A homogeneous disk of mass $m = 5$ kg rotates at the constant rate $\omega_1 = 8$ rad/s with respect to the bent axle ABC, which itself rotates at the constant rate $\omega_2 = 3$ rad/s about the y axis. Determine the angular momentum $\mathbf{H}_C$ of the disk about its center C.
- 18.149** A rod of uniform cross section is used to form the shaft shown. Denoting by m the total mass of the shaft and knowing that the shaft rotates with a constant angular velocity ω , determine (a) the angular momentum $\mathbf{H}_G$ of the shaft about its mass center G, (b) the angle formed by $\mathbf{H}_G$ and the axis AB, (c) the angular momentum of the shaft about point A.

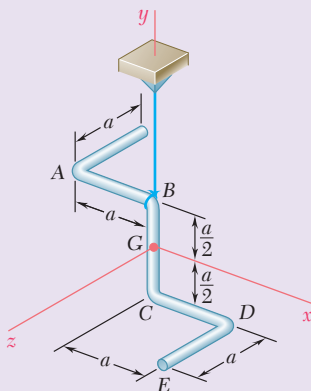


Fig. P18.150

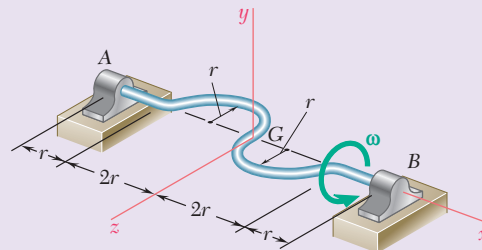


Fig. P18.149

- 18.150** A uniform rod of mass m and length $5a$ is bent into the shape shown and is suspended from a wire attached at point B. Knowing that the rod is hit at point A in the negative y direction and denoting the corresponding impulse by $-(F \Delta t)\mathbf{j}$, determine immediately after the impact (a) the velocity of the mass center G, (b) the angular velocity of the rod.

18.151 A four-bladed airplane propeller has a mass of 160 kg and a radius of gyration of 800 mm. Knowing that the propeller rotates at 1600 rpm as the airplane is traveling in a circular path of 600-m radius at 540 km/h, determine the magnitude of the couple exerted by the propeller on its shaft due to the rotation of the airplane.

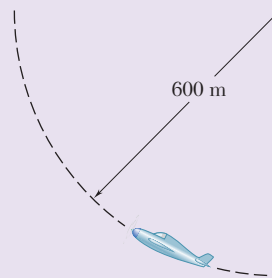


Fig. P18.151

18.152 A 2.4-kg piece of sheet steel with dimensions 160×640 mm was bent to form the component shown. The component is at rest ($\omega = 0$) when a couple $\mathbf{M}_0 = (0.8 \text{ N}\cdot\text{m})\mathbf{k}$ is applied to it. Determine (a) the angular acceleration of the component, (b) the dynamic reactions at A and B immediately after the couple is applied.

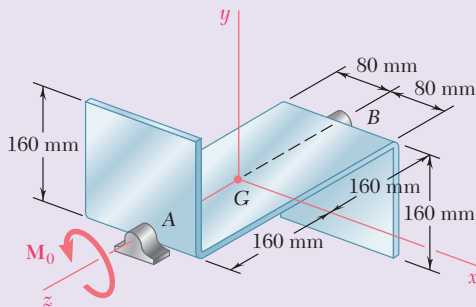


Fig. P18.152

18.153 A homogeneous disk of weight $W = 6$ lb rotates at the constant rate $\omega_1 = 16$ rad/s with respect to arm ABC, which is welded to a shaft DCE rotating at the constant rate $\omega_2 = 8$ rad/s. Determine the dynamic reactions at D and E.

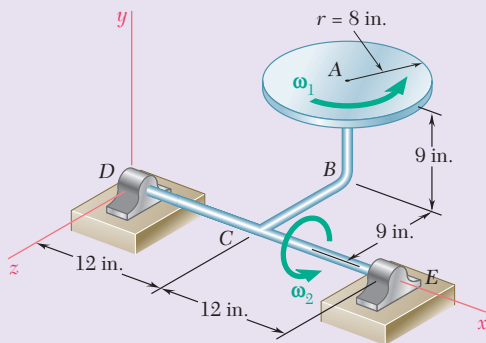


Fig. P18.153

18.154 A 48-kg advertising panel of length $2a = 2.4$ m and width $2b = 1.6$ m is kept rotating at a constant rate ω_1 about its horizontal axis by a small electric motor attached at A to frame ACB. This frame itself is kept rotating at a constant rate ω_2 about a vertical axis by a second motor attached at C to the column CD. Knowing that the panel and the frame complete a full revolution in 6 s and 12 s, respectively, express, as a function of the angle θ , the dynamic reaction exerted on column CD by its support at D.

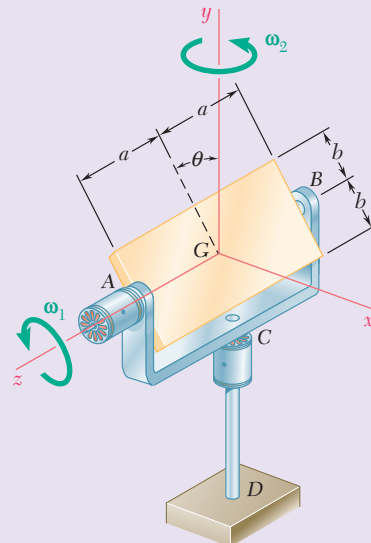


Fig. P18.154

- 18.155** A 2500-kg satellite is 2.4 m high and has octagonal bases of sides 1.2 m. The coordinate axes shown are the principal centroidal axes of inertia of the satellite, and its radii of gyration are $k_x = k_z = 0.90$ m and $k_y = 0.98$ m. The satellite is equipped with a main 500-N thruster E and four 20-N thrusters A , B , C , and D that can expel fuel in the positive y direction. The satellite is spinning at the rate of 36 rev/h about its axis of symmetry Gy , which maintains a fixed direction in space, when thrusters A and B are activated for 2 s. Determine (a) the precession axis of the satellite, (b) its rate of precession, (c) its rate of spin.

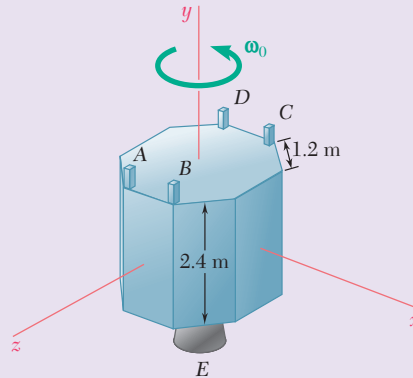


Fig. P18.155

- 18.156** The space capsule has no angular velocity when the jet at A is activated for 1 s in a direction parallel to the x axis. Knowing that the capsule has a mass of 1000 kg, that its radii of gyration are $k_z = \bar{k}_y = 1.00$ m and $\bar{k}_z = 1.25$ m, and that the jet at A produces a thrust of 50 N, determine the axis of precession and the rates of precession and spin after the jet has stopped.

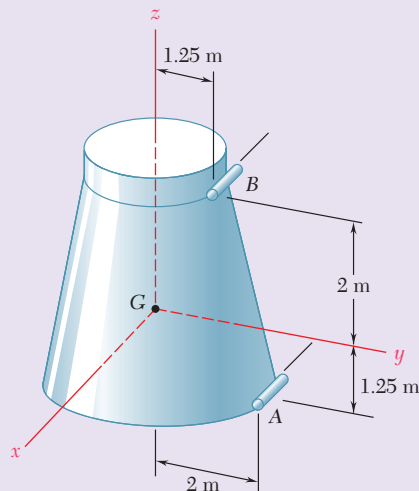


Fig. P18.156

- 18.157** A homogeneous disk of mass m is connected at A and B to a fork-ended shaft of negligible mass that is supported by a bearing at C . The disk is free to rotate about its horizontal diameter AB and the shaft is free to rotate about a vertical axis through C . Initially the disk lies in a vertical plane ($\theta_0 = 90^\circ$) and the shaft has an angular velocity $\dot{\phi}_0 = 8 \text{ rad/s}$. If the disk is slightly disturbed, determine for the ensuing motion (a) the minimum value of $\dot{\phi}$, (b) the maximum value of $\dot{\theta}$.

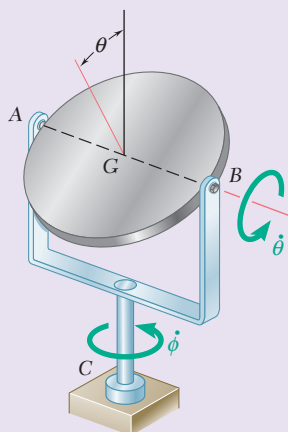


Fig. P18.157

- 18.158** The essential features of the gyrocompass are shown. The rotor spins at the rate $\dot{\psi}$ about an axis mounted in a single gimbal, which may rotate freely about the vertical axis AB . The angle formed by the axis of the rotor and the plane of the meridian is denoted by θ , and the latitude of the position on the earth is denoted by λ . We note that line OC is parallel to the axis of the earth, and we denote by ω_e the angular velocity of the earth about its axis.

(a) Show that the equations of motion of the gyrocompass are

$$I'\ddot{\theta} + I\omega_z\omega_e \cos \lambda \sin \theta - I'\omega_e^2 \cos^2 \lambda \sin \theta \cos \theta = 0$$

$$I\dot{\omega}_z = 0$$

where ω_z is the rectangular component of the total angular velocity ω along the axis of the rotor, and I and I' are the moments of inertia of the rotor with respect to its axis of symmetry and a transverse axis through O , respectively.

(b) Neglecting the term containing ω_e^2 , show that for small values of θ , we have

$$\ddot{\theta} + \frac{I\omega_z\omega_e \cos \lambda}{I'} \theta = 0$$

and that the axis of the gyrocompass oscillates about the north–south direction.

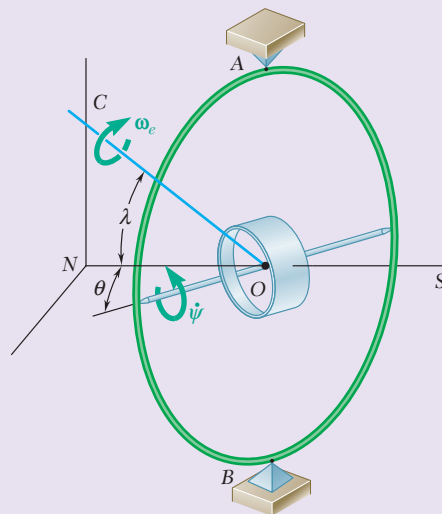


Fig. P18.158



19

Mechanical Vibrations

The Wind Damper inside of a building helps protect against typhoons and earthquakes by reducing the effects of wind and vibrations on the building. Mechanical systems may undergo *free vibrations* or they may be subject to *forced vibrations*. The vibrations are *damped* when there is energy dissipation and *undamped* otherwise. This chapter is an introduction to many fundamental concepts in vibration analysis.

Introduction

19.1 VIBRATIONS WITHOUT DAMPING

19.1A Simple Harmonic Motion and Free Vibrations of Particles

19.1B Simple Pendulum (Approximate Solution)

19.1C Simple Pendulum (Exact Solution)

19.2 FREE VIBRATIONS OF RIGID BODIES

19.3 APPLYING THE PRINCIPLE OF CONSERVATION OF ENERGY

19.4 FORCED VIBRATIONS

19.5 DAMPED VIBRATIONS

19.5A Damped Free Vibrations

19.5B Damped Forced Vibrations

19.5C Electrical Analogs

Objectives

- **Define, compare, and contrast** simple harmonic motion, undamped free and forced vibrations, and damped free and forced vibrations.
- Using Newton's second law, **determine** the differential equation of motion of a particle or a rigid body undergoing vibratory motion.
- Using the conservation of energy, **determine** the differential equation of motion of a particle or a rigid body undergoing vibratory motion.
- **Calculate** the natural circular frequency, period, and natural frequency for a system undergoing simple harmonic motion.
- **Calculate** the maximum amplitude and the magnification factor for a body undergoing forced vibrations.
- **Compare and contrast** the vibration responses of underdamped, critically damped, and overdamped systems.

Introduction

A **mechanical vibration** is the motion of a particle or body that oscillates about a position of equilibrium. Most vibrations in machines and structures are undesirable because of the increased stresses and energy losses that accompany them. Appropriate design therefore aims to eliminate or reduce vibrations as much as possible. The analysis of vibrations has become increasingly important in recent years owing to the current trend toward higher-speed machines and lighter structures. There is every reason to expect that this trend will continue and that an even greater need for vibration analysis will develop in the future.

The analysis of vibrations is a very extensive subject to which entire texts have been devoted. Our present study is limited to the simplest types of vibrations—namely, the vibrations of a body or a system of bodies with one degree of freedom.

A mechanical vibration generally results when a system is displaced from a position of stable equilibrium. The system tends to return to this position under the action of restoring forces (either elastic forces, as in the case of a mass attached to a spring, or gravitational forces, as in the case of a pendulum). But the system generally reaches its original position with an acquired velocity that carries it beyond that position. Since the process can be repeated indefinitely, the system keeps moving back and forth across its position of equilibrium. The time interval required for the system to complete a full cycle of motion is called the **period** of the vibration. The number of cycles per unit time defines the **frequency**, and the maximum displacement of the system from its position of equilibrium is called the **amplitude** of the vibration.

When the motion is maintained by the restoring forces only, the vibration is said to be a **free vibration**. When a periodic force is applied to the system, the resulting motion is described as a **forced vibration**. If we can neglect the effects of friction, the vibrations are said to be **undamped**. However, all vibrations are actually **damped** to some degree. If a free vibration is only slightly damped, its amplitude slowly decreases until, after a certain time, the motion comes to a stop. But if damping is large enough to prevent any true vibration, the system then slowly regains its original position. A damped forced vibration is maintained as long as the periodic force that produces the vibration is applied. The amplitude of the vibration, however, is affected by the magnitude of the damping forces.

In this chapter, we first examine vibrations without damping, studying vibrations of particles, rigid bodies, and forced vibrations. Then we will look at damped vibrations, including both free and forced vibrations.

19.1 VIBRATIONS WITHOUT DAMPING

The first step in analyzing vibrations is to formulate an equation of motion for the simple case of a particle in free vibration. We will modify this equation as we consider more complicated situations, such as damped and forced vibrations.

19.1A Simple Harmonic Motion and Free Vibrations of Particles

Consider a body with a mass m attached to a spring with a constant k (Fig. 19.1a). At the moment, we are concerned only with the motion of its mass center, so we will refer to this body as a particle. When the particle is in static equilibrium, the forces acting on it are its weight $\mathbf{W}$ and the force $\mathbf{T}$ exerted by the spring, which has a magnitude $T = k\delta_{st}$, where δ_{st} denotes the static elongation of the spring from its unstretched length. We therefore have

$$W = k\delta_{st}$$

Suppose now that the particle is displaced through a distance x_m from its equilibrium position and released with no initial velocity. If we have chosen x_m to be smaller than δ_{st} , the particle moves back and forth through its equilibrium position; a vibration with an amplitude x_m is generated. Note that we can also produce a vibration by imparting an initial velocity to the particle when it is in its equilibrium position $x = 0$ or, more generally, by starting the particle from any given position $x = x_0$ with a given initial velocity $\mathbf{v}_0$.

To analyze the vibration, let us consider the particle in a position P at some arbitrary time t (Fig. 19.1b). Denoting the displacement OP measured from the equilibrium position O (positive downward) by x , we note that the forces acting on the particle are its weight $\mathbf{W}$ and the force $\mathbf{T}$ exerted by the spring. In this position, the spring force has a magnitude $T = k(\delta_{st} + x)$. Recalling that $W = k\delta_{st}$, we find that the magnitude of the resultant $\mathbf{F}$ of the two forces (positive downward) is

$$F = W - k(\delta_{st} + x) = -kx \quad (19.1)$$

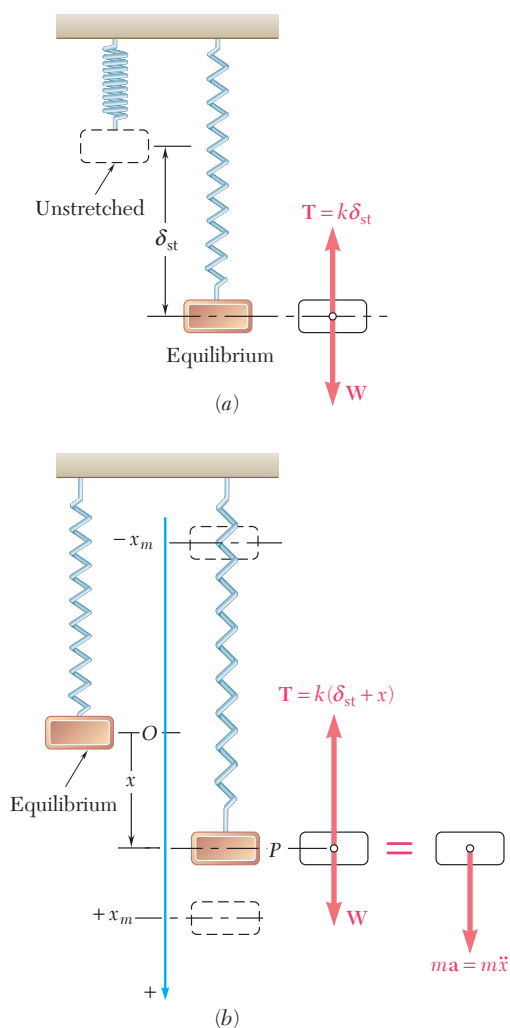


Fig. 19.1 (a) At the equilibrium position, the spring force is equal to the weight; (b) the block at position P with its free-body diagram and kinetic diagram.

Thus, the resultant of the forces exerted on the particle is proportional to the displacement OP **measured from the equilibrium position**. Recalling the sign convention, we note that $\mathbf{F}$ is always directed *toward* the equilibrium position O . Substituting for F into the fundamental equation $F = ma$ and recalling that a is the second derivative $\ddot{x}$ of x with respect to t , we have

Equation of motion for simple harmonic motion

$$m\ddot{x} + kx = 0 \quad (19.2)$$

Note that we use the same sign convention for the acceleration $\ddot{x}$ and for the displacement x , namely, positive downward. By measuring the displacement from the static equilibrium point, we get a homogeneous differential equation; that is, the right-hand side is equal to zero.

The motion defined by Eq. (19.2) is called **simple harmonic motion**. It is characterized by the fact that **the acceleration is proportional to the displacement and in the opposite direction**. We can verify that each of the functions

$$x_1 = \sin(\sqrt{k/m} t) \quad \text{and} \quad x_2 = \cos(\sqrt{k/m} t)$$

satisfies Eq. (19.2). These functions, therefore, constitute two *particular solutions* of the differential equation (19.2). We can obtain the *general solution* of Eq. (19.2) by multiplying each of the particular solutions by an arbitrary constant and adding. Thus, the general solution is

$$x = C_1 x_1 + C_2 x_2 = C_1 \sin\left(\sqrt{\frac{k}{m}} t\right) + C_2 \cos\left(\sqrt{\frac{k}{m}} t\right) \quad (19.3)$$

Note that x is a **periodic function** of the time t and therefore represents a vibration of the particle P . The coefficient of t in the expression we have obtained is referred to as the **natural circular frequency** of the vibration and is denoted by ω_n . We have

$$\text{Natural circular frequency} = \omega_n = \sqrt{\frac{k}{m}} \quad (19.4)$$

Substituting for $\sqrt{k/m}$ into Eq. (19.3) gives

$$x = C_1 \sin \omega_n t + C_2 \cos \omega_n t \quad (19.5)$$

This is the general solution of the differential equation

$$\ddot{x} + \omega_n^2 x = 0 \quad (19.6)$$

that we can obtain from Eq. (19.2) by dividing both terms by m and observing that $k/m = \omega_n^2$. Differentiating both sides of Eq. (19.5) twice with respect to t , we obtain the expressions for the velocity and the acceleration at time t as

$$v = \dot{x} = C_1 \omega_n \cos \omega_n t - C_2 \omega_n \sin \omega_n t \quad (19.7)$$

$$a = \ddot{x} = -C_1 \omega_n^2 \sin \omega_n t - C_2 \omega_n^2 \cos \omega_n t \quad (19.8)$$

The values of the constants C_1 and C_2 depend upon the *initial conditions* of the motion. For example, we have $C_1 = 0$ if the particle is displaced from its equilibrium position and released at $t = 0$ with no

initial velocity. Also, we have $C_2 = 0$ if the particle starts from O at $t = 0$ with a given initial velocity. In general, substituting $t = 0$ and the initial values x_0 and v_0 of the displacement and the velocity into Eqs. (19.5) and (19.7), we find that $C_1 = v_0/\omega_n$ and $C_2 = x_0$.

We can write these expressions for the displacement, velocity, and acceleration of a particle in a more compact form if we note that Eq. (19.5) says that the displacement $x = OP$ is the sum of the x components of two vectors $\mathbf{C}_1$ and $\mathbf{C}_2$, respectively, with magnitudes of C_1 and C_2 that are directed as shown in Fig. 19.2a. As t varies, both vectors $\overrightarrow{OQ}$ rotate clockwise; we also note that the magnitude of their resultant $\overrightarrow{OQ}$ is equal to the maximum displacement x_m . Thus, we can obtain the simple harmonic motion of P along the x axis by projecting on this axis the motion of a point Q describing an *auxiliary circle* of radius x_m with a constant angular velocity ω_n . This explains the name natural *circular* frequency given to ω_n . Denoting the angle formed by the vectors $\overrightarrow{OQ}$ and $\mathbf{C}_1$ by ϕ , we have

$$OP = OQ \sin(\omega_n t + \phi) \quad (19.9)$$

This leads to new expressions for the displacement, velocity, and acceleration of P :

$$x = x_m \sin(\omega_n t + \phi) \quad (19.10)$$

$$v = \dot{x} = x_m \omega_n \cos(\omega_n t + \phi) \quad (19.11)$$

$$a = \ddot{x} = -x_m \omega_n^2 \sin(\omega_n t + \phi) \quad (19.12)$$

The displacement-time curve is represented by a sine curve (Fig. 19.2b); the maximum value x_m of the displacement is called the **amplitude** of the vibration, and the angle ϕ that defines the initial position of Q on the circle is called the **phase angle**. As we can see from Fig. 19.2, a full cycle occurs every 2π rad. The corresponding value of t is denoted by τ_n . This is called the **period** of the free vibration and is measured in seconds. We have

$$\text{Period} = \tau_n = \frac{2\pi}{\omega_n} \quad (19.13)$$

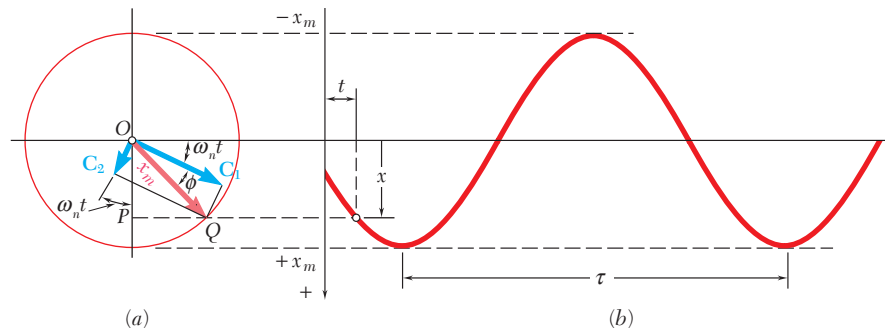


Fig. 19.2 (a) Auxiliary circle of simple harmonic motion: the resultant OQ rotates at constant angular velocity ω_n ; (b) the graph of displacement versus time is a sine curve.

The number of cycles described per unit of time is denoted by f_n and is known as the **natural frequency** of the vibration. We have

$$\text{Natural frequency} = f_n = \frac{1}{\tau_n} = \frac{\omega_n}{2\pi} \quad (19.14)$$

The unit of frequency is called a *hertz* (Hz). It also follows from Eq. (19.14) that a frequency of 1 s^{-1} or 1 Hz corresponds to a circular frequency of $2\pi \text{ rad/s}$. In problems involving angular velocities expressed in revolutions per minute (rpm), we have $1 \text{ rpm} = \frac{1}{60} \text{ s}^{-1} = \frac{1}{60} \text{ Hz}$, or $1 \text{ rpm} = (2\pi/60) \text{ rad/s}$.

Recall that we defined ω_n in Eq. (19.4) in terms of the constant k of the spring and the mass m of the particle. Thus, the period and the frequency are independent both of the initial conditions and of the amplitude of the vibration. Also, τ_n and f_n depend on the *mass* rather than on the *weight* of the particle and thus are independent of the value of g .

We can represent the velocity-time and acceleration-time curves using sine curves of the same period as the displacement-time curve—but with different amplitudes and different phase angles. From Eqs. (19.11) and (19.12), the maximum values of the magnitudes of the velocity and acceleration are

$$v_m = x_m \omega_n \quad a_m = x_m \omega_n^2 \quad (19.15)$$

The point Q describes the auxiliary circle with a radius x_m at the constant angular velocity ω_n , so its velocity and acceleration are equal, respectively, to the expressions of Eq. (19.15). Recalling Eqs. (19.11) and (19.12), we can find the velocity and acceleration of P at any instant by projecting vectors of magnitudes $v_m = x_m \omega_n$ and $a_m = x_m \omega_n^2$ on the x axis. These two vectors represent the velocity and acceleration of Q , respectively, at the same instant (Fig. 19.3).

These results are not limited to the solution of the problem of a mass attached to a spring. We can use them to analyze the rectilinear motion of a particle whenever the resultant $\mathbf{F}$ of the forces acting on the particle is proportional to the displacement x and directed toward O . In such a case, we can write the fundamental equation of motion $F = ma$ in the form of Eq. (19.6), which is characteristic of a simple harmonic motion. Observing that the coefficient of x must be equal to ω_n^2 , we can easily determine the natural circular frequency ω_n of the motion. Substituting the value obtained for ω_n into Eqs. (19.13) and (19.14), we then obtain the period τ_n and the natural frequency f_n of the motion.

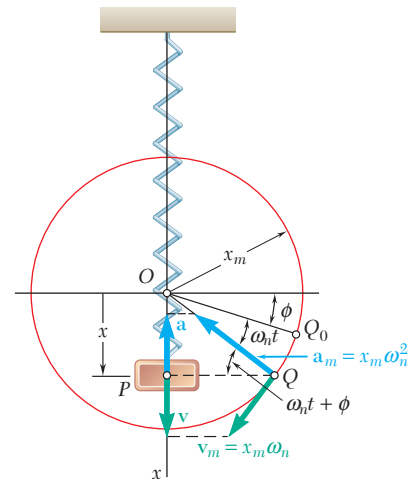


Fig. 19.3 Auxiliary circle of simple harmonic motion showing the maximum values of velocity and acceleration.

19.1B Simple Pendulum (Approximate Solution)

Many of the vibrations encountered in engineering applications can be represented using simple harmonic motion. Many others can be *approximated* by a simple harmonic motion—provided that their amplitude remains small. Consider, for example, a **simple pendulum** consisting of a bob with

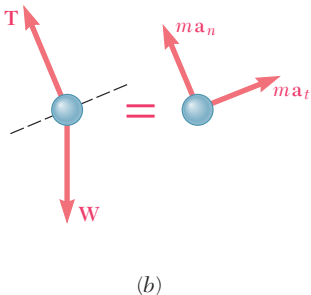
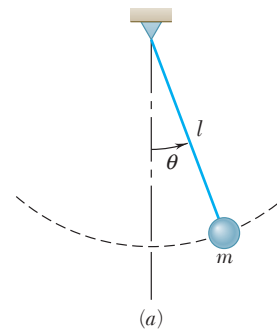


Fig. 19.4 (a) A simple pendulum consists of a bob of mass m at the end of a cord of length l ; (b) free-body diagram and kinetic diagram of the simple pendulum.

a mass m attached to a cord of length l that can oscillate in a vertical plane (Fig. 19.4a). At a given time t , the cord forms an angle θ with the vertical. The forces acting on the bob are its weight $\mathbf{W}$ and the force $\mathbf{T}$ exerted by the cord (Fig. 19.4b). Resolving the vector $m\mathbf{a}$ into tangential and normal components, with ma_t directed to the right (i.e., in the direction corresponding to increasing values of θ), and observing that $a_t = l\alpha = l\ddot{\theta}$, we have

$$\Sigma F_t = ma_t; \quad -W \sin \theta = ml\ddot{\theta}$$

Noting that $W = mg$ and dividing through by ml , we obtain

$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0 \quad (19.16)$$

For oscillations of small amplitude, we can replace $\sin \theta$ by θ , which is expressed in radians, obtaining

$$\ddot{\theta} + \frac{g}{l} \theta = 0 \quad (19.17)$$

Comparison with Eq. (19.6) shows that the differential equation (19.17) is that of a simple harmonic motion with a natural circular frequency ω_n equal to $(g/l)^{1/2}$. Thus, we can express the general solution of Eq. (19.17) as

$$\theta = \theta_m \sin(\omega_n t + \phi)$$

where θ_m is the amplitude of the oscillations and ϕ is a phase angle. Substituting the value obtained for ω_n into Eq. (19.13), we get the expression for the period of the small oscillations of a pendulum of length l as

$$\tau_n = \frac{2\pi}{\omega_n} = 2\pi\sqrt{\frac{l}{g}} \quad (19.18)$$

*19.1C Simple Pendulum (Exact Solution)

Formula (19.18) is only approximate. To obtain an exact expression for the period of the oscillations of a simple pendulum, we must return to Eq. (19.16). Multiplying both terms by $2\dot{\theta}$ and integrating from an initial position corresponding to the maximum deflection (that is, $\theta = \theta_m$ and $\dot{\theta} = 0$), we have

$$\left(\frac{d\theta}{dt}\right)^2 = \frac{2g}{l}(\cos \theta - \cos \theta_m)$$

We replace $\theta \cos \theta$ by $1 - 2 \sin^2(\theta/2)$ and $\cos \theta_m$ by a similar expression, solve for dt , and integrate over a quarter period from $t = 0$, $\theta = 0$ to $t = \tau_n/4$, $\theta = \theta_m$. This gives

$$\tau_n = 2\sqrt{\frac{l}{g}} \int_0^{\theta_m} \frac{d\theta}{\sqrt{\sin^2(\theta_m/2) - \sin^2(\theta/2)}}$$

The integral on the right-hand side is known as an *elliptic integral*; it cannot be expressed in terms of the usual algebraic or trigonometric functions. However, setting

$$\sin(\theta/2) = \sin(\theta_m/2) \sin \phi$$

we can write

$$\tau_n = 4\sqrt{\frac{l}{g}} \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - \sin^2(\theta_m/2) \sin^2 \phi}} \quad (19.19)$$

We can calculate this integral, commonly denoted by K , by using a numerical method of integration. It also can be found using computer programs such as Maple, Mathematica, or Matlab or in *tables of elliptic integrals* for various values of $\theta_m/2$.[†]

In order to compare this result with that of the preceding section, we write Eq. (19.19) in the form

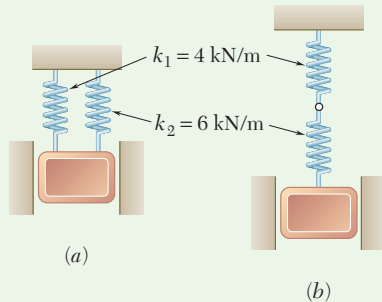
$$\tau_n = \frac{2K}{\pi} \left(2\pi \sqrt{\frac{l}{g}} \right) \quad (19.20)$$

Formula (19.20) shows that we can obtain the actual value of the period of a simple pendulum by multiplying the approximate value given in Eq. (19.18) by the correction factor $2K/\pi$. Values of the correction factor are given in Table 19.1 for various values of the amplitude θ_m . Note that for ordinary engineering computations, the correction factor can be omitted as long as the amplitude does not exceed 10° .

Table 19.1 Correction Factor for the Period of a Simple Pendulum

θ_m	0°	10°	20°	30°	60°	90°	120°	150°	180°
K	1.571	1.574	1.583	1.598	1.686	1.854	2.157	2.768	∞
$2K/\pi$	1.000	1.002	1.008	1.017	1.073	1.180	1.373	1.762	∞

[†]See, for example, *Standard Mathematical Tables and Formulae*, CRC Press, Cleveland, Ohio.



Sample Problem 19.1

A 50-kg block moves between vertical guides as shown. The block is pulled 40 mm down from its equilibrium position and released. For each spring arrangement, determine the period of the vibration, the maximum velocity of the block, and the maximum acceleration of the block.

STRATEGY: You first need to calculate the equivalent spring constant for each arrangement of the springs. Then you can use the information in this section to determine the motion.

MODELING and ANALYSIS:

a. Springs Attached in Parallel. First determine the constant k of a single spring equivalent to the two springs by finding the magnitude of the force $\mathbf{P}$ required to cause a given deflection δ (Fig. 1). Since for a deflection δ the magnitudes of the forces exerted by the springs are, respectively, $k_1\delta$ and $k_2\delta$, you have

$$P = k_1\delta + k_2\delta = (k_1 + k_2)\delta$$

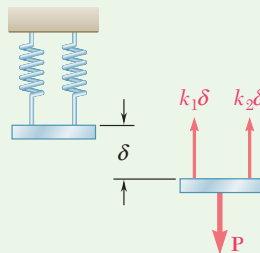


Fig. 1 Springs in parallel elongated a distance δ .

Thus, the constant k of the single equivalent spring is

$$k = \frac{P}{\delta} = k_1 + k_2 = 4 \text{ kN/m} + 6 \text{ kN/m} = 10 \text{ kN/m} = 10^4 \text{ N/m}$$

Since $m = 50 \text{ kg}$, Eq. (19.4) yields

Period of Vibration:

$$\omega_n^2 = \frac{k}{m} = \frac{10^4 \text{ N/m}}{50 \text{ kg}} \quad \omega_n = 14.14 \text{ rad/s}$$

$$\tau_n = 2\pi/\omega_n \quad \tau_n = 0.444 \text{ s} \quad \blacktriangleleft$$

Maximum Velocity:

$$v_m = x_m\omega_n = (0.040 \text{ m})(14.14 \text{ rad/s})$$

$$v_m = 0.566 \text{ m/s} \quad \mathbf{v}_m = 0.566 \text{ m/s} \quad \blacktriangleup \quad \blacktriangleleft$$

Maximum Acceleration:

$$a_m = x_m\omega_n^2 = (0.040 \text{ m})(14.14 \text{ rad/s})^2$$

$$a_m = 8.00 \text{ m/s}^2 \quad \mathbf{a}_m = 8.00 \text{ m/s}^2 \quad \blacktriangleup \quad \blacktriangleleft$$

b. Springs Attached in Series. In this case, determine the constant k of a single spring equivalent to the two springs by finding the total elongation δ of the springs under a given static load $\mathbf{P}$ (Fig. 2).

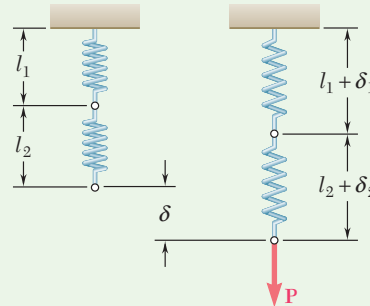


Fig. 2 Springs in series elongated a distance δ .

To facilitate the computation, you can use an arbitrary static load with a magnitude of $P = 12$ kN (this number is chosen since it has four and six as divisors). You obtain

$$\delta = \delta_1 + \delta_2 = \frac{P}{k_1} + \frac{P}{k_2} = \frac{12 \text{ kN}}{4 \text{ kN/m}} + \frac{12 \text{ kN}}{6 \text{ kN/m}} = 5 \text{ m}$$

$$k = \frac{P}{\delta} = \frac{12 \text{ kN}}{5 \text{ m}} = 2.4 \text{ kN/m} = 2400 \text{ N/m}$$

Period of Vibration:

$$\omega_n^2 = \frac{k}{m} = \frac{2400 \text{ N/m}}{50 \text{ kg}} \quad \omega_n = 6.93 \text{ rad/s}$$

$$\tau_n = \frac{2\pi}{\omega_n} \quad \tau_n = 0.907 \text{ s} \quad \blacktriangleleft$$

Maximum Velocity:

$$v_m = x_m \omega_n = (0.040 \text{ m})(6.93 \text{ rad/s})$$

$$v_m = 0.277 \text{ m/s} \quad \mathbf{v}_m = 0.277 \text{ m/s} \updownarrow \quad \blacktriangleleft$$

Maximum Acceleration:

$$a_m = x_m \omega_n^2 = (0.040 \text{ m})(6.93 \text{ rad/s})^2$$

$$a_m = 1.920 \text{ m/s}^2 \quad \mathbf{a}_m = 1.920 \text{ m/s}^2 \updownarrow \quad \blacktriangleleft$$

REFLECT and THINK: The problem did not ask you to determine the expression for combining springs in series, but from this analysis, it is clear that $\delta = \frac{P}{k_1} + \frac{P}{k_2} = \frac{P}{k}$ or $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$. Thus, for springs in series, $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$, and for springs in parallel, $k = k_1 + k_2$.

SOLVING PROBLEMS ON YOUR OWN

This chapter deals with **mechanical vibrations**, i.e., with the motion of a particle or body oscillating about a position of equilibrium. In this first section, we saw that a **free vibration** of a particle occurs when the particle is subjected to a force proportional to its displacement and in the opposite direction, such as the force exerted by a spring (Fig. 19.1). The resulting motion, called **simple harmonic motion**, is characterized by the differential equation as

$$m\ddot{x} + kx = 0 \quad (19.2)$$

where x is the displacement of the particle from the equilibrium point, $\ddot{x}$ is its acceleration, m is its mass, and k is the constant of the spring. We found the solution of this differential equation to be

$$x = x_m \sin (\omega_n t + \phi) \quad (19.10)$$

where x_m = amplitude of the vibration

$$\omega_n = \sqrt{k/m} = \text{natural circular frequency (rad/s)}$$

$$\phi = \text{phase angle (rad)}$$

We defined the **period** of the vibration as the time $\tau_n = 2\pi/\omega_n$ needed for the particle to complete one cycle. The **natural frequency** is the number of cycles per second, $f_n = 1/\tau_n = \omega_n/2\pi$, expressed in Hz or s^{-1} . Differentiating Eq. (19.10) twice yields the velocity and the acceleration of the particle at any time. We found the maximum values of the velocity and acceleration to be

$$v_m = x_m \omega_n \quad a_m = x_m \omega_n^2 \quad (19.15)$$

To determine the parameters in Eq. (19.10), you can follow these steps:

1. Draw a free-body diagram showing the forces exerted on the particle when the particle is at a distance x from its position of equilibrium. The resultant of these forces is proportional to x , and its direction is opposite to the positive direction of x [Eq. (19.1)].

2. Write the differential equation of motion by equating $m\ddot{x}$ to the resultant of the forces found in Step 1. Note that once you have chosen a positive direction for x , you should use the same sign convention for the acceleration $\ddot{x}$. After transposition, you will obtain an equation of the form of Eq. (19.2).

3. Determine the natural circular frequency ω_n by dividing the coefficient of x by the coefficient of $\ddot{x}$ in this equation and taking the square root of the result. Make sure that ω_n is expressed in rad/s.

4. Determine the amplitude x_m and the phase angle ϕ by substituting the value obtained for ω_n and the initial values of x and $\dot{x}$ into Eq. (19.10) and the equation obtained by differentiating Eq. (19.10) with respect to t .

You can now use Eq. (19.10) and the two equations obtained by differentiating Eq. (19.10) twice with respect to t to find the displacement, velocity, and acceleration of the particle at any time. Equations (19.15) yield the maximum velocity v_m and the maximum acceleration a_m .

5. For the small oscillations of a simple pendulum, the angle θ that the cord of the pendulum forms with the vertical satisfies the differential equation

$$\ddot{\theta} + \frac{g}{l}\theta = 0 \quad (19.17)$$

where l is the length of the cord and θ is expressed in radians [Sec. 19.1B]. This equation defines again a simple harmonic motion, and its solution is of the same form as Eq. (19.10) as

$$\theta = \theta_m \sin (\omega_n t + \phi)$$

where the natural circular frequency $\omega_n = \sqrt{g/l}$ is expressed in rad/s. The determination of the various constants in this expression is carried out in a manner similar to that described previously. Remember that the velocity of the bob is tangent to the path and that its magnitude is $v = l\dot{\theta}$, whereas the acceleration of the bob has a tangential component $\mathbf{a}_t$ with a magnitude of $a_t = l\ddot{\theta}$ and a normal component $\mathbf{a}_n$ directed toward the center of the path and with a magnitude of $a_n = l\dot{\theta}^2$.

Problems

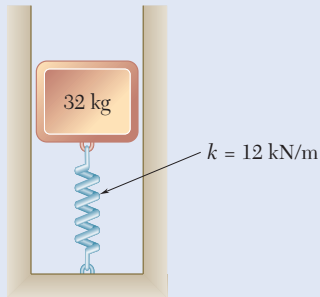


Fig. P19.4

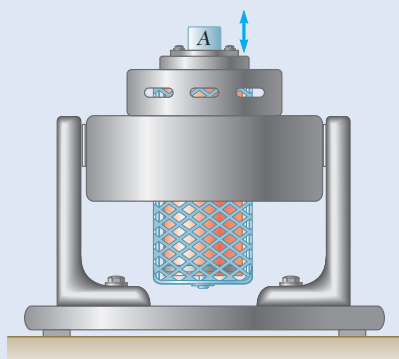


Fig. P19.6

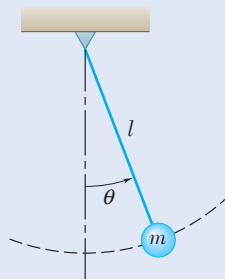


Fig. P19.7 and P19.8

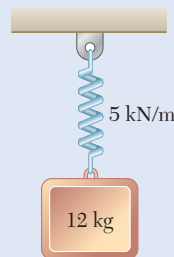


Fig. P19.5

- 19.1** A particle moves in simple harmonic motion. Knowing that the maximum velocity is 200 mm/s and the maximum acceleration is 4 m/s^2 , determine the amplitude and frequency of the motion.
- 19.2** A particle moves in simple harmonic motion. Knowing that the amplitude is 15 in. and the maximum acceleration is 15 ft/s^2 , determine the maximum velocity of the particle and the frequency of its motion.
- 19.3** Determine the amplitude and maximum acceleration of a particle that moves in simple harmonic motion with a maximum velocity of 4 ft/s and a frequency of 6 Hz .
- 19.4** A 32-kg block is attached to a spring and can move without friction in a slot as shown. The block is in its equilibrium position when it is struck by a hammer that imparts to the block an initial velocity of 250 mm/s . Determine (a) the period and frequency of the resulting motion, (b) the amplitude of the motion and the maximum acceleration of the block.
- 19.5** A 12-kg block is supported by the spring shown. If the block is moved vertically downward from its equilibrium position and released, determine (a) the period and frequency of the resulting motion, (b) the maximum velocity and acceleration of the block if the amplitude of its motion is 50 mm .

- 19.6** An instrument package A is bolted to a shaker table as shown. The table moves vertically in simple harmonic motion at the same frequency as the variable-speed motor that drives it. The package is to be tested at a peak acceleration of 150 ft/s^2 . Knowing that the amplitude of the shaker table is 2.3 in. , determine (a) the required speed of the motor in rpm, (b) the maximum velocity of the table.
- 19.7** A simple pendulum consisting of a bob attached to a cord oscillates in a vertical plane with a period of 1.3 s . Assuming simple harmonic motion and knowing that the maximum velocity of the bob is 0.4 m/s , determine (a) the amplitude of the motion in degrees, (b) the maximum tangential acceleration of the bob.
- 19.8** A simple pendulum consisting of a bob attached to a cord of length $l = 800 \text{ mm}$ oscillates in a vertical plane. Assuming simple harmonic motion and knowing that the bob is released from rest when $\theta = 6^\circ$, determine (a) the frequency of oscillation, (b) the maximum velocity of the bob.

- 19.9** A 10-lb block *A* rests on a 40-lb plate *B* that is attached to an unstretched spring with a constant of $k = 60$ lb/ft. Plate *B* is slowly moved 2.4 in. to the left and released from rest. Assuming that block *A* does not slip on the plate, determine (a) the amplitude and frequency of the resulting motion, (b) the corresponding smallest allowable value of the coefficient of static friction.

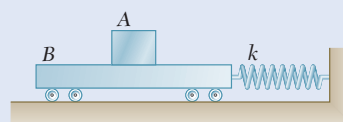


Fig. P19.9

- 19.10** A 5-kg fragile glass vase is surrounded by packing material in a cardboard box of negligible weight. The packing material has negligible damping and a force-deflection relationship as shown. Knowing that the box is dropped from a height of 1 m and the impact with the ground is perfectly plastic, determine (a) the amplitude of vibration for the vase, (b) the maximum acceleration the vase experiences in g's.

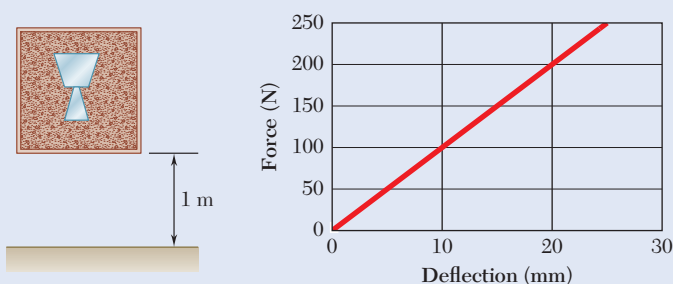


Fig. P19.10

- 19.11** A 3-lb block is supported as shown by a spring of constant $k = 2$ lb/in. that can act in tension or compression. The block is in its equilibrium position when it is struck from below by a hammer that imparts to the block an upward velocity of 90 in./s. Determine (a) the time required for the block to move 3 in. upward, (b) the corresponding velocity and acceleration of the block.

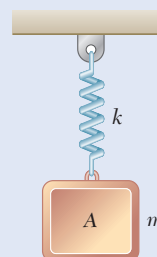


Fig. P19.11

- 19.12** In Prob. 19.11, determine the position, velocity, and acceleration of the block 0.90 s after it has been struck by the hammer.

- 19.13** The bob of a simple pendulum of length $l = 40$ in. is released from rest when $\theta = 5^\circ$. Assuming simple harmonic motion, determine 1.6 s after release (a) the angle θ , (b) the magnitudes of the velocity and acceleration of the bob.

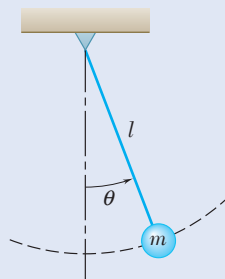


Fig. P19.13

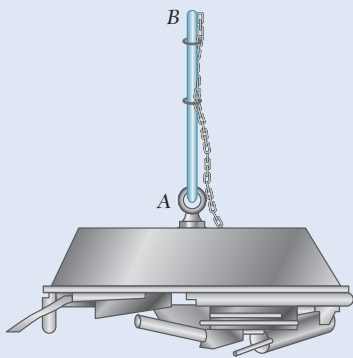


Fig. P19.14

19.14 A 150-kg electromagnet is at rest and is holding 100 kg of scrap steel when the current is turned off and the steel is dropped. Knowing that the cable and the supporting crane have a total stiffness equivalent to a spring of constant 200 kN/m, determine (a) the frequency, the amplitude, and the maximum velocity of the resulting motion, (b) the minimum tension that will occur in the cable during the motion, (c) the velocity of the magnet 0.03 s after the current is turned off.

19.15 A 5-kg collar C is released from rest in the position shown and slides without friction on a vertical rod until it hits a spring with a constant of $k = 720$ N/m that it compresses. The velocity of the collar is reduced to zero, and the collar reverses the direction of its motion and returns to its initial position. The cycle is then repeated. Determine (a) the period of the motion of the collar, (b) the velocity of the collar 0.4 s after it was released. (Note: This is a periodic motion, but it is not simple harmonic motion.)

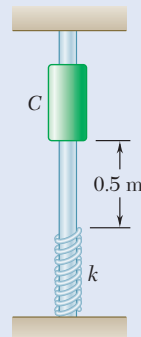


Fig. P19.15

19.16 A small bob is attached to a cord of length 1.2 m and is released from rest when $\theta_A = 5^\circ$. Knowing that $d = 0.6$ m, determine (a) the time required for the bob to return to point A, (b) the amplitude θ_C .

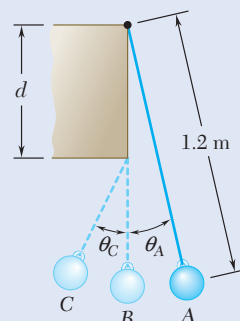


Fig. P19.16

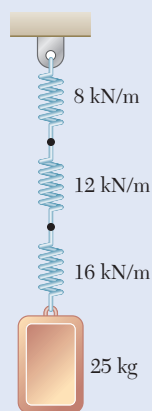


Fig. P19.17

19.17 A 25-kg block is supported by the spring arrangement shown. If the block is moved vertically downward from its equilibrium position and released, determine (a) the period and frequency of the resulting motion, (b) the maximum velocity and acceleration of the block if the amplitude of the motion is 30 mm.

19.18 A 11-lb block is attached to the lower end of a spring whose upper end is fixed and vibrates with a period of 7.2 s. Knowing that the constant k of a spring is inversely proportional to its length (e.g., if you cut a 10 lb/in. spring in half, the remaining two springs each have a spring constant of 20 lb/in.), determine the period of a 7-lb block that is attached to the center of the same spring if the upper and lower ends of the spring are fixed.

19.19 Block A has a mass m and is supported by the spring arrangement as shown. Knowing that the mass of the pulley is negligible and that the block is moved vertically downward from its equilibrium position and released, determine the frequency of the motion.

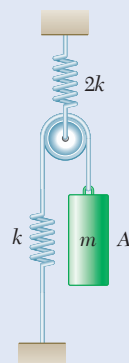


Fig. P19.19

19.20 A 13.6-kg block is supported by the spring arrangement shown. If the block is moved from its equilibrium position 44 mm vertically downward and released, determine (a) the period and frequency of the resulting motion, (b) the maximum velocity and acceleration of the block.

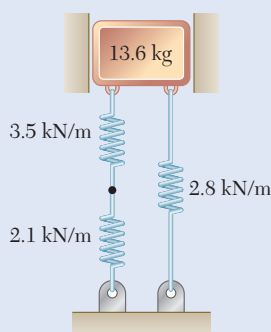


Fig. P19.20

19.21 and 19.22 A 50-kg block is supported by the spring arrangement shown. The block is moved vertically downward from its equilibrium position and released. Knowing that the amplitude of the resulting motion is 60 mm, determine (a) the period and frequency of the motion, (b) the maximum velocity and maximum acceleration of the block.

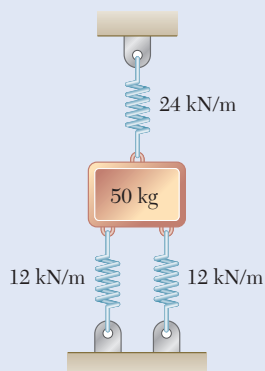


Fig. P19.21

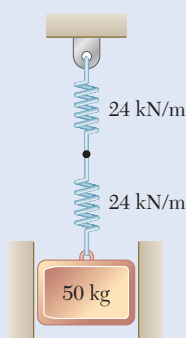


Fig. P19.22

19.23 Two springs with constants k_1 and k_2 are connected in series to a block A that vibrates in simple harmonic motion with a period of 5 s. When the same two springs are connected in parallel to the same block, the block vibrates with a period of 2 s. Determine the ratio k_1/k_2 of the two spring constants.

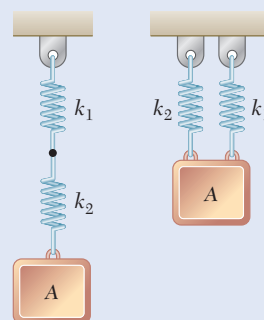


Fig. P19.23

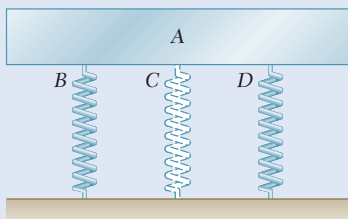


Fig. P19.25



Fig. P19.26

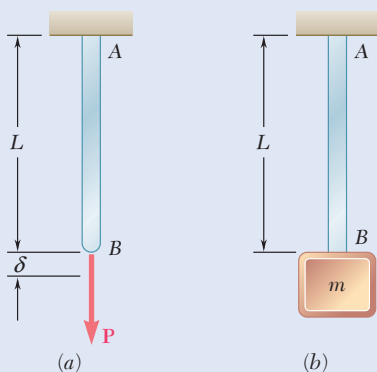


Fig. P19.28

- 19.24** The period of vibration of the system shown is observed to be 0.8 s. If block A is removed, the period is observed to be 0.7 s. Determine (a) the mass of block C, (b) the period of vibration when both blocks A and B have been removed.

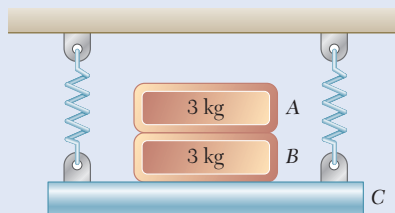


Fig. P19.24

- 19.25** The 100-lb platform A is attached to springs B and D, each of which has a constant $k = 120$ lb/ft. Knowing that the frequency of vibration of the platform is to remain unchanged when an 80-lb block is placed on it and a third spring C is added between springs B and D, determine the required constant of spring C.

- 19.26** The period of vibration for a barrel floating in salt water is found to be 0.58 s when the barrel is empty and 1.8 s when it is filled with 55 gallons of crude oil. Knowing that the density of the oil is 900 kg/m^3 , determine (a) the mass of the empty barrel, (b) the density of the salt water, ρ_{sw} . [Hint: The force of the water on the bottom of the barrel can be modeled as a spring with constant $k = \rho_{\text{sw}}gA$.]

- 19.27** From mechanics of materials, it is known that for a simply supported beam of uniform cross section, a static load $\mathbf{P}$ applied at the center will cause a deflection of $\delta_A = PL^3/48EI$, where L is the length of the beam, E is the modulus of elasticity, and I is the moment of inertia of the cross-sectional area of the beam. Knowing that $L = 15$ ft, $E = 30 \times 10^6$ psi, and $I = 2 \times 10^{-3} \text{ ft}^4$, determine (a) the equivalent spring constant of the beam, (b) the frequency of vibration of a 1500-lb block attached to the center of the beam. Neglect the mass of the beam and assume that the load remains in contact with the beam.

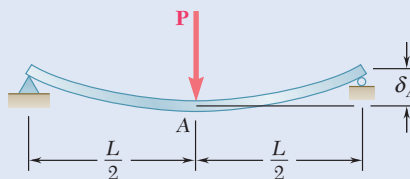


Fig. P19.27

- 19.28** From mechanics of materials it is known that when a static load $\mathbf{P}$ is applied at the end B of a uniform metal rod fixed at end A, the length of the rod will increase by an amount $\delta = PL/AE$, where L is the length of the undeformed rod, A is its cross-sectional area, and E is the modulus of elasticity of the metal. Knowing that $L = 450$ mm and $E = 200$ GPa and that the diameter of the rod is 8 mm, and neglecting the mass of the rod, determine (a) the equivalent spring constant of the rod, (b) the frequency of the vertical vibrations of a block of mass $m = 8$ kg attached to end B of the same rod.

- 19.29** Denoting by δ_{st} the static deflection of a beam under a given load, show that the frequency of vibration of the load is

$$f = \frac{1}{2\pi} \sqrt{\frac{g}{\delta_{st}}}$$

Neglect the mass of the beam, and assume that the load remains in contact with the beam.

- 19.30** A 40-mm deflection of the second floor of a building is measured directly under a newly installed 3500-kg piece of rotating machinery that has a slightly unbalanced rotor. Assuming that the deflection of the floor is proportional to the load it supports, determine (a) the equivalent spring constant of the floor system, (b) the speed in rpm of the rotating machinery that should be avoided if it is not to coincide with the natural frequency of the floor-machinery system.
- 19.31** If $h = 700$ mm and $d = 500$ mm and each spring has a constant $k = 600$ N/m, determine the mass m for which the period of small oscillations is (a) 0.50 s, (b) infinite. Neglect the mass of the rod and assume that each spring can act in either tension or compression.

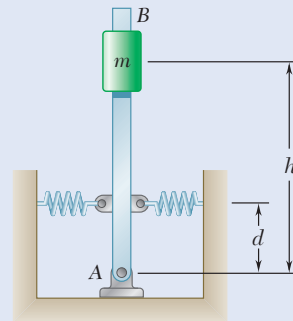


Fig. P19.31

- 19.32** The force–deflection equation for a nonlinear spring fixed at one end is $F = 1.5x^{1/2}$ where F is the force, expressed in newtons, applied at the other end and x is the deflection expressed in meters. (a) Determine the deflection x_0 if a 4-oz block is suspended from the spring and is at rest. (b) Assuming that the slope of the force–deflection curve at the point corresponding to this loading can be used as an equivalent spring constant, determine the frequency of vibration of the block if it is given a very small downward displacement from its equilibrium position and released.
- *19.33** Expanding the integrand in Eq. (19.19) of Sec. 19.1C into a series of even powers of $\sin \phi$ and integrating, show that the period of a simple pendulum of length l may be approximated by the formula

$$\tau = 2\pi \sqrt{\frac{l}{g}} \left(1 + \frac{1}{4} \sin^2 \frac{\theta_m}{2} \right)$$

where θ_m is the amplitude of the oscillations.

- *19.34** Using the formula given in Prob. 19.33, determine the amplitude θ_m for which the period of a simple pendulum is $\frac{1}{2}$ percent longer than the period of the same pendulum for small oscillations.
- *19.35** Using the data of Table 19.1, determine the period of a simple pendulum of length $l = 750$ mm (a) for small oscillations, (b) for oscillations of amplitude $\theta_m = 60^\circ$, (c) for oscillations of amplitude $\theta_m = 90^\circ$.
- *19.36** Using the data of Table 19.1, determine the length in inches of a simple pendulum that oscillates with a period of 2 s and an amplitude of 90° .

19.2 FREE VIBRATIONS OF RIGID BODIES

The analysis of the vibrations of a rigid body (or of a system of rigid bodies) possessing a single degree of freedom is similar to the analysis of the vibrations of a particle. We choose an appropriate variable, such as a distance x or an angle θ , to define the position of the body or system of bodies and write an equation relating this variable and its second derivative with respect to t . If the equation is of the same form as Eq. (19.6), i.e., if we have

$$\ddot{x} + \omega_n^2 x = 0 \quad \text{or} \quad \ddot{\theta} + \omega_n^2 \theta = 0 \quad (19.21)$$

the vibration is a simple harmonic motion. We can obtain the period and natural frequency of the vibration by identifying ω_n and substituting its value into Eqs. (19.13) and (19.14).

In general, a simple way to obtain one of Eqs. (19.21) is to use Newton's second law. To do this, first draw free-body and kinetic diagrams for the system displaced in the positive direction. The acceleration in your kinetic diagram needs to be in the same positive direction you defined for the displacement. From your drawn diagrams, it is straightforward to write the appropriate equation of motion. Recall that the goal should be the determination of the coefficient of the variable x or θ —not the determination of the variable itself or of the derivative $\dot{x}$ or $\dot{\theta}$. Setting this coefficient equal to ω_n^2 , we obtain the natural circular frequency ω_n from which we can determine τ_n and f_n .

This method can be used to analyze vibrations that are truly represented by a simple harmonic motion or by vibrations of small amplitude that can be *approximated* by a simple harmonic motion. As an example, let us determine the period of the small oscillations of a square plate with a side $2b$ that is suspended from the midpoint O of one of its sides (Fig. 19.5a). We consider the plate in an arbitrary position defined by the angle θ that the line OG forms with the vertical. Then we draw free-body and kinetic diagrams to express that the weight $\mathbf{W}$ of the plate and the components $\mathbf{R}_x$ and $\mathbf{R}_y$ of the reaction at O are equivalent to the vectors $m\mathbf{a}_t$ and $m\mathbf{a}_n$ and to the couple $\bar{I}\alpha$ (Fig. 19.5b). Since the angular velocity and angular acceleration of the plate are equal to $\dot{\theta}$ and $\ddot{\theta}$, respectively, the magnitudes of the two vectors $m\mathbf{a}_t$ and $m\mathbf{a}_n$ are $mb\dot{\theta}$ and $mb\dot{\theta}^2$, respectively, and the moment of the couple is $\bar{I}\ddot{\theta}$. In previous applications of this method (Chap. 16), we tried whenever possible to assume the correct sense for the acceleration. Here, however, we must assume the same positive sense for θ and $\ddot{\theta}$ in order to obtain an equation of the form in Eq. (19.21). Consequently, we assume the angular acceleration $\ddot{\theta}$ is positive counterclockwise—even though this assumption is obviously unrealistic. Equating moments about O , we have

$$-W(b \sin \theta) = (mb\ddot{\theta})b + \bar{I}\ddot{\theta}$$

Noting that

$$\bar{I} = \frac{1}{12}m[(2b)^2 + (2b)^2] = \frac{2}{3}mb^2 \quad \text{and} \quad W = mg$$

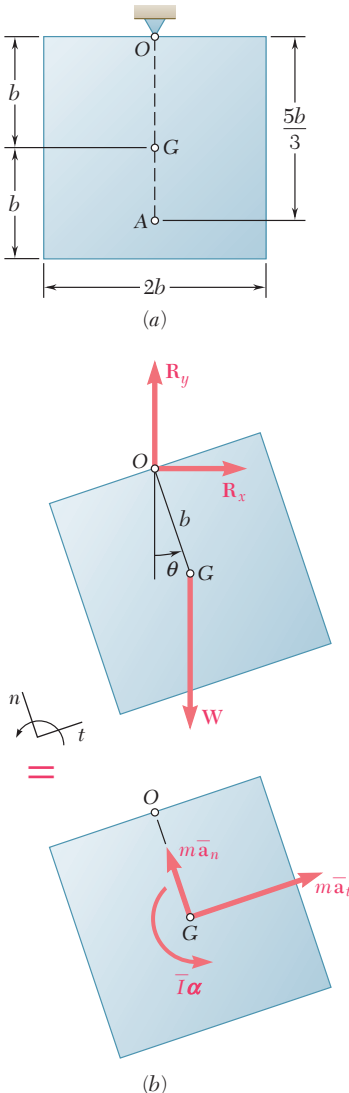


Fig. 19.5 (a) A square plate of side $2b$ suspended from the midpoint of one of its sides; (b) free-body diagram and kinetic diagram for the plate.

we obtain

$$\ddot{\theta} + \frac{3g}{5b} \sin \theta = 0 \quad (19.22)$$

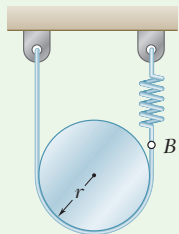
For oscillations of small amplitude, we can replace $\sin \theta$ by θ , expressed in radians, which gives

$$\ddot{\theta} + \frac{3g}{5b} \theta = 0 \quad (19.23)$$

Comparison with Eq. (19.21) shows that this equation is that of a simple harmonic motion and that the natural circular frequency ω_n of the oscillations is equal to $(3g/5b)^{1/2}$. Substituting into Eq. (19.13), we find that the period of the oscillations is

$$\tau_n = \frac{2\pi}{\omega_n} = 2\pi\sqrt{\frac{5b}{3g}} \quad (19.24)$$

This result is valid only for oscillations of small amplitude. A more accurate description of the motion of the plate is obtained by comparing Eqs. (19.16) and (19.22). Note that the two equations are identical if we choose l equal to $5b/3$. This means that the plate oscillates as a simple pendulum with a length of $l = 5b/3$, and we can use the results of Sec. 19.1C to correct the value of the period given in Eq. (19.24). Point A of the plate located on line OG at a distance $l = 5b/3$ from O is defined as the **center of oscillation** corresponding to O (Fig. 19.5a).



Sample Problem 19.2

A cylinder with weight W and radius r is suspended from a looped cord as shown. One end of the cord is attached directly to a rigid support, and the other end is attached to a spring with a constant k . Determine the period and natural frequency of the vibrations of the cylinder.

STRATEGY: First choose a coordinate to describe the motion, and then use Newton's second law to determine the equations of motion.

MODELING: Choose the cylinder to be your system, and model it as a rigid body. The system of external forces acting on the cylinder consists of the weight $\mathbf{W}$ and of the forces $\mathbf{T}_1$ and $\mathbf{T}_2$ exerted by the cord. Draw free-body and kinetic diagrams (Fig. 1) to express that this system is equivalent to the system represented by the vector $m\bar{\mathbf{a}}$ attached at G and the couple $\bar{I}\alpha$.

ANALYSIS:

Kinematics of Motion. Express the linear displacement and the acceleration of the cylinder in terms of the angular displacement θ . Choosing the positive sense clockwise and measuring the displacements from the equilibrium position (Fig. 2), you have

$$\begin{aligned}\bar{x} &= r\theta & \delta &= 2\bar{x} = 2r\theta \\ \alpha &= \ddot{\theta} \downarrow & \bar{a} &= r\alpha = r\ddot{\theta} & \bar{a} &= r\ddot{\theta} \downarrow\end{aligned}\quad (1)$$

Equations of Motion. Newton's second law gives you (Fig. 1)

$$+\downarrow \Sigma M_A = m\bar{a}d_{\perp} + \bar{I}\alpha: \quad Wr - T_2(2r) = m\bar{a}r + \bar{I}\alpha \quad (2)$$

When the cylinder is in its position of equilibrium, the tension in the cord is $T_0 = \frac{1}{2}W$. Note that for an angular displacement θ , the magnitude of $\mathbf{T}_2$ is

$$T_2 = T_0 + k\delta = \frac{1}{2}W + k\delta = \frac{1}{2}W + k(2r\theta) \quad (3)$$

Substituting from Eqs. (1) and (3) into Eq. (2) and recalling that $\bar{I} = \frac{1}{2}mr^2$, you have

$$\begin{aligned}Wr - \left(\frac{1}{2}W + 2kr\theta\right)(2r) &= m(r\ddot{\theta})r + \frac{1}{2}mr^2\ddot{\theta} \\ \ddot{\theta} + \frac{8k}{3m}\theta &= 0\end{aligned}$$

The motion is simple harmonic, and you have

$$\begin{aligned}\omega_n^2 &= \frac{8k}{3m} & \omega_n &= \sqrt{\frac{8k}{3m}} \\ \tau_n &= \frac{2\pi}{\omega_n} & \tau_n &= 2\pi\sqrt{\frac{3m}{8k}} \\ f_n &= \frac{\omega_n}{2\pi} & f_n &= \frac{1}{2\pi}\sqrt{\frac{8k}{3m}}\end{aligned}$$

REFLECT and THINK: If the cylinder had been smooth, it would not have rotated when displaced downward. Also note that the answers you obtained are independent of r .

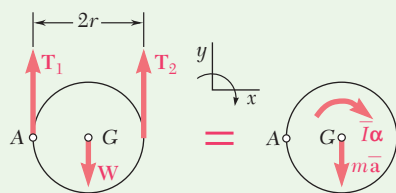


Fig. 1 Free-body diagram and kinetic diagram for the cylinder.

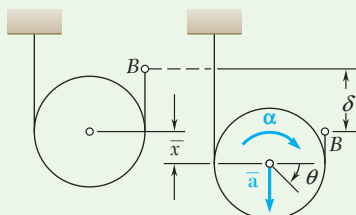
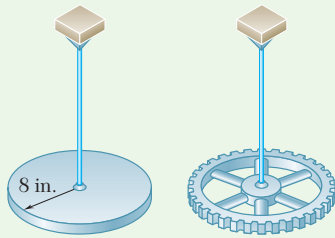


Fig. 2 Linear and angular displacements and linear and angular accelerations of the cylinder.

Sample Problem 19.3



A circular disk weighs 20 lb, has a radius of 8 in., and is suspended from a wire as shown. The disk is rotated (thus twisting the wire) and then released; the period of the torsional vibration is observed to be 1.13 s. A gear is then suspended from the same wire, and the period of torsional vibration for the gear is observed to be 1.93 s. Assuming that the moment of the couple exerted by the wire is proportional to the angle of twist, determine (a) the torsional spring constant of the wire, (b) the centroidal moment of inertia of the gear, (c) the maximum angular velocity reached by the gear if it is rotated through 90° and released.

STRATEGY: Use Newton's second law to obtain the equation of motion. From this, you can find the circular natural frequency in terms of the torsional spring constant and the centroidal moment of inertia. You can determine the torsional spring constant for the wire from the analysis of the disk. Then you can use that to describe the motion of the gear.

MODELING: Choose the disk (or gear) as your system, and model it as a rigid body. The kinematic variables are shown in Fig. 1, and the free-body and kinetic diagrams are shown in Fig. 2.

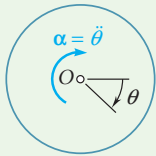


Fig. 1 Angular displacement and acceleration for the disk (or gear).

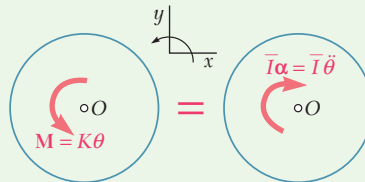


Fig. 2 Free-body diagram and kinetic diagram for the disk (or gear).

ANALYSIS:

a. Vibration of Disk. Denoting the angular displacement of the disk by θ (Fig. 1), you can express that the magnitude of the couple exerted by the wire is $M = K\theta$, where K is the torsional spring constant of the wire. Applying Newton's second law, you have

$$+\circlearrowleft \Sigma M_O = \bar{I}\alpha: \quad +K\theta = -\bar{I}\ddot{\theta}$$

$$\ddot{\theta} + \frac{K}{\bar{I}}\theta = 0$$

The motion is simple harmonic, so you have

$$\omega_n^2 = \frac{K}{\bar{I}} \quad \tau_n = \frac{2\pi}{\omega_n} \quad \tau_n = 2\pi\sqrt{\frac{\bar{I}}{K}} \quad (1)$$

For the disk,

$$\tau_n = 1.13 \text{ s} \quad \bar{I} = \frac{1}{2}mr^2 = \frac{1}{2}\left(\frac{20 \text{ lb}}{32.2 \text{ ft/s}^2}\right)\left(\frac{8}{12} \text{ ft}\right)^2 = 0.138 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$$

(continued)

Substituting into Eq. (1), you obtain

$$1.13 = 2\pi\sqrt{\frac{0.138}{K}} \quad K = 4.27 \text{ lb}\cdot\text{ft}/\text{rad} \quad \blacktriangleleft$$

b. Vibration of Gear. The period of vibration of the gear is 1.93 s and $K = 4.27 \text{ lb}\cdot\text{ft}/\text{rad}$, so Eq. (1) yields

$$1.93 = 2\pi\sqrt{\frac{\bar{I}}{4.27}} \quad \bar{I}_{\text{gear}} = 0.403 \text{ lb}\cdot\text{ft}\cdot\text{s}^2 \quad \blacktriangleleft$$

c. Maximum Angular Velocity of Gear. Because it is simple harmonic motion, you have

$$\theta = \theta_m \sin \omega_n t \quad \omega = \theta_m \omega_n \cos \omega_n t \quad \omega_m = \theta_m \omega_n$$

Recalling that $\theta_m = 90^\circ = 1.571 \text{ rad}$ and $\tau = 1.93 \text{ s}$, you have

$$\omega_m = \theta_m \omega_n = \theta_m \left(\frac{2\pi}{\tau} \right) = (1.571 \text{ rad}) \left(\frac{2\pi}{1.93 \text{ s}} \right) \quad \omega_m = 5.11 \text{ rad/s} \quad \blacktriangleleft$$

REFLECT and THINK: A torsional spring is often used experimentally to measure the mass moment of inertia of different objects. It is common engineering practice to use one situation to determine the dynamic characteristics of a system and then to use those parameters to analyze a slightly different situation.

SOLVING PROBLEMS ON YOUR OWN

In this section, you saw that a rigid body, or a system of rigid bodies, whose position can be defined by a single coordinate x or θ , executes a simple harmonic motion if the differential equation obtained by applying Newton's second law is of the form

$$\ddot{x} + \omega_n^2 x = 0 \quad \text{or} \quad \ddot{\theta} + \omega_n^2 \theta = 0 \quad (19.21)$$

Your goal should be to determine ω_n , from which you can obtain the period τ_n and the natural frequency f_n . Taking into account the initial conditions, you can then write an equation of the form

$$x = x_m \sin(\omega_n t + \phi) \quad (19.10)$$

where you should replace x by θ if a rotation is involved. To solve the problems in this section, you should follow these steps:

1. Choose a coordinate that measures the displacement of the body from its equilibrium position. You will find that many of the problems in this section involve the rotation of a body about a fixed axis and that the angle measuring the rotation of the body from its equilibrium position is the most convenient coordinate to use. In problems involving the general plane motion of a body, where a coordinate x (and possibly a coordinate y) is used to define the position of the mass center G of the body and a coordinate θ is used to measure its rotation about G , kinematic relations will allow you to express x (and y) in terms of θ [Sample Prob. 19.2].

2. Draw a free-body diagram and a kinetic diagram to express that the system of the external forces is equivalent to the vector $m\bar{a}$ and the couple $\bar{I}\alpha$ where $\bar{a} = \ddot{x}$ and $\alpha = \ddot{\theta}$. Be sure that each applied force or couple is drawn in a direction consistent with the assumed displacement and that the senses of $\bar{a}$ and α are those in which the coordinates x and θ are increasing.

3. Write the differential equations of motion by equating the sums of the components of the external forces and the inertial terms in the x and y directions and the sums of their moments about a given point. If necessary, use the kinematic relations developed in Step 1 to obtain equations involving only the coordinate θ . If θ is a small angle, replace $\sin \theta$ by θ and $\cos \theta$ by 1 if these functions appear in your equations. Eliminating any unknown reactions, you will obtain an equation of the type of Eqs. (19.21). Note that, in problems involving a body rotating about a fixed axis, you can immediately obtain such an equation by equating the moments of the external forces and inertial terms about the fixed axis.

(continued)

4. Comparing the equation you have obtained with one of Eqs. (19.21), you can identify ω_n^2 and thus determine the natural circular frequency ω_n . Remember that the object of your analysis is *not to solve* the differential equation you have obtained *but to identify* ω_n^2 .

5. Determine the amplitude and the phase angle ϕ by substituting the value obtained for ω_n and the initial values of the coordinate and its first derivative into Eq. (19.10) and the equation obtained by differentiating Eq. (19.10) with respect to t . From Eq. (19.10) and the two equations obtained by differentiating Eq. (19.10) twice with respect to t and using the kinematic relations developed in Step 1, you will be able to determine the position, velocity, and acceleration of any point of the body at any given time.

6. In problems involving torsional vibrations, the torsional spring constant K is expressed in N·m/rad or lb·ft/rad. The product of K and the angle of twist θ , where θ is expressed in radians, yields the moment of the restoring couple, which should be equated to the inertial terms in the system. [Sample Prob. 19.3].

Problems

- 19.37** The uniform rod shown has mass 6 kg and is attached to a spring of constant $k = 700 \text{ N/m}$. If end B of the rod is depressed 10 mm and released, determine (a) the period of vibration, (b) the maximum velocity of end B .

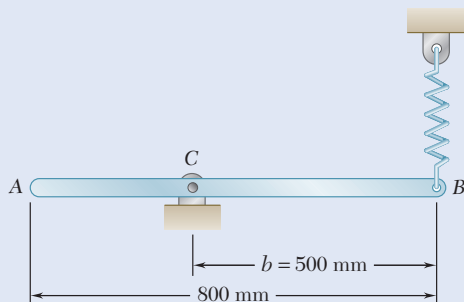


Fig. P19.37

- 19.38** A belt is placed around the rim of a 500-lb flywheel and attached as shown to two springs, each of constant $k = 85 \text{ lb/in.}$ If end C of the belt is pulled 1.5 in. down and released, the period of vibration of the flywheel is observed to be 0.5 s. Knowing that the initial tension in the belt is sufficient to prevent slipping, determine (a) the maximum angular velocity of the flywheel, (b) the centroidal radius of gyration of the flywheel.

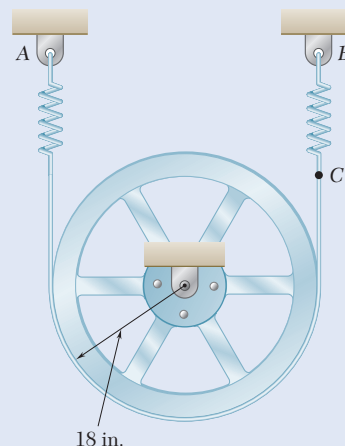


Fig. P19.38

- 19.39** A 6-kg uniform cylinder can roll without sliding on a horizontal surface and is attached by a pin at point C to the 4-kg horizontal bar AB . The bar is attached to two springs, each having a constant of $k = 5 \text{ kN/m}$, as shown. Knowing that the bar is moved 12 mm to the right of the equilibrium position and released, determine (a) the period of vibration of the system, (b) the magnitude of the maximum velocity of bar AB .

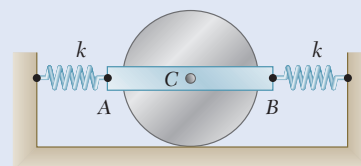


Fig. P19.39 and P19.40

- 19.40** A 6-kg uniform cylinder is assumed to roll without sliding on a horizontal surface and is attached by a pin at point C to the 4-kg horizontal bar AB . The bar is attached to two springs, each having a constant of $k = 3.5 \text{ kN/m}$, as shown. Knowing that the coefficient of static friction between the cylinder and the surface is 0.5, determine the maximum amplitude of the motion of point C that is compatible with the assumption of rolling.

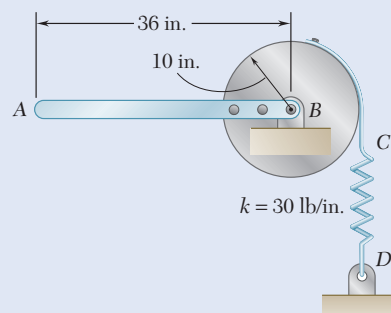


Fig. P19.41

- 19.41** A 15-lb slender rod AB is riveted to a 12-lb uniform disk as shown. A belt is attached to the rim of the disk and to a spring that holds the rod at rest in the position shown. If end A of the rod is moved 0.75 in. down and released, determine (a) the period of vibration, (b) the maximum velocity of end A .

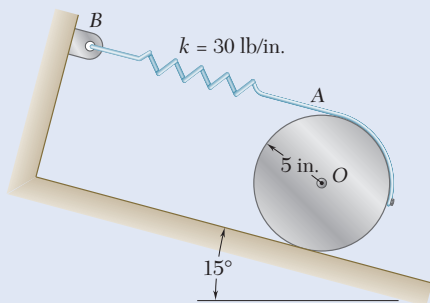


Fig. P19.42

19.42 A 30-lb uniform cylinder can roll without sliding on a 15° incline. A belt is attached to the rim of the cylinder, and a spring holds the cylinder at rest in the position shown. If the center of the cylinder is moved 2 in. down the incline and released, determine (a) the period of vibration, (b) the maximum acceleration of the center of the cylinder.

19.43 A square plate of mass m is held by eight springs, each of constant k . Knowing that each spring can act in either tension or compression, determine the frequency of the resulting vibration if (a) the plate is given a small vertical displacement and released, (b) the plate is rotated through a small angle about G and released.

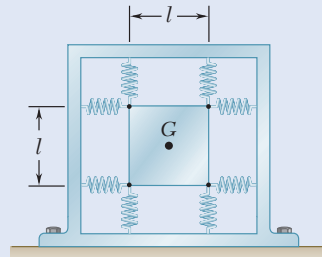


Fig. P19.43

19.44 Two small weights w are attached at A and B to the rim of a uniform disk of radius r and weight W . Denoting by τ_0 the period of small oscillations when $\beta = 0$, determine the angle β for which the period of small oscillations is $2\tau_0$.

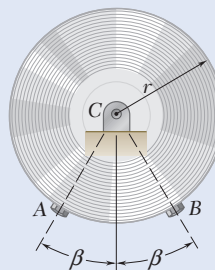


Fig. P19.44 and P19.45

19.45 Two 40-g weights are attached at A and B to the rim of a 1.5-kg uniform disk of radius $r = 100$ mm. Determine the frequency of small oscillations when $\beta = 60^\circ$.

19.46 A three-blade wind turbine used for research is supported on a shaft so that it is free to rotate about O . One technique to determine the centroidal mass moment of inertia of an object is to place a known weight at a known distance from the axis of rotation and to measure the frequency of oscillations after releasing it from rest with a small initial angle. In this case, a weight of $W_{add} = 50$ lb is attached to one of the blades at a distance $R = 20$ ft from the axis of rotation. Knowing that when the blade with the added weight is displaced slightly from the vertical axis, and the system is found to have a period of 7.6 s, determine the centroidal mass moment of inertia of the three-blade rotor.

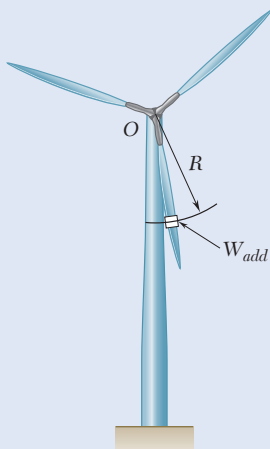


Fig. P19.46

- 19.47** A connecting rod is supported by a knife-edge at point A ; the period of its small oscillations is observed to be 0.87 s. The rod is then inverted and supported by a knife edge at point B and the period of its small oscillations is observed to be 0.78 s. Knowing that $r_a + r_b = 10$ in., determine (a) the location of the mass center G , (b) the centroidal radius of gyration $\bar{k}$.

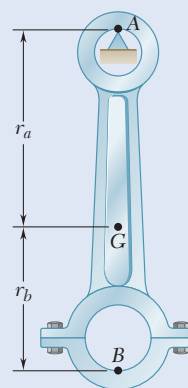


Fig. P19.47

- 19.48** A semicircular hole is cut in a uniform square plate that is attached to a frictionless pin at its geometric center O . Determine (a) the period of small oscillations of the plate, (b) the length of a simple pendulum that has the same period.

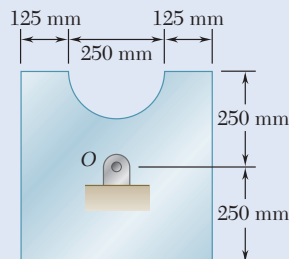


Fig. P19.48

- 19.49** A uniform disk of radius $r = 250$ mm is attached at A to a 650-mm rod AB of negligible mass that can rotate freely in a vertical plane about B . Determine the period of small oscillations (a) if the disk is free to rotate in a bearing at A , (b) if the rod is riveted to the disk at A .

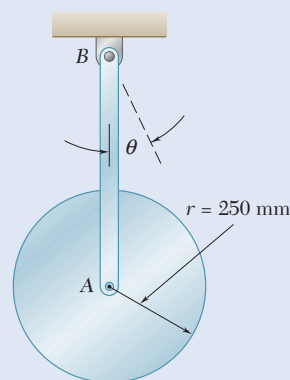


Fig. P19.49

- 19.50** A small collar of mass 1 kg is rigidly attached to a 3-kg uniform rod of length $L = 750$ mm. Determine (a) the distance d to maximize the frequency of oscillation when the rod is given a small initial displacement, (b) the corresponding period of oscillation.

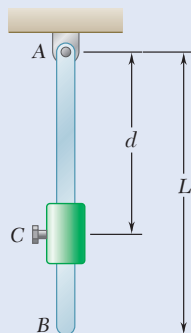


Fig. P19.50

- 19.51** A thin homogeneous wire is bent into the shape of an isosceles triangle of sides b , b , and $1.6b$. Determine the period of small oscillations if the wire (a) is suspended from point A as shown, (b) is suspended from point B .

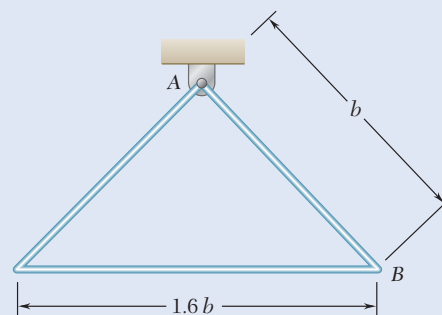


Fig. P19.51

- 19.52** A *compound pendulum* is defined as a rigid body that oscillates about a fixed point O , called the center of suspension. Show that the period of oscillation of a compound pendulum is equal to the period of a simple pendulum of length OA , where the distance from A to the mass center G is $GA = \bar{k}^2/\bar{r}$. Point A is defined as the center of oscillation and coincides with the center of percussion defined in Prob. 17.66.

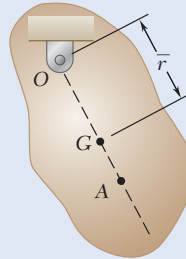


Fig. P19.52 and P19.53

- 19.53** A rigid slab oscillates about a fixed point O . Show that the smallest period of oscillation occurs when the distance $\bar{r}$ from point O to the mass center G is equal to $\bar{k}$.
- 19.54** Show that if the compound pendulum of Prob. 19.52 is suspended from A instead of O , the period of oscillation is the same as before and the new center of oscillation is located at O .
- 19.55** The 8-kg uniform bar AB is hinged at C and is attached at A to a spring of constant $k = 500$ N/m. If end A is given a small displacement and released, determine (a) the frequency of small oscillations, (b) the smallest value of the spring constant k for which oscillations will occur.

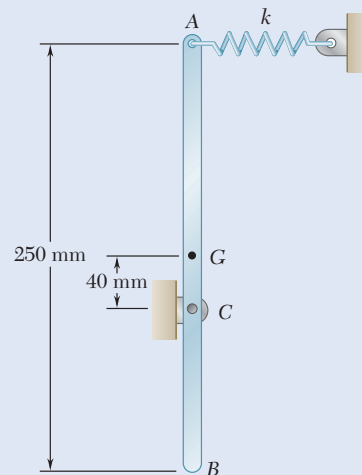


Fig. P19.55

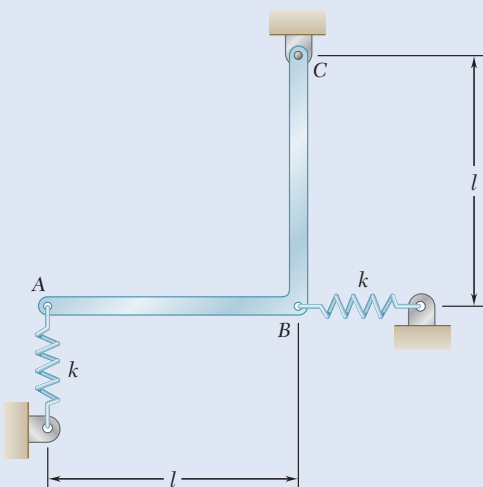


Fig. P19.56

- 19.56** Two uniform rods each have a mass m and length l and are welded together to form an L-shaped assembly. The assembly is constrained by two springs, each with a constant k , and is in equilibrium in a vertical plane in the position shown. Determine the frequency of small oscillations of the system.

19.57 A uniform disk with radius r and mass m can roll without slipping on a cylindrical surface and is attached to bar ABC with a length L and negligible mass. The bar is attached at point A to a spring with a constant k and can rotate freely about point B in the vertical plane. Knowing that end A is given a small displacement and released, determine the frequency of the resulting vibration in terms of m , L , k , and g .

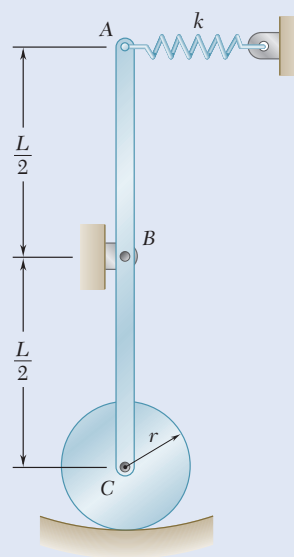


Fig. P19.57

19.58 A 1300-kg sports car has a center of gravity G located a distance h above a line connecting the front and rear axles. The car is suspended from cables that are attached to the front and rear axles as shown. Knowing that the periods of oscillation are 4.04 s when $L = 4$ m and 3.54 s when $L = 3$ m, determine h and the centroidal radius of gyration.

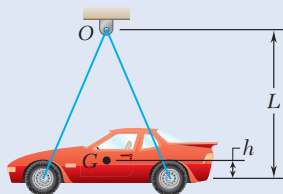


Fig. P19.58

19.59 A 6-lb slender rod is suspended from a steel wire that is known to have a torsional spring constant $K = 1.5$ ft-lb/rad. If the rod is rotated through 180° about the vertical and released, determine (a) the period of oscillation, (b) the maximum velocity of end A of the rod.

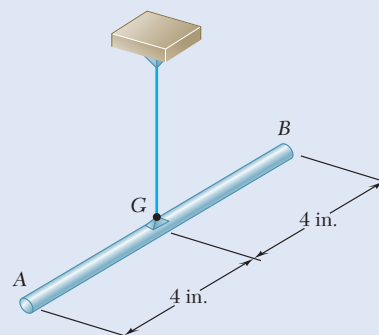


Fig. P19.59

19.60 A uniform disk of radius $r = 250$ mm is attached at A to a 650-mm rod AB of negligible mass that can rotate freely in a vertical plane about B . If the rod is displaced 2° from the position shown and released, determine the magnitude of the maximum velocity of point A , assuming that the disk is (a) free to rotate in a bearing at A , (b) riveted to the rod at A .

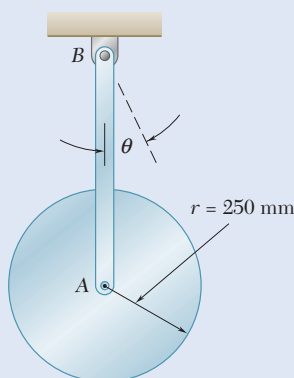


Fig. P19.60

19.61 Two uniform rods, each of mass m and length l , are welded together to form the T-shaped assembly shown. Determine the frequency of small oscillations of the assembly.

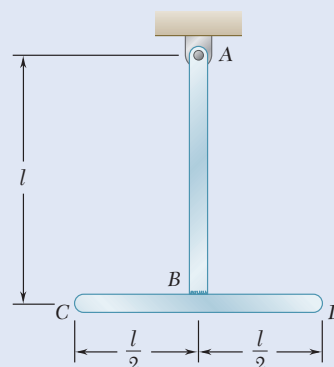


Fig. P19.61

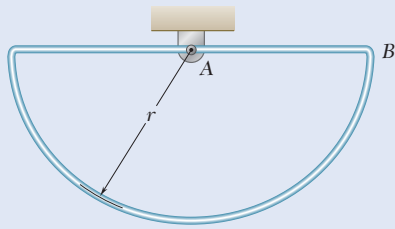


Fig. P19.62

19.62 A homogeneous wire bent to form the figure shown is attached to a pin support at A . Knowing that $r = 220$ mm and that point B is pushed down 20 mm and released, determine the magnitude of the velocity of B , 8 s later.

19.63 A horizontal platform P is held by several rigid bars that are connected to a vertical wire. The period of oscillation of the platform is found to be 2.2 s when the platform is empty and 3.8 s when an object A of unknown moment of inertia is placed on the platform with its mass center directly above the center of the plate. Knowing that the wire has a torsional constant $K = 27$ N·m/rad, determine the centroidal moment of inertia of object A .

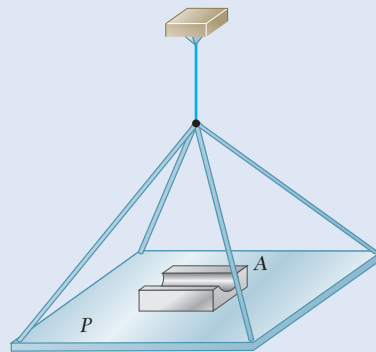


Fig. P19.63

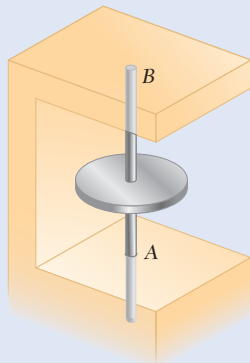


Fig. P19.64

19.64 A uniform disk of radius $r = 120$ mm is welded at its center to two elastic rods of equal length with fixed ends at A and B . Knowing that the disk rotates through an 8° angle when a 500-mN·m couple is applied to the disk and that it oscillates with a period of 1.3 s when the couple is removed, determine (a) the mass of the disk, (b) the period of vibration if one of the rods is removed.

19.65 A 5-kg uniform rod CD of length $l = 0.7$ m is welded at C to two elastic rods, which have fixed ends at A and B and are known to have a combined torsional spring constant $K = 24$ N·m/rad. Determine the period of small oscillations, if the equilibrium position of CD is (a) vertical as shown, (b) horizontal.

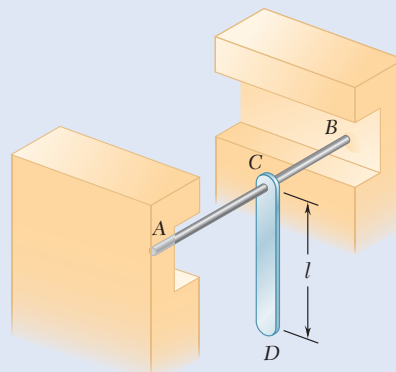


Fig. P19.65

- 19.66** A uniform equilateral triangular plate with a side b is suspended from three vertical wires of the same length l . Determine the period of small oscillations of the plate when (a) it is rotated through a small angle about a vertical axis through its mass center G , (b) it is given a small horizontal displacement in a direction perpendicular to AB .

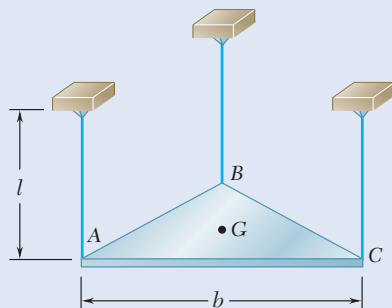


Fig. P19.66

- 19.67** A period of 6.00 s is observed for the angular oscillations of a 4-oz gyroscope rotor suspended from a wire as shown. Knowing that a period of 3.80 s is obtained when a 1.25-in.-diameter steel sphere is suspended in the same fashion, determine the centroidal radius of gyration of the rotor. (Specific weight of steel = 490 lb/ft³.)

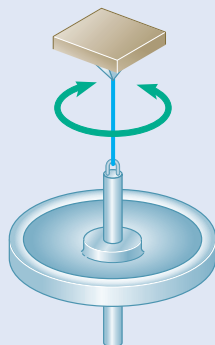


Fig. P19.67

- 19.68** The centroidal radius of gyration $\bar{k}_y$ of an airplane is determined by suspending the airplane by two 12-ft-long cables as shown. The airplane is rotated through a small angle about the vertical through G and then released. Knowing that the observed period of oscillation is 3.3 s, determine the centroidal radius of gyration $\bar{k}_y$.

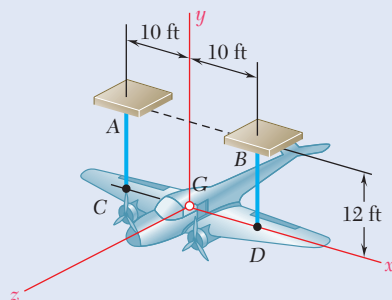


Fig. P19.68

19.3 APPLYING THE PRINCIPLE OF CONSERVATION OF ENERGY

Conservation of energy provides an alternative method to determine the natural frequency of a system. Usually, velocity kinematics are easier than acceleration kinematics, so using energy is sometimes easier than using Newton's second law directly. We saw in Sec. 19.1A that, when a particle with mass m is in simple harmonic motion, the resultant $\mathbf{F}$ of the forces exerted on the particle has a magnitude proportional to the displacement x measured from the position of equilibrium O and is directed toward O ; we have $F = -kx$. Referring to Sec. 13.2A, we note that $\mathbf{F}$ is a *conservative force* and that the corresponding potential energy is $V = \frac{1}{2}kx^2$, where V is assumed equal to zero in the equilibrium position $x = 0$. The velocity of the particle is equal to $\dot{x}$, so its kinetic energy is $T = \frac{1}{2}m\dot{x}^2$. We can state that the total energy of the particle is conserved by writing

$$T + V = \text{constant} \quad \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2 = \text{constant}$$

Dividing through by $m/2$ and recalling from Sec. 19.1A that $k/m = \omega_n^2$, where ω_n is the natural circular frequency of the vibration, we have

$$\dot{x}^2 + \omega_n^2 x^2 = \text{constant} \quad (19.25)$$

Equation (19.25) is characteristic of a simple harmonic motion, since we can obtain it from Eq. (19.6) by multiplying both terms by $2\dot{x}$ and integrating.

Once we have established that the motion of the system is a simple harmonic motion or that it can be approximated by a simple harmonic motion, the principle of conservation of energy provides a convenient way for determining the period of vibration of a rigid body or of a system of rigid bodies possessing a single degree of freedom. Choosing an appropriate variable, such as a distance x or an angle θ , we consider two particular positions of the system:

1. **The displacement of the system is maximum.** We have $T_1 = 0$, and we can express V_1 in terms of the amplitude x_m or θ_m (choosing $V = 0$ in the equilibrium position).
2. **The system passes through its equilibrium position.** We have $V_2 = 0$, and we can express T_2 in terms of the maximum velocity $\dot{x}_m$ or the maximum angular velocity $\dot{\theta}_m$.

We then express that the total energy of the system is conserved and write $T_1 + V_1 = T_2 + V_2$. Recalling from Eq. (19.15) that for simple harmonic motion the maximum velocity is equal to the product of the amplitude and of the natural circular frequency ω_n , we find that we can solve this equation for ω_n .

As an example, let us consider again the square plate of Sec. 19.2 and determine the period of its motion with this new approach. In the position of maximum displacement (Fig. 19.6a), we have

$$T_1 = 0 \quad V_1 = W(b - b \cos \theta_m) = Wb(1 - \cos \theta_m)$$

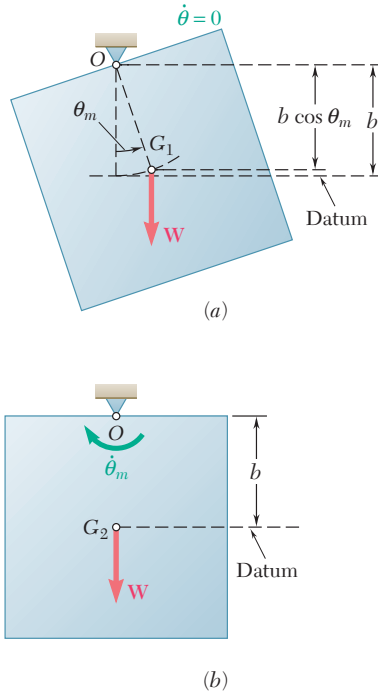


Fig. 19.6 A square plate: (a) in the position of maximum displacement; (b) as it passes through its equilibrium position.

or since $1 - \cos \theta_m = 2 \sin^2(\theta_m/2) \approx 2(\theta_m/2)^2 = \theta_m^2/2$ for oscillations of small amplitude,

$$T_1 = 0 \quad V_1 = \frac{1}{2} W b \theta_m^2 \quad (19.26)$$

As the plate passes through its position of equilibrium (Fig. 19.6b), its velocity is maximum, and we have

$$T_2 = \frac{1}{2} m \bar{v}_m^2 + \frac{1}{2} \bar{I} \omega_m^2 = \frac{1}{2} m b^2 \dot{\theta}_m^2 + \frac{1}{2} \bar{I} \dot{\theta}_m^2 \quad V_2 = 0$$

or recalling from Sec. 19.2 that $\bar{I} = \frac{2}{3} m b^2$,

$$T_2 = \frac{1}{2} \left(\frac{5}{3} m b^2 \right) \dot{\theta}_m^2 \quad V_2 = 0 \quad (19.27)$$

Substituting from Eqs. (19.26) and (19.27) into $T_1 + V_1 = T_2 + V_2$ and noting that the maximum velocity $\dot{\theta}_m$ is equal to the product $\theta_m \omega_n$, we have

$$\frac{1}{2} W b \theta_m^2 = \frac{1}{2} \left(\frac{5}{3} m b^2 \right) \theta_m^2 \omega_n^2 \quad (19.28)$$

This gives us $\omega_n^2 = 3g/5b$ and

$$\tau_n = \frac{2\pi}{\omega_n} = 2\pi \sqrt{\frac{5b}{3g}} \quad (19.29)$$

as obtained earlier in Sec. 19.2.

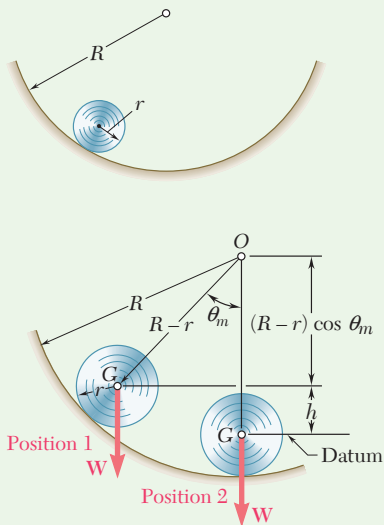


Fig. 1 The cylinder in positions 1 and 2.

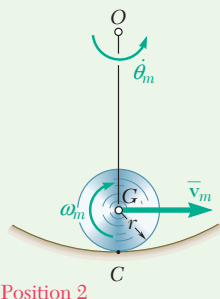


Fig. 2 Kinematic quantities to describe the motion of the disk.

Sample Problem 19.4

Determine the period of small oscillations of a cylinder with radius r that rolls without slipping inside a curved surface with radius R .

STRATEGY: Since the cylinder rolls without slipping, you can apply the principle of conservation of energy between position 1, where $\theta = \theta_m$, and position 2, where $\theta = 0$.

MODELING: Choose the cylinder to be your system and model it as a rigid body. Denote the angle that line OG forms with the vertical by θ (Fig. 1).

ANALYSIS: Position 1.

Kinetic Energy. The velocity of the cylinder is zero, so $T_1 = 0$.

Potential Energy. Choose a datum as shown in Fig. 1 and denote the weight of the cylinder by W . Then you have

$$V_1 = Wh = W(R - r)(1 - \cos \theta)$$

For small oscillations, $(1 - \cos \theta) = 2 \sin^2(\theta/2) \approx \theta^2/2$, so you have

$$V_1 = W(R - r) \frac{\theta_m^2}{2}$$

Position 2. Denote the angular velocity of line OG as the cylinder passes through position 2 by θ_m , and observe that point C is the instantaneous center of rotation of the cylinder (Fig. 2). Then

$$\bar{v}_m = (R - r)\dot{\theta}_m \quad \omega_m = \frac{\bar{v}_m}{r} = \frac{R - r}{r} \dot{\theta}_m$$

Kinetic Energy.

$$\begin{aligned} T_2 &= \frac{1}{2} m \bar{v}_m^2 + \frac{1}{2} I \omega_m^2 \\ &= \frac{1}{2} m (R - r)^2 \dot{\theta}_m^2 + \frac{1}{2} \left(\frac{1}{2} m r^2 \right) \left(\frac{R - r}{r} \right)^2 \dot{\theta}_m^2 \\ &= \frac{3}{4} m (R - r)^2 \dot{\theta}_m^2 \end{aligned}$$

Potential Energy.

$$V_2 = 0$$

Conservation of Energy.

$$T_1 + V_1 = T_2 + V_2$$

$$0 + W(R - r) \frac{\theta_m^2}{2} = \frac{3}{4} m (R - r)^2 \dot{\theta}_m^2 + 0$$

Since $\dot{\theta}_m = \omega_n \theta_m$ and $W = mg$, you have

$$mg(R - r) \frac{\theta_m^2}{2} = \frac{3}{4} m (R - r)^2 (\omega_n \theta_m)^2 \quad \omega_n^2 = \frac{2}{3} \frac{g}{R - r}$$

$$\tau_n = \frac{2\pi}{\omega_n}$$

$$\tau_n = 2\pi \sqrt{\frac{3}{2} \frac{R - r}{g}} \quad \blacktriangleleft$$

REFLECT and THINK: This answer makes sense, because as the radius R increases, the period also increases. In the limit as R goes to infinity, the period also goes to infinity, that is, the system would not oscillate. This is the case of a cylinder on a horizontal surface. The small angle approximation, $(1 - \cos \theta) = 2 \sin^2(\theta/2) \approx \theta^2/2$, is often used in problems like this one.

SOLVING PROBLEMS ON YOUR OWN

In the problems that follow, you will be asked to use the *principle of conservation of energy* to determine the period or natural frequency of the simple harmonic motion of a particle or rigid body. Assuming that you choose an angle θ to define the position of the system (with $\theta = 0$ in the equilibrium position), as you will in most of the problems in this section, you will express that the total energy of the system is conserved using $T_1 + V_1 = T_2 + V_2$ between position 1 of maximum displacement ($\theta_1 = \theta_m, \dot{\theta}_1 = 0$) and position 2 of maximum velocity ($\dot{\theta}_2 = \dot{\theta}_m, \theta_2 = 0$). It follows that T_1 and V_2 are both zero, and the energy equation reduces to $V_1 = T_2$, where V_1 and T_2 are homogeneous quadratic expressions in θ_m and $\dot{\theta}_m$, respectively. Recalling that for a simple harmonic motion, $\dot{\theta}_m = \theta_m \omega_n$, and substituting this product into the energy equation, after reduction you obtain an equation that you can solve for ω_n^2 . Once you have determined the natural circular frequency ω_n , you can obtain the period τ_n and the natural frequency f_n of the vibration.

The steps that you should take are as follows:

1. Calculate the potential energy V_1 of the system in its position of maximum displacement. Draw a sketch of the system in its position of maximum displacement and express the potential energy of all the forces involved in terms of the maximum displacement x_m or θ_m .

a. The potential energy associated with the weight W of a body is $V_g = Wy$, where y is the elevation of the center of gravity G of the body above its equilibrium position. If the problem you are solving involves the oscillation of a rigid body about a horizontal axis through a point O located at a distance b from G (Fig. 19.6), express y in terms of the angle θ that the line OG forms with the vertical: $y = b(1 - \cos \theta)$. For small values of θ , you can replace this expression with $y = \frac{1}{2}b\theta^2$ [Sample Prob. 19.4]. Therefore, when θ reaches its maximum value θ_m and for oscillations of small amplitude, you can express V_g as

$$V_g = \frac{1}{2}Wb\theta_m^2$$

Note that if G is located above O in its equilibrium position (instead of below O , as we have assumed), the vertical displacement y is negative and should be approximated as $y = -\frac{1}{2}b\theta^2$, which results in a negative value for V_g . In the absence of other forces, the equilibrium position is unstable, and the system does not oscillate. (See, for instance, Prob. 19.89.)

(continued)

b. The potential energy associated with the elastic force exerted by a spring is $V_e = \frac{1}{2}kx^2$, where k is the constant of the spring and x is its deflection. In problems involving the rotation of a body about an axis, you generally have $x = a\theta$, where a is the distance from the axis of rotation to the point of the body where the spring is attached and θ is the angle of rotation. Therefore, when x reaches its maximum value x_m and θ reaches its maximum value θ_m , you can express V_e as

$$V_e = \frac{1}{2}kx_m^2 = \frac{1}{2}ka^2\theta_m^2$$

c. The potential energy V_1 of the system in its position of maximum displacement is obtained by adding the various potential energies that you have computed. It is equal to the product of a constant and θ_m^2 .

2. Calculate the kinetic energy T_2 of the system in its position of maximum velocity. Note that this position is also the equilibrium position of the system.

a. If the system consists of a single rigid body, the kinetic energy T_2 of the system is the sum of the kinetic energy associated with the motion of the mass center G of the body and the kinetic energy associated with the rotation of the body about G . Therefore, you can write

$$T_2 = \frac{1}{2}m\bar{v}_m^2 + \frac{1}{2}\bar{I}\omega_m^2$$

Assuming that the position of the body has been defined by an angle θ , express $\bar{v}_m$ and ω_m in terms of the rate of change $\dot{\theta}_m$ of θ as the body passes through its equilibrium position. The kinetic energy of the body is thus expressed as the product of a constant and $\dot{\theta}_m^2$. Note that if θ measures the rotation of the body about its mass center, as was the case for the plate of Fig. 19.6, then $\omega_m = \dot{\theta}_m$. In other cases, however, the kinematics of the motion should be used to derive a relation between ω_m and $\dot{\theta}_m$ [Sample Prob. 19.4].

b. If the system consists of several rigid bodies, repeat the previous computation for each of the bodies using the same coordinate θ and add the results.

3. Equate the potential energy V_1 of the system to its kinetic energy T_2 ,

$$V_1 = T_2$$

and recalling the first of Eqs. (19.15), replace $\dot{\theta}_m$ in the right-hand term with the product of the amplitude θ_m and the circular frequency ω_n . Since both terms now contain the factor θ_m^2 , you can cancel this factor and solve the resulting equation for the circular frequency ω_n .

Problems

19.69 Two blocks each have a mass 1.5 kg and are attached to links that are pin-connected to bar BC as shown. The masses of the links and bar are negligible, and the blocks can slide without friction. Block D is attached to a spring of constant $k = 720 \text{ N/m}$. Knowing that block A is at rest when it is struck horizontally with a mallet and given an initial velocity of 250 mm/s , determine the magnitude of the maximum displacement of block D during the resulting motion.

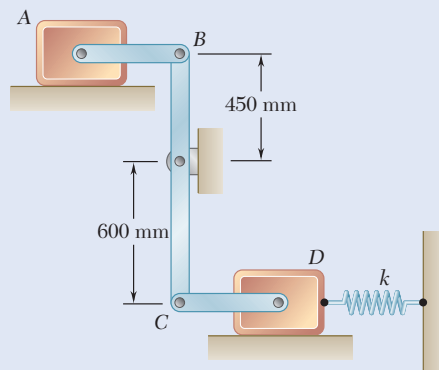


Fig. P19.69

19.70 Two small spheres, A and C , each have a mass m and are attached to rod AB that is supported by a pin and bracket at B and by a spring CD with constant k . Knowing that the mass of the rod is negligible and that the system is in equilibrium when the rod is horizontal, determine the frequency of the small oscillations of the system.

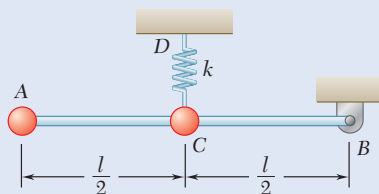


Fig. P19.70

19.71 A 14-oz sphere A and a 10-oz sphere C are attached to the ends of a rod AC of negligible weight that can rotate in a vertical plane about an axis at B . Determine the period of small oscillations of the rod.

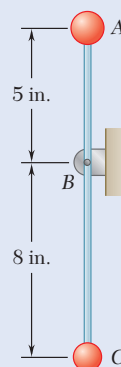


Fig. P19.71

19.72 Determine the period of small oscillations of a small particle that moves without friction inside a cylindrical surface of radius R .

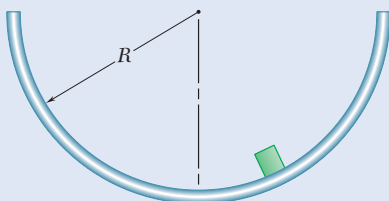


Fig. P19.72

19.73 The inner rim of an 85-lb flywheel is placed on a knife edge, and the period of its small oscillations is found to be 1.26 s . Determine the centroidal moment of inertia of the flywheel.

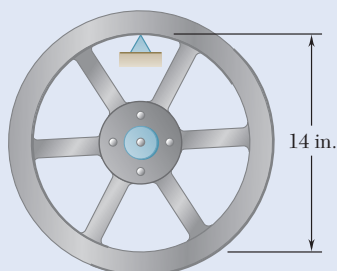


Fig. P19.73

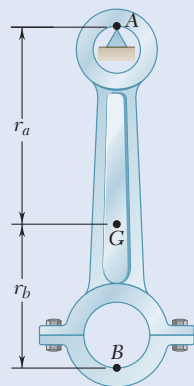


Fig. P19.74

19.74 A connecting rod is supported by a knife edge at point A; the period of its small oscillations is observed to be 1.03 s. Knowing that the distance r_a is 6 in., determine the centroidal radius of gyration of the connecting rod.

19.75 A uniform rod AB can rotate in a vertical plane about a horizontal axis at C located at a distance c above the mass center G of the rod. For small oscillations determine the value of c for which the frequency of the motion will be maximum.

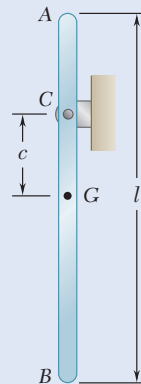


Fig. P19.75

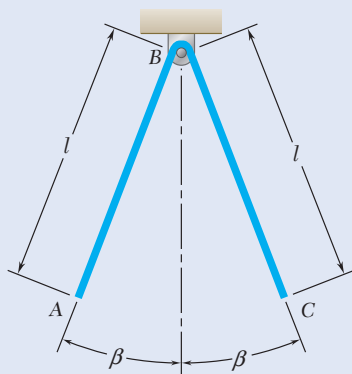


Fig. P19.76

19.76 A homogeneous wire of length $2l$ is bent as shown and allowed to oscillate about a frictionless pin at B . Denoting by τ_0 the period of small oscillations when $\beta = 0$, determine the angle β for which the period of small oscillations is $2\tau_0$.

19.77 A uniform disk of radius r and mass m can roll without slipping on a cylindrical surface and is attached to bar ABC of length L and negligible mass. The bar is attached to a spring of constant k and can rotate freely in the vertical plane about point B . Knowing that end A is given a small displacement and released, determine the frequency of the resulting oscillations in terms of m , L , k , and g .

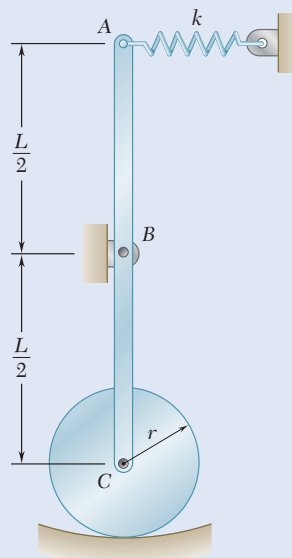


Fig. P19.77

19.78 Two uniform rods, each of weight $W = 1.2$ lb and length $l = 8$ in., are welded together to form the assembly shown. Knowing that the constant of each spring is $k = 0.6$ lb/in. and that end A is given a small displacement and released, determine the frequency of the resulting motion.

19.79 A 15-lb uniform cylinder can roll without sliding on an incline and is attached to a spring AB as shown. If the center of the cylinder is moved 0.4 in. down the incline and released, determine (a) the period of vibration, (b) the maximum velocity of the center of the cylinder.

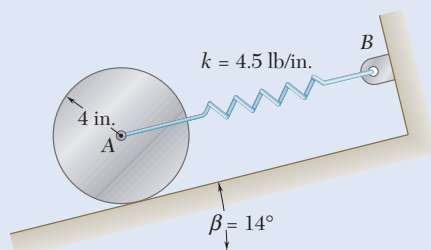


Fig. P19.79

19.80 A 3-kg slender rod AB is bolted to a 5-kg uniform disk. A spring of constant 280 N/m is attached to the disk and is unstretched in the position shown. If end B of the rod is given a small displacement and released, determine the period of vibration of the system.

19.81 A slender 10-kg bar AB with a length of $l = 0.6$ m is connected to two collars of negligible weight. Collar A is attached to a spring with a constant of $k = 1.5$ kN/m and can slide on a horizontal rod, while collar B can slide freely on a vertical rod. Knowing that the system is in equilibrium when bar AB is vertical and that collar A is given a small displacement and released, determine the period of the resulting vibrations.

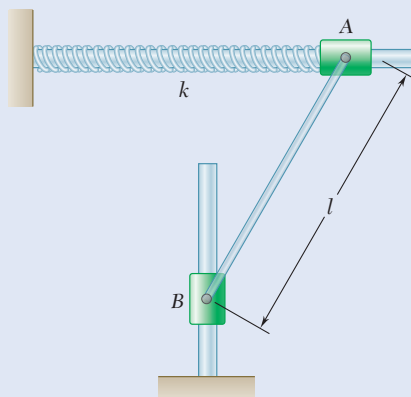


Fig. P19.81 and P19.82

19.82 A slender 5-kg bar AB with a length of $l = 0.6$ m is connected to two collars, each of mass 2.5 kg. Collar A is attached to a spring with a constant of $k = 1.5$ kN/m and can slide on a horizontal rod, while collar B can slide freely on a vertical rod. Knowing that the system is in equilibrium when bar AB is vertical and that collar A is given a small displacement and released, determine the period of the resulting vibrations.

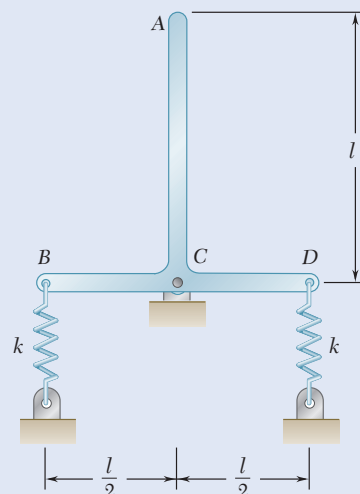


Fig. P19.78

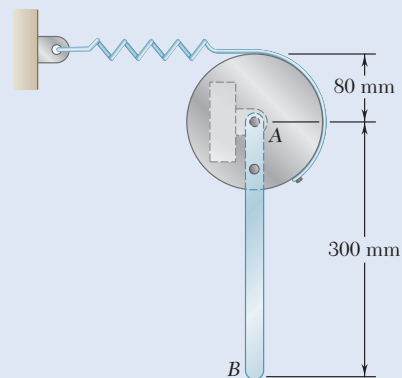


Fig. P19.80

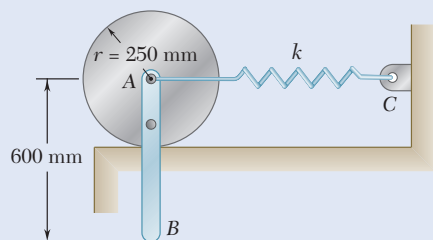


Fig. P19.83

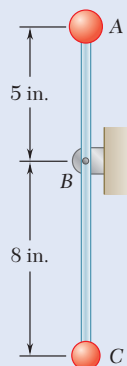


Fig. P19.85

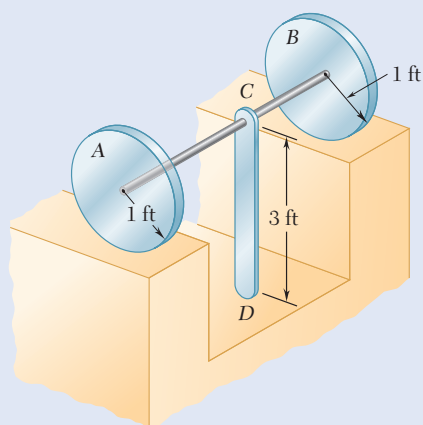


Fig. P19.86

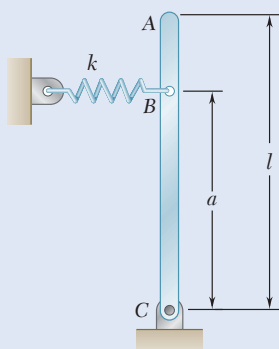


Fig. P19.89

19.83 An 800-g rod AB is bolted to a 1.2-kg disk. A spring of constant $k = 12 \text{ N/m}$ is attached to the center of the disk at A and to the wall at C . Knowing that the disk rolls without sliding, determine the period of small oscillations of the system.

19.84 Three identical 3.6-kg uniform slender bars are connected by pins as shown and can move in a vertical plane. Knowing that bar BC is given a small displacement and released, determine the period of vibration of the system.

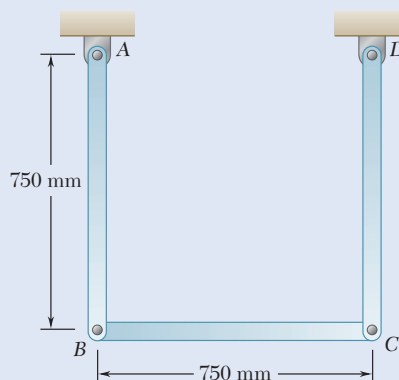


Fig. P19.84

19.85 A 14-oz sphere A and a 10-oz sphere C are attached to the ends of a 20-oz rod AC that can rotate in a vertical plane about an axis at B . Determine the period of small oscillations of the rod.

19.86 A 10-lb uniform rod CD is welded at C to a shaft of negligible mass that is welded to the centers of two 20-lb uniform disks A and B . Knowing that the disks roll without sliding, determine the period of small oscillations of the system.

19.87 and 19.88 Two uniform rods AB and CD , each of length l and mass m , are attached to gears as shown. Knowing that the mass of gear C is m and that the mass of gear A is $4m$, determine the period of small oscillations of the system.

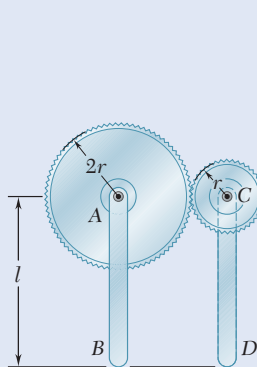


Fig. P19.87

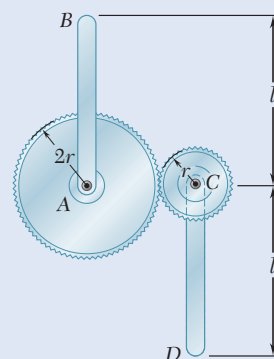


Fig. P19.88

19.89 An inverted pendulum consisting of a rigid bar ABC of length l and mass m is supported by a pin and bracket at C . A spring of constant k is attached to the bar at B and is undeformed when the bar is in the vertical position shown. Determine (a) the frequency of small oscillations, (b) the smallest value of a for which these oscillations will occur.

- 19.90** Two 12-lb uniform disks are attached to the 20-lb rod AB as shown. Knowing that the constant of the spring is 30 lb/in. and that the disks roll without sliding, determine the frequency of vibration of the system.

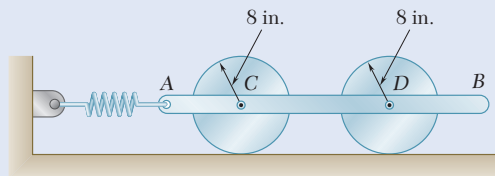


Fig. P19.90

- 19.91** The 20-lb rod AB is attached to two 8-lb disks as shown. Knowing that the disks roll without sliding, determine the frequency of small oscillations of the system.

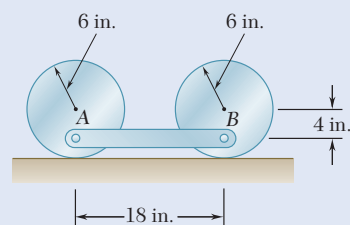


Fig. P19.91

- 19.92** A half section of a uniform cylinder of radius r and mass m rests on two casters A and B , each of which is a uniform cylinder of radius $r/4$ and mass $m/8$. Knowing that the half cylinder is rotated through a small angle and released and that no slipping occurs, determine the frequency of small oscillations.

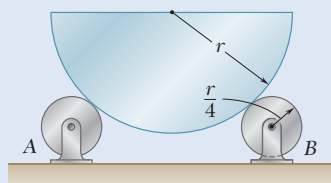


Fig. P19.92

- 19.93** The motion of the uniform rod AB is guided by the cord BC and by the small roller at A . Determine the frequency of oscillation when the end B of the rod is given a small horizontal displacement and released.

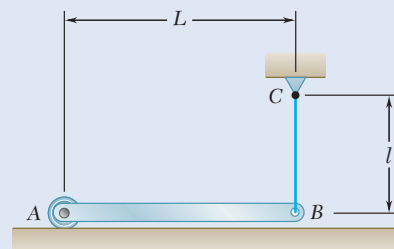


Fig. P19.93

- 19.94** A uniform rod of length L is supported by a ball-and-socket joint at A and by a vertical wire CD . Derive an expression for the period of oscillation of the rod if end B is given a small horizontal displacement and then released.

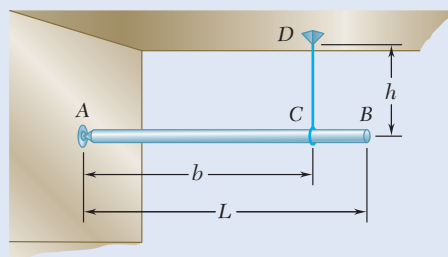


Fig. P19.94

- 19.95** A section of uniform pipe is suspended from two vertical cables attached at A and B . Determine the frequency of oscillation when the pipe is given a small rotation about the centroidal axis OO' and released.

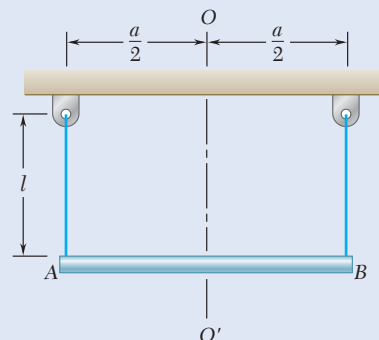


Fig. P19.95

- 19.96** Three collars each have a mass m and are connected by pins to bars AC and BC , each having length l and negligible mass. Collars A and B can slide without friction on a horizontal rod and are connected by a spring of constant k . Collar C can slide without friction on a vertical rod and the system is in equilibrium in the position shown. Knowing that collar C is given a small displacement and released, determine the frequency of the resulting motion of the system.

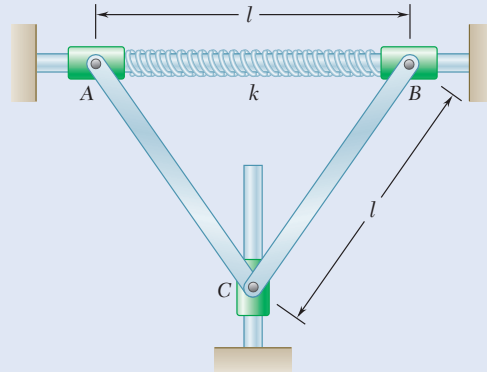


Fig. P19.96

- *19.97** A thin plate of length l rests on a half cylinder of radius r . Derive an expression for the period of small oscillations of the plate.

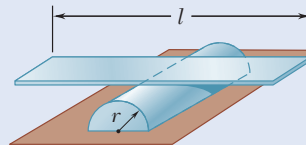


Fig. P19.97

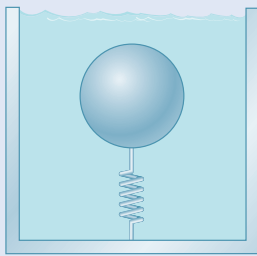


Fig. P19.98

- *19.98** As a submerged body moves through a fluid, the particles of the fluid flow around the body and thus acquire kinetic energy. In the case of a sphere moving in an ideal fluid, the total kinetic energy acquired by the fluid is $\frac{1}{4}\rho Vv^2$, where ρ is the mass density of the fluid, V is the volume of the sphere, and v is the velocity of the sphere. Consider a 500-g hollow spherical shell of radius 80 mm that is held submerged in a tank of water by a spring of constant 500 N/m. (a) Neglecting fluid friction, determine the period of vibration of the shell when it is displaced vertically and then released. (b) Solve part a, assuming that the tank is accelerated upward at the constant rate of 8 m/s^2 .

19.4 FORCED VIBRATIONS

From the point of view of engineering applications, the most important vibrations are the **forced vibrations** of a system. These vibrations occur when a system is subjected to a periodic force or when it is elastically connected to a support that has an alternating motion.

Consider first the case of a body of mass m suspended from a spring and subjected to a periodic force $\mathbf{P}$ with a magnitude of $P = P_m \sin \omega_f t$, where ω_f is the circular frequency of $\mathbf{P}$ and is referred to as the **forced circular frequency** of the motion (Fig. 19.7). This force may be an actual external force applied to the body, or it may be a result of the rotation of some unbalanced part of the body (see Sample Prob. 19.5). Denoting the displacement of the body measured from its equilibrium position by x , the equation of motion is obtained from the free-body diagram and kinetic diagram in Fig. 19.7 as

$$+\downarrow \Sigma F = ma: \quad P_m \sin \omega_f t + W - k(\delta_{st} + x) = m\ddot{x}$$

Recalling that $W = k\delta_{st}$, we have

$$m\ddot{x} + kx = P_m \sin \omega_f t \quad (19.30)$$

Next we consider the case of a body with a mass m suspended from a spring attached to a moving support whose displacement δ is equal to $\delta_m \sin \omega_f t$ (Fig. 19.8). Measuring the displacement x of the body from the position of static equilibrium corresponding to $\omega_f t = 0$, we find that the total elongation of the spring at time t is $\delta_{st} + x - \delta_m \sin \omega_f t$. The equation of motion is thus

$$+\downarrow \Sigma F = ma: \quad W - k(\delta_{st} + x - \delta_m \sin \omega_f t) = m\ddot{x}$$

Again recalling that $W = k\delta_{st}$, we have

$$m\ddot{x} + kx = k\delta_m \sin \omega_f t \quad (19.31)$$

Note that Eqs. (19.30) and (19.31) are of the same form and that a solution of the first equation will satisfy the second if we set $P_m = k\delta_m$.

A differential equation such as Eq. (19.30) or (19.31), possessing a right-hand side different from zero, is said to be *nonhomogeneous*. We can obtain its general solution by adding a particular solution of the given equation to the general solution of the corresponding *homogeneous* equation (with the right-hand side equal to zero). We can obtain a *particular solution* of Eq. (19.30) or (19.31) by trying a solution of the form

$$x_{\text{part}} = x_m \sin \omega_f t \quad (19.32)$$

Substituting x_{part} for x into Eq. (19.30), we find

$$-m\omega_f^2 x_m \sin \omega_f t + kx_m \sin \omega_f t = P_m \sin \omega_f t$$

We can solve this equation for the amplitude as

$$x_m = \frac{P_m}{k - m\omega_f^2}$$

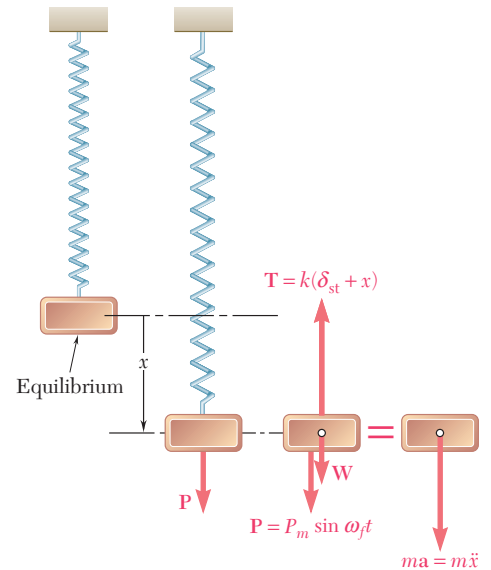


Fig. 19.7 Free-body diagram and kinetic diagram of a block suspended from a spring and subjected to a periodic force.

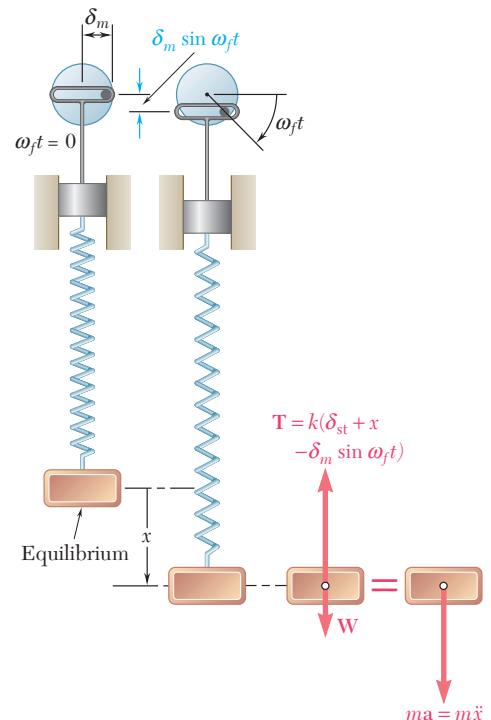


Fig. 19.8 Free-body diagram and kinetic diagram of a block suspended from a spring attached to a harmonically moving support.

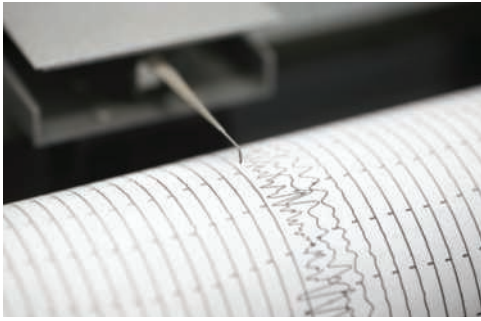


Photo 19.1 A seismometer operates by measuring the amount of electrical energy needed to keep a mass centered in the housing in the presence of strong ground vibration.

Recall from Eq. (19.4) that $k/m = \omega_n^2$, where ω_n is the natural circular frequency of the system. Then we have

$$x_m = \frac{P_m/k}{1 - (\omega_f/\omega_n)^2} \quad (19.33)$$

If we define the frequency ratio, r , as $r = \omega_f/\omega_n$, we can write this equation as

$$x_m = \frac{P_m/k}{1 - r^2}$$

In a similar way, substituting from Eq. (19.32) into Eq. (19.31), we obtain

$$x_m = \frac{\delta_m}{1 - (\omega_f/\omega_n)^2} \quad (19.33')$$

or

$$x_m = \frac{\delta_m}{1 - r^2}$$

The homogeneous equation corresponding to Eq. (19.30) or (19.31) is Eq. (19.2), which defines the free vibration of the body. We found its general solution, called the *complementary function*, in Sec. 19.1A:

$$x_{\text{comp}} = C_1 \sin \omega_n t + C_2 \cos \omega_n t \quad (19.34)$$

Adding the particular solution of Eq. (19.32) to the complementary function of Eq. (19.34), we obtain the **general solution** of Eqs. (19.30) and (19.31) as

$$x = C_1 \sin \omega_n t + C_2 \cos \omega_n t + x_m \sin \omega_f t \quad (19.35)$$

Note that this vibration consists of two superposed vibrations. The first two terms in Eq. (19.35) represent a free vibration of the system. The frequency of this vibration is the *natural frequency* of the system, which depends only upon the constant k of the spring and the mass m of the body, and the constants C_1 and C_2 can be determined from the initial conditions. This free vibration is also called a **transient** vibration, since in actual practice, it is soon damped out by friction forces (Sec. 19.5B).

The last term in Eq. (19.35) represents the **steady-state** vibration produced and maintained by the impressed force or impressed support movement. Its frequency is the **forced frequency** imposed by this force or movement, and its amplitude x_m , defined by Eq. (19.33) or (19.33'), depends upon the **frequency ratio** $r = \omega_f/\omega_n$. Dividing the amplitude x_m of the steady-state vibration by P_m/k in the case of a periodic force, or by δ_m in the case of an oscillating support, we obtain the **magnification factor**. From Eqs. (19.33) and (19.33'), we obtain

$$\text{Magnification factor} = \frac{x_m}{P_m/k} = \frac{x_m}{\delta_m} = \frac{1}{1 - (\omega_f/\omega_n)^2} \quad (19.36)$$

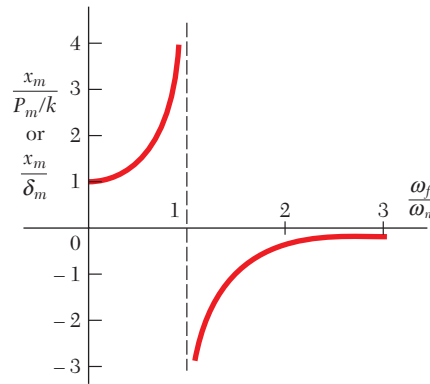
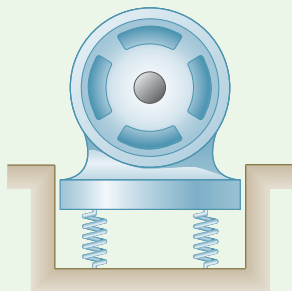


Fig. 19.9 For an undamped system, the magnification factor becomes infinite at a forcing frequency equal to the natural frequency.

In Fig. 19.9, we have plotted the magnification factor against the frequency ratio ω_f/ω_n . Note that when $\omega_f = \omega_n$, the amplitude of the forced vibration becomes infinite. The impressed force or impressed support movement is said to be in **resonance** with the given system. Actually, the amplitude of the vibration remains finite because of damping forces (Sec. 19.5B); nevertheless, such a situation should be avoided, and the forced frequency should not be chosen too close to the natural frequency of the system. Also note that for $\omega_f < \omega_n$, the coefficient of $\sin \omega_f t$ in Eq. (19.35) is positive, whereas for $\omega_f > \omega_n$, this coefficient is negative. In the first case, the forced vibration is *in phase* with the impressed force or impressed support movement, while in the second case, it is 180° *out of phase*.

Finally, observe that we can obtain the velocity and acceleration of the steady-state vibration by differentiating the last term of Eq. (19.35) twice with respect to t . The maximum values are given by expressions similar to those of Eqs. (19.15) of Sec. 19.1A, except that these expressions now involve the amplitude and the circular frequency of the forced vibration:

$$v_m = x_m \omega_f \quad a_m = x_m \omega_f^2 \quad (19.37)$$



Sample Problem 19.5

A motor weighing 350 lb is supported by four springs, each having a constant of 750 lb/in. The unbalance of the rotor is equivalent to a weight of 1 oz located 6 in. from the axis of rotation. The motor is constrained to move vertically. Determine (a) the speed in rpm at which resonance will occur, (b) the amplitude of the vibration of the motor at a speed of 1200 rpm.

STRATEGY: You can determine the resonance speed directly from the given data since you know $\omega_n = \sqrt{k/m}$. To find the vibration amplitude at a speed of 1200 rpm, you can use Eq. (19.33).

MODELING: Choose the motor to be your system, and model it as a single degree-of-freedom particle undergoing forced oscillation.

ANALYSIS:

a. Resonance Speed. The resonance speed is equal to the natural circular frequency ω_n (in rpm) of the free vibration of the motor. The mass of the motor, M , and the equivalent constant of the supporting springs are

$$M = \frac{350 \text{ lb}}{32.2 \text{ ft/s}^2} = 10.87 \text{ lb}\cdot\text{s}^2/\text{ft}$$

$$k = 4(750 \text{ lb/in.}) = 3000 \text{ lb/in.} = 36,000 \text{ lb/ft}$$

$$\omega_n = \sqrt{\frac{k}{M}} = \sqrt{\frac{36,000}{10.87}} = 57.5 \text{ rad/s} = 549 \text{ rpm}$$

Resonance speed = 549 rpm ◀

b. Amplitude of Vibration at 1200 rpm. The angular velocity of the motor and the mass m of the equivalent 1-oz weight are

$$\omega = 1200 \text{ rpm} = 125.7 \text{ rad/s}$$

$$m = (1 \text{ oz}) \frac{1 \text{ lb}}{16 \text{ oz}} \frac{1}{32.2 \text{ ft/s}^2} = 0.001941 \text{ lb}\cdot\text{s}^2/\text{ft}$$

To find the equivalent of an applied force, you can draw a free-body diagram and kinetic diagram (Fig. 1).

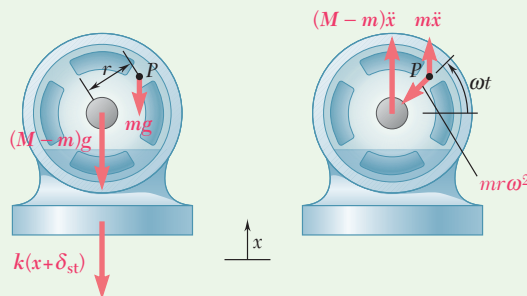


Fig. 1 Free-body diagram and kinetic diagram for the system.

Applying Newton's second law in the vertical direction gives

$$-(M - m)g - mg - k(x + \delta_{st}) = (M - m)\ddot{x} + m\ddot{x} - mr\omega^2 \sin \omega t$$

Recognizing that $Mg = k\delta_{st}$, this equation simplifies to

$$M\ddot{x} + kx = mr\omega^2 \sin \omega t$$

Thus, the rotating unbalanced mass is equivalent to an applied force

$$P_m = mr\omega^2 = (0.001941 \text{ lb}\cdot\text{s}^2/\text{ft})(\frac{6}{12} \text{ ft})(125.7 \text{ rad/s})^2 = 15.33 \text{ lb}$$

The static deflection that would be caused by a constant load P_m is

$$\frac{P_m}{k} = \frac{15.33 \text{ lb}}{3000 \text{ lb/in.}} = 0.00511 \text{ in.}$$

The forced circular frequency ω_f of the motion is the angular velocity of the motor,

$$\omega_f = \omega = 125.7 \text{ rad/s}$$

Substituting the values of P_m/k , ω_f , and ω_n into Eq. (19.33), we obtain

$$x_m = \frac{P_m/k}{1 - (\omega_f/\omega_n)^2} = \frac{0.00511 \text{ in.}}{1 - (125.7/57.5)^2} = -0.001352 \text{ in.}$$

$$x_m = 0.001352 \text{ in. (out of phase)} \quad \blacktriangleleft$$

REFLECT and THINK: In problems involving an unbalanced mass, the result of the imbalance is equivalent to an applied force of $P_m = mr\omega^2$. In this problem, since $\omega_f > \omega_n$, the vibration is 180° out of phase with the force due to the unbalance of the rotor. For example, when the unbalanced mass is directly below the axis of rotation, the position of the motor is $x_m = 0.001352 \text{ in.}$ above the position of equilibrium.

SOLVING PROBLEMS ON YOUR OWN

In this section, we analyzed the **forced vibrations** of a mechanical system. These vibrations occur either when the system is subjected to a periodic force **P** (Fig. 19.7) or when it is elastically connected to a support that has an alternating motion (Fig. 19.8). In the first case, the motion of the system is defined by the differential equation

$$m\ddot{x} + kx = P_m \sin \omega_f t \quad (19.30)$$

where the right-hand side represents the magnitude of the force **P** at a given instant. In the second case, the motion is defined by the differential equation

$$m\ddot{x} + kx = k\delta_m \sin \omega_f t \quad (19.31)$$

where the right-hand side is the product of the spring constant k and the displacement of the support at a given instant.

You will be concerned only with the **steady-state** motion of the system, which is defined by a *particular solution* of Eqs. (19.30) and (19.31), of the form

$$x_{\text{part}} = x_m \sin \omega_f t \quad (19.32)$$

1. If the forced vibration is caused by a periodic force **P** with an amplitude P_m and circular frequency ω_f , the amplitude of the vibration is

$$x_m = \frac{P_m/k}{1 - (\omega_f/\omega_n)^2} \quad (19.33)$$

where ω_n is the *natural circular frequency* of the system, $\omega_n = \sqrt{k/m}$, and k is the spring constant. Note that the circular frequency of the vibration is ω_f and that the amplitude x_m does not depend upon the initial conditions. For $\omega_f = \omega_n$, the denominator in Eq. (19.33) is zero, and x_m is infinite (Fig. 19.9); the impressed force **P** is said to be in **resonance** with the system. Also, for $\omega_f < \omega_n$, x_m is positive and the vibration is *in phase* with **P**, whereas for $\omega_f > \omega_n$, x_m is negative and the vibration is *out of phase*.

a. In the problems that follow, you may be asked to determine one of the parameters in Eq. (19.33) when the others are known. We suggest that you keep Fig. 19.9 in front of you when solving these problems. For example, if you are asked to find the frequency at which the amplitude of a forced vibration has a given value, but you do not know whether the vibration is in or out of phase with respect to the impressed force, you should note from Fig. 19.9 that there can be two frequencies satisfying this requirement. One frequency corresponds to a positive value of x_m and to a vibration in phase with the impressed force, and the other corresponds to a negative value of x_m and to a vibration out of phase with the impressed force.

b. Once you have obtained the amplitude x_m of the motion of a component of the system from Eq. (19.33), you can use Eqs. (19.37) to determine the maximum values of the velocity and acceleration of that component:

$$v_m = x_m \omega_f \quad a_m = x_m \omega_f^2 \quad (19.37)$$

c. When the impressed force P is due to the unbalance of the rotor of a motor, its maximum value is $P_m = mr\omega_f^2$, where m is the mass of the rotor, r is the distance between its mass center and the axis of rotation, and ω_f is equal to the angular velocity ω of the rotor expressed in rad/s [Sample Prob. 19.5].

2. If the forced vibration is caused by the simple harmonic motion of a support with an amplitude δ_m and a circular frequency ω_f , the amplitude of the vibration is

$$x_m = \frac{\delta_m}{1 - (\omega_f/\omega_n)^2} \quad (19.33')$$

where ω_n is the *natural circular frequency* of the system and $\omega_n = \sqrt{k/m}$. Again, note that the circular frequency of the vibration is ω_f and that the amplitude x_m does not depend upon the initial conditions.

a. Be sure to read our comments in paragraphs 1, 1a, and 1b, since they apply equally well to a vibration caused by the motion of a support.

b. If the maximum acceleration a_m of the support is specified, rather than its maximum displacement δ_m , remember that, since the motion of the support is a simple harmonic motion, you can use the relation $a_m = \delta_m \omega_f^2$ to determine δ_m ; then substitute this value into Eq. (19.33').

Problems

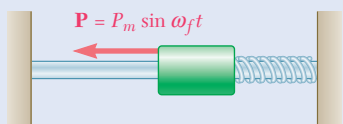


Fig. P19.99, P19.100 and P19.101

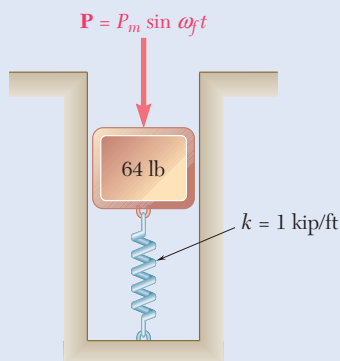


Fig. P19.102

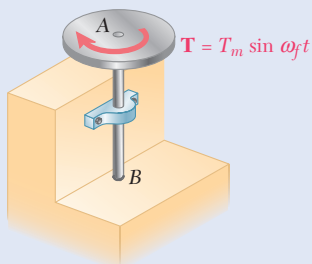


Fig. P19.104

19.99 A 4-kg collar can slide on a frictionless horizontal rod and is attached to a spring with a constant of 450 N/m. It is acted upon by a periodic force with a magnitude of $P = P_m \sin \omega_f t$, where $P_m = 13$ N. Determine the amplitude of the motion of the collar if (a) $\omega_f = 5$ rad/s, (b) $\omega_f = 10$ rad/s.

19.100 A 4-kg collar can slide on a frictionless horizontal rod and is attached to a spring with constant k . It is acted upon by a periodic force of magnitude $P = P_m \sin \omega_f t$, where $P_m = 9$ N and $\omega_f = 5$ rad/s. Determine the value of the spring constant k knowing that the motion of the collar has an amplitude of 150 mm and is (a) in phase with the applied force, (b) out of phase with the applied force.

19.101 A collar with mass m that slides on a frictionless horizontal rod is attached to a spring with constant k and is acted upon by a periodic force with a magnitude of $P = P_m \sin \omega_f t$. Determine the range of values of ω_f for which the amplitude of the vibration exceeds three times the static deflection caused by a constant force with a magnitude of P_m .

19.102 A 64-lb block is attached to a spring with a constant of $k = 1$ kip/ft and can move without friction in a vertical slot as shown. It is acted upon by a periodic force with a magnitude of $P = P_m \sin \omega_f t$, where $\omega_f = 10$ rad/s. Knowing that the amplitude of the motion is 0.75 in., determine P_m .

19.103 A small 20-kg block A is attached to the rod BC of negligible mass that is supported at B by a pin and bracket and at C by a spring of constant $k = 2$ kN/m. The system can move in a vertical plane and is in equilibrium when the rod is horizontal. The rod is acted upon at C by a periodic force P of magnitude $P = P_m \sin \omega_f t$, where $P_m = 6$ N. Knowing that $b = 200$ mm, determine the range of values of ω_f for which the amplitude of vibration of block A exceeds 3.5 mm.

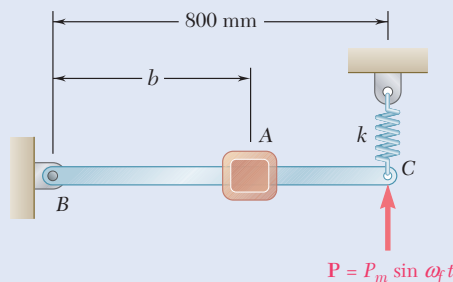


Fig. P19.103

19.104 An 8-kg uniform disk of radius 200 mm is welded to a vertical shaft with a fixed end at B. The disk rotates through an angle of 3° when a static couple of magnitude 50 N·m is applied to it. If the disk is acted upon by a periodic torsional couple of magnitude $T = T_m \sin \omega_f t$, where $T_m = 60$ N·m, determine the range of values of ω_f for which the amplitude of the vibration is less than the angle of rotation caused by a static couple of magnitude T_m .

- 19.105** An 18-lb block A slides in a vertical frictionless slot and is connected to a moving support B by means of a spring AB of constant $k = 10$ lb/in. Knowing that the displacement of the support is $\delta = \delta_m \sin \omega_f t$, where $\delta_m = 6$ in., determine the range of values of ω_f for which the amplitude of the fluctuating force exerted by the spring on the block is less than 30 lb.

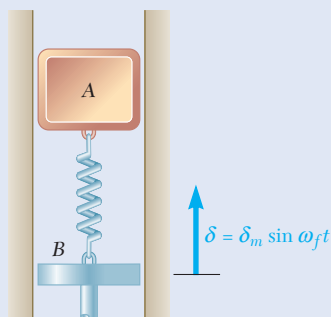


Fig. P19.105

- 19.106** A beam ABC is supported by a pin connection at A and by rollers at B . A 120-kg block placed on the end of the beam causes a static deflection of 15 mm at C . Assuming that the support at A undergoes a vertical periodic displacement $\delta = \delta_m \sin \omega_f t$, where $\delta_m = 10$ mm and $\omega_f = 18$ rad/s, and the support at B does not move, determine the maximum acceleration of the block at C . Neglect the weight of the beam and assume that the block does not leave the beam.

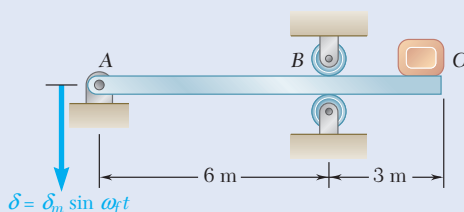


Fig. P19.106

- 19.107** A small 2-kg sphere B is attached to the bar AB of negligible mass that is supported at A by a pin and bracket and connected at C to a moving support D by means of a spring of constant $k = 3.6$ kN/m. Knowing that support D undergoes a vertical displacement $\delta = \delta_m \sin \omega_f t$, where $\delta_m = 3$ mm and $\omega_f = 15$ rad/s, determine (a) the magnitude of the maximum angular velocity of bar AB , (b) the magnitude of the maximum acceleration of sphere B .

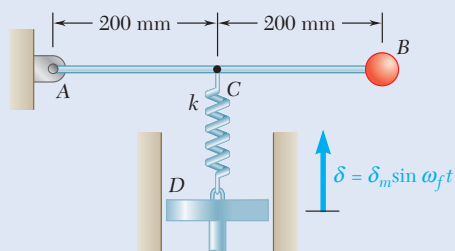


Fig. P19.107

- 19.108** The crude-oil pumping rig shown is driven at 20 rpm. The inside diameter of the well pipe is 2 in., and the diameter of the pump rod is 0.75 in. The length of the pump rod and the length of the column of oil lifted during the stroke are essentially the same, and equal to 6000 ft. During the downward stroke, a valve at the lower end of the pump rod opens to let a quantity of oil into the well pipe, and the column of oil is then lifted to obtain a discharge into the connecting pipeline. Thus, the amount of oil pumped in a given time depends upon the stroke of the lower end of the pump rod. Knowing that the upper end of the rod at D is essentially sinusoidal with a stroke of 45 in. and the specific weight of crude oil is 56.2 lb/ft^3 , determine (a) the output of the well in ft^3/min if the shaft is rigid, (b) the output of the well in ft^3/min if the stiffness of the rod is 2210 N/m , the equivalent mass of the oil and shaft is 290 kg , and damping is negligible.

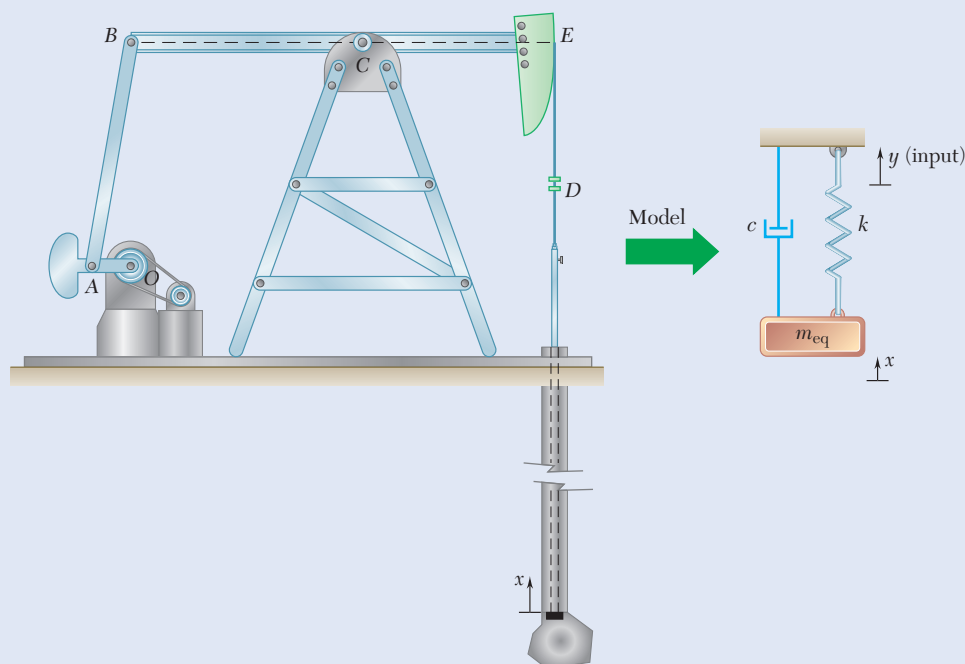


Fig. P19.108

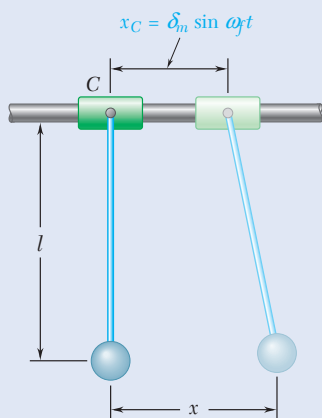


Fig. P19.109 and P19.110

- 19.109** A simple pendulum of length l is suspended from collar C that is forced to move horizontally according to the relation $x_C = \delta_m \sin \omega_f t$. Determine the range of values of ω_f for which the amplitude of the motion of the bob is less than δ_m . (Assume that δ_m is small compared with the length l of the pendulum.)
- 19.110** The 2.75-lb bob of a simple pendulum of length $l = 24 \text{ in.}$ is suspended from a 3-lb collar C . The collar is forced to move according to the relation $x_C = \delta_m \sin \omega_f t$, with an amplitude $\delta_m = 0.4 \text{ in.}$ and a frequency $f_f = 0.5 \text{ Hz}$. Determine (a) the amplitude of the motion of the bob, (b) the force that must be applied to collar C to maintain the motion.

- 19.111** An 18-lb block *A* slides in a vertical frictionless slot and is connected to a moving support *B* by means of a spring *AB* of constant $k = 8$ lb/ft. Knowing that the acceleration of the support is $a = a_m \sin \omega_f t$, where $a_m = 5$ ft/s² and $\omega_f = 6$ rad/s, determine (a) the maximum displacement of block *A*, (b) the amplitude of the fluctuating force exerted by the spring on the block.

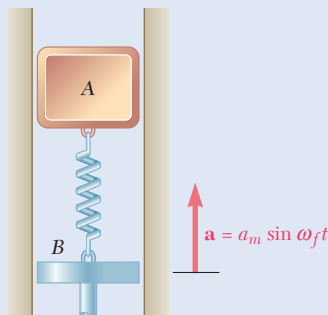


Fig. P19.111

- 19.112** A variable-speed motor is rigidly attached to a beam *BC*. When the speed of the motor is less than 600 rpm or more than 1200 rpm, a small object placed at *A* is observed to remain in contact with the beam. For speeds between 600 and 1200 rpm, the object is observed to “dance” and actually to lose contact with the beam. Determine the speed at which resonance will occur.

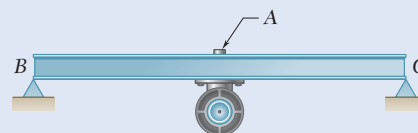


Fig. P19.112

- 19.113** A motor of mass *M* is supported by springs with an equivalent spring constant *k*. The unbalance of its rotor is equivalent to a mass *m* located at a distance *r* from the axis of rotation. Show that when the angular velocity of the motor is ω_f , the amplitude x_m of the motion of the motor is

$$x_m = \frac{r(m/M)(\omega_f/\omega_n)^2}{1 - (\omega_f/\omega_n)^2}$$

where $\omega_n = \sqrt{k/M}$.

- 19.114** As the rotational speed of a spring-supported 100-kg motor is increased, the amplitude of the vibration due to the unbalance of its 15-kg rotor first increases and then decreases. It is observed that as very high speeds are reached, the amplitude of the vibration approaches 3.3 mm. Determine the distance between the mass center of the rotor and its axis of rotation. (*Hint*: Use the formula derived in Prob. 19.113.)

- 19.115** A motor of weight 40 lb is supported by four springs, each of constant 225 lb/in. The motor is constrained to move vertically, and the amplitude of its motion is observed to be 0.05 in. at a speed of 1200 rpm. Knowing that the weight of the rotor is 9 lb, determine the distance between the mass center of the rotor and the axis of the shaft.

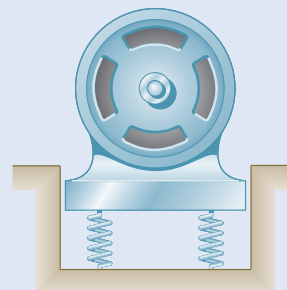


Fig. P19.115

- 19.116** A motor weighing 400 lb is supported by springs having a total constant of 1200 lb/in. The unbalance of the rotor is equivalent to a 1-oz weight located 8 in. from the axis of rotation. Determine the range of allowable values of the motor speed if the amplitude of the vibration is not to exceed 0.06 in.



Fig. P19.117

19.117 A 180-kg motor is bolted to a light horizontal beam. The unbalance of its rotor is equivalent to a 28-g mass located 150 mm from the axis of rotation, and the static deflection of the beam due to the weight of the motor is 12 mm. The amplitude of the vibration due to the unbalance can be decreased by adding a plate to the base of the motor. If the amplitude of vibration is to be less than $60\text{ }\mu\text{m}$ for motor speeds above 300 rpm, determine the required mass of the plate.

19.118 The unbalance of the rotor of a 400-lb motor is equivalent to a 3-oz weight located 6 in. from the axis of rotation. In order to limit to 0.2 lb the amplitude of the fluctuating force exerted on the foundation when the motor is run at speeds of 100 rpm and above, a pad is to be placed between the motor and the foundation. Determine (a) the maximum allowable spring constant k of the pad, (b) the corresponding amplitude of the fluctuating force exerted on the foundation when the motor is run at 200 rpm.

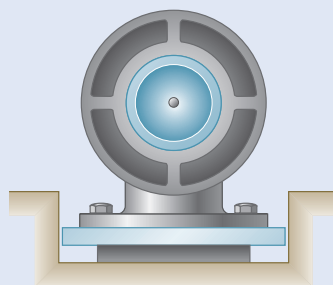


Fig. P19.118

19.119 A counter-rotating eccentric mass exciter consisting of two rotating 100-g masses describing circles of radius r at the same speed but in opposite senses is placed on a machine element to induce a steady-state vibration of the element. The total mass of the system is 300 kg, the constant of each spring is $k = 600\text{ kN/m}$, and the rotational speed of the exciter is 1200 rpm. Knowing that the amplitude of the total fluctuating force exerted on the foundation is 160 N, determine the radius r .

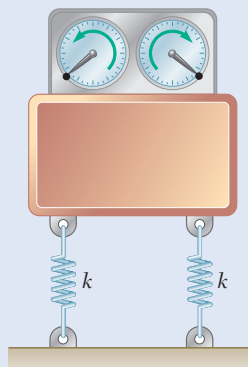


Fig. P19.119

19.120 A 360-lb motor is supported by springs of total constant 12.5 kips/ft. The unbalance of the rotor is equivalent to a 0.9-oz weight located 7.5 in. from the axis of rotation. Determine the range of speeds of the motor for which the amplitude of the fluctuating force exerted on the foundation is less than 5 lb.

- 19.121** Figures (1) and (2) show how springs can be used to support a block in two different situations. In Fig. (1), they help decrease the amplitude of the fluctuating force transmitted by the block to the foundation. In Fig. (2), they help decrease the amplitude of the fluctuating displacement transmitted by the foundation to the block. The ratio of the transmitted force to the impressed force or the ratio of the transmitted displacement to the impressed displacement is called the *transmissibility*. Derive an equation for the transmissibility for each situation. Give your answer in terms of the ratio ω_f/ω_n of the frequency ω_f of the impressed force or impressed displacement to the natural frequency ω_n of the spring-mass system. Show that in order to cause any reduction in transmissibility, the ratio ω_f/ω_n must be greater than $\sqrt{2}$.

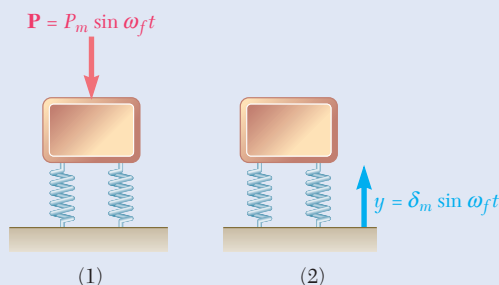


Fig. P19.121

- 19.122** A vibrometer used to measure the amplitude of vibrations consists essentially of a box containing a mass-spring system with a known natural frequency of 120 Hz. The box is rigidly attached to a surface that is moving according to the equation $y = \delta_m \sin \omega_f t$. If the amplitude z_m of the motion of the mass relative to the box is used as a measure of the amplitude δ_m of the vibration of the surface, determine (a) the percent error when the frequency of the vibration is 600 Hz, (b) the frequency at which the error is zero.

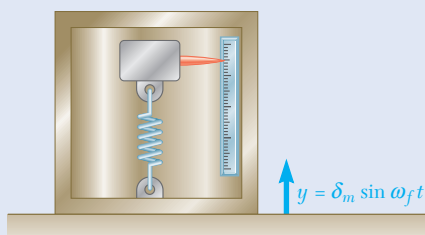


Fig. P19.122 and P19.123

- 19.123** A certain accelerometer consists essentially of a box containing a mass-spring system with a known natural frequency of 2200 Hz. The box is rigidly attached to a surface that is moving according to the equation $y = \delta_m \sin \omega_f t$. If the amplitude z_m of the motion of the mass relative to the box times a scale factor ω_n^2 is used as a measure of the maximum acceleration $\alpha_m = \delta_m \omega_f^2$ of the vibrating surface, determine the percent error when the frequency of the vibration is 600 Hz.

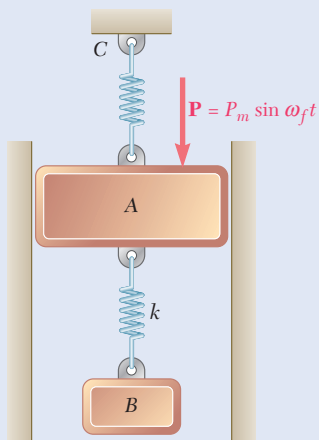


Fig. P19.124

19.124 Block A can move without friction in the slot as shown and is acted upon by a vertical periodic force of magnitude $P = P_m \sin \omega_f t$, where $\omega_f = 2 \text{ rad/s}$ and $P_m = 20 \text{ N}$. A spring of constant k is attached to the bottom of block A and to a 22-kg block B. Determine (a) the value of the constant k that will prevent a steady-state vibration of block A, (b) the corresponding amplitude of the vibration of block B.

19.125 A 60-lb disk is attached with an eccentricity $e = 0.006 \text{ in.}$ to the midpoint of a vertical shaft AB that revolves at a constant angular velocity ω_f . Knowing that the spring constant k for horizontal movement of the disk is 40,000 lb/ft, determine (a) the angular velocity ω_f at which resonance will occur, (b) the deflection r of the shaft when $\omega_f = 1200 \text{ rpm}$.

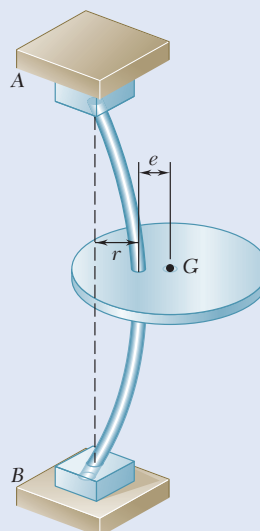


Fig. P19.125

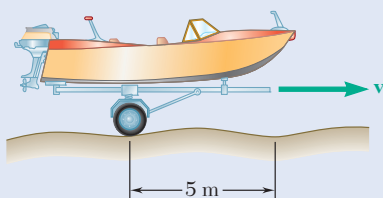


Fig. P19.126

19.126 A small trailer and its load have a total mass of 250 kg. The trailer is supported by two springs, each of constant 10 kN/m, and is pulled over a road, the surface of which can be approximated by a sine curve with an amplitude of 40 mm and a wavelength of 5 m (i.e., the distance between successive crests is 5 m and the vertical distance from crest to trough is 80 mm). Determine (a) the speed at which resonance will occur, (b) the amplitude of the vibration of the trailer at a speed of 50 km/h.

19.5 DAMPED VIBRATIONS

The vibrating systems considered in the first part of this chapter were assumed free of damping. Actually, all vibrations are damped to some degree by friction forces. These forces can be caused by *dry friction*, or *Coulomb friction*, between rigid bodies; by *fluid friction* when a rigid body moves in a fluid; or by *internal friction* between the molecules of a seemingly elastic body. A type of damping of special interest is the *viscous damping* caused by fluid friction at low and moderate speeds. We will first consider free vibrations with viscous damping and then examine the effect of viscous damping on forced vibrations.

*19.5A Damped Free Vibrations

Viscous damping is characterized by the fact that the friction force is *directly proportional and opposite in direction to the velocity* of the moving body. As an example, let us again consider a body with mass m suspended from a spring of constant k , assuming that the body is attached to the plunger of a dashpot (Fig. 19.10). The magnitude of the friction force exerted on the plunger by the surrounding fluid is equal to $c\dot{x}$, where the constant c , expressed in N·s/m or lb·s/ft and known as the *coefficient of viscous damping*, depends upon the physical properties of the fluid and the construction of the dashpot. Examining the free-body and kinetic diagrams, the equation of motion is

$$+\downarrow \Sigma F = ma: \quad W - k(\delta_{st} + x) - c\dot{x} = m\ddot{x}$$

Recalling that $W = k\delta_{st}$, we have

$$m\ddot{x} + c\dot{x} + kx = 0 \quad (19.38)$$

If we substitute $x = e^{\lambda t}$ into Eq. (19.38) and divide through by $e^{\lambda t}$, we obtain

Characteristic equation

$$m\lambda^2 + c\lambda + k = 0 \quad (19.39)$$

and obtain the roots

$$\lambda = -\frac{c}{2m} \pm \sqrt{\left(\frac{c}{2m}\right)^2 - \frac{k}{m}} \quad (19.40)$$

Defining the *critical damping coefficient* c_c as the value of c that makes the radical in Eq. (19.40) equal to zero, we have

$$\left(\frac{c_c}{2m}\right)^2 - \frac{k}{m} = 0 \quad c_c = 2m\sqrt{\frac{k}{m}} = 2m\omega_n \quad (19.41)$$

where ω_n is the natural circular frequency of the system in the absence of damping. We can distinguish three different cases of damping, depending upon the value of the coefficient c .

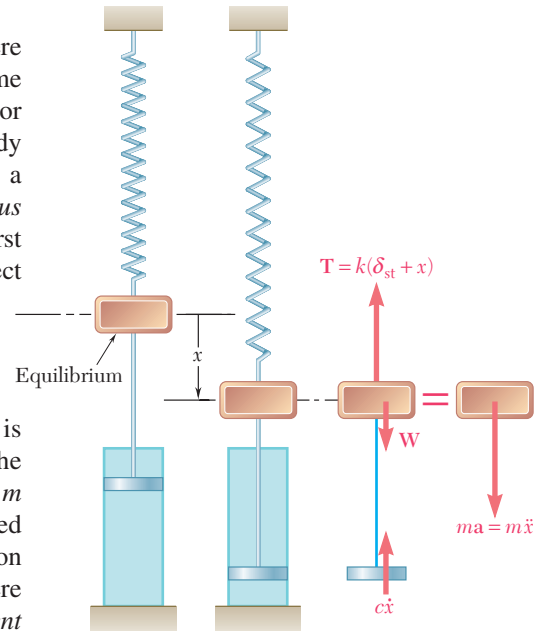


Fig. 19.10 Free-body diagram and kinetic diagram of a spring-mass-damper system.

1. **Overdamped:** $c > c_c$. The roots λ_1 and λ_2 of the characteristic equation (19.39) are real and distinct, and the general solution of the differential equation (19.38) is

$$x = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} \quad (19.42)$$

This solution corresponds to a nonvibratory motion. Since λ_1 and λ_2 are both negative, x approaches zero as t increases indefinitely. However, the system actually regains its equilibrium position after a finite time.

2. **Critically damped:** $c = c_c$. The characteristic equation has a double root $\lambda = -c_c/2m = -\omega_n$, and the general solution of Eq. (19.38) is

$$x = (C_1 + C_2 t) e^{-\omega_n t} \quad (19.43)$$

This motion is again nonvibratory. Critically damped systems are of special interest in engineering applications because they regain their equilibrium position in the shortest possible time without oscillation.

3. **Underdamped:** $c < c_c$. The roots of Eq. (19.39) are complex and conjugate, and the general solution of Eq. (19.38) is of the form

$$x = e^{-(c/2m)t} (C_1 \sin \omega_d t + C_2 \cos \omega_d t) \quad (19.44)$$

where ω_d is defined by the relation

$$\omega_d^2 = \frac{k}{m} - \left(\frac{c}{2m}\right)^2$$

Substituting $k/m = \omega_n^2$ and recalling Eq. (19.41), we have

$$\omega_d = \omega_n \sqrt{1 - \left(\frac{c}{c_c}\right)^2} \quad (19.45)$$

where the constant c/c_c is known as the **damping factor** or the **damping ratio**. This quantity is often denoted by ζ . Even though the motion does not actually repeat itself, the constant ω_d is commonly referred to as the *damped circular frequency*. In terms of the damping ratio, the damped circular frequency is

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \quad (19.45')$$

A substitution similar to the one used in Sec. 19.1A enables us to write the general solution of Eq. (19.38) in the form

$$x = x_0 e^{-(c/2m)t} \sin(\omega_d t + \phi) \quad (19.46)$$

or

$$x = x_0 e^{-\zeta \omega_n t} \sin(\omega_d t + \phi) \quad (19.46')$$

The motion defined by Eq. (19.46) is vibratory with diminishing amplitude (Fig. 19.11). The time interval $\tau_d = 2\pi/\omega_d$ separating two successive points where the curve defined by Eq. (19.46) touches one of the limiting curves shown in Fig. 19.11 is commonly referred to as the *period of the damped vibration*. Recalling Eq. (19.45), we observe that $\omega_d < \omega_n$ and, thus, that τ_d is larger than the period of vibration τ_n of the corresponding undamped system.

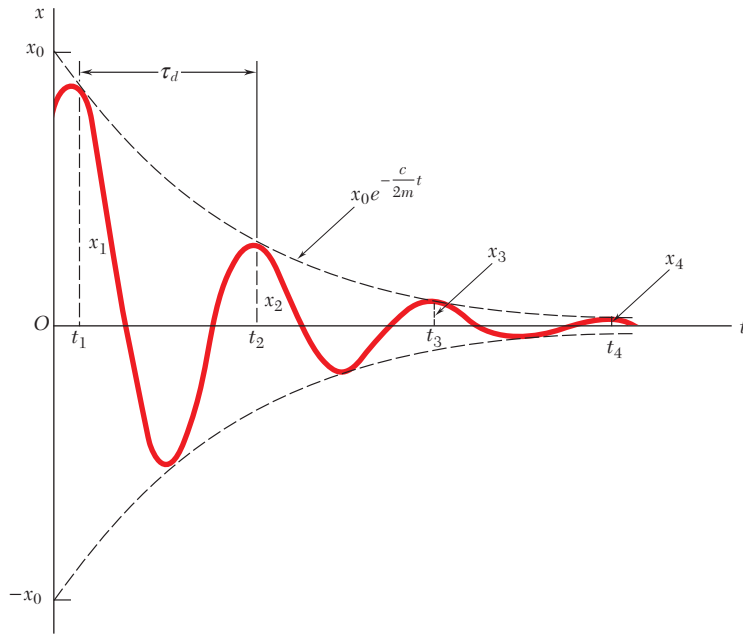


Fig. 19.11 The free response of a viscously damped system decays exponentially and oscillates with a frequency ω_d .

*19.5B Damped Forced Vibrations

If the system considered in the preceding section is subjected to a periodic force $\mathbf{P}$ of magnitude $P = P_m \sin \omega_f t$, the equation of motion becomes

$$m\ddot{x} + c\dot{x} + kx = P_m \sin \omega_f t \quad (19.47)$$

We can obtain the general solution of Eq. (19.47) by adding a particular solution of Eq. (19.47) to the complementary function or general solution of the homogeneous equation (19.38). The complementary function is given by Eq. (19.42), (19.43), or (19.44), depending upon the type of damping considered. It represents a *transient* motion that is eventually damped out.

Our interest in this section is centered on the steady-state vibration represented by a particular solution of Eq. (19.47) of the form

$$x_{\text{part}} = x_m \sin (\omega_f t - \phi) \quad (19.48)$$

Substituting x_{part} for x into Eq. (19.47), we obtain

$$\begin{aligned} -m\omega_f^2 x_m \sin (\omega_f t - \phi) + c\omega_f x_m \cos (\omega_f t - \phi) + kx_m \sin (\omega_f t - \phi) \\ = P_m \sin \omega_f t \end{aligned}$$

Making $\omega_f t - \phi$ successively equal to 0 and to $\pi/2$ gives

$$c\omega_f x_m = P_m \sin \phi \quad (19.49)$$

$$(k - m\omega_f^2)x_m = P_m \cos \phi \quad (19.50)$$



Photo 19.2 The automobile suspension shown consists essentially of a spring and a shock absorber, which will cause the body of the car to undergo *damped forced vibrations* when the car is driven over an uneven road.



Photo 19.3 This truck is experiencing damped forced vibration in the vehicle dynamics test.

Squaring both sides of Eqs. (19.49) and (19.50) and adding, we have

$$[(k - m\omega_f^2)^2 + (c\omega_f)^2]x_m^2 = P_m^2 \quad (19.51)$$

Solving Eq. (19.51) for x_m and dividing Eqs. (19.49) and (19.50) by the result, we obtain, respectively,

$$x_m = \frac{P_m}{\sqrt{(k - m\omega_f^2)^2 + (c\omega_f)^2}} \quad \tan \phi = \frac{c\omega_f}{k - m\omega_f^2} \quad (19.52)$$

Recalling from Eq. (19.4) that $k/m = \omega_n^2$, where ω_n is the circular frequency of the undamped free vibration, and from Eq. (19.41) that $2m\omega_n = c_c$, where c_c is the critical damping coefficient of the system, we have

$$\frac{x_m}{P_m/k} = \frac{x_m}{\delta_m} = \frac{1}{\sqrt{[1 - (\omega_f/\omega_n)^2]^2 + [2(c/c_c)(\omega_f/\omega_n)]^2}} \quad (19.53)$$

$$\tan \phi = \frac{2(c/c_c)(\omega_f/\omega_n)}{1 - (\omega_f/\omega_n)^2} \quad (19.54)$$

Defining the frequency ratio $r = \omega_f/\omega_n$, we can write the steady-state response of a viscously damped system in terms of the frequency ratio and the damping ratio as

$$\frac{x_m}{P_m/k} = \frac{x_m}{\delta_{st}} = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}} \quad (19.53')$$

$$\tan \phi = \frac{2\zeta r}{1 - r^2} \quad (19.54')$$

We can use these equations to determine the amplitude of the steady-state vibration produced by an impressed force of magnitude $P = P_m \sin \omega_f t$ or by an impressed support movement $\delta = \delta_m \sin \omega_f t$. Using these same parameters, Eq. (19.54) defines the *phase difference* ϕ between the impressed force or impressed support movement and the resulting steady-state vibration of the damped system. The magnification factor has been plotted against the frequency ratio in Fig. 19.12 for various values of the damping ratio. Note that we can keep the amplitude of a forced vibration small by choosing a large coefficient of viscous damping c or by keeping the natural and forced frequencies far apart.

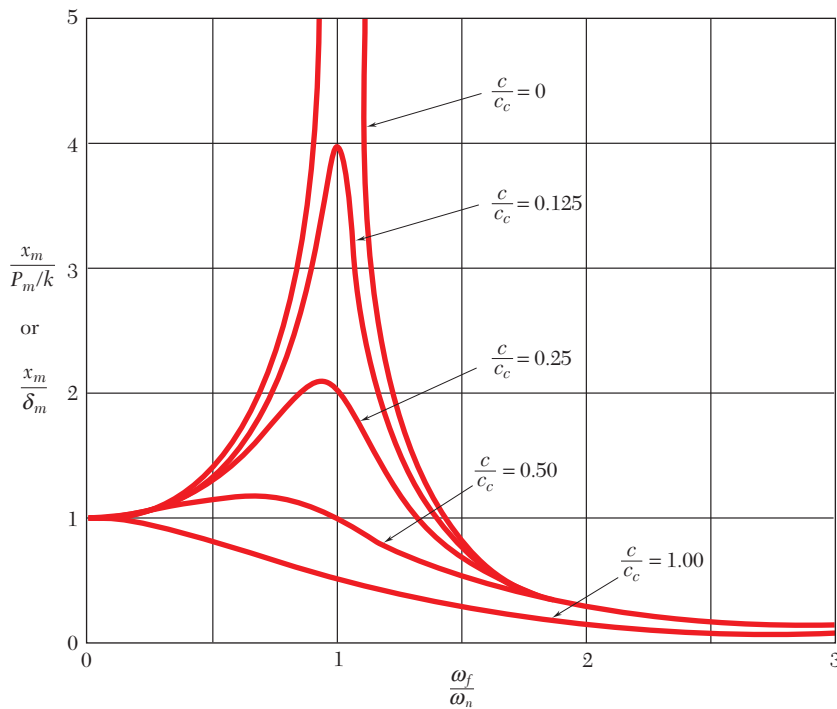


Fig. 19.12 Graph of magnification factor as a function of frequency ratio for several values of the damping ratio.

*19.5C Electrical Analogs

Oscillating electrical circuits are characterized by differential equations of the same type as those just discussed. Their analysis is therefore similar to that of a mechanical system, and the results obtained for a given vibrating system can be readily extended to the equivalent circuit. Conversely, any result obtained for an electrical circuit also applies to the corresponding mechanical system.

Consider an electrical circuit consisting of an inductor of inductance L , a resistor of resistance R , and a capacitor of capacitance C , connected in series with a source of alternating voltage $E = E_m \sin \omega_f t$ (Fig. 19.13). Elementary circuit theory[†] says that if i denotes the current in the circuit and q denotes the electric charge on the capacitor, the drop in potential is $L(di/dt)$ across the inductor, Ri across the resistor, and q/C across the capacitor. The algebraic sum of the applied voltage and of the drops in potential around the circuit loop must be zero, so we have

$$E_m \sin \omega_f t - L \frac{di}{dt} - Ri - \frac{q}{C} = 0 \quad (19.55)$$

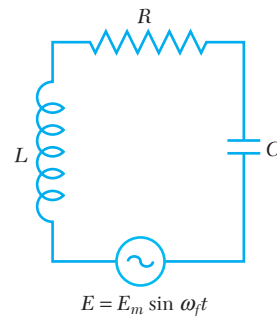


Fig. 19.13 An electrical circuit with inductance L , resistance R , capacitance C , and a source of alternating voltage E .

[†]See C. R. Paul, S. A. Nasar, and L. E. Unnewehr, *Introduction to Electrical Engineering*, 2nd ed., McGraw-Hill, New York, 1992.

Rearranging the terms and recalling that at any instant the current i is equal to the rate of change $\dot{q}$ of the charge q , we have

$$L\ddot{q} + R\dot{q} + \frac{1}{C}q = E_m \sin \omega_f t$$

(19.56)

We can verify that Eq. (19.56), which defines the oscillations of the electrical circuit of Fig. 19.13, is of the same type as Eq. (19.47), which characterizes the damped forced vibrations of the mechanical system of Fig. 19.10. By comparing the two equations, we can construct a table of the analogous mechanical and electrical expressions.

Table 19.2 can be used to extend the results obtained earlier for various mechanical systems to their electrical analogs. For instance, we can determine the amplitude i_m of the current in the circuit of Fig. 19.13 by noting that it corresponds to the maximum value v_m of the velocity in the analogous mechanical system. Recalling from the first of Eqs. (19.37) that $v_m = x_m \omega_f$, substituting for x_m from Eq. (19.52), and replacing the constants of the mechanical system by the corresponding electrical expressions, we have

$$i_m = \frac{\omega_f E_m}{\sqrt{\left(\frac{1}{C} - L\omega_f^2\right)^2 + (R\omega_f)^2}}$$
$$i_m = \frac{E_m}{\sqrt{R^2 + \left(L\omega_f - \frac{1}{C\omega_f}\right)^2}}$$

(19.57)

The radical term in this expression is known as the *impedance* of the electrical circuit.

The analogy between mechanical systems and electrical circuits holds for transient as well as steady-state oscillations. The oscillations of the circuit shown in Fig. 19.14, for instance, are analogous to the damped free vibrations of the system of Fig. 19.10. As far as the initial conditions are concerned, we should note that closing the switch S when the charge on the capacitor is $q = q_0$ is equivalent to releasing the mass of the mechanical system with no initial velocity from the position $x = x_0$. Also note that, if a battery of constant voltage E is introduced in the electrical circuit of Fig. 19.14, closing the switch S is equivalent to suddenly applying a force of constant magnitude P to the mass of the mechanical system of Fig. 19.10.

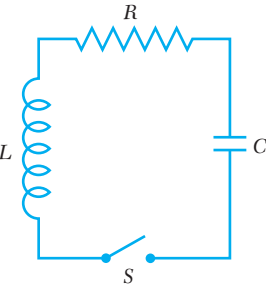


Fig. 19.14 An LRC circuit with switch S .

Table 19.2 Characteristics of a Mechanical System and of Its Electrical Analog

Mechanical System		Electrical Circuit	
m	Mass	L	Inductance
c	Coefficient of viscous damping	R	Resistance
k	Spring constant	$1/C$	Reciprocal of capacitance
x	Displacement	q	Charge
v	Velocity	i	Current
P	Applied force	E	Applied voltage

This discussion would be of questionable value if its only result were to make it possible for mechanics students to analyze electrical circuits without learning the elements of circuit theory. We hope that this discussion will instead encourage students to apply to the solution of problems in mechanical vibrations the mathematical techniques they may learn in later courses in circuit theory. The chief value of the concept of electrical analogs, however, resides in its application to *experimental methods* for determining the characteristics of a given mechanical system. Indeed, an electrical circuit is much more easily constructed than is a mechanical model, and the fact that we can modify its characteristics by varying the inductance, resistance, or capacitance of its various components makes the use of the electrical analog particularly convenient.

To determine the electrical analog of a given mechanical system, we focus our attention on each moving mass in the system and observe which springs, dashpots, or external forces are applied directly to it. We can then construct an equivalent electrical loop to match each of these mechanical units; the various loops obtained in this way will together form the desired circuit. Consider, for instance, the mechanical system of Fig. 19.15. The mass m_1 is acted upon by two springs with constants k_1 and k_2 and by two dashpots characterized by the coefficients of viscous damping c_1 and c_2 . The electrical circuit should therefore include a loop consisting of an inductor of inductance L_1 proportional to m_1 ; of two capacitors of capacitance C_1 and C_2 inversely proportional to k_1 and k_2 , respectively; and of two resistors of resistance R_1 and R_2 , proportional to c_1 and c_2 , respectively. Since the mass m_2 is acted upon by the spring k_2 and the dashpot c_2 , as well as by the force $P = P_m \sin \omega_f t$, the circuit should also include a loop containing the capacitor C_2 , the resistor R_2 , the new inductor L_2 , and the voltage source $E = E_m \sin \omega_f t$ (Fig. 19.16).

To check that the mechanical system of Fig. 19.15 and the electrical circuit of Fig. 19.16 actually satisfy the same differential equations, we first derive the equations of motion for m_1 and m_2 . Denoting the displacements of m_1 and m_2 from their equilibrium positions by x_1 and x_2 , respectively, we observe that the elongation of the spring k_1 (measured from the equilibrium position) is equal to x_1 , while the elongation of the spring k_2 is equal to the relative displacement $x_2 - x_1$ of m_2 with respect to m_1 . The equations of motion for m_1 and m_2 are therefore

$$m_1 \ddot{x}_1 + c_1 \dot{x}_1 + c_2(\dot{x}_1 - \dot{x}_2) + k_1 x_1 + k_2(x_1 - x_2) = 0 \quad (19.58)$$

$$m_2 \ddot{x}_2 + c_2(\dot{x}_2 - \dot{x}_1) + k_2(x_2 - x_1) = P_m \sin \omega_f t \quad (19.59)$$

Now consider the electrical circuit of Fig. 19.16; we denote the current in the first and second loops by i_1 and i_2 , respectively, and by q_1 and q_2 the integrals $\int i_1 dt$ and $\int i_2 dt$. Noting that the charge on the capacitor C_1 is q_1 and the charge on C_2 is $q_1 - q_2$, we can state that the sum of the potential differences in each loop is zero and obtain

$$L_1 \ddot{q}_1 + R_1 \dot{q}_1 + R_2(\dot{q}_1 - \dot{q}_2) + \frac{q_1}{C_1} + \frac{q_1 - q_2}{C_2} = 0 \quad (19.60)$$

$$L_2 \ddot{q}_2 + R_2(\dot{q}_2 - \dot{q}_1) + \frac{q_2 - q_1}{C_2} = E_m \sin \omega_f t \quad (19.61)$$

We easily check that Eqs. (19.60) and (19.61) reduce to Eqs. (19.58) and (19.59), respectively, after performing the substitutions indicated in Table 19.2.

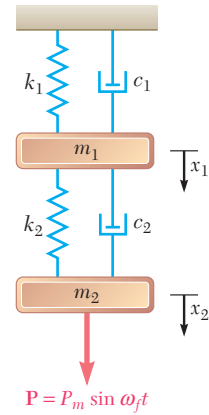


Fig. 19.15 Model of a two-degree-of-freedom harmonically excited system.

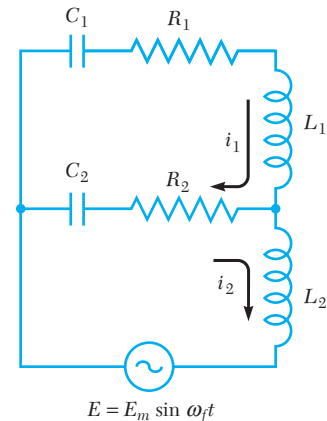


Fig. 19.16 An electrical circuit analogous to the mechanical system in Fig. 19.15.

SOLVING PROBLEMS ON YOUR OWN

In this section, we developed a more realistic model of a vibrating system by including the effect of the **viscous damping** caused by fluid friction. We represented viscous damping in Fig. 19.10 by the force exerted on the moving body by a plunger moving in a dashpot. This force is equal in magnitude to $c\dot{x}$, where the constant c , expressed in N·s/m or lb·s/ft, is known as the *coefficient of viscous damping*. Keep in mind that the same sign convention should be used for x , $\dot{x}$, and $\ddot{x}$.

1. Damped free vibrations. The differential equation defining this motion was found to be

$$m\ddot{x} + c\dot{x} + kx = 0 \quad (19.38)$$

To obtain the solution of this equation, calculate the *critical damping coefficient* c_c , using the formula

$$c_c = 2m\sqrt{k/m} = 2m\omega_n \quad (19.41)$$

where ω_n is the natural circular frequency of the undamped system.

a. If $c > c_c$ (overdamped), the solution of Eq. (19.38) is

$$x = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} \quad (19.42)$$

where

$$\lambda_{1,2} = -\frac{c}{2m} \pm \sqrt{\left(\frac{c}{2m}\right)^2 - \frac{k}{m}} \quad (19.40)$$

and where the constants C_1 and C_2 can be determined from the initial conditions $x(0)$ and $\dot{x}(0)$. This solution corresponds to a nonvibratory motion.

b. If $c = c_c$ (critically damped), the solution of Eq. (19.38) is

$$x = (C_1 + C_2 t) e^{-\omega_n t} \quad (19.43)$$

which also corresponds to a nonvibratory motion. Critically damped systems are of special interest in engineering applications because they regain their equilibrium position in the shortest possible time without oscillation.

c. If $c < c_c$ (underdamped), the solution of Eq. (19.38) is

$$x = x_0 e^{-(c/2m)t} \sin(\omega_d t + \phi) \quad (19.46)$$

or in terms of the damping ratio $\zeta = c/c_{cr}$,

$$x = x_0 e^{-\zeta \omega_n t} \sin(\omega_d t + \phi) \quad (19.46')$$

where

$$\omega_d = \omega_n \sqrt{1 - \left(\frac{c}{c_c}\right)^2} \quad (19.45)$$

or

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \quad (19.45')$$

and where x_0 and ϕ can be determined from the initial conditions $x(0)$ and $\dot{x}(0)$. This solution corresponds to oscillations of decreasing amplitude and of period $\tau_d = 2\pi/\omega_d$ (Fig. 19.11).

2. Damped forced vibrations. These vibrations occur when a system with viscous damping is subjected to a periodic force $\mathbf{P}$ with a magnitude of $P = P_m \sin \omega_f t$ or when it is elastically connected to a support with an alternating motion of $\delta = \delta_m \sin \omega_f t$. In the first case, the motion is defined by the differential equation

$$m\ddot{x} + c\dot{x} + kx = P_m \sin \omega_f t \quad (19.47)$$

and in the second case, by a similar equation obtained by replacing P_m with $k\delta_m$. You will be concerned only with the *steady-state* motion of the system, which is defined by a *particular solution* of these equations of the form

$$x_{\text{part}} = x_m \sin (\omega_f t - \phi) \quad (19.48)$$

where

$$\frac{x_m}{P_m/k} = \frac{x_m}{\delta_m} = \frac{1}{\sqrt{[1 - (\omega_f/\omega_n)^2]^2 + [2(c/c_c)(\omega_f/\omega_n)]^2}} \quad (19.53)$$

and

$$\tan \phi = \frac{2(c/c_c)(\omega_f/\omega_n)}{1 - (\omega_f/\omega_n)^2} \quad (19.54)$$

The expression given in Eq. (19.53) is referred to as the *magnification factor* and has been plotted against the frequency ratio ω_f/ω_n in Fig. 19.12 for various values of the damping ratio c/c_c . Eqs. (19.53) and (19.54) can be written in terms of the damping ratio ζ and frequency ratio r as shown in Eqs. (19.53') and (19.54'). In the problems that follow, you may be asked to determine one of the parameters in Eqs. (19.53) and (19.54) when the others are known.

Problems

19.127 Show that in the case of heavy damping ($c > c_c$), a body never passes through its position of equilibrium O if it is (a) released with no initial velocity from an arbitrary position, (b) started from O with an arbitrary initial velocity.

19.128 Show that in the case of heavy damping ($c > c_c$), a body released from an arbitrary position with an arbitrary initial velocity cannot pass more than once through its equilibrium position.

19.129 In the case of light damping ($c < c_c$), the displacements x_1, x_2, x_3 , shown in Fig. 19.11 may be assumed equal to the maximum displacements. Show that the ratio of any two successive maximum displacements x_n and x_{n+1} is a constant and that the natural logarithm of this ratio, called the *logarithmic decrement*, is

$$\ln \frac{x_n}{x_{n+1}} = \frac{2\pi(c/c_c)}{\sqrt{1 - (c/c_c)^2}}$$

19.130 In practice, it is often difficult to determine the logarithmic decrement of a system with light damping defined in Prob. 19.129 by measuring two successive maximum displacements. Show that the logarithmic decrement can also be expressed as $(1/k) \ln(x_n/x_{n+k})$, where k is the number of cycles between readings of the maximum displacement.

19.131 In an underdamped system ($c < c_c$), the period of vibration is commonly defined as the time interval $\tau_d = 2\pi/\omega_d$ corresponding to two successive points where the displacement-time curve touches one of the limiting curves shown in Fig. 19.11. Show that the interval of time (a) between a maximum positive displacement and the following maximum negative displacement is $\frac{1}{2}\tau_d$, (b) between two successive zero displacements is $\frac{1}{2}\tau_d$, (c) between a maximum positive displacement and the following zero displacement is greater than $\frac{1}{4}\tau_d$.

19.132 A loaded railroad car weighing 30,000 lb is rolling at a constant velocity v_0 when it couples with a spring and dashpot bumper system (Fig. 1). The recorded displacement-time curve of the loaded railroad car after coupling is as shown (Fig. 2). Determine (a) the damping constant, (b) the spring constant. (*Hint*: Use the definition of logarithmic decrement given in 19.129.)

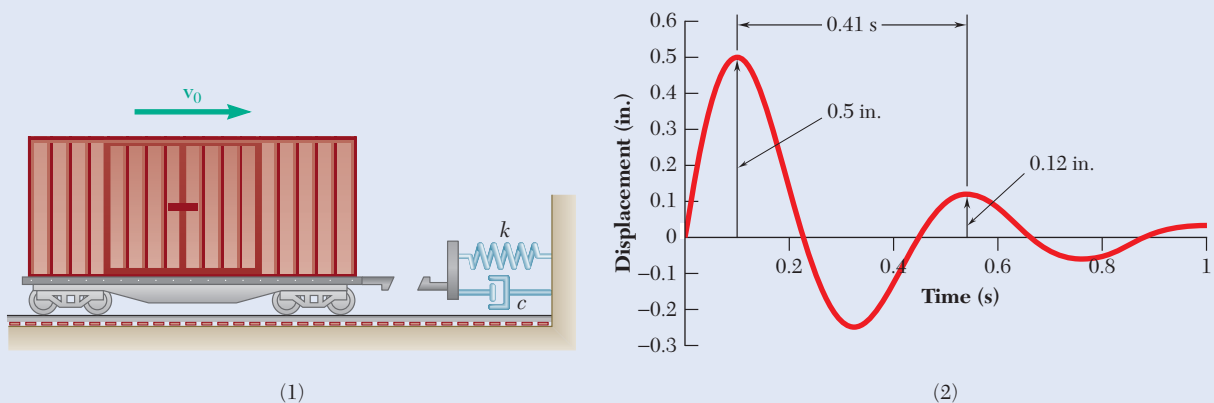


Fig. P19.132

19.133 A torsional pendulum has a centroidal mass moment of inertia of $0.3 \text{ kg}\cdot\text{m}^2$ and when given an initial twist and released is found to have a frequency of oscillation of 200 rpm. Knowing that when this pendulum is immersed in oil and when given the same initial condition it is found to have a frequency of oscillation of 180 rpm, determine the damping constant for the oil.

19.134 The barrel of a field gun weighs 1500 lb and is returned into firing position after recoil by a recuperator of constant $c = 1100 \text{ lb}\cdot\text{s}/\text{ft}$. Determine (a) the constant k that should be used for the recuperator to return the barrel into firing position in the shortest possible time without any oscillation, (b) the time needed for the barrel to move back two-thirds of the way from its maximum-recoil position to its firing position.

19.135 A 2-kg block is supported by a spring with a constant of $k = 128 \text{ N/m}$ and a dashpot with a coefficient of viscous damping of $c = 0.6 \text{ N}\cdot\text{s}/\text{m}$. The block is in equilibrium when it is struck from below by a hammer that imparts to the block an upward velocity of 0.4 m/s . Determine (a) the logarithmic decrement, (b) the maximum upward displacement of the block from equilibrium after two cycles.

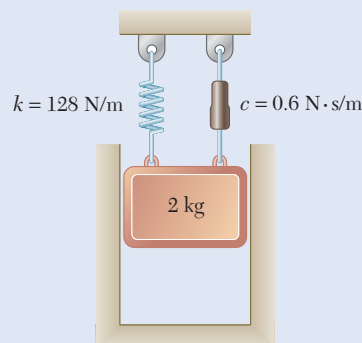


Fig. P19.135

19.136 A 4-kg block A is dropped from a height of 800 mm onto a 9-kg block B that is at rest. Block B is supported by a spring of constant $k = 1500 \text{ N/m}$ and is attached to a dashpot of damping coefficient $c = 230 \text{ N}\cdot\text{s}/\text{m}$. Knowing that there is no rebound, determine the maximum distance the blocks will move after the impact.

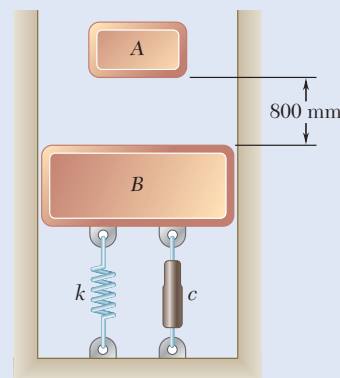


Fig. P19.136

19.137 A 0.9-kg block B is connected by a cord to a 2.4-kg block A that is suspended as shown from two springs, each with a constant of $k = 180 \text{ N/m}$, and a dashpot with a damping coefficient of $c = 7.5 \text{ N}\cdot\text{s}/\text{m}$. Knowing that the system is at rest when the cord connecting A and B is cut, determine the minimum tension that will occur in each spring during the resulting motion.

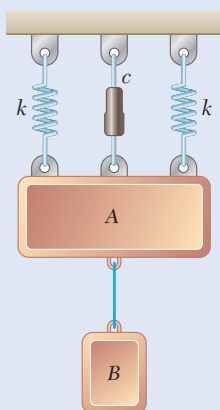


Fig. P19.137 and P19.138

19.138 A 0.9-kg block B is connected by a cord to a 2.4-kg block A that is suspended as shown from two springs, each with a constant of $k = 180 \text{ N/m}$, and a dashpot with a damping coefficient of $c = 60 \text{ N}\cdot\text{s}/\text{m}$. Knowing that the system is at rest when the cord connecting A and B is cut, determine the velocity of block A after 0.1 s.

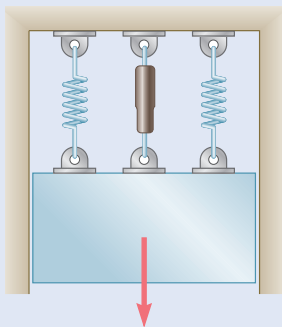


Fig. P19.139

19.139 A machine element weighing 800 lb is supported by two springs, each having a constant of 200 lb/in. A periodic force of maximum value 30 lb is applied to the element with a frequency of 2.5 cycles per second. Knowing that the coefficient of damping is 8 lb·s/in., determine the amplitude of the steady-state vibration of the element.

19.140 In Prob. 19.139, determine the required value of the coefficient of damping if the amplitude of the steady-state vibration of the element is to be 0.15 in.

19.141 In the case of the forced vibration of a system, determine the range of values of the damping factor c/c_c for which the magnification factor will always decrease as the frequency ratio ω_f/ω_n increases.

19.142 Show that for a small value of the damping factor c/c_c , the maximum amplitude of a forced vibration occurs when $\omega_f \approx \omega_n$ and that the corresponding value of the magnification factor is $\frac{1}{2}(c/c_c)$.

19.143 A counter-rotating eccentric mass exciter consisting of two rotating 14-oz weights describing circles of 6-in. radius at the same speed but in opposite senses is placed on a machine element to induce a steady-state vibration of the element and to determine some of the dynamic characteristics of the element. At a speed of 1200 rpm, a stroboscope shows the eccentric masses to be exactly under their respective axes of rotation and the element to be passing through its position of static equilibrium. Knowing that the amplitude of the motion of the element at that speed is 0.6 in. and that the total weight of the system is 300 lb, determine (a) the combined spring constant k , (b) the damping factor c/c_c .

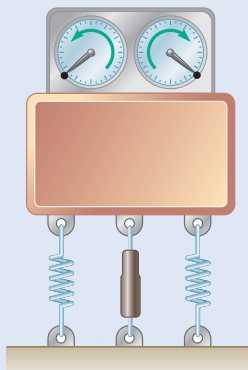


Fig. P19.143

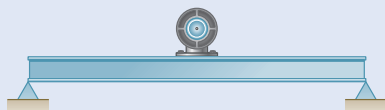


Fig. P19.144 and P19.145

19.144 A 36-lb motor is bolted to a light horizontal beam that has a static deflection of 0.075 in. due to the weight of the motor. Knowing that the unbalance of the rotor is equivalent to a weight of 0.64 oz located 6.25 in. from the axis of rotation, determine the amplitude of the vibration of the motor at a speed of 900 rpm, assuming (a) that no damping is present, (b) that the damping factor c/c_c is equal to 0.055.

19.145 A 45-kg motor is bolted to a light horizontal beam that has a static deflection of 6 mm due to the weight of the motor. The unbalance of the motor is equivalent to a mass of 110 g located 75 mm from the axis of rotation. Knowing that the amplitude of the vibration of the motor is 0.25 mm at a speed of 300 rpm, determine (a) the damping factor c/c_c , (b) the coefficient of damping c .

- 19.146** The unbalance of the rotor of a 180-kg motor is equivalent to a mass of 85 g located 150 mm from the axis of rotation. The pad that is placed between the motor and the foundation is equivalent to a spring with a constant of $k = 7.5 \text{ kN/m}$ in parallel with a dashpot with constant c . Knowing that the magnitude of the maximum acceleration of the motor is 9 mm/s^2 at a speed of 100 rpm, determine the damping factor c/c_c .

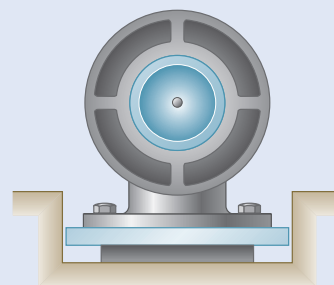


Fig. P19.146

- 19.147** A machine element is supported by springs and is connected to a dashpot as shown. Show that if a periodic force of magnitude $P = P_m \sin \omega_f t$ is applied to the element, the amplitude of the fluctuating force transmitted to the foundation is

$$F_m = P_m \sqrt{\frac{1 + [2(c/c_c)(\omega_f/\omega_n)]^2}{[1 - (\omega_f/\omega_n)^2]^2 + [2(c/c_c)(\omega_f/\omega_n)]^2}}$$

- 19.148** A 91-kg machine element supported by four springs, each of constant $k = 175 \text{ N/m}$, is subjected to a periodic force of frequency 0.8 Hz and amplitude 89 N. Determine the amplitude of the fluctuating force transmitted to the foundation if (a) a dashpot with a coefficient of damping $c = 365 \text{ N}\cdot\text{s/m}$ is connected to the machine element and to the ground, (b) the dashpot is removed.

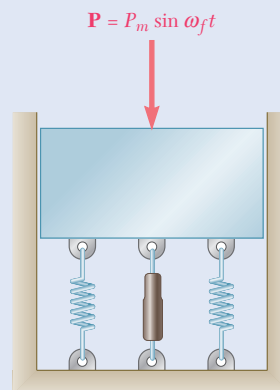


Fig. P19.147 and P19.148

- 19.149** A simplified model of a washing machine is shown. A bundle of wet clothes forms a weight w_b of 20 lb in the machine and causes a rotating unbalance. The rotating weight is 40 lb (including w_b) and the radius of the washer basket e is 9 in. Knowing the washer has an equivalent spring constant $k = 70 \text{ lb/ft}$ and damping ratio $\zeta = c/c_c = 0.05$ and during the spin cycle the drum rotates at 250 rpm, determine the amplitude of the motion and the magnitude of the force transmitted to the sides of the washing machine.

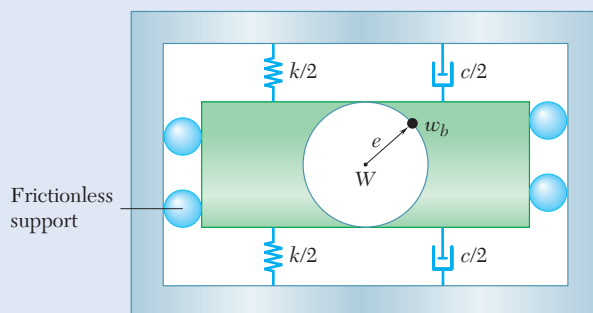


Fig. P19.149

- *19.150** For a steady-state vibration with damping under a harmonic force, show that the mechanical energy dissipated per cycle by the dashpot is $E = \pi c x_m^2 \omega_f$, where c is the coefficient of damping, x_m is the amplitude of the motion, and ω_f is the circular frequency of the harmonic force.

- *19.151** The suspension of an automobile can be approximated by the simplified spring-and-dashpot system shown. (a) Write the differential equation defining the vertical displacement of the mass m when the system moves at a speed v over a road with a sinusoidal cross section of amplitude δ_m and wave length L . (b) Derive an expression for the amplitude of the vertical displacement of the mass m .

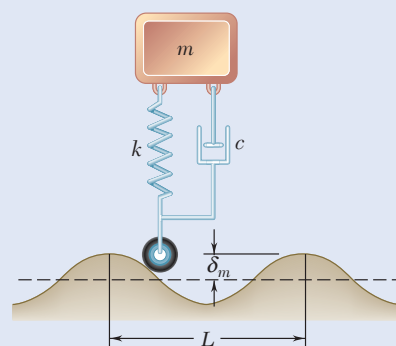


Fig. P19.151

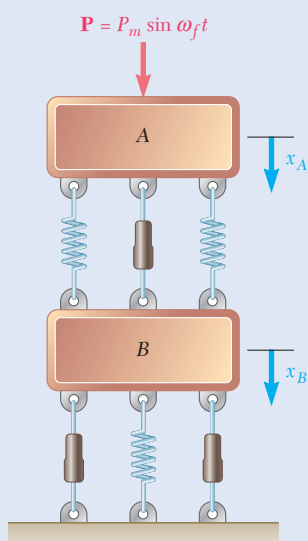


Fig. P19.152

***19.152** Two blocks A and B , each of mass m , are supported as shown by three springs of the same constant k . Blocks A and B are connected by a dashpot and block B is connected to the ground by two dashpots, each dashpot having the same coefficient of damping c . Block A is subjected to a force of magnitude $P = P_m \sin \omega_f t$. Write the differential equations defining the displacements x_A and x_B of the two blocks from their equilibrium positions.

19.153 Express in terms of L , C , and E the range of values of the resistance R for which oscillations will take place in the circuit shown when switch S is closed.

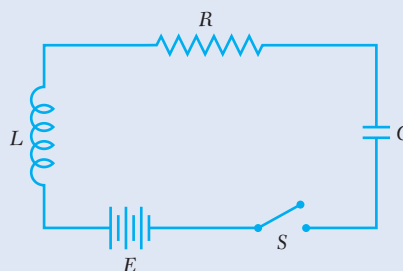


Fig. P19.153

19.154 Consider the circuit of Prob. 19.153 when the capacitor C is removed. If switch S is closed at time $t = 0$, determine (a) the final value of the current in the circuit, (b) the time t at which the current will have reached $(1 - 1/e)$ times its final value. (The desired value of t is known as the *time constant* of the circuit.)

19.155 and 19.156 Draw the electrical analogue of the mechanical system shown. (Hint: Draw the loops corresponding to the free bodies m and A .)

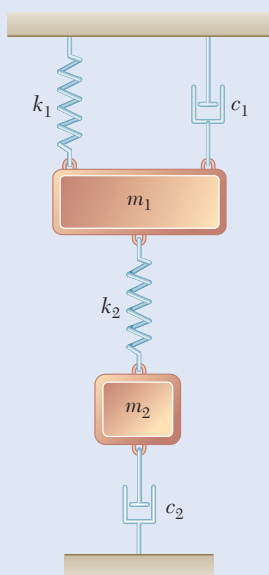


Fig. P19.156 and P19.158

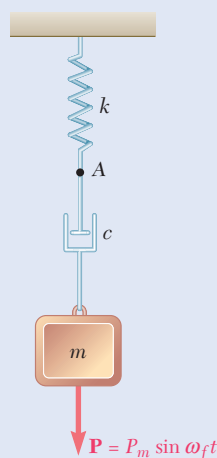


Fig. P19.155 and P19.157

19.157 and 19.158 Write the differential equations defining (a) the displacements of the mass m and of the point A , (b) the charges on the capacitors of the electrical analogue.

Review and Summary

This chapter was devoted to the study of **mechanical vibrations**, i.e., to the analysis of the motion of particles and rigid bodies oscillating about a position of equilibrium. In the first part of the chapter [Secs. 19.1 through 19.4], we considered *vibrations without damping*, while the second part was devoted to *damped vibrations* [Sec. 19.5].

Free Vibrations of a Particle

In Sec. 19.1, we considered the **free vibrations of a particle**, i.e., the motion of a particle P subjected to a restoring force proportional to the displacement of the particle—such as the force exerted by a spring. If the displacement x of the particle P is measured from its equilibrium position O (Fig. 19.17), the resultant $\mathbf{F}$ of the forces acting on P (including its weight) has a magnitude kx and is directed toward O . Applying Newton's second law $F = ma$ and recalling that $a = \ddot{x}$, we wrote the differential equation

$$m\ddot{x} + kx = 0 \quad (19.2)$$

or, setting $\omega_n^2 = k/m$,

$$\ddot{x} + \omega_n^2 x = 0 \quad (19.6)$$

The motion defined by this equation is called **simple harmonic motion**.

The solution of Eq. (19.6), which represents the displacement of the particle P , was expressed as

$$x = x_m \sin(\omega_n t + \phi) \quad (19.10) \quad \text{Fig. 19.17}$$

where x_m = amplitude of the vibration

$\omega_n = \sqrt{k/m}$ = natural circular frequency

ϕ = phase angle

The **period of the vibration** (i.e., the time required for a full cycle) and its **natural frequency** (i.e., the number of cycles per second) were expressed as

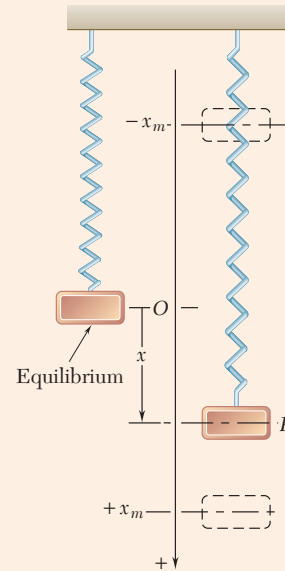
$$\text{Period} = \tau_n = \frac{2\pi}{\omega_n} \quad (19.13)$$

$$\text{Natural frequency} = f_n = \frac{1}{\tau_n} = \frac{\omega_n}{2\pi} \quad (19.14)$$

We obtained the velocity and acceleration of the particle by differentiating Eq. (19.10), and their maximum values were found to be

$$v_m = x_m \omega_n \quad a_m = x_m \omega_n^2 \quad (19.15)$$

Since all of the above parameters depend directly upon the natural circular frequency ω_n and thus upon the ratio k/m , it is essential in any given problem to calculate the value of the constant k . This can be done by determining the



relation between the restoring force and the corresponding displacement of the particle [Sample Prob. 19.1].

It was also shown that we can represent the oscillatory motion of particle P by the projection on the x axis of the motion of a point Q describing an auxiliary circle of radius x_m with the constant angular velocity ω_n (Fig. 19.18). Then we can obtain the instantaneous values of the velocity and acceleration of P by projecting on the x axis the vectors $\mathbf{v}_m$ and $\mathbf{a}_m$ representing, respectively, the velocity and acceleration of Q .

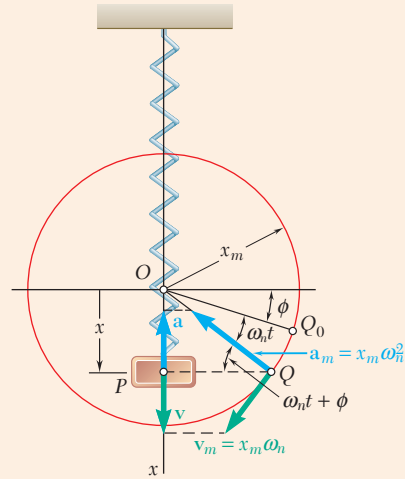


Fig. 19.18

Simple Pendulum

Although the motion of a **simple pendulum** is not truly a simple harmonic motion, we can use the formulas given previously with $\omega_n^2 = g/l$ to calculate the period and natural frequency of the *small oscillations* of a simple pendulum [Sec. 19.1B]. Large-amplitude oscillations of a simple pendulum were discussed in Sec. 19.1C.

Free Vibrations of a Rigid Body

We can analyze the **free vibrations of a rigid body** by choosing an appropriate variable, such as a distance x or an angle θ , to define the position of the body. We then draw a free-body diagram and kinetic diagram to express the equivalence of the external forces and inertial terms and write an equation relating the selected variable and its second derivative [Sec. 19.2]. If the equation obtained is of the form

$$\ddot{x} + \omega_n^2 x = 0 \quad \text{or} \quad \ddot{\theta} + \omega_n^2 \theta = 0 \quad (19.21)$$

the vibration considered is a simple harmonic motion, and its period and natural frequency can be obtained by *identifying* ω_n and substituting its value into Eqs. (19.13) and (19.14) [Sample Probs. 19.2 and 19.3].

Using the Principle of Conservation of Energy

We can use the *principle of conservation of energy* as an alternative method for determining the period and natural frequency of the simple harmonic motion of a particle or rigid body [Sec. 19.3]. Choosing again an appropriate

variable, such as θ , to define the position of the system, we express that the total energy of the system is conserved, using $T_1 + V_1 = T_2 + V_2$, between the position of maximum displacement ($\theta_1 = \theta_m$) and the position of maximum velocity ($\dot{\theta}_2 = \dot{\theta}_m$). If the motion considered is simple harmonic, the two sides of the equation obtained consist of homogeneous quadratic expressions in θ_m and $\dot{\theta}_m$, respectively. Substituting $\dot{\theta}_m = \theta_m \omega_n$ in this equation, we can factor out θ_m^2 and solve for the circular frequency ω_n [Sample Prob. 19.4]. It is important to note that if the motion can be approximated only by a simple harmonic motion, such as for the small oscillations of a body under gravity, we must approximate the potential energy by a quadratic expression in θ_m [Sample Prob. 19.4].

Forced Vibrations

In Sec. 19.4, we considered the **forced vibrations** of a mechanical system. These vibrations occur when the system is subjected to a periodic force (Fig. 19.19) or when it is elastically connected to a support that has an alternating motion (Fig. 19.20). Denoting the forced circular frequency by ω_f , we found that in the first case, the motion of the system was defined by the differential equation

$$m\ddot{x} + kx = P_m \sin \omega_f t \quad (19.30)$$

and that in the second case, it was defined by the differential equation

$$m\ddot{x} + kx = k\delta_m \sin \omega_f t \quad (19.31)$$

We can obtain the general solution of these equations by adding a particular solution of the form

$$x_{\text{part}} = x_m \sin \omega_f t \quad (19.32)$$

to the general solution of the corresponding homogeneous equation. The particular solution of Eq. (19.32) represents a **steady-state vibration** of the

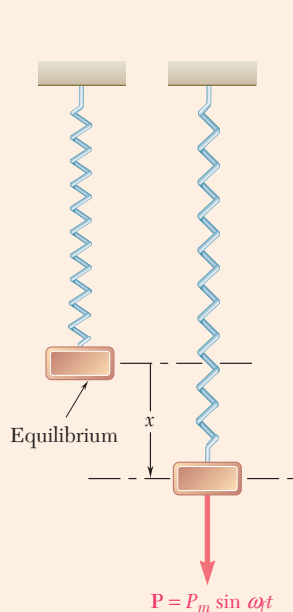


Fig. 19.19

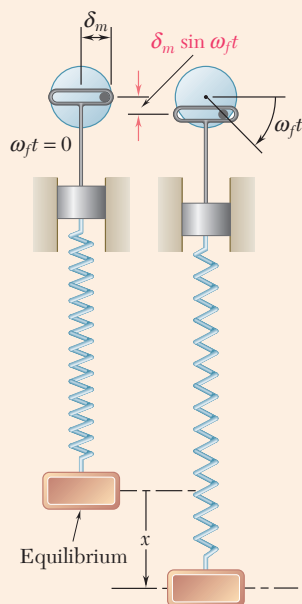


Fig. 19.20

system, whereas the solution of the homogeneous equation represents a **transient free vibration** that can generally be neglected.

Dividing the amplitude x_m of the steady-state vibration by P_m/k in the case of a periodic force or by δ_m in the case of an oscillating support, we defined the **magnification factor** of the vibration and found that

$$\text{Magnification factor} = \frac{x_m}{P_m/k} = \frac{x_m}{\delta_m} = \frac{1}{1 - (\omega_f/\omega_n)^2} \quad (19.36)$$

According to Eq. (19.36), the amplitude x_m of the forced vibration becomes infinite when $\omega_f = \omega_n$, i.e., when the forced frequency is equal to the natural frequency of the system. The impressed force or impressed support movement is then said to be in **resonance** with the system [Sample Prob. 19.5]. (Actually, the amplitude of the vibration remains finite, due to damping forces.)

Damped Free Vibrations

In Sec. 19.5, we considered the **damped vibrations** of a mechanical system. First, we analyzed the damped free vibrations of a system with **viscous damping** [Sec. 19.5A]. We found that the motion of such a system was defined by the differential equation

$$m\ddot{x} + c\dot{x} + kx = 0 \quad (19.38)$$

where c is a constant called the *coefficient of viscous damping*. Defining the *critical damping coefficient* c_c as

$$c_c = 2m\sqrt{\frac{k}{m}} = 2m\omega_n \quad (19.41)$$

where ω_n is the natural circular frequency of the system in the absence of damping, we distinguished three different cases of damping, namely, (1) *overdamped*, when $c > c_c$; (2) *critically damped*, when $c = c_c$; and (3) *underdamped*, when $c < c_c$. In the first two cases, the system when disturbed tends to regain its equilibrium position without any oscillation. In the third case, the motion is vibratory with diminishing amplitude. For an underdamped system, the transient response is

$$x = x_0 e^{-(c/2m)t} \sin(\omega_d t + \phi) \quad (19.46)$$

where

$$\omega_d = \omega_n \sqrt{1 - \left(\frac{c}{c_c}\right)^2} \quad (19.45)$$

Damped Forced Vibrations

In Sec. 19.5B, we considered the **damped forced vibrations** of a mechanical system. These vibrations occur when a system with viscous damping is subjected to a periodic force $\mathbf{P}$ of magnitude $P = P_m \sin \omega_f t$ or when it is elastically connected to a support with an alternating motion of $\delta = \delta_m \sin \omega_f t$. In the first case, the motion of the system was defined by the differential equation

$$m\ddot{x} + c\dot{x} + kx = P_m \sin \omega_f t \quad (19.47)$$

and in the second case, by a similar equation obtained by replacing P_m by $k\delta_m$ in (19.47).

The *steady-state vibration* of the system is represented by a particular solution of Eq. (19.47) of the form

$$x_{\text{part}} = x_m \sin (\omega_f t - \phi) \quad (19.48)$$

Dividing the amplitude x_m of the steady-state vibration by P_m/k in the case of a periodic force or by δ_m in the case of an oscillating support, we obtained the following expression for the magnification factor as

$$\frac{x_m}{P_m/k} = \frac{x_m}{\delta_m} = \frac{1}{\sqrt{[1 - (\omega_f/\omega_n)^2]^2 + [2(c/c_c)(\omega_f/\omega_n)]^2}} \quad (19.53)$$

or

$$\frac{x_m}{P_m/k} = \frac{x_m}{\delta_{st}} = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}$$

where $\omega_n = \sqrt{k/m}$ = natural circular frequency of undamped system

$c_c = 2m\omega_n$ = critical damping coefficient

$c/c_c = \zeta$ = damping ratio

$r = \omega/\omega_n$ = frequency ratio

We also found that the *phase difference* ϕ between the impressed force or support movement and the resulting steady-state vibration of the damped system was defined by the relation

$$\tan \phi = \frac{2(c/c_c)(\omega_f/\omega_n)}{1 - (\omega_f/\omega_n)^2} \quad (19.54)$$

or

$$\tan \phi = \frac{2\zeta r}{1 - r^2} \quad (19.54')$$

Electrical Analogs

This chapter ended with a discussion of *electrical analogs* [Sec. 19.5C] in which we showed that the vibrations of mechanical systems and the oscillations of electrical circuits are defined by the same differential equations. Electrical analogs of mechanical systems therefore can be used to study or predict the behavior of these systems.

Review Problems

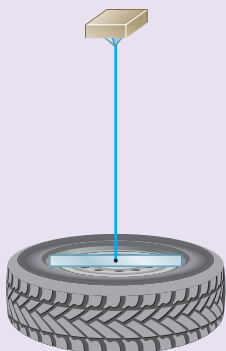


Fig. P19.159

19.159 An automobile wheel-and-tire assembly of total weight 47 lb is attached to a mounting plate of negligible weight that is suspended from a steel wire. The torsional spring constant of the wire is known to be $K = 0.40$ lb-in/rad. The wheel is rotated through 90° about the vertical and then released. Knowing that the period of oscillation is observed to be 30 s, determine the centroidal mass moment of inertia and the centroidal radius of gyration of the wheel-and-tire assembly.

19.160 The period of vibration of the system shown is observed to be 0.6 s. After cylinder B has been removed, the period is observed to be 0.5 s. Determine (a) the weight of cylinder A , (b) the constant of the spring.

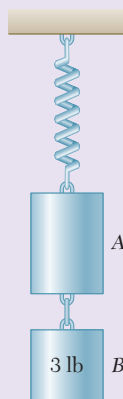


Fig. P19.160

19.161 Disks A and B weigh 30 lb and 12 lb, respectively, and a small 5-lb block C is attached to the rim of disk B . Assuming that no slipping occurs between the disks, determine the period of small oscillations of the system.

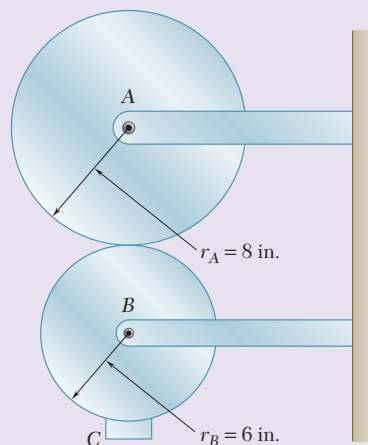


Fig. P19.161

19.162 The block shown is depressed 1.2 in. from its equilibrium position and released. Knowing that after 10 cycles the maximum displacement of the block is 0.5 in., determine (a) the damping factor c/c_c , (b) the value of the coefficient of viscous damping. (*Hint*: See Problems 19.129 and 19.130.)

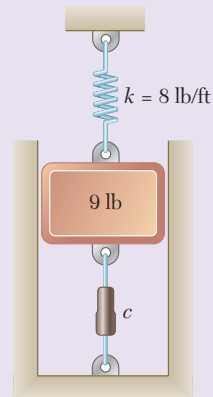


Fig. P19.162

19.163 An 0.8-lb ball is connected to a paddle by means of an elastic cord AB of constant $k = 5$ lb/ft. Knowing that the paddle is moved vertically according to the relation $\delta = \delta_m \sin \omega_f t$, where $\delta_m = 8$ in., determine the maximum allowable circular frequency ω_f if the cord is not to become slack.

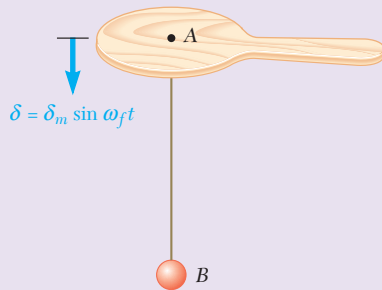


Fig. P19.163

19.164 A 3-kg slender rod AB is bolted to a 5-kg uniform disk. A dashpot with a damping coefficient of $c = 9$ N·s/m is attached to the disk as shown. Determine (a) the differential equation of motion for small oscillations, (b) the damping factor c/c_c .

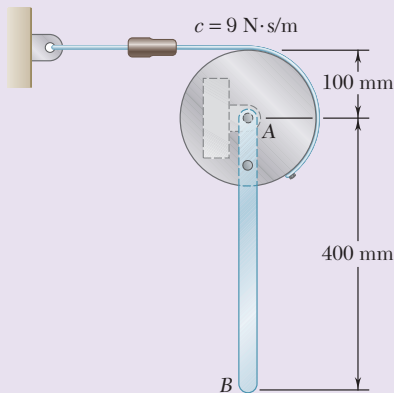


Fig. P19.164

19.165 A 4-lb uniform rod is supported by a pin at O and a spring at A and is connected to a dashpot at B . Determine (a) the differential equation of motion for small oscillations, (b) the angle that the rod will form with the horizontal 5 s after end B has been pushed 0.9 in. down and released.

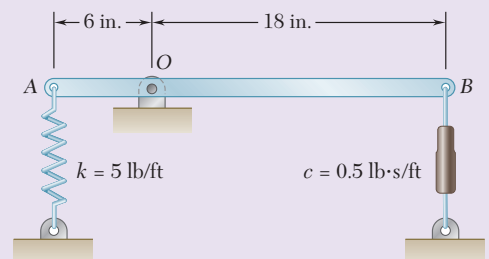


Fig. P19.165

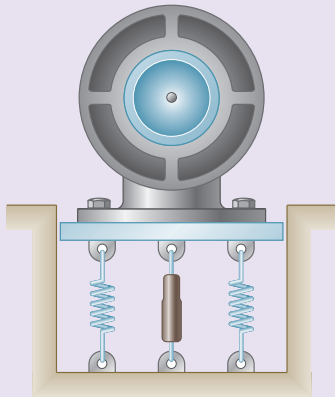


Fig. P19.166

19.166 A 400-kg motor supported by four springs, each of constant 150 kN/m , and a dashpot of constant $c = 6500 \text{ N}\cdot\text{s/m}$ is constrained to move vertically. Knowing that the unbalance of the rotor is equivalent to a 23-g mass located at a distance of 100 mm from the axis of rotation, determine for a speed of 800 rpm (a) the amplitude of the fluctuating force transmitted to the foundation, (b) the amplitude of the vertical motion of the motor.

19.167 The compressor shown has a mass of 250 kg and operates at 2000 rpm. At this operating condition, the force transmitted to the ground is excessively high and is found to be $m r \omega_f^2$, where $m r$ is the unbalance and ω_f is the forcing frequency. To fix this problem, it is proposed to isolate the compressor by mounting it on a square concrete block separated from the rest of the floor as shown. The density of concrete is 2400 kg/m^3 and the spring constant for the soil is found to be $80 \times 10^6 \text{ N/m}$. The geometry of the compressor leads to choosing a block that is 1.5 m by 1.5 m. Determine the depth h that will reduce the force transmitted to the ground by 75 percent.

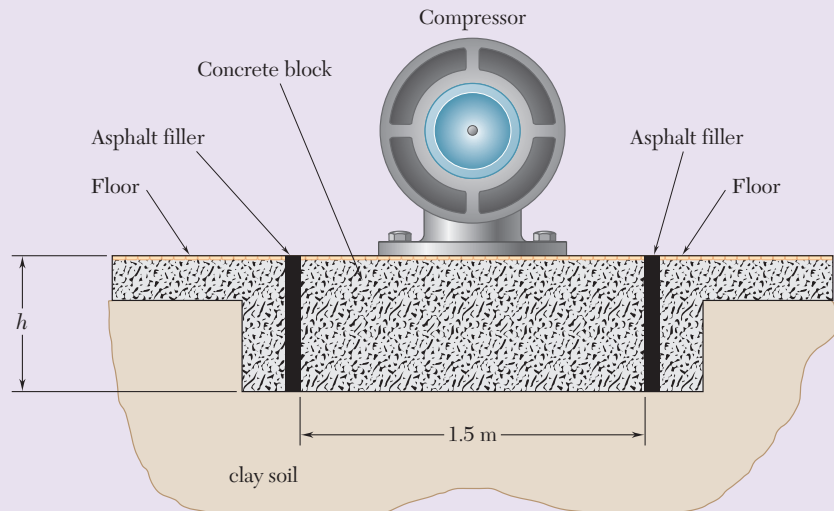


Fig. P19.167

19.168 A small ball of mass m attached at the midpoint of a tightly stretched elastic cord of length l can slide on a horizontal plane. The ball is given a small displacement in a direction perpendicular to the cord and released. Assuming the tension T in the cord to remain constant, (a) write the differential equation of motion of the ball, (b) determine the period of vibration.

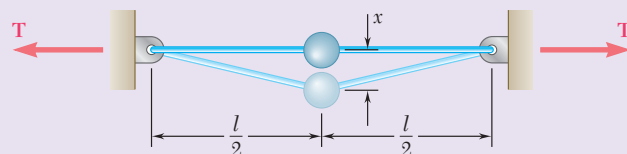


Fig. P19.168

- 19.169** A certain vibrometer used to measure vibration amplitudes consists essentially of a box containing a slender rod to which a mass m is attached; the natural frequency of the mass–rod system is known to be 5 Hz. When the box is rigidly attached to the casing of a motor rotating at 600 rpm, the mass is observed to vibrate with an amplitude of 0.06 in. relative to the box. Determine the amplitude of the vertical motion of the motor.

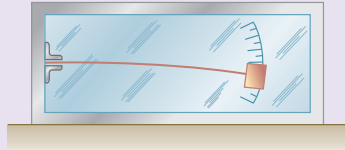


Fig. P19.169

- 19.170** If either a simple or a compound pendulum is used to determine experimentally the acceleration of gravity g , difficulties are encountered. In the case of the simple pendulum, the string is not truly weightless, while in the case of the compound pendulum, the exact location of the mass center is difficult to establish. In the case of a compound pendulum, the difficulty can be eliminated by using a reversible, or Kater, pendulum. Two knife edges A and B are placed so that they are obviously not at the same distance from the mass center G , and the distance l is measured with great precision. The position of a counterweight D is then adjusted so that the period of oscillation τ is the same when either knife edge is used. Show that the period τ obtained is equal to that of a true simple pendulum of length l and that $g = 4\pi^2 l / \tau^2$.

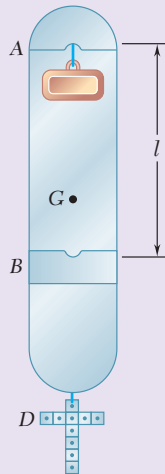


Fig. P19.170

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Fundamentals of Engineering Examination

Engineers are required to be licensed when their work directly affects the public health, safety, and welfare. The intent is to ensure that engineers have met minimum qualifications involving competence, ability, experience, and character. The licensing process involves an initial exam, called the *Fundamentals of Engineering Examination*; professional experience; and a second exam, called the *Principles and Practice of Engineering*. Those who successfully complete these requirements are licensed as a *Professional Engineer*. The exams are developed under the auspices of the *National Council of Examiners for Engineering and Surveying*.

The first exam, the *Fundamentals of Engineering Examination*, can be taken just before or after graduation from a four-year accredited engineering program. The exam stresses subject material in a typical undergraduate engineering program, including statics. The topics included in the exam cover much of the material in this book. The following is a list of the main topic areas, with references to the appropriate sections in this book. Also included are problems that can be solved to review this material.

Concurrent Force Systems (2.1–2.2; 2.4)

Problems: 2.31, 2.35, 2.36, 2.37, 2.77, 2.83, 2.92, 2.94, 2.97

Vector Forces (3.1–3.2)

Problems: 3.17, 3.18, 3.26, 3.33, 3.37, 3.39

Equilibrium in Two Dimensions (2.3; 4.1–4.2)

Problems: 4.1, 4.13, 4.14, 4.17, 4.31, 4.33, 4.67, 4.77

Equilibrium in Three Dimensions (2.5; 4.3)

Problems: 4.99, 4.101, 4.103, 4.108, 4.115, 4.117, 4.127, 4.129, 4.135

Centroids of Areas and Volumes (5.1–5.2; 5.4)

Problems: 5.9, 5.16, 5.30, 5.35, 5.41, 5.55, 5.62, 5.96, 5.102, 5.103, 5.125

Analysis of Trusses (6.1–6.2)

Problems: 6.3, 6.4, 6.32, 6.43, 6.44, 6.53

Equilibrium of Two-Dimensional Frames (6.3)

Problems: 6.75, 6.81, 6.85, 6.93, 6.94

Shear and Bending Moment (7.1–7.3)

Problems: 7.22, 7.30, 7.36, 7.41, 7.45, 7.49, 7.71, 7.79

Friction (8.1–8.2; 8.4)

Problems: 8.11, 8.18, 8.19, 8.30, 8.49, 8.52, 8.103, 8.104, 8.105

Moments of Inertia (9.1–9.4)

Problems: 9.6, 9.31, 9.32, 9.33, 9.72, 9.74, 9.80, 9.83, 9.98, 9.103

Kinematics (11.1–11.2; 11.4–11.5; 15.1–15.4)

Problems: 11.3, 11.10, 11.34, 11.35, 11.97, 11.102, 15.4, 15.6, 15.28, 15.39, 15.61, 15.63, 15.82, 15.111, 15.112

Force, Mass, and Acceleration (12.1; 16.1–16.2)

Problems: 12.5, 12.6, 12.11, 12.23, 12.36, 12.44, 12.45, 12.50, 16.1, 16.3, 16.9, 16.26, 16.27, 16.50, 16.60, 16.63, 16.78, 16.84

Work and Energy (13.1–13.2; 13.8; 17.1)

Problems: 13.3, 13.6, 13.13, 13.17, 13.40, 13.42, 13.47, 13.64, 13.66, 13.68, 17.1, 17.2, 17.16, 17.20

Impulse and Momentum (13.3–13.4; 17.2–17.3)

Problems: 13.119, 13.120, 13.129, 13.134, 13.146, 13.155, 13.163, 13.169, 17.53, 17.58, 17.70, 17.72, 17.96, 17.97, 17.104

Vibration (19.1; 19.2–19.4)

Problems: 19.1, 19.2, 19.11, 19.18, 19.23, 19.28, 19.50, 19.55, 19.64, 19.79, 19.99, 19.101, 19.105, 19.116

Friction (Problems involving friction occur in each of the above subjects.)

Answers to Problems

CHAPTER 2

- 2.1** 1391 N $\nearrow$ 47.8°.
2.2 906 lb $\nearrow$ 26.6°.
2.4 8.03 kips $\nearrow$ 3.8°.
2.5 (a) 101.4 N. (b) 196.6 N.
2.6 (a) 853 lb. (b) 567 lb.
2.8 (a) $T_{AC} = 2.60$ kN. (b) $R = 4.26$ kN.
2.9 (a) $T_{AC} = 2.66$ kN $\swarrow$ 34.3°.
2.10 (a) 37.1°. (b) 73.2 N.
2.11 (a) 392 lb. (b) 346 lb.
2.13 (a) 368 lb $\rightarrow$. (b) 213 lb.
2.14 (a) 21.1 N $\downarrow$. (b) 45.3 N.
2.15 414 lb $\swarrow$ 72.0°.
2.16 1391 N $\nearrow$ 47.8°.
2.17 8.03 kips $\nearrow$ 3.8°.
2.19 104.4 N $\searrow$ 86.7°.
2.21 (29 lb) 21.0 lb, 20.0 lb; (50 lb) 14.00 lb, 48.0 lb; (51 lb) 24.0 lb, 45.0 lb.
2.23 (80 N) 61.3 N, 51.4 N; (120 N) 41.0 N, 112.8 N; (150 N) 122.9 N, 86.0 N.
2.24 (40 lb) 20.0 lb, 34.6 lb; (50 lb) 38.3 lb, 32.1 lb; (60 lb) 54.4 lb, 25.4 lb.
2.26 (a) 523 lb. (b) 428 lb.
2.27 (a) 621 N. (b) 160.8 N.
2.28 (a) 610 lb. (b) 500 lb.
2.29 (a) 2190 N. (b) 2060 N.
2.31 38.6 lb $\nearrow$ 36.6°.
2.32 251 N $\searrow$ 85.3°.
2.34 654 N $\swarrow$ 21.5°.
2.35 309 N $\nearrow$ 86.6°.
2.36 474 N $\swarrow$ 32.5°.
2.37 203 lb $\nearrow$ 8.46°.
2.39 (a) 21.7°. (b) 229 N.
2.40 (a) 26.5 N. (b) 623 N.
2.42 (a) 56.3°. (b) 204 lb.
2.43 (a) 352 lb. (b) 261 lb.
2.44 (a) 5.22 kN. (b) 3.45 kN.
2.46 (a) 305 N. (b) 514 N.
2.48 (a) 1244 lb. (b) 115.4 lb.
2.49 $T_{CA} = 134.6$ N; $T_{CB} = 110.4$ N.
2.50 179.3 N $< P < 669$ N.
2.51 $F_A = 1303$ lb; $F_B = 420$ lb.
2.53 $F_C = 6.40$ kN; $F_D = 4.80$ kN.
2.54 $F_B = 15.00$ kN; $F_C = 8.00$ kN.
2.55 (a) $T_{ACB} = 269$ lb. (b) $T_{CD} = 37.0$ lb.
2.57 (a) $\alpha = 35.0^\circ$; $T_{AC} = 4.91$ kN; $T_{BC} = 3.44$ kN, (b) $\alpha = 55.0^\circ$; $T_{AC} = T_{BC} = 3.66$ kN.
2.58 (a) 784 N. (b) $\alpha = 71.0^\circ$.
2.59 (a) $\alpha = 5.00^\circ$. (b) 104.6 lb.
2.61 1.250 m.
2.62 75.6 mm.
2.63 (a) 10.98 lb. (b) 30.0 lb.
2.65 $27.4^\circ \leq \alpha \leq 222.6^\circ$.
2.67 (a) 300 lb. (b) 300 lb. (c) 200 lb. (d) 200 lb. (e) 150.0 lb.
2.68 (a) 200 lb. (b) 150.0 lb.
2.69 (a) 1293 N. (b) 2220 N.
2.71 (a) 220 N, 544 N, 126.8 N. (b) 68.5°, 25.0°, 77.8°.
2.72 (a) 237 N, 258 N, 282 N. (b) 121.8°, 55.0°, 51.1°.
2.73 (a) 175.8 N, 257 N, 251 N. (b) 116.1°, 130.0°, 51.1°.
2.74 (a) 350 N, 169.0 N, 93.8 N. (b) 28.9°, 115.0°, 76.4°.
2.75 (a) 20.5 lb, 43.3 lb, 14.33 lb. (b) 114.2°, 30.0°, 106.7°.
2.77 (a) 1861 lb, 3360 lb, 677 lb. (b) 118.5°, 30.5°, 80.0°.
2.79 (a) 770 N; 71.8°; 110.5°; 28.0°.
2.81 (a) 140.3°. (b) $F_x = 79.9$ lb, $F_z = 120.1$ lb; $F = 226$ lb.
2.82 (a) 118.2°. (b) $F_x = 36.0$ lb, $F_y = -90.0$ lb; $F = 110.0$ lb.
2.84 (a) $F_x = 507$ N, $F_y = 919$ N, $F_z = 582$ N. (b) 61.0°.
2.85 240 N; 255 N; 160.0 N.
2.87 1.260 kips; 1.213 kips; 0.970 kips.
2.88 0.820 kips; 0.978 kips; 0.789 kips.
2.89 192.0 N; 288 N; 216 N.
2.91 515 N; $\theta_x = 70.2^\circ$; $\theta_y = 27.6^\circ$; $\theta_z = 71.5^\circ$.
2.92 515 N; $\theta_x = 79.8^\circ$; $\theta_y = 33.4^\circ$; $\theta_z = 58.6^\circ$.
2.94 913 lb; $\theta_x = 50.6^\circ$; $\theta_y = 117.6^\circ$; $\theta_z = 51.8^\circ$.
2.95 748 N; $\theta_x = 120.1^\circ$; $\theta_y = 52.5^\circ$; $\theta_z = 128.0^\circ$.
2.96 $T_{AB} = 490$ N; $T_{AD} = 515$ N.
2.97 130.0 lb.
2.99 13.98 kN.
2.101 926 N $\uparrow$.
2.103 $T_{DA} = 14.42$ lb; $T_{DB} = T_{DC} = 13.00$ lb.
2.104 $T_{DA} = 14.42$ lb; $T_{DB} = T_{DC} = 13.27$ lb.
2.106 $T_{AB} = 571$ lb; $T_{AC} = 830$ lb; $T_{AD} = 528$ lb.
2.107 960 N.
2.108 $0 \leq Q < 300$ N.
2.109 845 N.
2.110 768 N.
2.112 2000 lb.
2.113 $T_{AB} = 30.8$ lb; $T_{AC} = 62.5$ lb.
2.115 $T_{AB} = 510$ N; $T_{AC} = 56.2$ N; $T_{AD} = 536$ N.
2.116 $T_{AB} = 1340$ N; $T_{AC} = 1025$ N; $T_{AD} = 915$ N.
2.117 $T_{AB} = 1431$ N; $T_{AC} = 1560$ N; $T_{AD} = 183.0$ N.
2.118 $T_{AB} = 1249$ N; $T_{AC} = 490$ N; $T_{AD} = 1647$ N.
2.119 $T_{AB} = 974$ lb; $T_{AC} = 531$ lb; $T_{AD} = 533$ lb.
2.121 378 N.
2.123 $T_{BAC} = 76.7$ lb; $T_{AD} = 26.9$ lb; $T_{AE} = 49.2$ lb.
2.124 (a) 305 lb. (b) $T_{BAC} = 117.0$ lb; $T_{AD} = 40.9$ lb.
2.125 (a) 1155 N. (b) 1012 N.
2.127 21.8 kN $\swarrow$ 73.4°.
2.128 (102 lb) 48.0 lb, 90.0 lb; (106 lb) 56.0 lb, 90.0 lb; (200 lb) 160.0 lb, 120.0 lb.
2.130 (a) 172.7 lb. (b) 231 lb.
2.131 (a) 312 N. (b) 144.0 N.
2.133 (a) 56.4 lb; 103.9 lb; 20.5 lb. (b) 62.0°, 150.0°, 99.8°.
2.135 940 N; 65.7°, 28.2°, 76.4°.
2.136 $P = 131.2$ N; $Q = 29.6$ N.
2.137 (a) 125.0 lb. (b) 45.0 lb.

CHAPTER 3

- 3.1** (a) $196.2 \text{ N}\cdot\text{m} \downarrow$. (b) $199.0 \text{ N} \searrow 59.5^\circ$.
3.2 (a) $196.2 \text{ N}\cdot\text{m} \downarrow$. (b) $321 \text{ N} \nearrow 35.0^\circ$. (c) $231 \text{ N} \uparrow$ at point D .
3.4 (a) $41.7 \text{ N}\cdot\text{m} \uparrow$. (b) $147.4 \text{ N} \searrow 45.0^\circ$.
3.5 (a) $41.7 \text{ N}\cdot\text{m} \uparrow$. (b) 334 N . (c) $176.8 \text{ N} \searrow 58.0^\circ$.
3.6 $115.7 \text{ lb}\cdot\text{in}$.
3.7 $115.7 \text{ lb}\cdot\text{in}$.
3.9 (a) $292 \text{ N}\cdot\text{m} \downarrow$. (b) $292 \text{ N}\cdot\text{m} \downarrow$.
3.11 $116.2 \text{ lb}\cdot\text{ft} \uparrow$.
3.12 $128.2 \text{ lb}\cdot\text{ft} \uparrow$.
3.13 $140.0 \text{ N}\cdot\text{m} \uparrow$.
3.17 (a) $\lambda = -0.677\mathbf{i} - 0.369\mathbf{j} - 0.636\mathbf{k}$.
 (b) $\lambda = -0.0514\mathbf{i} + 0.566\mathbf{j} + 0.823\mathbf{k}$.
3.18 1.184 m .
3.20 (a) $9\mathbf{i} + 22\mathbf{j} + 21\mathbf{k}$. (b) $22\mathbf{i} + 11\mathbf{k}$. (c) 0 .
3.22 $(2400 \text{ lb}\cdot\text{ft})\mathbf{j} + (1440 \text{ lb}\cdot\text{ft})\mathbf{k}$.
3.23 $(7.50 \text{ N}\cdot\text{m})\mathbf{i} - (6.00 \text{ N}\cdot\text{m})\mathbf{j} - (10.39 \text{ N}\cdot\text{m})\mathbf{k}$.
3.25 $(-25.4 \text{ lb}\cdot\text{ft})\mathbf{i} - (12.60 \text{ lb}\cdot\text{ft})\mathbf{j} - (12.60 \text{ lb}\cdot\text{ft})\mathbf{k}$.
3.26 $(1200 \text{ N}\cdot\text{m})\mathbf{i} - (1500 \text{ N}\cdot\text{m})\mathbf{j} - (900 \text{ N}\cdot\text{m})\mathbf{k}$.
3.27 7.37 ft .
3.28 100.8 mm .
3.29 144.8 mm .
3.30 5.17 ft .
3.32 2.36 m .
3.33 1.491 m .
3.35 $\mathbf{P}\cdot\mathbf{Q} = -5$; $\mathbf{P}\cdot\mathbf{S} = +5$; $\mathbf{Q}\cdot\mathbf{S} = -38$.
3.37 77.9° .
3.39 (a) 59.0° . (b) 144.0 lb .
3.40 (a) 70.5° . (b) 60.0 lb .
3.41 (a) 52.9° . (b) 326 N .
3.43 26.8° .
3.44 33.3° .
3.45 (a) 67.0 . (b) 111.0 .
3.46 2 .
3.47 $M_x = 78.9 \text{ kN}\cdot\text{m}$, $M_y = 13.15 \text{ kN}\cdot\text{m}$, $M_z = -9.86 \text{ kN}\cdot\text{m}$.
3.48 3.04 kN .
3.49 $\phi = 24.6^\circ$; $d = 34.6 \text{ in}$.
3.51 1.252 m .
3.52 1.256 m .
3.53 283 lb .
3.55 $+207 \text{ lb}\cdot\text{ft}$.
3.57 $-90.0 \text{ N}\cdot\text{m}$.
3.58 $-111.0 \text{ N}\cdot\text{m}$.
3.59 $+2.28 \text{ N}\cdot\text{m}$.
3.60 $-9.50 \text{ N}\cdot\text{m}$.
3.61 $a P/\sqrt{2}$.
3.64 13.06 in .
3.65 12.69 in .
3.67 0.249 m .
3.68 0.1198 m .
3.70 (a) $7.33 \text{ N}\cdot\text{m} \uparrow$. (b) 91.6 mm .
3.71 $6.19 \text{ N}\cdot\text{m} \downarrow$.
3.73 1.125 in .
3.74 (a) 26.7 N . (b) 50.0 N . (c) 23.5 N .
3.76 $M = 604 \text{ lb}\cdot\text{in}$; $\theta_x = 72.8^\circ$, $\theta_y = 27.3^\circ$, $\theta_z = 110.5^\circ$.
3.77 $M = 1170 \text{ lb}\cdot\text{in}$; $\theta_x = 81.2^\circ$, $\theta_y = 13.70^\circ$, $\theta_z = 100.4^\circ$.
3.78 $M = 3.22 \text{ N}\cdot\text{m}$; $\theta_x = 90.0^\circ$, $\theta_y = 53.1^\circ$, $\theta_z = 36.9^\circ$.
3.79 $M = 2.72 \text{ N}\cdot\text{m}$; $\theta_x = 134.9^\circ$, $\theta_y = 58.0^\circ$, $\theta_z = 61.9^\circ$.
3.80 $M = 2150 \text{ lb}\cdot\text{ft}$; $\theta_x = 113.0^\circ$, $\theta_y = 92.7^\circ$, $\theta_z = 23.2^\circ$.
3.82 (a) $\mathbf{F}_A = 560 \text{ lb} \searrow 20.0^\circ$; $\mathbf{M}_A = 7720 \text{ lb}\cdot\text{ft} \downarrow$.
 (b) $\mathbf{F}_B = 560 \text{ lb} \searrow 20.0^\circ$; $\mathbf{M}_B = 4290 \text{ lb}\cdot\text{ft} \downarrow$.
3.83 $\mathbf{F}_A = 389 \text{ N} \searrow 60.0^\circ$; $\mathbf{F}_C = 651 \text{ N} \searrow 60.0^\circ$.

- 3.84** (a) $\mathbf{F} = 30.0 \text{ lb} \downarrow$; $\mathbf{M} = 150.0 \text{ lb}\cdot\text{in} \uparrow$. (b) $\mathbf{B} = 50.0 \text{ lb} \leftarrow$; $\mathbf{C} = 50.0 \text{ lb} \rightarrow$.
3.86 $\mathbf{F}_A = 168.0 \text{ N} \nearrow 50.0^\circ$; $\mathbf{F}_C = 192.0 \text{ N} \nearrow 50.0^\circ$.
3.87 $\mathbf{F} = 900 \text{ N} \downarrow$; $x = 50.0 \text{ mm}$.
3.89 (a) $\mathbf{F} = 48.0 \text{ lb} \searrow 65.0^\circ$; $\mathbf{M} = 490 \text{ lb}\cdot\text{in} \downarrow$.
 (b) $\mathbf{F} = 48.0 \text{ lb} \searrow 65.0^\circ$ applied 17.78 in . to the left of B .
3.90 (a) 48.0 N intersecting line AB 144.0 mm to the right of A .
 (b) 77.7° or -15.72° .
3.91 $(0.227 \text{ lb})\mathbf{i} + (0.1057 \text{ lb})\mathbf{k}$; 63.6 in . to the right of B .
3.93 $\mathbf{F} = -(250 \text{ kN})\mathbf{j}$; $\mathbf{M} = (15.00 \text{ kN}\cdot\text{m})\mathbf{i} + (7.50 \text{ kN}\cdot\text{m})\mathbf{k}$.
3.95 $\mathbf{F} = -(122.9 \text{ N})\mathbf{j} - (86.0 \text{ N})\mathbf{k}$; $\mathbf{M} = (22.6 \text{ N}\cdot\text{m})\mathbf{i} + (15.49 \text{ N}\cdot\text{m})\mathbf{j} - (22.1 \text{ N}\cdot\text{m})\mathbf{k}$.
3.96 $\mathbf{F} = (5.00 \text{ N})\mathbf{i} + (150.0 \text{ N})\mathbf{j} - (90.0 \text{ N})\mathbf{k}$; $\mathbf{M} = (77.4 \text{ N}\cdot\text{m})\mathbf{i} + (61.5 \text{ N}\cdot\text{m})\mathbf{j} + (106.8 \text{ N}\cdot\text{m})\mathbf{k}$.
3.97 $\mathbf{F} = (36.0 \text{ lb})\mathbf{i} - (28.0 \text{ lb})\mathbf{j} - (6.00 \text{ lb})\mathbf{k}$; $\mathbf{M} = -(157.0 \text{ lb}\cdot\text{ft})\mathbf{i} + (22.5 \text{ lb}\cdot\text{ft})\mathbf{j} - (240 \text{ lb}\cdot\text{ft})\mathbf{k}$.
3.98 $\mathbf{F} = -(28.5 \text{ N})\mathbf{i} + (106.3 \text{ N})\mathbf{k}$; $\mathbf{M} = (12.35 \text{ N}\cdot\text{m})\mathbf{i} - (19.16 \text{ N}\cdot\text{m})\mathbf{j} - (5.13 \text{ N}\cdot\text{m})\mathbf{k}$.
3.99 $\mathbf{F} = -(128.0 \text{ lb})\mathbf{i} - (256 \text{ lb})\mathbf{j} + (32.0 \text{ lb})\mathbf{k}$; $\mathbf{M} = (4.10 \text{ kip}\cdot\text{ft})\mathbf{i} + (16.38 \text{ kip}\cdot\text{ft})\mathbf{k}$.
3.101 (a) Loading a: $500 \text{ N} \downarrow$; $1000 \text{ N}\cdot\text{m} \downarrow$.
 Loading b: $500 \text{ N} \uparrow$; $500 \text{ N}\cdot\text{m} \uparrow$.
 Loading c: $500 \text{ N} \downarrow$; $500 \text{ N}\cdot\text{m} \downarrow$.
 Loading d: $500 \text{ N} \downarrow$; $1100 \text{ N}\cdot\text{m} \downarrow$.
 Loading e: $500 \text{ N} \downarrow$; $1000 \text{ N}\cdot\text{m} \downarrow$.
 Loading f: $500 \text{ N} \downarrow$; $200 \text{ N}\cdot\text{m} \downarrow$.
 Loading g: $500 \text{ N} \downarrow$; $2300 \text{ N}\cdot\text{m} \uparrow$.
 Loading h: $500 \text{ N} \downarrow$; $600 \text{ N}\cdot\text{m} \uparrow$.
 (b) Loadings a and e are equivalent.
3.102 Equivalent to case f of problem 3.101.
3.104 Equivalent force-couple system at D .
3.105 (a) 2.00 ft to the right of C . (b) 2.31 ft to the right of C .
3.106 (a) 39.6 in . to the right of D . (b) 33.1 in .
3.108 $44.7 \text{ lb} \searrow 26.6^\circ$; 10.61 in . to the left of C and 5.30 in . below C .
3.110 (a) $224 \text{ N} \searrow 63.4^\circ$. (b) 130.0 mm to the left of B and 260 mm below B .
3.111 (a) $269 \text{ N} \searrow 68.2^\circ$. (b) 120.0 mm to the left of B and 300 mm below B .
3.113 $773 \text{ lb} \nearrow 79.0^\circ$; 9.54 ft to the right of A .
3.114 (a) $29.9 \text{ lb} \searrow 23.0^\circ$. (b) AB : 10.30 in . to the left of B ;
 BC : 4.36 in . below B .
3.115 (a) $60.2 \text{ lb}\cdot\text{in} \uparrow$. (b) $200 \text{ lb}\cdot\text{in} \uparrow$. (c) $20.0 \text{ lb}\cdot\text{in} \downarrow$.
3.116 (a) 0.365 m above G . (b) 0.227 m to the right of G .
3.117 (a) 0.299 m above G . (b) 0.259 m to the right of G .
3.118 (a) $\mathbf{R} = F \searrow \tan^{-1}(a^2/2bx)$; $\mathbf{M} = 2Fb^2(x - x^3/a^2)/\sqrt{a^4 + 4b^2x^2} \uparrow$.
 (b) 0.369 m .
3.119 $\mathbf{R} = -(300 \text{ N})\mathbf{i} - (240 \text{ N})\mathbf{j} + (25.0 \text{ N})\mathbf{k}$;
 $\mathbf{M} = -(3.00 \text{ N}\cdot\text{m})\mathbf{i} + (13.50 \text{ N}\cdot\text{m})\mathbf{j} + (9.00 \text{ N}\cdot\text{m})\mathbf{k}$.
3.120 $\mathbf{R} = (420 \text{ N})\mathbf{j} - (339 \text{ N})\mathbf{k}$; $\mathbf{M} = (1.125 \text{ N}\cdot\text{m})\mathbf{i} + (163.9 \text{ N}\cdot\text{m})\mathbf{j} - (109.9 \text{ N}\cdot\text{m})\mathbf{k}$.
3.122 (a) 60.0° . (b) $(20.0 \text{ lb})\mathbf{i} - (34.6 \text{ lb})\mathbf{j}$; $(520 \text{ lb}\cdot\text{in})\mathbf{i}$.
3.124 $\mathbf{R} = -(420 \text{ N})\mathbf{i} - (50.0 \text{ N})\mathbf{j} - (250 \text{ N})\mathbf{k}$; $\mathbf{M} = (30.8 \text{ N}\cdot\text{m})\mathbf{j} - (22.0 \text{ N}\cdot\text{m})\mathbf{k}$.
3.125 (a) $\mathbf{B} = -(75.0 \text{ N})\mathbf{k}$, $\mathbf{C} = -(25.0 \text{ N})\mathbf{i} + (37.5 \text{ N})\mathbf{k}$.
 (b) $R_y = 0$, $R_z = -37.5 \text{ N}$. (c) when the slot is vertical.
3.126 $\mathbf{A} = (1.600 \text{ lb})\mathbf{i} - (36.0 \text{ lb})\mathbf{j} + (2.00 \text{ lb})\mathbf{k}$,
 $\mathbf{B} = -(9.60 \text{ lb})\mathbf{i} + (36.0 \text{ lb})\mathbf{j} + (2.00 \text{ lb})\mathbf{k}$.
3.127 1035 N ; 2.57 m from OG and 3.05 m from OE .
3.128 2.32 m from OG and 1.165 m from OE .
3.129 405 lb ; 12.60 ft to the right of AB and 2.94 ft below BC .
3.130 $a = 0.722 \text{ ft}$; $b = 20.6 \text{ ft}$.
3.133 (a) $P\sqrt{3}$; $\theta_x = \theta_y = \theta_z = 54.7^\circ$. (b) $-a$
 (c) Axis of the wrench is diagonal OA .

- 3.134** (a) P ; $\theta_x = 90.0^\circ$, $\theta_y = 90.0^\circ$, $\theta_z = 0$. (b) $5a/2$.
(c) Axis of the wrench is parallel to the z -axis at $x = a$, $y = -a$.
- 3.136** (a) $-(21.0 \text{ lb})\mathbf{j}$. (b) 0.571 in . (c) At $x = 0$, $z = 1.667 \text{ in}$; and is parallel to the y axis.
- 3.137** (a) $-(84.0 \text{ N})\mathbf{j} - (80.0 \text{ N})\mathbf{k}$. (b) 0.477 m . (c) $x = 0.526 \text{ m}$, $y = 0$, $z = -0.1857 \text{ m}$.
- 3.140** (a) $3P(2\mathbf{i} - 20\mathbf{j} - \mathbf{k})/25$. (b) $-0.0988a$.
(c) $x = 2.00a$, $y = 0$, $z = -1.990a$.
- 3.141** $\mathbf{R} = (20.0 \text{ N})\mathbf{i} + (30.0 \text{ N})\mathbf{j} - (10.00 \text{ N})\mathbf{k}$; $y = -0.540 \text{ m}$, $z = -0.420 \text{ m}$.
- 3.143** $\mathbf{F}_A = (M/b)\mathbf{i} + R[1 + (a/b)]\mathbf{k}$; $\mathbf{F}_B = -(M/b)\mathbf{i} - (aR/b)\mathbf{k}$.
- 3.147** (a) $20.5 \text{ N}\cdot\text{m}$. (b) 68.4 mm .
- 3.148** $760 \text{ N}\cdot\text{m}$.
- 3.150** 43.6° .
- 3.151** $23.0 \text{ N}\cdot\text{m}$.
- 3.153** $\mathbf{M} = 4.50 \text{ N}\cdot\text{m}$; $\theta_x = 90.0^\circ$, $\theta_y = 177.1^\circ$, $\theta_z = 87.1^\circ$.
- 3.154** $\mathbf{F} = 260 \text{ lb} \searrow 67.4^\circ$; $M_c = 200 \text{ lb}\cdot\text{in}$.
- 3.156** (a) 135.0 mm . (b) $\mathbf{F}_2 = (42.0 \text{ N})\mathbf{i} + (42.0 \text{ N})\mathbf{j} - (49.0 \text{ N})\mathbf{k}$;
 $\mathbf{M}_2 = -(25.9 \text{ N}\cdot\text{m})\mathbf{i} + (21.2 \text{ N}\cdot\text{m})\mathbf{j}$
- 3.158** (a) $\mathbf{B} = (2.50 \text{ lb})\mathbf{i}$, $\mathbf{C} = (0.1000 \text{ lb})\mathbf{i} - (2.47 \text{ lb})\mathbf{j} - (0.700 \text{ lb})\mathbf{k}$.
(b) $\mathbf{R}_y = -2.47 \text{ lb}$; $\mathbf{M}_x = 1.360 \text{ lb}\cdot\text{ft}$.

CHAPTER 4

- 4.1** 42.0 N $\uparrow$.
- 4.2** 0.264 m .
- 4.4** (a) 245 lb $\uparrow$. (b) 140.0 lb .
- 4.5** (a) 34.0 kN $\uparrow$. (b) 4.96 kN $\uparrow$.
- 4.6** (a) 81.1 kN . (b) 134.1 kN $\uparrow$.
- 4.7** (a) $\mathbf{A} = 20.0 \text{ lb}$ $\downarrow$; $\mathbf{B} = 150.0 \text{ lb}$ $\uparrow$. (b) $\mathbf{A} = 10.00 \text{ lb}$ $\downarrow$;
 $\mathbf{B} = 140.0 \text{ lb}$ $\uparrow$.
- 4.9** $1.250 \text{ kN} \leq Q \leq 27.5 \text{ kN}$.
- 4.12** $6.00 \text{ kips} \leq P \leq 27.0 \text{ kips}$.
- 4.13** $150.0 \text{ mm} \leq d \leq 400 \text{ mm}$.
- 4.14** $2.00 \text{ in.} \leq a \leq 10.00 \text{ in.}$
- 4.15** (a) 600 N . (b) 1253 N $\nearrow 69.8^\circ$.
- 4.17** (a) 80.8 lb $\downarrow$. (b) 216 lb $\nearrow 22.0^\circ$.
- 4.18** 232 lb .
- 4.19** (a) 2.00 kN . (b) 2.32 kN $\nearrow 46.4^\circ$.
- 4.22** (a) 400 N . (b) 458 N $\nearrow 49.1^\circ$.
- 4.23** (a) $\mathbf{A} = 44.7 \text{ lb}$ $\searrow 26.6^\circ$; $\mathbf{B} = 30.0 \text{ lb}$ $\uparrow$.
(b) $\mathbf{A} = 30.2 \text{ lb}$ $\searrow 41.4^\circ$; $\mathbf{B} = 34.6 \text{ lb}$ $\searrow 60.0^\circ$.
- 4.24** (a) $\mathbf{A} = 20.0 \text{ lb}$ $\uparrow$; $\mathbf{B} = 50.0 \text{ lb}$ $\searrow 36.9^\circ$.
(b) $\mathbf{A} = 23.1 \text{ lb}$ $\nearrow 60.0^\circ$; $\mathbf{B} = 59.6 \text{ lb}$ $\searrow 30.2^\circ$.
- 4.25** (a) 190.9 N . (b) 142.3 N $\nearrow 18.43^\circ$.
- 4.26** (a) 324 N . (b) 270 N $\rightarrow$.
- 4.28** (a) $\mathbf{A} = 225 \text{ N}$ $\uparrow$; $\mathbf{C} = 641 \text{ N}$ $\nearrow 20.6^\circ$.
(b) $\mathbf{A} = 365 \text{ N}$ $\nearrow 60.0^\circ$; $\mathbf{C} = 884 \text{ N}$ $\nearrow 22.0^\circ$.
- 4.31** $T = 2P/3$; $\mathbf{C} = 0.577P$ $\rightarrow$.
- 4.32** $T = 0.586P$; $\mathbf{C} = 0.414P$ $\rightarrow$.
- 4.33** (a) 117.0 lb . (b) 129.8 lb $\searrow 56.3^\circ$.
- 4.34** (a) 195.0 lb . (b) 225 lb $\searrow 45.0^\circ$.
- 4.35** (a) 1432 N . (b) 1100 N $\uparrow$. (c) 1400 N $\leftarrow$.
- 4.36** $T_{BE} = 196.2 \text{ N}$; $\mathbf{A} = 73.6 \text{ N}$ $\rightarrow$; $\mathbf{D} = 73.6 \text{ N}$ $\leftarrow$.
- 4.39** (a) 600 N . (b) $\mathbf{A} = 4.00 \text{ kN}$ $\leftarrow$; $\mathbf{B} = 4.00 \text{ kN}$ $\rightarrow$.
- 4.40** (a) 105.1 N . (b) $\mathbf{A} = 147.2 \text{ N}$ $\uparrow$; $\mathbf{B} = 105.1 \text{ N}$ $\leftarrow$.
- 4.41** (a) $\mathbf{A} = 20.2 \text{ lb}$ $\uparrow$; $\mathbf{B} = 30.0 \text{ lb}$ $\searrow 60.0^\circ$. (b) 16.21 lb $\downarrow$.
- 4.42** $5.44 \text{ lb} \leq P \leq 17.23 \text{ lb}$.
- 4.43** (a) $\mathbf{E} = 8.80 \text{ kips}$ $\uparrow$; $\mathbf{M}_E = 36.0 \text{ kip}\cdot\text{ft}$ $\downarrow$.
(b) $\mathbf{E} = 4.80 \text{ kips}$ $\uparrow$; $\mathbf{M}_E = 51.0 \text{ kip}\cdot\text{ft}$ $\downarrow$.
- 4.45** $T_{\max} = 2240 \text{ N}$; $T_{\min} = 1522 \text{ N}$.
- 4.46** $\mathbf{C} = 1951 \text{ N}$ $\searrow 88.5^\circ$; $\mathbf{M}_C = 75.0 \text{ N}\cdot\text{m}$ $\downarrow$.
- 4.47** $1.232 \text{ kN} \leq T \leq 1.774 \text{ kN}$.
- 4.48** (a) $\mathbf{D} = 20.0 \text{ lb}$ $\downarrow$; $\mathbf{M}_D = 20.0 \text{ lb}\cdot\text{ft}$ $\uparrow$.
(b) $\mathbf{D} = 10.00 \text{ lb}$ $\downarrow$; $\mathbf{M}_D = 30.0 \text{ lb}\cdot\text{ft}$ $\downarrow$.
- 4.50** (a) $\mathbf{A} = 78.5 \text{ N}$ $\uparrow$; $\mathbf{M}_A = 125.6 \text{ N}\cdot\text{m}$ $\uparrow$.
(b) $\mathbf{A} = 111.0 \text{ N}$ $\uparrow$; $\mathbf{M}_A = 125.6 \text{ N}\cdot\text{m}$ $\uparrow$.
(c) $\mathbf{A} = 157.0 \text{ N}$ $\uparrow$; $\mathbf{M}_A = 251 \text{ N}\cdot\text{m}$ $\uparrow$.
- 4.51** $\theta = \sin^{-1}(2M \cot \alpha / Wl)$.
- 4.52** $\theta = \tan^{-1}(Q/3P)$.
- 4.53** (a) $T = (W/2)/(1 - \tan \theta)$. (b) 39.8° .
- 4.54** (a) $\theta = 2 \cos^{-1}[\frac{1}{4}(\frac{W}{P} \pm \sqrt{\frac{W^2}{P^2} + 8})]$. (b) 65.1° .
- 4.55** (a) $\theta = 2 \sin^{-1}(W/2P)$. (b) 29.0° .
- 4.57** 141.1° .
- 4.59** (1) completely constrained; determinate; $\mathbf{A} = \mathbf{C} = 196.2 \text{ N}$ $\uparrow$.
(2) completely constrained; determinate; $\mathbf{B} = 0$,
 $\mathbf{C} = \mathbf{D} = 196.2 \text{ N}$ $\uparrow$.
(3) completely constrained; indeterminate; $\mathbf{A}_x = 294 \text{ N}$ $\rightarrow$;
 $\mathbf{D}_x = 294$ $\leftarrow$.
(4) improperly constrained; indeterminate; no equilibrium.
(5) partially constrained; determinate; equilibrium;
 $\mathbf{C} = \mathbf{D} = 196.2 \text{ N}$ $\uparrow$.
(6) completely constrained; determinate;
 $\mathbf{B} = 294 \text{ N}$ $\rightarrow$, $\mathbf{D} = 491 \text{ N}$ $\searrow 53.1^\circ$.
(7) partially constrained; no equilibrium.
(8) completely constrained; indeterminate; $\mathbf{B} = 196.2 \text{ N}$ $\uparrow$,
 $\mathbf{D}_y = 196.2 \text{ N}$ $\uparrow$.
- 4.61** $T = 289 \text{ lb}$; $\mathbf{A} = 577 \text{ lb}$ $\nearrow 60.0^\circ$.
- 4.62** $\mathbf{A} = 400 \text{ N}$ $\uparrow$; $\mathbf{B} = 500 \text{ N}$ $\searrow 53.1^\circ$.
- 4.63** $a \geq 138.6 \text{ mm}$.
- 4.65** $\mathbf{B} = 501 \text{ N}$ $\searrow 56.3^\circ$; $\mathbf{C} = 324 \text{ N}$ $\searrow 31.0^\circ$.
- 4.66** $\mathbf{A} = 82.5 \text{ lb}$ $\nearrow 14.04^\circ$; $T = 100.0 \text{ lb}$.
- 4.67** $\mathbf{B} = 888 \text{ N}$ $\searrow 41.3^\circ$; $\mathbf{D} = 943 \text{ N}$ $\searrow 45.0^\circ$.
- 4.69** (a) 499 N . (b) 457 N $\searrow 26.6^\circ$.
- 4.71** (a) 5.63 kips . (b) 4.52 kips $\nearrow 4.76^\circ$.
- 4.72** (a) 24.9 lb $\nearrow 30.0^\circ$. (b) 15.34 lb $\nearrow 30.0^\circ$.
- 4.73** $\mathbf{A} = 778 \text{ N}$ $\downarrow$; $\mathbf{C} = 1012 \text{ N}$ $\searrow 77.9^\circ$.
- 4.75** $\mathbf{A} = 170.0 \text{ N}$ $\searrow 33.9^\circ$; $\mathbf{C} = 160.0 \text{ N}$ $\nearrow 28.1^\circ$.
- 4.77** $T = 100.0 \text{ lb}$; $\mathbf{B} = 111.1 \text{ lb}$ $\searrow 30.3^\circ$.
- 4.78** (a) 400 N . (b) 458 N $\nearrow 49.1^\circ$.
- 4.79** (a) $2P$ $\searrow 60.0^\circ$. (b) $1.239P$ $\searrow 36.2^\circ$.
- 4.80** (a) $1.55 P$ $\searrow 30.0^\circ$. (b) $1.086 P$ $\nearrow 22.9^\circ$.
- 4.81** $\mathbf{A} = 163.1 \text{ N}$ $\searrow 74.1^\circ$; $\mathbf{B} = 258 \text{ N}$ $\searrow 65.0^\circ$.
- 4.83** 60.0 mm .
- 4.84** $\tan \theta = 2 \tan \beta$.
- 4.85** (a) 49.1° . (b) $\mathbf{A} = 45.3 \text{ N}$ $\leftarrow$; $\mathbf{B} = 90.6 \text{ N}$ $\nearrow 60.0^\circ$.
- 4.86** 32.5° .
- 4.88** (a) 225 mm . (b) 23.1 N . (c) 12.21 N $\rightarrow$.
- 4.90** (a) 59.4° . (b) $\mathbf{A} = 8.45 \text{ lb}$ $\rightarrow$; $\mathbf{B} = 13.09 \text{ lb}$ $\searrow 49.8^\circ$.
- 4.91** $\mathbf{A} = (120.0 \text{ N})\mathbf{j} + (133.3 \text{ N})\mathbf{k}$; $\mathbf{D} = (60.0 \text{ N})\mathbf{j} + (166.7 \text{ N})\mathbf{k}$.
- 4.93** (a) 96.0 lb . (b) $\mathbf{A} = (2.40 \text{ lb})\mathbf{j}$; $\mathbf{B} = (214 \text{ lb})\mathbf{j}$.
- 4.94** $\mathbf{A} = (22.9 \text{ lb})\mathbf{i} + (8.50 \text{ lb})\mathbf{j}$; $\mathbf{B} = (22.9 \text{ lb})\mathbf{i} + (25.5 \text{ lb})\mathbf{j}$;
 $\mathbf{C} = -(45.8 \text{ lb})\mathbf{i}$.
- 4.95** (a) 78.5 N . (b) $\mathbf{A} = -(27.5 \text{ N})\mathbf{i} + (58.9 \text{ N})\mathbf{j}$; $\mathbf{B} = (106.0 \text{ N})\mathbf{i} + (58.9 \text{ N})\mathbf{j}$.
- 4.97** $T_A = 21.0 \text{ lb}$; $T_B = T_C = 17.50 \text{ lb}$.
- 4.99** (a) 121.9 N . (b) -46.2 N . (c) 100.9 N .
- 4.100** (a) 95.6 N . (b) -7.36 N . (c) 88.3 N .
- 4.101** $T_A = 23.5 \text{ N}$; $T_C = 11.77 \text{ N}$; $T_D = 105.9 \text{ N}$.
- 4.102** (a) 0.480 in . (b) $T_A = 23.5 \text{ N}$; $T_C = 0$; $T_D = 117.7 \text{ N}$.
- 4.103** (a) $T_A = 6.00 \text{ lb}$; $T_B = T_C = 9.00 \text{ lb}$. (b) 15.00 in .
- 4.105** $T_{BD} = 1100 \text{ lb}$; $T_{BE} = 1100 \text{ lb}$; $\mathbf{A} = (1200 \text{ lb})\mathbf{i} - (560 \text{ lb})\mathbf{j}$.
- 4.106** $T_{BD} = 780 \text{ N}$; $T_{BE} = 390 \text{ N}$; $\mathbf{A} = -(195.0 \text{ N})\mathbf{i} + (1170 \text{ N})\mathbf{j} + (130.0 \text{ N})\mathbf{k}$.
- 4.107** $T_{BD} = 525 \text{ N}$; $T_{BE} = 105.0 \text{ N}$; $\mathbf{A} = -(105.0 \text{ N})\mathbf{i} + (840 \text{ N})\mathbf{j} + (140.0 \text{ N})\mathbf{k}$.

- 4.108** $T_{AD} = 2.60 \text{ kN}$; $T_{AE} = 2.80 \text{ kN}$; $\mathbf{C} = (1.800 \text{ kN})\mathbf{j} + (4.80 \text{ kN})\mathbf{k}$.
4.109 $T_{AD} = 5.20 \text{ kN}$; $T_{AE} = 5.60 \text{ kN}$; $\mathbf{C} = (9.60 \text{ kN})\mathbf{k}$.
4.110 $T_{BD} = T_{BE} = 176.8 \text{ lb}$; $\mathbf{C} = -(50.0 \text{ lb})\mathbf{j} + (216.5 \text{ lb})\mathbf{k}$.
4.113 $F_{CD} = 19.62 \text{ N}$; $\mathbf{A} = -(19.22 \text{ N})\mathbf{i} + (45.1 \text{ N})\mathbf{j}$;
 $\mathbf{B} = (49.1 \text{ N})\mathbf{j}$.
4.115 $\mathbf{A} = -(56.3 \text{ lb})\mathbf{i}$; $\mathbf{B} = -(56.2 \text{ lb})\mathbf{i} + (150.0 \text{ lb})\mathbf{j} - (75.0 \text{ lb})\mathbf{k}$;
 $F_{CE} = 202 \text{ lb}$.
4.116 (a) 116.6 lb . (b) $\mathbf{A} = -(72.7 \text{ lb})\mathbf{j} - (38.1 \text{ lb})\mathbf{k}$;
 $\mathbf{B} = (37.5 \text{ lb})\mathbf{j}$.
4.117 (a) 345 N . (b) $\mathbf{A} = (114.4 \text{ N})\mathbf{i} + (377 \text{ N})\mathbf{j} + (141.5 \text{ N})\mathbf{k}$;
 $\mathbf{B} = (113.2 \text{ N})\mathbf{j} + (185.5 \text{ N})\mathbf{k}$.
4.119 $F_{CD} = 19.62 \text{ N}$; $\mathbf{B} = -(19.22 \text{ N})\mathbf{i} + (94.2 \text{ N})\mathbf{j}$;
 $\mathbf{M}_B = -(40.6 \text{ N}\cdot\text{m})\mathbf{i} - (17.30 \text{ N}\cdot\text{m})\mathbf{j}$.
4.120 $\mathbf{A} = -(112.5 \text{ lb})\mathbf{i} + (150.0 \text{ lb})\mathbf{j} - (75.0 \text{ lb})\mathbf{k}$;
 $\mathbf{M}_A = (600 \text{ lb}\cdot\text{ft})\mathbf{i} + (225 \text{ lb}\cdot\text{ft})\mathbf{j}$; $F_{CE} = 202 \text{ lb}$.
4.121 (a) 5.00 lb . (b) $\mathbf{C} = -(5.00 \text{ lb})\mathbf{i} + (6.00 \text{ lb})\mathbf{j} - (5.00 \text{ lb})\mathbf{k}$;
 $\mathbf{M}_C = (8.00 \text{ lb}\cdot\text{in.})\mathbf{j} - (12.00 \text{ lb}\cdot\text{in.})\mathbf{k}$.
4.122 $T_{CF} = 200 \text{ N}$; $T_{DE} = 450 \text{ N}$; $\mathbf{A} = (160.0 \text{ N})\mathbf{i} + (270 \text{ N})\mathbf{k}$;
 $\mathbf{M}_A = -(16.20 \text{ N}\cdot\text{m})\mathbf{i}$.
4.123 $T_{BD} = 2.18 \text{ kN}$; $T_{BE} = 3.96 \text{ kN}$; $T_{CD} = 1.500 \text{ kN}$.
4.124 $T_{BD} = 0$; $T_{BE} = 3.96 \text{ kN}$; $T_{CD} = 3.00 \text{ kN}$.
4.127 $\mathbf{A} = (120.0 \text{ lb})\mathbf{j} - (150.0 \text{ lb})\mathbf{k}$; $\mathbf{B} = (180.0 \text{ lb})\mathbf{i} + (150.0 \text{ lb})\mathbf{k}$;
 $\mathbf{C} = -(180.0 \text{ lb})\mathbf{i} + (120.0 \text{ lb})\mathbf{j}$.
4.128 $\mathbf{A} = (20.0 \text{ lb})\mathbf{j} + (25.0 \text{ lb})\mathbf{k}$; $\mathbf{B} = (30.0 \text{ lb})\mathbf{i} - (25.0 \text{ lb})\mathbf{k}$;
 $\mathbf{C} = -(30.0 \text{ lb})\mathbf{i} - (20.0 \text{ lb})\mathbf{j}$.
4.129 $T_{BE} = 975 \text{ N}$; $T_{CF} = 600 \text{ N}$; $T_{DG} = 625 \text{ N}$;
 $\mathbf{A} = (2100 \text{ N})\mathbf{i} + (175.0 \text{ N})\mathbf{j} - (375 \text{ N})\mathbf{k}$.
4.131 $T_B = -0.366 P$; $T_C = 1.219 P$; $T_D = -0.853 P$;
 $F = -0.345 P\mathbf{i} + P\mathbf{j} - 0.862 P\mathbf{k}$.
4.133 360 N .
4.135 85.3 lb .
4.136 181.7 lb .
4.137 $(45.0 \text{ lb})\mathbf{j}$.
4.138 343 N .
4.140 (a) $x = 4.00 \text{ ft}$, $y = 8.00 \text{ ft}$. (b) 10.73 lb .
4.141 (a) $x = 0$, $y = 16.00 \text{ ft}$. (b) 11.31 lb .
4.142 (a) $1761 \text{ lb}\uparrow$. (b) $689 \text{ lb}\uparrow$.
4.143 (a) 150.0 lb . (b) $225 \text{ lb} \searrow 32.3^\circ$.
4.145 (a) 130.0 N . (b) $224 \searrow 2.05^\circ$.
4.146 $T = 80.0 \text{ N}$; $\mathbf{A} = 160.0 \text{ N} \searrow 30.0^\circ$; $\mathbf{C} = 160.0 \text{ N} \searrow 30.0^\circ$.
4.148 $\mathbf{A} = 680 \text{ N} \searrow 28.1^\circ$; $\mathbf{B} = 600 \text{ N} \leftarrow$.
4.149 $\mathbf{A} = 63.6 \text{ lb} \searrow 45.0^\circ$; $\mathbf{C} = 87.5 \text{ lb} \searrow 59.0^\circ$.
4.151 $T_A = 5.63 \text{ lb}$; $T_B = 16.88 \text{ lb}$; $T_C = 22.5 \text{ lb}$.
4.153 (a) $\mathbf{A} = 0.745 P \searrow 63.4^\circ$; $\mathbf{C} = 0.471 P \searrow 45.0^\circ$.
(b) $\mathbf{A} = 0.812 P \searrow 60.0^\circ$; $\mathbf{C} = 0.503 P \searrow 36.2^\circ$.
(c) $\mathbf{A} = 0.448 P \searrow 60.0^\circ$; $\mathbf{C} = 0.652 P \searrow 69.9^\circ$.
(d) improperly constrained; no equilibrium.

CHAPTER 5

- 5.1** $\bar{X} = 42.2 \text{ mm}$, $\bar{Y} = 24.2 \text{ mm}$.
5.2 $\bar{X} = 1.045 \text{ in.}$, $\bar{Y} = 3.59 \text{ in.}$
5.3 $\bar{X} = 2.84 \text{ mm}$, $\bar{Y} = 24.8 \text{ mm}$.
5.4 $\bar{X} = 52.0 \text{ mm}$, $\bar{Y} = 65.0 \text{ mm}$.
5.5 $\bar{X} = 3.27 \text{ in.}$, $\bar{Y} = 2.82 \text{ in.}$
5.6 $\bar{X} = -10.00 \text{ mm}$, $\bar{Y} = 87.5 \text{ mm}$.
5.9 $\bar{X} = \bar{Y} = 16.75 \text{ mm}$.
5.10 $\bar{X} = 10.11 \text{ in.}$, $\bar{Y} = 3.88 \text{ in.}$
5.11 $\bar{X} = 30.0 \text{ mm}$, $\bar{Y} = 64.8 \text{ mm}$.
5.13 $\bar{X} = 3.20 \text{ in.}$, $\bar{Y} = 2.00 \text{ in.}$
5.14 $\bar{X} = 0$, $\bar{Y} = 1.372 \text{ m}$.
5.16 $\bar{Y} = \left(\frac{2}{3} \right) \left(\frac{r_2^3 - r_1^3}{r_2^2 - r_1^2} \right) \left(\frac{2 \cos \alpha}{\pi - 2\alpha} \right)$.

- 5.17** $\bar{Y} = (r_1 + r_2)(\cos \alpha)/(\pi - 2\alpha)$.
5.19 0.520 .
5.20 459 N .
5.21 0.235 in^3 for A_1 , -0.235 in^3 for A_2 .
5.23 (a) $b(c^2 - y^2)/2$. (b) $y = 0$; $Q_x = bc^2/2$.
5.24 $\bar{X} = 40.9 \text{ mm}$, $\bar{Y} = 25.3 \text{ mm}$.
5.26 $\bar{X} = 3.38 \text{ in.}$, $\bar{Y} = 2.93 \text{ in.}$
5.29 (a) 125.3 N . (b) $137.0 \text{ N} \searrow 56.7^\circ$.
5.30 120.0 mm .
5.31 99.5 mm .
5.32 (a) $0.513a$. (b) $0.691a$.
5.34 $\bar{x} = \frac{2}{3}a$, $\bar{y} = \frac{2}{3}h$.
5.35 $\bar{x} = a/2$, $\bar{y} = 2h/5$.
5.37 $\bar{x} = a(3 - 4 \sin \alpha)/6$ ($1 - \alpha$), $\bar{y} = 0$.
5.39 $\bar{x} = 2a/3(4 - \pi)$, $\bar{y} = 2b/3(4 - \pi)$.
5.40 $\bar{x} = a/4$, $\bar{y} = 3b/10$.
5.41 $\bar{x} = 3a/5$, $\bar{y} = 12b/35$.
5.43 $\bar{x} = 17a/130$, $\bar{y} = 11b/26$.
5.44 $\bar{x} = a$, $\bar{y} = 17b/35$.
5.45 $2a/5$.
5.46 $-2\sqrt{2}r/3\pi$.
5.48 $\bar{x} = -9.27a$, $\bar{y} = 3.09a$.
5.49 $\bar{x} = L/\pi$, $\bar{y} = \pi a/8$.
5.51 $\bar{x} = \bar{y} = 1.027 \text{ in.}$
5.52 (a) $V = 401 \times 10^3 \text{ mm}^3$; $A = 34.1 \times 10^3 \text{ mm}^2$.
(b) $V = 492 \times 10^3 \text{ mm}^3$; $A = 41.9 \times 10^3 \text{ mm}^2$.
5.53 (a) $V = 248 \text{ in}^3$; $A = 547 \text{ in}^2$.
(b) $V = 72.3 \text{ in}^3$; $A = 169.6 \text{ in}^2$.
5.54 (a) $V = 2.26 \times 10^6 \text{ mm}^3$; $A = 116.3 \times 10^3 \text{ mm}^2$.
(b) $V = 1.471 \times 10^6 \text{ mm}^3$; $A = 116.3 \times 10^3 \text{ mm}^2$.
5.55 $V = 3470 \text{ mm}^3$; $A = 2320 \text{ mm}^2$.
5.58 308 in^2 .
5.60 31.9 liters .
5.62 $V = 3.96 \text{ in}^3$, $W = 1.211 \text{ lb}$.
5.63 14.52 in^2 .
5.64 0.0305 kg .
5.66 (a) $\mathbf{R} = 6000 \text{ N} \downarrow$, $\bar{x} = 3.60 \text{ m}$.
(b) $\mathbf{A} = 6000 \text{ N} \uparrow$, $\mathbf{M}_A = 21.6 \text{ kN}\cdot\text{m} \uparrow$.
5.67 (a) $\mathbf{R} = 7.60 \text{ kN} \downarrow$, $\bar{x} = 2.57 \text{ m}$.
(b) $\mathbf{A} = 4.35 \text{ kN} \uparrow$, $\mathbf{B} = 3.25 \text{ kN} \uparrow$.
5.69 $\mathbf{A} = 900 \text{ lb} \uparrow$; $\mathbf{M}_A = 9200 \text{ lb}\cdot\text{in.} \uparrow$.
5.70 $\mathbf{B} = 1360 \text{ lb} \uparrow$; $\mathbf{C} = 2360 \text{ lb} \uparrow$.
5.71 $\mathbf{A} = 105.0 \text{ N} \uparrow$; $\mathbf{B} = 270 \text{ N} \uparrow$.
5.73 $\mathbf{A} = 3.00 \text{ kN} \uparrow$; $\mathbf{M}_A = 12.60 \text{ kN}\cdot\text{m} \uparrow$.
5.74 (a) 0.536 m . (b) $\mathbf{A} = \mathbf{B} = 761 \text{ N} \uparrow$.
5.76 $\mathbf{B} = 3770 \text{ lb} \uparrow$; $\mathbf{C} = 429 \text{ lb} \uparrow$.
5.77 (a) 900 lb/ft . (b) $7200 \text{ lb} \uparrow$.
5.78 $w_A = 10.00 \text{ kN/m}$; $w_B = 50 \text{ kN/m}$.
5.80 (a) $\mathbf{H} = 10.11 \text{ kips} \rightarrow$, $\mathbf{V} = 37.8 \text{ kips} \uparrow$.
(b) 10.48 ft to the right of A .
(c) $\mathbf{R} = 10.66 \text{ kips} \searrow 18.43^\circ$.
5.81 (a) $\mathbf{H} = 44.1 \text{ kN} \rightarrow$, $\mathbf{V} = 228 \text{ kN} \uparrow$.
(b) 1.159 m to the right of A .
(c) $\mathbf{R} = 59.1 \text{ kN} \searrow 41.6^\circ$.
5.82 6.98% .
5.84 12.00 in.
5.85 4.00 in.
5.87 $\mathbf{T} = 6.72 \text{ kN} \leftarrow$; $\mathbf{A} = 141.2 \text{ kN} \leftarrow$.
5.88 $\mathbf{A} = 1197 \text{ N} \searrow 53.1^\circ$; $\mathbf{B} = 1511 \text{ N} \searrow 53.1^\circ$.
5.89 3570 N .
5.90 6.00 ft .
5.92 0.683 m .

- 5.93** 0.0711 m.
5.94 208 lb.
5.96 (a) 0.548 L. (b) $2\sqrt{3}$.
5.97 (a) $b/10$ to the left of base of cone.
 (b) $0.1136b$ to the right of base of cone.
5.98 (a) $-0.402 a$. (b) $h/a = 2/5$ or $2/3$.
5.99 27.8 mm above base of cone.
5.100 18.28 mm.
5.102 -0.0656 in.
5.103 2.57 in.
5.104 -19.02 mm.
5.106 $\bar{X} = 125.0$ mm, $\bar{Y} = 167.0$ mm, $\bar{Z} = 33.5$ mm.
5.107 $\bar{X} = 0.295$ m, $\bar{Y} = 0.423$ m, $\bar{Z} = 1.703$ m.
5.109 $\bar{X} = \bar{Z} = 4.21$ in., $\bar{Y} = 7.03$ in.
5.110 $\bar{X} = 180.2$ mm, $\bar{Y} = 38.0$ mm, $\bar{Z} = 193.5$ mm.
5.111 $\bar{X} = 17.00$ in., $\bar{Y} = 15.68$ in., $\bar{Z} = 14.16$ in.
5.113 $\bar{X} = 46.5$ mm, $\bar{Y} = 27.2$ mm, $\bar{Z} = 30.0$ mm.
5.114 $\bar{X} = 0.909$ m, $\bar{Y} = 0.1842$ m, $\bar{Z} = 0.884$ m.
5.116 $\bar{X} = 0.410$ m, $\bar{Y} = 0.510$ m, $\bar{Z} = 0.1500$ m.
5.117 $\bar{X} = 0$, $\bar{Y} = 10.05$ in., $\bar{Z} = 5.15$ in.
5.118 $\bar{X} = 61.1$ mm from the end of the handle.
5.119 $\bar{Y} = 0.526$ in. above the base.
5.121 $\bar{Y} = 421$ mm above the floor.
5.122 $(\bar{x}_1) = 21a/88$; $(\bar{x}_2) = 27a/40$.
5.123 $(\bar{x}_1) = 21h/88$; $(\bar{x}_2) = 27h/40$.
5.124 $(\bar{x}_1) = 2h/9$; $(\bar{x}_2) = 2 h/3$.
5.125 $\bar{x} = 2.34$ m; $\bar{y} = \bar{z} = 0$.
5.128 $\bar{x} = 1.297a$; $\bar{y} = \bar{z} = 0$.
5.129 $\bar{x} = \bar{z} = 0$; $\bar{y} = 0.374b$.
5.132 (a) $\bar{x} = \bar{z} = 0$, $\bar{y} = -121.9$ mm. (b) $\bar{x} = \bar{z} = 0$,
 $\bar{y} = -90.2$ mm.
5.134 $\bar{x} = 0$, $\bar{y} = 5h/16$, $\bar{z} = -b/4$.
5.135 $\bar{x} = a/2$, $\bar{y} = 8h/25$, $\bar{z} = b/2$.
5.136 $V = 688$ ft³; $\bar{x} = 15.91$ ft.
5.137 $\bar{X} = 5.67$ in., $\bar{Y} = 5.17$ in.
5.138 $\bar{X} = 92.0$ mm, $\bar{Y} = 23.3$ mm.
5.139 (a) 5.09 lb. (b) 9.48 lb $\nless 57.5^\circ$.
5.141 $\bar{x} = 2L/5$, $\bar{y} = 12h/25$.
5.143 **A** = 2860 lb $\uparrow$; **B** = 740 lb $\uparrow$.
5.144 $w_{BC} = 2810$ N/m; $w_{DE} = 3150$ N/m.
5.146 $-(2h^2 - 3b^2)/2 (4h - 3b)$.
5.148 $\bar{X} = \bar{Z} = 0$, $\bar{Y} = 83.3$ mm above the base.

CHAPTER 6

- 6.1** $F_{AB} = 900$ lb T ; $F_{AC} = 780$ lb C ; $F_{BC} = 720$ lb T .
6.2 $F_{AB} = 1.700$ kN T ; $F_{AC} = 2.00$ kN T ; $F_{BC} = 2.50$ kN T .
6.3 $F_{AB} = 720$ lb T ; $F_{AC} = 1200$ lb C ; $F_{BC} = 780$ lb C .
6.4 $F_{AB} = F_{BC} = 0$; $F_{AD} = F_{CF} = 7.00$ kN C ; $F_{BD} = F_{BF} = 34.0$ kN C ; $F_{BE} = 8.00$ kN T ; $F_{DE} = F_{EF} = 30.0$ kN T .
6.6 $F_{AC} = 80.0$ kN T ; $F_{CE} = 45.0$ kN T ; $F_{DE} = 51.0$ kN C ;
 $F_{BD} = 51.0$ kN C ; $F_{CD} = 48.0$ kN T ; $F_{BC} = 19.00$ kN C .
6.8 $F_{AB} = 20.0$ kN T ; $F_{AD} = 20.6$ kN C ; $F_{BC} = 30.0$ kN T ;
 $F_{BD} = 11.18$ kN C ; $F_{CD} = 10.00$ kN T .
6.9 $F_{AB} = F_{DE} = 8.00$ kN C ; $F_{AF} = F_{FG} = F_{GH} = F_{EH} = 6.93$ kN T ;
 $F_{BC} = F_{CD} = F_{BG} = F_{DG} = 4.00$ kN C ;
 $F_{BF} = F_{DH} = F_{CG} = 4.00$ kN T .
6.11 $F_{AB} = F_{FH} = 1500$ lb C ; $F_{AC} = F_{CE} = F_{EG} = F_{GH} = 1200$ lb T ;
 $F_{BC} = F_{FG} = 0$; $F_{BD} = F_{DF} = 1200$ lb C ; $F_{BE} = F_{EF} = 60.0$ lb C ; $F_{DE} = 72.0$ lb T .
6.12 $F_{AB} = F_{FH} = 1500$ lb C ; $F_{AC} = F_{CE} = F_{EG} = F_{GH} = 1200$ lb T ;
 $F_{BC} = F_{FG} = 0$; $F_{BD} = F_{DF} = 1000$ lb C ;
 $F_{BE} = F_{EF} = 500$ lb C ; $F_{DE} = 600$ lb T .
6.13 $F_{AB} = 6.24$ kN C ; $F_{AC} = 2.76$ kN T ; $F_{BC} = 2.50$ kN C ;
 $F_{BD} = 4.16$ kN C ; $F_{CD} = 1.867$ kN T ; $F_{CE} = 2.88$ kN T ;
 $F_D = 3.75$ kN C ; $F_{DF} = 0$; $F_{EF} = 1.200$ kN C .
6.15 $F_{AB} = F_{FG} = 7.50$ kips C ; $F_{AC} = F_{EG} = 4.50$ kips T ;
 $F_{BC} = F_{EF} = 7.50$ kips T ; $F_{BD} = F_{DF} = 9.00$ kips C ;
 $F_{CD} = F_{DE} = 0$; $F_{CE} = 9.00$ kips T .
6.17 $F_{AB} = 47.2$ kN C ; $F_{AC} = 44.6$ kN T ; $F_{BC} = 10.50$ kN C ;
 $F_{BD} = 47.2$ kN C ; $F_{CD} = 17.50$ kN T ; $F_{CE} = 30.6$ kN T ;
 $F_{DE} = 0$.
6.18 $F_{AB} = 2250$ N C ; $F_{AC} = 1200$ N T ; $F_{BC} = 750$ N T ;
 $F_{BD} = 1700$ N C ; $F_{BE} = 400$ N C ; $F_{CE} = 850$ N C ;
 $F_{CF} = 1600$ N T ; $F_{DE} = 1500$ N T ; $F_{EF} = 2250$ N T .
6.19 $F_{AB} = F_{FH} = 7.50$ kips C ; $F_{AC} = F_{GH} = 4.50$ kips T ;
 $F_{BC} = F_{FG} = 4.00$ kips T ; $F_{BD} = F_{DF} = 6.00$ kips C ;
 $F_{BE} = F_{EF} = 2.50$ kips T ; $F_{CE} = F_{EG} = 4.50$ kips T ; $F_{DE} = 0$.
6.21 $F_{AB} = 9.90$ kN C ; $F_{AC} = 7.83$ kN T ; $F_{BC} = 0$; $F_{BD} = 7.07$ kN C ;
 $F_{BE} = 2.00$ kN C ; $F_{CE} = 7.83$ kN T ; $F_{DE} = 1.000$ kN T ;
 $F_{DF} = 5.03$ kN C ; $F_{DG} = 0.559$ kN C ; $F_{EG} = 5.59$ kN T .
6.22 $F_{AB} = 3610$ lb C ; $F_{AC} = 4110$ lb T ; $F_{BC} = 768$ lb C ;
 $F_{BD} = 3840$ lb C ; $F_{CD} = 1371$ lb T ; $F_{CE} = 2740$ lb T ;
 $F_{DE} = 1536$ lb C .
6.23 $F_{DF} = 4060$ lb C ; $F_{DG} = 1371$ lb T ; $F_{EG} = 2740$ lb T ;
 $F_{FG} = 768$ lb C ; $F_{FH} = 4290$ lb C ; $F_{GH} = 4110$ lb T .
6.24 $F_{AB} = F_{DF} = 2.29$ kN T ; $F_{AC} = F_{EF} = 2.29$ kN C ; $F_{BC} = F_{DE} = 0.600$ kN C ; $F_{BD} = 2.21$ kN T ; $F_{BE} = F_{EH} = 0$;
 $F_{CE} = 2.21$ kN C ; $F_{CH} = F_{EJ} = 1.200$ kN C .
6.27 $F_{AB} = F_{BC} = F_{CD} = 36.0$ kips T ; $F_{AE} = 57.6$ kips T ;
 $F_{AF} = 45.0$ kips C ; $F_{BF} = F_{BG} = F_{CG} = F_{CH} = 0$; $F_{DH} = F_{FG} = F_{GH} = 39.0$ kips C ; $F_{EF} = 36.0$ kips C .
6.28 $F_{AB} = 128.0$ kN T ; $F_{AC} = 136.7$ kN C ; $F_{BD} = F_{DF} = F_{FH} = 128.0$ kN T ;
 $F_{CE} = F_{EG} = 136.7$ kN C ; $F_{GH} = 192.7$ kN C ;
 $F_{BC} = F_{BE} = F_{DE} = F_{DG} = F_{FG} = 0$.
6.29 Truss of Prob. 6.33a is the only simple truss.
6.30 Trusses of Prob. 6.32b and Prob. 6.33b are simple trusses.
6.32 (a) *AI, BJ, CK, DI, EI, FK, GK*.
 (b) *FK, IO*.
6.34 (a) *BC, HI, IJ, JK*. (b) *BF, BG, CG, CH*.
6.35 $F_{AB} = F_{AD} = 244$ lb C ; $F_{AC} = 1040$ lb T ; $F_{BC} = F_{CD} = 500$ lb C ; $F_{BD} = 280$ lb T .
6.36 $F_{AB} = F_{AD} = 861$ N C ; $F_{AC} = 676$ N C ; $F_{BC} = F_{CD} = 162.5$ N T ; $F_{BD} = 244$ N T .
6.37 $F_{AB} = F_{AD} = 2810$ N T ; $F_{AC} = 5510$ N C ; $F_{BC} = F_{CD} = 1325$ N T ; $F_{BD} = 1908$ N C .
6.38 $F_{AB} = F_{AC} = 1061$ lb C ; $F_{AD} = 2500$ lb T ; $F_{BC} = 2100$ lb T ;
 $F_{BD} = F_{CD} = 1250$ lb C ; $F_{BE} = F_{CE} = 1250$ lb C ;
 $F_{DE} = 1500$ lb T .
6.39 $F_{AB} = 840$ N C ; $F_{AC} = 110.6$ N C ; $F_{AD} = 394$ N C ;
 $F_{AE} = 0$; $F_{BC} = 160.0$ N T ; $F_{BE} = 200$ N T ; $F_{CD} = 225$ N T ;
 $F_{CE} = 233$ N C ; $F_{DE} = 120.0$ N T .
6.40 $F_{AB} = F_{AE} = F_{BC} = 0$; $F_{AC} = 995$ N T ; $F_{AD} = 1181$ N C ;
 $F_{BE} = 600$ N T ; $F_{CD} = 375$ N T ; $F_{CE} = 700$ N C ; $F_{DE} = 360$ N T .
6.43 $F_{DF} = 5.45$ kN C ; $F_{DG} = 1.000$ kN T ; $F_{EG} = 4.65$ kN T .
6.44 $F_{GI} = 4.65$ kN T ; $F_{HI} = 1.800$ kN C ; $F_{HJ} = 4.65$ kN C .
6.45 $F_{BD} = 36.0$ kips C ; $F_{CD} = 45.0$ kips C .
6.46 $F_{DF} = 60.0$ kips C ; $F_{DG} = 15.00$ kips C .
6.49 $F_{CD} = 20.0$ kN C ; $F_{DF} = 52.0$ kN C .
6.50 $F_{CE} = 36.0$ kN T ; $F_{EF} = 15.00$ kN C .
6.51 $F_{DE} = 25.0$ kips T ; $F_{DF} = 13.00$ kips C .

- 6.52** $F_{EG} = 16.00$ kips T ; $F_{EF} = 6.40$ kips C .
6.53 $F_{DF} = 91.4$ kN T ; $F_{DE} = 38.6$ kN C .
6.54 $F_{CD} = 64.2$ kN T ; $F_{CE} = 92.1$ kN C .
6.55 $F_{CE} = 7.20$ kN T ; $F_{DE} = 1.047$ kN C ; $F_{DF} = 6.39$ kN C .
6.56 $F_{EG} = 3.46$ kN T ; $F_{GH} = 3.78$ kN C ; $F_{HJ} = 3.55$ kN C .
6.59 $F_{AD} = 3.38$ kips C ; $F_{CD} = 0$; $F_{CE} = 14.03$ kips T .
6.60 $F_{DG} = 18.75$ kips C ; $F_{FG} = 14.03$ kips T ; $F_{FH} = 17.43$ kips T .
6.61 $F_{DG} = 3.75$ kN T ; $F_{FI} = 3.75$ kN C .
6.62 $F_{GJ} = 11.25$ kN T ; $F_{IK} = 11.25$ kN C .
6.65 (a) CJ . (b) 1.026 kN T .
6.66 (a) IO . (b) 2.05 kN T .
6.67 $F_{BE} = 10.00$ kips T ; $F_{DE} = 0$; $F_{EF} = 5.00$ kips T .
6.68 $F_{BE} = 2.50$ kips T ; $F_{DE} = 1.500$ kips C ; $F_{DG} = 2.50$ kips T .
6.69 (a) improperly constrained. (b) completely constrained, determinate. (c) completely constrained, indeterminate.
6.70 (a) completely constrained, determinate. (b) partially constrained. (c) improperly constrained.
6.71 (a) completely constrained, determinate. (b) completely constrained, indeterminate. (c) improperly constrained.
6.72 (a) partially constrained. (b) completely constrained, determinate. (c) completely constrained, indeterminate.
6.75 $F_{BD} = 375$ N C ; $C_x = 205$ N $\leftarrow$; $C_y = 360$ N $\downarrow$.
6.76 $F_{BD} = 780$ lb T ; $C_x = 720$ lb $\leftarrow$; $C_y = 140.0$ lb $\downarrow$.
6.77 (a) 125.0 N $\searrow 36.9^\circ$. (b) 125.0 N $\nearrow 36.9^\circ$.
6.78 $A_x = 120.0$ lb $\rightarrow$; $A_y = 30.0$ lb $\uparrow$; $B_x = 120.0$ lb $\leftarrow$, $B_y = 80.0$ lb $\downarrow$; $C = 30.0$ lb $\downarrow$; $D = 80.0$ lb $\uparrow$.
6.79 $A_x = 18.00$ kN $\leftarrow$; $A_y = 20.0$ kN $\downarrow$; $B = 9.00$ kN $\rightarrow$; $C_x = 9.00$ kN $\rightarrow$; $C_y = 20.0$ kN $\uparrow$.
6.80 $A = 20.0$ kN $\downarrow$, $B = 18.00$ kN $\leftarrow$; $C_x = 18.00$ kN $\rightarrow$, $C_y = 20.0$ kN $\uparrow$.
6.81 $A = 150.0$ lb $\rightarrow$; $B_x = 150.0$ lb $\leftarrow$, $B_y = 60.0$ lb $\uparrow$; $C = 20.0$ lb $\uparrow$; $D = 80.0$ lb $\downarrow$.
6.83 (a) $A_x = 2700$ N $\rightarrow$, $A_y = 200$ N $\uparrow$; $E_x = 2700$ N $\leftarrow$, $E_y = 600$ N $\uparrow$.
 (b) $A_x = 300$ N $\rightarrow$, $A_y = 200$ N $\uparrow$; $E_x = 300$ N $\leftarrow$, $E_y = 600$ N $\uparrow$.
6.85 (a) $A_x = 300$ N $\leftarrow$, $A_y = 660$ N $\uparrow$; $E_x = 300$ N $\rightarrow$, $E_y = 90.0$ N $\uparrow$. (b) $A_x = 300$ N $\leftarrow$, $A_y = 150.0$ N $\uparrow$; $E_x = 300$ N $\rightarrow$, $E_y = 600$ N $\uparrow$.
6.87 (a) $A_x = 80.0$ lb $\leftarrow$, $A_y = 40.0$ lb $\uparrow$; $B_x = 80.0$ lb $\rightarrow$, $B_y = 60.0$ lb $\uparrow$. (b) $A_x = 0$, $A_y = 40.0$ lb $\uparrow$; $B_x = 0$, $B_y = 60.0$ lb $\uparrow$.
6.88 (a) and (c) $B_x = 32.0$ lb $\rightarrow$, $B_y = 10.00$ lb $\uparrow$; $F_x = 32.0$ lb $\leftarrow$, $F_y = 38.0$ lb $\uparrow$. (b) $B_x = 32.0$ lb $\leftarrow$, $B_y = 34.0$ lb $\uparrow$; $F_x = 32.0$ lb $\rightarrow$, $F_y = 14.00$ lb $\uparrow$.
6.89 (a) and (c) $B_x = 24.0$ lb $\leftarrow$, $B_y = 7.50$ lb $\downarrow$; $F_x = 24.0$ lb $\rightarrow$, $F_y = 7.50$ lb $\uparrow$. (b) $B_x = 24.0$ lb $\leftarrow$, $B_y = 10.50$ lb $\uparrow$; $F_x = 24.0$ lb $\rightarrow$, $F_y = 10.50$ lb $\downarrow$.
6.91 $D_x = 13.60$ kN $\rightarrow$, $D_y = 7.50$ kN $\uparrow$; $E_x = 13.60$ kN $\leftarrow$, $E_y = 2.70$ kN $\downarrow$.
6.92 $A_x = 45.0$ N $\leftarrow$, $A_y = 30.0$ N $\downarrow$; $B_x = 45.0$ N $\rightarrow$, $B_y = 270$ N $\uparrow$.
6.93 (a) $E_x = 2.00$ kips $\leftarrow$, $E_y = 2.25$ kips $\uparrow$.
 (b) $C_x = 4.00$ kips $\leftarrow$, $C_y = 5.75$ kips $\uparrow$.
6.94 (a) $E_x = 3.00$ kips $\leftarrow$, $E_y = 1.500$ kips $\uparrow$.
 (b) $C_x = 3.00$ kips $\leftarrow$, $C_y = 6.50$ kips $\uparrow$.
6.95 (a) $A = 982$ lb $\uparrow$; $B = 935$ lb $\uparrow$; $C = 733$ lb $\uparrow$.
 (b) $\Delta B = +291$ lb; $\Delta C = -72.7$ lb.
6.96 (a) 572 lb. (b) $A = 1070$ lb $\uparrow$; $B = 709$ lb $\uparrow$; $C = 870$ lb $\uparrow$.
6.99 $B = 152.0$ lb $\downarrow$; $C_x = 60.0$ lb $\leftarrow$, $C_y = 200$ lb $\uparrow$; $D_x = 60.0$ lb $\rightarrow$, 42.0 lb $\uparrow$.
6.100 $B = 108.0$ lb $\downarrow$; $C_x = 90.0$ lb $\leftarrow$, $C_y = 150.0$ lb $\uparrow$; $D_x = 90.0$ lb $\rightarrow$, $D_y = 18.00$ lb $\uparrow$.
6.101 $A_x = 13.00$ kN $\leftarrow$, $A_y = 4.00$ kN $\downarrow$; $B_x = 36.0$ kN $\rightarrow$, $B_y = 6.00$ kN $\uparrow$; $E_x = 23.0$ kN $\leftarrow$, $E_y = 2.00$ kN $\downarrow$.
6.102 $A_x = 2025$ N $\leftarrow$, $A_y = 1800$ kN $\downarrow$; $B_x = 4050$ N $\rightarrow$, $B_y = 1200$ N $\uparrow$; $E_x = 2025$ N $\leftarrow$, $E_y = 600$ N $\uparrow$.
6.103 $A_x = 1110$ lb $\leftarrow$, $A_y = 600$ lb $\uparrow$; $B_x = 1110$ lb $\leftarrow$, $B_y = 800$ lb $\downarrow$; $D_x = 2220$ lb $\rightarrow$, $D_y = 200$ lb $\uparrow$.
6.104 $A_x = 660$ lb $\leftarrow$, $A_y = 240$ lb $\uparrow$; $B_x = 660$ lb $\leftarrow$, $B_y = 320$ lb $\downarrow$; $D_x = 1320$ lb $\rightarrow$, $D_y = 80.0$ lb $\uparrow$.
6.107 (a) $A_x = 200$ kN $\rightarrow$, $A_y = 122.0$ kN $\uparrow$.
 (b) $B_x = 200$ kN $\leftarrow$, $B_y = 10.00$ kN $\downarrow$.
6.108 (a) $A_x = 205$ kN $\rightarrow$, $A_y = 134.5$ kN $\uparrow$.
 (b) $B_x = 205$ kN $\leftarrow$, $B_y = 5.50$ kN $\uparrow$.
6.109 $B = 98.5$ lb $\nearrow 24.0^\circ$; $C = 90.6$ lb $\searrow 6.34^\circ$.
6.110 $B = 25.0$ lb $\uparrow$; $C = 79.1$ lb $\searrow 18.43^\circ$.
6.112 $F_{AF} = P/4$ C ; $F_{BG} = F_{DG} = P/\sqrt{2}$ C ; $F_{EH} = P/4$ T .
6.113 $F_{AG} = \sqrt{2}P/6$ C ; $F_{BF} = 2\sqrt{2}P/3$ C ; $F_{DI} = \sqrt{2}P/3$ C ; $F_{EH} = \sqrt{2}P/6$ T .
6.115 $F_{AF} = M_0/4a$ C ; $F_{BG} = F_{DG} = M_0/\sqrt{2}a$ T ; $F_{EH} = 3M_0/4a$ C .
6.116 $F_{AF} = M_0/6a$ T ; $F_{BG} = \sqrt{2}M_0/6a$ T ; $F_{DG} = \sqrt{2}M_0/3a$ T ; $F_{EH} = M_0/6a$ C .
6.117 $A = P/15$ $\uparrow$; $D = 2P/15$ $\uparrow$; $E = 8P/15$ $\uparrow$; $H = 4P/15$ $\uparrow$.
6.118 $E = P/5$ $\downarrow$; $F = 8P/5$ $\uparrow$; $G = 4P/5$ $\downarrow$; $H = 2P/5$ $\uparrow$.
6.120 (a) $A = 2.06P \nearrow 14.04^\circ$; $B = 2.06 \searrow 14.04^\circ$; frame is rigid.
 (b) Frame is not rigid. (c) $A = 1.25P \searrow 36.9^\circ$, $B = 1.031P \nearrow 14.04^\circ$; frame is rigid.
6.122 (a) 2860 N $\downarrow$. (b) 2700 N $\nearrow 68.5^\circ$.
6.123 564 lb $\rightarrow$.
6.124 275 lb $\rightarrow$.
6.125 764 N $\leftarrow$.
6.127 (a) 746 N $\downarrow$. (b) 565 N $\searrow 61.3^\circ$.
6.129 832 lb·in. $\uparrow$.
6.130 360 lb·in. $\uparrow$.
6.131 195.0 kN·m $\downarrow$.
6.132 40.5 kN·m $\uparrow$.
6.133 (a) 160.8 N·m $\uparrow$. (b) 155.9 N·m $\uparrow$.
6.134 (a) 117.8 N·m $\uparrow$. (b) 47.9 N·m $\uparrow$.
6.137 18.43 N·m $\downarrow$.
6.138 208 N·m $\downarrow$.
6.139 $F_{AE} = 800$ N T ; $F_{DG} = 100.0$ N C .
6.140 $P = 120.0$ N $\downarrow$; $Q = 110.0$ N $\leftarrow$.
6.141 $F = 3290$ lb $\searrow 15.12^\circ$; $D = 4550$ lb $\leftarrow$.
6.143 $D = 30.0$ kN $\leftarrow$; $F = 37.5$ kN $\searrow 36.9^\circ$.
6.144 $D = 150.0$ kN $\leftarrow$; $F = 96.4$ kN $\searrow 13.50^\circ$.
6.145 (a) 475 lb. (b) 528 lb $\searrow 63.3^\circ$.
6.147 44.8 kN.
6.148 8.45 kN.
6.149 140.0 N.
6.151 315 lb.
6.152 (a) 312 lb. (b) 135.0 lb·in. $\downarrow$.
6.153 (a) 4.91 kips C . (b) 10.69 kips C .
6.154 (a) 2.86 kips C . (b) 9.43 kips C .
6.155 (a) 9.29 kN $\searrow 44.4^\circ$. (b) 8.04 kN $\searrow 34.4^\circ$.
6.159 (a) $(90.0 \text{ N}\cdot\text{m})\mathbf{i}$. (b) $A = 0$; $M_A = -(48.0 \text{ N}\cdot\text{m})\mathbf{i}$, $B = 0$; $M_B = -(72.0 \text{ N}\cdot\text{m})\mathbf{i}$.
6.160 (a) 27.0 mm. (b) 40.0 N·m $\downarrow$.
6.163 $E_x = 100.0$ kN $\rightarrow$, $E_y = 154.9$ kN $\uparrow$; $F_x = 26.5$ kN $\rightarrow$, $F_y = 118.1$ kN $\downarrow$; $H_x = 126.5$ kN $\leftarrow$, $H_y = 36.8$ kN $\downarrow$.
6.164 $F_{AB} = 4.00$ kN T ; $F_{AD} = 15.00$ kN T ; $F_{BD} = 9.00$ kN C ; $F_{BE} = 5.00$ kN T ; $F_{CD} = 16.00$ kN C ; $F_{DE} = 4.00$ kN C .

- 6.165** $F_{AB} = 7.83 \text{ kN C}$; $F_{AC} = 7.00 \text{ kN T}$; $F_{BC} = 1.886 \text{ kN C}$;
 $F_{BD} = 6.34 \text{ kN C}$; $F_{CD} = 1.491 \text{ kN T}$; $F_{CE} = 5.00 \text{ kN T}$;
 $F_{DE} = 2.83 \text{ kN C}$; $F_{DF} = 3.35 \text{ kN C}$; $F_{EF} = 2.75 \text{ kN T}$;
 $F_{EG} = 1.061 \text{ kN C}$; $F_{EH} = 3.75 \text{ kN T}$; $F_{FG} = 4.24 \text{ kN C}$;
 $F_{GH} = 5.30 \text{ kN C}$.
- 6.166** $F_{AB} = 8.20 \text{ kips T}$; $F_{AG} = 4.50 \text{ kips T}$; $F_{FG} = 11.60 \text{ kips C}$.
- 6.168** $A_x = 900 \text{ lb } \leftarrow$; $A_y = 75.0 \text{ lb } \uparrow$; $B = 825 \text{ lb } \downarrow$;
 $D_x = 900 \text{ lb } \rightarrow$; $D_y = 750 \text{ lb } \uparrow$.
- 6.170** $B_x = 700 \text{ N } \leftarrow$; $B_y = 200 \text{ N } \downarrow$; $E_x = 700 \text{ N } \rightarrow$;
 $E_y = 500 \text{ N } \uparrow$.
- 6.171** $C_x = 78.0 \text{ lb } \rightarrow$; $C_y = 28.0 \text{ lb } \uparrow$; $F_x = 78.0 \text{ lb } \leftarrow$;
 $F_y = 12.00 \text{ lb } \uparrow$.
- 6.172** $A = 327 \text{ lb } \rightarrow$; $B = 827 \text{ lb } \leftarrow$; $D = 621 \text{ lb } \uparrow$; $E = 246 \text{ lb } \uparrow$.
- 6.174** (a) $21.0 \text{ kN } \leftarrow$. (b) $52.5 \text{ kN } \leftarrow$.

CHAPTER 7

- 7.1** $F = 720 \text{ lb } \rightarrow$; $V = 140.0 \text{ lb } \uparrow$; $M = 1120 \text{ lb}\cdot\text{in.}$ $\uparrow$ (On JC).
- 7.2** $F = 120.0 \text{ lb } \leftarrow$; $V = 30.0 \text{ lb } \downarrow$; $M = 120.0 \text{ lb}\cdot\text{in.}$ $\uparrow$.
- 7.3** $F = 125.0 \text{ N } \nearrow 67.4^\circ$; $V = 300 \text{ N } \searrow 22.6^\circ$;
 $M = 156.0 \text{ N}\cdot\text{m.}$ $\downarrow$.
- 7.4** $F = 2330 \text{ N } \nearrow 67.4^\circ$; $V = 720 \text{ N } \searrow 22.6^\circ$; $M = 374 \text{ N}\cdot\text{m.}$ $\downarrow$.
- 7.7** $F = 23.6 \text{ lb } \nearrow 76.0^\circ$; $V = 29.1 \text{ lb } \searrow 14.04^\circ$;
 $M = 540 \text{ lb}\cdot\text{in.}$ $\downarrow$.
- 7.8** (a) 30.0 lb at C . (b) 33.5 lb at B and D . (c) $960 \text{ lb}\cdot\text{in.}$ at C .
- 7.9** $F = 103.9 \text{ N } \searrow 60.0^\circ$; $V = 60.0 \text{ N } \searrow 30.0^\circ$;
 $M = 18.71 \text{ N}\cdot\text{m}$ $\downarrow$ (On AJ).
- 7.10** $F = 60.0 \text{ N } \nearrow 30.0^\circ$; $V = 103.9 \text{ N } \searrow 60.0^\circ$;
 $M = 10.80 \text{ N}\cdot\text{m}$ $\uparrow$ (On BK).
- 7.11** $F = 194.6 \text{ N } \searrow 60.0^\circ$; $V = 257 \text{ N } \nearrow 30.0^\circ$;
 $M = 24.7 \text{ N}\cdot\text{m}$ $\downarrow$ (On AJ).
- 7.12** $45.2 \text{ N}\cdot\text{m}$ for $\theta = 82.9^\circ$.
- 7.15** $F = 250 \text{ N } \searrow 36.9^\circ$; $V = 120.0 \text{ N } \nearrow 53.1^\circ$;
 $M = 120.0 \text{ N}\cdot\text{m}$ $\uparrow$ (On BJ).
- 7.16** $F = 560 \text{ N } \leftarrow$; $V = 90.0 \text{ N } \downarrow$; $M = 72.0 \text{ N}\cdot\text{m}$ $\downarrow$ (On AK).
- 7.17** $150.0 \text{ lb}\cdot\text{in.}$ at D .
- 7.18** $105.0 \text{ lb}\cdot\text{in.}$ at E .
- 7.19** $F = 200 \text{ N } \searrow 36.9^\circ$; $V = 120.0 \text{ N } \nearrow 53.1^\circ$;
 $M = 120.0 \text{ N}\cdot\text{m}$ $\uparrow$ (On BJ).
- 7.20** $F = 520 \text{ N } \leftarrow$; $V = 120.0 \text{ N } \downarrow$; $M = 96.0 \text{ N}\cdot\text{m}$ $\downarrow$ (On AK).
- 7.23** 0.0557 Wr (On AJ).
- 7.24** 0.1009 Wr for $\theta = 57.3^\circ$.
- 7.25** 0.289 Wr (On BJ).
- 7.26** 0.417 Wr (On BJ).
- 7.29** (b) $|V|_{\max} = wL/4$; $|M|_{\max} = 3wL^2/32$.
- 7.30** (b) $|V|_{\max} = w_0L/2$; $|M|_{\max} = w_0L^2/6$.
- 7.31** (b) $|V|_{\max} = 2P/3$; $|M|_{\max} = 2PL/9$.
- 7.32** (b) $|V|_{\max} = 2P$; $|M|_{\max} = 3Pa$.
- 7.35** (b) $|V|_{\max} = 40.0 \text{ kN}$; $|M|_{\max} = 55.0 \text{ kN}\cdot\text{m}$.
- 7.36** (b) $|V|_{\max} = 50.5 \text{ kN}$; $|M|_{\max} = 39.8 \text{ kN}\cdot\text{m}$.
- 7.39** (b) $|V|_{\max} = 64.0 \text{ kN}$; $|M|_{\max} = 92.0 \text{ kN}\cdot\text{m}$.
- 7.40** (b) $|V|_{\max} = 40.0 \text{ kN}$; $|M|_{\max} = 40.0 \text{ kN}\cdot\text{m}$.
- 7.41** (b) $|V|_{\max} = 18.00 \text{ kips}$; $|M|_{\max} = 48.5 \text{ kip}\cdot\text{ft}$.
- 7.42** (b) $|V|_{\max} = 15.30 \text{ kips}$; $|M|_{\max} = 46.8 \text{ kip}\cdot\text{ft}$.
- 7.45** (b) $|V|_{\max} = 6.00 \text{ kips}$; $|M|_{\max} = 12.00 \text{ kip}\cdot\text{ft}$.
- 7.46** (b) $|V|_{\max} = 4.00 \text{ kips}$; $|M|_{\max} = 6.00 \text{ kip}\cdot\text{ft}$.
- 7.47** (b) $|V|_{\max} = 6.00 \text{ kN}$; $|M|_{\max} = 9.00 \text{ kN}\cdot\text{m}$.
- 7.48** (b) $|V|_{\max} = 6.00 \text{ kN}$; $|M|_{\max} = 9.00 \text{ kN}\cdot\text{m}$.
- 7.49** $|V|_{\max} = 180.0 \text{ N}$; $|M|_{\max} = 36.0 \text{ N}\cdot\text{m}$.
- 7.50** $|V|_{\max} = 800 \text{ N}$; $|M|_{\max} = 180.0 \text{ N}\cdot\text{m}$.
- 7.51** $|V|_{\max} = 90.0 \text{ lb}$; $|M|_{\max} = 1400 \text{ lb}\cdot\text{in.}$
- 7.52** $|V|_{\max} = 165.0 \text{ lb}$; $|M|_{\max} = 1625 \text{ lb}\cdot\text{in.}$
- 7.55** (a) 54.5° . (b) $675 \text{ N}\cdot\text{m}$.
- 7.56** (a) 1.236 . (b) 0.1180 wa^2 .
- 7.57** (a) 40.0 mm . (b) $1.600 \text{ N}\cdot\text{m}$.
- 7.58** (a) 0.840 m . (b) $1.680 \text{ N}\cdot\text{m}$.
- 7.59** 0.207 L .
- 7.62** (a) 0.414 wL ; 0.0858 wL^2 . (b) 0.250 wL ; 0.250 wL^2 .
- 7.69** (a) $|V|_{\max} = 15.00 \text{ kN}$; $|M|_{\max} = 42.0 \text{ kN}\cdot\text{m}$.
- 7.70** (b) $|V|_{\max} = 17.00 \text{ kN}$; $|M|_{\max} = 17.00 \text{ kN}\cdot\text{m}$.
- 7.77** (b) $75.0 \text{ kN}\cdot\text{m}$, 4.00 m from A .
- 7.78** (b) $1.378 \text{ kN}\cdot\text{m}$, 1.050 m from A .
- 7.79** (b) $26.4 \text{ kN}\cdot\text{m}$, 2.05 m from A .
- 7.80** (b) $5.76 \text{ kN}\cdot\text{m}$, 2.40 m from A .
- 7.81** (b) $14.40 \text{ kip}\cdot\text{ft}$, 6.00 ft from A .
- 7.82** (b) $16.20 \text{ kip}\cdot\text{ft}$, 13.50 ft from A .
- 7.86** (a) $V = (w_0/6L)(L^2 - 3x^2)$; $M = (w_0/6L)(L^2x - x^3)$.
 (b) 0.0642 w_0L^2A $\nearrow 0.577L$.
- 7.87** (a) $V = (w_0L/4)[3(x/L)^2 - 4(x/L) + 1]$;
 $M = (w_0L^2/4)[(x/L)^3 - 2(x/L)^2 + (x/L)]$.
 (b) $w_0L^2/27$, at $x = L/3$.
- 7.89** (a) $P = 4.00 \text{ kN } \downarrow$; $Q = 6.00 \text{ kN } \downarrow$. (b) $M_C = -900 \text{ N}\cdot\text{m}$.
- 7.90** (a) $P = 2.50 \text{ kN } \downarrow$; $Q = 7.50 \text{ kN } \downarrow$. (b) $M_C = -900 \text{ N}\cdot\text{m}$.
- 7.91** (a) $P = 1.350 \text{ kips } \downarrow$; $Q = 0.450 \text{ kips } \downarrow$. (b) $V_{\max} = 2.70 \text{ kips}$
 at A ; $M_{\max} = 6.345 \text{ kip}\cdot\text{ft}$, 5.40 ft from A .
- 7.92** (a) $P = 0.540 \text{ kips } \downarrow$; $Q = 1.860 \text{ kips } \downarrow$.
 (b) $V_{\max} = 3.14 \text{ kips}$ at B ; $M_{\max} = 7.00 \text{ kip}\cdot\text{ft}$, 6.88 ft from A .
- 7.93** (a) $E_x = 10.00 \text{ kN } \rightarrow$, $E_y = 7.00 \text{ kN } \uparrow$. (b) 12.21 kN .
- 7.94** 1.667 m .
- 7.95** (a) $838 \text{ lb } \searrow 17.35^\circ$. (b) $971 \text{ lb } \nearrow 34.5^\circ$.
- 7.96** (a) $2670 \text{ lb } \nearrow 2.10^\circ$. (b) $2810 \text{ lb } \searrow 18.65^\circ$.
- 7.97** (a) $d_B = 1.733 \text{ m}$; $d_D = 4.20 \text{ m}$. (b) $21.5 \text{ kN } \searrow 3.81^\circ$.
- 7.98** (a) 2.80 m . (b) $A = 32.0 \text{ kN } \searrow 38.7^\circ$; $E = 25.0 \text{ kN } \rightarrow$.
- 7.101** 196.2 N .
- 7.102** 157.0 N .
- 7.103** (a) 240 lb . (b) 9.00 ft .
- 7.104** $a = 7.50 \text{ ft}$; $b = 17.50 \text{ ft}$.
- 7.107** (a) 1775 N . (b) 60.1 m .
- 7.109** (a) $50,200 \text{ kips}$. (b) 3580 ft .
- 7.110** 3.75 ft .
- 7.111** (a) $56,400 \text{ kips}$. (b) 4284 ft .
- 7.112** (a) 6.75 m . (b) $T_{AB} = 615 \text{ N}$; $T_{BC} = 600 \text{ N}$.
- 7.114** (a) $\sqrt{3}L\Delta/8$. (b) 12.25 ft .
- 7.115** $h = 27.6 \text{ mm}$; $\theta_A = 25.5^\circ$; $\theta_C = 27.6^\circ$.
- 7.116** (a) 4.05 m . (b) 16.41 m . (c) $A_x = 5890 \text{ N } \leftarrow$, $A_y = 5300 \text{ N } \uparrow$.
- 7.117** (a) $58,900 \text{ kips}$. (b) 29.2° .
- 7.118** (a) 16.00 ft to the left of B . (b) 2000 lb .
- 7.125** $Y = h[1 - \cos(\pi x/L)]$; $T_{\min} = w_0L^2/h\pi^2$;
 $T_{\max} = (w_0L/\pi)\sqrt{(L^2/h^2\pi^2) + 1}$
- 7.127** (a) 12.36 ft . (b) 15.38 lb .
- 7.128** (a) 412 ft . (b) 875 lb .
- 7.129** (a) 35.6 m . (b) 49.2 kg .
- 7.130** 49.86 ft .
- 7.133** (a) 5.89 m . (b) $10.89 \text{ N } \rightarrow$.
- 7.134** 10.05 ft .
- 7.135** (a) 56.3 ft . (b) 2.36 lb/ft .
- 7.136** (a) 30.2 m . (b) 56.6 kg .
- 7.139** 31.8 N .
- 7.140** 29.8 N .
- 7.143** (a) $a = 79.0 \text{ ft}$; $b = 60.0 \text{ ft}$. (b) 103.9 ft .
- 7.144** (a) $a = 65.8 \text{ ft}$; $b = 50.0 \text{ ft}$. (b) 86.6 ft .
- 7.145** $119.1 \text{ N } \rightarrow$.
- 7.146** $177.6 \text{ N } \rightarrow$.
- 7.147** 3.50 ft .

- 7.148** 5.71 ft.
7.151 0.394 m and 10.97 m.
7.152 0.1408.
7.153 (a) 0.338. (b) 56.5° ; 0.755 wL .
7.154 (On AJ) $\mathbf{F} = 750 \text{ N } \uparrow$; $\mathbf{V} = 400 \text{ N } \leftarrow$; $\mathbf{M} = 130.0 \text{ N}\cdot\text{m } \curvearrowright$.
7.156 (On BJ) $\mathbf{F} = 12.50 \text{ lb } \nearrow 30.0^\circ$; $\mathbf{V} = 21.7 \text{ lb } \searrow 60.0^\circ$; $\mathbf{M} = 75.0 \text{ lb}\cdot\text{in. } \downarrow$.
7.157 (a) (On AJ) $\mathbf{F} = 500 \text{ N } \leftarrow$; $\mathbf{V} = 500 \text{ N } \uparrow$; $\mathbf{M} = 300 \text{ N}\cdot\text{m } \downarrow$.
 (b) (On AK) $\mathbf{F} = 970 \text{ N } \uparrow$; $\mathbf{V} = 171.0 \text{ N } \leftarrow$; $\mathbf{M} = 446 \text{ N}\cdot\text{m } \downarrow$.
7.158 (a) 40.0 kips. (b) 40.0 kip-ft.
7.161 (a) 18.00 kip-ft, 3.00 ft from A.
 (b) 34.1 kip-ft, 2.25 ft from A.
7.163 (a) 2.28 m. (b) $\mathbf{D}_x = 13.67 \text{ kN } \rightarrow$; $\mathbf{D}_y = 7.80 \text{ kN } \uparrow$.
 (c) 15.94 kN.
7.164 (a) 138.1 m. (b) 602 N.
7.165 (a) 4.22 ft. (b) 80.3° .

CHAPTER 8

- 8.1** Block is in equilibrium, $\mathbf{F} = 30.1 \text{ N } \searrow 20.0^\circ$.
8.2 Block moves up, $\mathbf{F} = 151.7 \text{ N } \searrow 20.0^\circ$.
8.3 Block moves, $\mathbf{F} = 36.1 \text{ lb } \searrow 30.0^\circ$.
8.4 Block is in equilibrium, $\mathbf{F} = 36.3 \text{ lb } \searrow 30.0^\circ$.
8.5 (a) 83.2 lb. (b) 66.3 lb.
8.7 (a) $29.7 \text{ N } \leftarrow$. (b) $20.9 \text{ N } \rightarrow$.
8.9 74.5 N.
8.10 $17.91^\circ \leq \theta \leq 66.4^\circ$.
8.11 31.0° .
8.12 46.4° .
8.13 Package C does not move; $\mathbf{F}_C = 10.16 \text{ N } \nearrow$.
 Package A and B move; $\mathbf{F}_A = 7.58 \text{ N } \nearrow$; $\mathbf{F}_B = 3.03 \text{ N } \nearrow$.
8.14 All packages move; $\mathbf{F}_A = \mathbf{F}_C = 7.58 \text{ N } \nearrow$; $\mathbf{F}_B = 3.03 \text{ N } \nearrow$.
8.17 (a) 75.0 lb. (b) Pipe will slide.
8.18 (a) $P = 36.0 \text{ lb } \rightarrow$. (b) $h_{\max} = 40.0 \text{ in.}$
8.19 $P = 8.34 \text{ lb.}$
8.20 $P = 7.50 \text{ lb.}$
8.21 (a) 0.300 Wr . (b) 0.349 Wr .
8.22 $M = Wr\mu_s(1 + \mu_s)/(1 + \mu_s^2)$.
8.23 (a) 136.4° . (b) 0.928 W .
8.25 0.208.
8.27 664 N $\downarrow$.
8.29 (a) Plate in equilibrium. (b) Plate moves downward.
8.30 $10.00 \text{ lb} < P < 36.7 \text{ lb.}$
8.32 0.860.
8.34 0.0533.
8.35 (a) 1.333. (b) 1.192. (c) 0.839.
8.36 (b) 2.69 lb.
8.37 (a) 2.94 N. (b) 4.41 N.
8.39 $30.6 \text{ N}\cdot\text{m } \curvearrowright$.
8.40 $18.90 \text{ N}\cdot\text{m } \curvearrowright$.
8.41 135.0 lb.
8.43 (a) System slides; $P = 62.8 \text{ N.}$ (b) System rotates about B ; $P = 73.2 \text{ N.}$
8.44 35.8° .
8.45 20.5° .
8.46 1.225 W .
8.47 $46.4^\circ \leq \theta \leq 52.4^\circ$ and $67.6^\circ \leq \theta \leq 79.4^\circ$.
8.48 (a) $283 \text{ N } \leftarrow$. (b) $\mathbf{B}_x = 413 \text{ N } \leftarrow$; $\mathbf{B}_y = 480 \text{ N } \downarrow$.
8.49 (a) $107.0 \text{ N } \leftarrow$. (b) $\mathbf{B}_x = 611 \text{ N } \leftarrow$; $\mathbf{B}_y = 480 \text{ N } \downarrow$.
8.52 (a) 15.26 kips. (b) 5.40 kips.
8.53 (a) 6.88 kips. (b) 5.40 kips.
8.54 $9.86 \text{ kN } \leftarrow$.
8.55 $9.13 \text{ N } \leftarrow$.
8.56 (a) 28.1° . (b) $728 \text{ N } \nearrow 14.04^\circ$.
8.57 (a) $50.4 \text{ lb } \downarrow$. (b) $50.4 \text{ lb } \downarrow$.
8.59 143.4 N.
8.60 1.400 lb.
8.62 (a) $197.0 \text{ lb } \rightarrow$. (b) Base will not move.
8.63 (a) $280 \text{ lb } \leftarrow$. (b) Base moves.
8.64 (b) $283 \text{ N } \leftarrow$.
8.65 0.442.
8.66 0.1103.
8.67 0.1013.
8.71 $693 \text{ lb}\cdot\text{ft.}$
8.72 $35.8 \text{ N}\cdot\text{m.}$
8.73 $9.02 \text{ N}\cdot\text{m.}$
8.74 (a) Screw A . (b) $14.06 \text{ lb}\cdot\text{in.}$
8.77 0.226.
8.78 4.70 kips.
8.79 450 N.
8.80 412 N.
8.81 334 N.
8.82 376 N.
8.84 $T_{AB} = 77.5 \text{ lb; } T_{CD} = 72.5 \text{ lb. } T_{EF} = 67.8 \text{ lb.}$
8.86 (a) 4.80 kN. (b) 1.375° .
8.88 $22.0 \text{ lb } \leftarrow$.
8.89 $1.948 \text{ lb } \downarrow$.
8.90 $18.01 \text{ lb } \leftarrow$.
8.92 0.1670.
8.93 3.75 lb.
8.98 10.87 lb.
8.99 0.0600 in.
8.100 154.4 N.
8.101 300 mm.
8.102 (a) 1.288 kN. (b) 1.058 kN.
8.103 2.34 ft.
8.104 (a) 0.329. (b) 2.67 turns.
8.105 $14.23 \text{ kg} \leq m \leq 175.7 \text{ kg.}$
8.106 (a) 0.292. (b) 310 N.
8.109 $31.8 \text{ N}\cdot\text{m } \curvearrowright$.
8.110 (a) $T_A = 8.40 \text{ lb; } T_B = 19.60 \text{ lb.}$ (b) 0.270.
8.111 (a) $T_A = 11.13 \text{ lb; } T_B = 20.9 \text{ lb.}$ (b) $91.3 \text{ lb}\cdot\text{in. } \downarrow$.
8.112 $35.1 \text{ N}\cdot\text{m.}$
8.113 (a) $27.0 \text{ N}\cdot\text{m.}$ (b) 675 N.
8.114 (a) $39.0 \text{ N}\cdot\text{m.}$ (b) 844 N.
8.117 4.49 in.
8.118 (a) 11.66 kg. (b) 38.6 kg. (c) 34.4 kg.
8.119 (a) 9.46 kg. (b) 167.2 kg. (c) 121.0 kg.
8.120 (a) 10.39 lb. (b) 58.5 lb.
8.121 (a) 28.9 lb. (b) 28.9 lb.
8.124 5.97 N.
8.125 9.56 N.
8.126 0.350.
8.128 (a) $30.3 \text{ lb}\cdot\text{in. } \curvearrowright$. (b) $3.78 \text{ lb } \downarrow$.
8.129 (a) $17.23 \text{ lb}\cdot\text{in. } \downarrow$. (b) $2.15 \text{ lb } \uparrow$.
8.133 (a) $51.0 \text{ N}\cdot\text{m.}$ (b) 875 N.
8.134 (a) $353 \text{ N } \leftarrow$. (b) $196.2 \text{ N } \leftarrow$.
8.136 (a) $136.0 \text{ lb } \rightarrow$. (b) $30.0 \text{ lb } \rightarrow$. (c) $12.86 \text{ lb } \rightarrow$.
8.137 $6.35 \leq L/a \leq 10.81$.
8.138 $151.5 \text{ N}\cdot\text{m.}$
8.140 0.225.
8.141 313 lb $\rightarrow$.
8.143 $6.44 \text{ N}\cdot\text{m.}$
8.144 (a) 0.238. (b) $218 \text{ N } \downarrow$.

CHAPTER 9

- 9.1 $a^3b/30$.
 9.2 $3a^3b/10$.
 9.3 $b^3h/12$.
 9.4 $a^3b/6$.
 9.6 $ab^3/6$.
 9.8 $3ab^3/10$.
 9.9 $ab^3/15$.
 9.10 $ab^3/15$.
 9.11 $0.1056 ab^3$.
 9.12 $3.43 a^3b$.
 9.15 $3a^3/35$; $b\sqrt{9/35}$.
 9.16 $0.0945ah^3$; $0.402h$.
 9.17 $3a^3b/35$; $a\sqrt{9/35}$.
 9.18 $31a^3h/20$; $a\sqrt{93/35}$.
 9.21 $20a^4$; $1.826a$.
 9.22 $4ab(a^2 + 4b^2)/3$; $\sqrt{(a^2 + 4b^2)/3}$.
 9.23 $64a^4/15$; $1.265a$.
 9.25 $(\pi/2)(R_2^4 - R_1^4)$; $(\pi/4)(R_2^4 - R_1^4)$.
 9.26 (b) for $t/R_m = 1$, -10.56% ; for $t/R_m = 1/2$, -2.99% ; for $t/R_m = 1/10$, -0.1250% .
 9.28 $bh(12h^2 + b^2)/48$; $\sqrt{(12h^2 + b^2)/24}$.
 9.31 $390 \times 10^3 \text{ mm}^4$; 21.9 mm .
 9.32 46.0 in^4 ; 1.599 in .
 9.33 $64.3 \times 10^3 \text{ mm}^4$; 8.87 mm .
 9.34 46.5 in^4 ; 1.607 in .
 9.37 $J_B = 1800 \text{ in}^4$; $J_D = 3600 \text{ in}^4$.
 9.39 3000 mm^2 ; $325 \times 10^3 \text{ mm}^4$.
 9.40 $24.6 \times 10^6 \text{ mm}^4$.
 9.41 $\bar{I}_x = 13.89 \times 10^6 \text{ mm}^4$; $\bar{I}_y = 20.9 \times 10^6 \text{ mm}^4$.
 9.42 $\bar{I}_x = 479 \times 10^3 \text{ mm}^4$; $\bar{I}_y = 149.7 \times 10^3 \text{ mm}^4$.
 9.43 $\bar{I}_x = 191.3 \text{ in}^4$; $\bar{I}_y = 75.2 \text{ in}^4$.
 9.44 $\bar{I}_x = 18.13 \text{ in}^4$; $\bar{I}_y = 4.51 \text{ in}^4$.
 9.47 (a) $11.57 \times 10^6 \text{ mm}^4$. (b) $7.81 \times 10^6 \text{ mm}^4$.
 9.48 (a) $12.16 \times 10^6 \text{ mm}^4$. (b) $9.73 \times 10^6 \text{ mm}^4$.
 9.49 $\bar{I}_x = 186.7 \times 10^6 \text{ mm}^4$; $\bar{k}_x = 118.6 \text{ mm}$; $\bar{I}_y = 167.7 \times 10^6 \text{ mm}^4$.
 $\bar{k}_y = 112.4 \text{ mm}$.
 9.50 $\bar{I}_x = 44.5 \text{ in}^4$; $\bar{k}_x = 2.16 \text{ in}$; $\bar{I}_y = 27.7 \text{ in}^4$; $\bar{k}_y = 1.709 \text{ in}$.
 9.51 $\bar{I}_x = 250 \text{ in}^4$; $\bar{k}_x = 4.10 \text{ in}$; $\bar{I}_y = 141.9 \text{ in}^4$; $\bar{k}_y = 3.09 \text{ in}$.
 9.52 $\bar{I}_x = 260 \times 10^6 \text{ mm}^4$; $\bar{k}_x = 144.6 \text{ mm}$; $\bar{I}_y = 17.53 \text{ mm}^4$; $\bar{k}_y = 37.6 \text{ mm}$.
 9.54 $\bar{I}_x = 745 \times 10^6 \text{ mm}^4$; $\bar{I}_y = 91.3 \times 10^6 \text{ mm}^4$.
 9.55 $\bar{I}_x = 3.55 \times 10^6 \text{ mm}^4$; $\bar{I}_y = 49.8 \times 10^6 \text{ mm}^4$.
 9.57 $h/2$.
 9.58 $15h/14$.
 9.59 $3\pi r/16$.
 9.60 $4h/7$.
 9.63 $5a/8$.
 9.64 80.0 mm .
 9.67 $a^4/2$.
 9.68 $b^2h^2/4$.
 9.69 $a^2b^2/6$.
 9.71 $-1.760 \times 10^6 \text{ mm}^4$.
 9.72 $2.40 \times 10^6 \text{ mm}^4$.
 9.74 -0.380 in^4 .
 9.75 $471 \times 10^3 \text{ mm}^4$.
 9.76 -9010 in^4 .
 9.78 $2.54 \times 10^6 \text{ mm}^4$.
 9.79 (a) $\bar{I}_{x'} = 0.482a^4$; $\bar{I}_{y'} = 1.482a^4$; $\bar{I}_{x'y'} = -0.589a^4$.
 (b) $\bar{I}_{x'} = 1.120a^4$; $\bar{I}_{y'} = 0.843a^4$; $\bar{I}_{x'y'} = 0.760a^4$.
 9.80 $\bar{I}_{x'} = 2.12 \times 10^6 \text{ mm}^4$; $\bar{I}_{y'} = 8.28 \times 10^6 \text{ mm}^4$; $\bar{I}_{x'y'} = -0.532 \times 10^6 \text{ mm}^4$.
 9.81 $\bar{I}_{x'} = 1033 \text{ in}^4$; $\bar{I}_{y'} = 2020 \text{ in}^4$; $\bar{I}_{x'y'} = -873 \text{ in}^4$.
 9.83 $\bar{I}_{x'} = 0.236 \text{ in}^4$; $\bar{I}_{y'} = 1.244 \text{ in}^4$; $\bar{I}_{x'y'} = 0.1132 \text{ in}^4$.
 9.85 20.2° and 110.2° ; $1.754a^4$; $0.209a^4$.
 9.86 25.1° and 115.1° ; $\bar{I}_{\max} = 8.32 \times 10^6 \text{ mm}^4$; $\bar{I}_{\min} = 2.08 \times 10^6 \text{ mm}^4$.
 9.87 29.7° and 119.7° ; 2530 in^4 ; 524 in^4 .
 9.89 -23.7° and 66.3° ; 1.257 in^4 ; 0.224 in^4 .
 9.91 (a) $\bar{I}_{x'} = 0.482a^4$; $\bar{I}_{y'} = 1.482a^4$; $\bar{I}_{x'y'} = -0.589a^4$.
 (b) $\bar{I}_{x'} = 1.120a^4$; $\bar{I}_{y'} = 0.843a^4$; $0.760a^4$.
 9.92 $\bar{I}_{x'} = 2.12 \times 10^6 \text{ mm}^4$; $\bar{I}_{y'} = 8.28 \times 10^6 \text{ mm}^4$; $\bar{I}_{x'y'} = -0.532 \times 10^6 \text{ mm}^4$.
 9.93 $\bar{I}_{x'} = 1033 \text{ in}^4$; $\bar{I}_{y'} = 2020 \text{ in}^4$; $\bar{I}_{x'y'} = -873 \text{ in}^4$.
 9.95 $\bar{I}_{x'} = 0.236 \text{ in}^4$; $\bar{I}_{y'} = 1.244 \text{ in}^4$; $\bar{I}_{x'y'} = 0.1132 \text{ in}^4$.
 9.97 20.2° ; $1.754a^4$; $0.209a^4$.
 9.98 23.9° ; $8.33 \times 10^6 \text{ mm}^4$; $1.465 \times 10^6 \text{ mm}^4$.
 9.99 33.4° ; $22.1 \times 10^3 \text{ in}^4$; 2490 in^4 .
 9.100 29.7° ; 2530 in^4 ; 524 in^4 .
 9.103 (a) -1.146 in^4 . (b) 29.1° clockwise. (c) 3.39 in^4 .
 9.104 23.8° clockwise; $0.524 \times 10^6 \text{ mm}^4$; $0.0917 \times 10^6 \text{ mm}^4$.
 9.105 19.54° counterclockwise; $4.34 \times 10^6 \text{ mm}^4$; $0.647 \times 10^6 \text{ mm}^4$.
 9.106 (a) 25.3° . (b) 1459 in^4 ; 40.5 in^4 .
 9.107 (a) $88.0 \times 10^6 \text{ mm}^4$. (b) $96.3 \times 10^6 \text{ mm}^4$; $39.7 \times 10^6 \text{ mm}^4$.
 9.111 (a) $\bar{I}_{AA'} = \bar{I}_{BB'} = ma^2/24$. (b) $ma^2/12$.
 9.112 (a) $m(r_1^2 + r_2^2)/4$. (b) $m(r_1^2 + r_2^2)/2$.
 9.113 (a) $0.0699 mb^2$. (b) $m(a^2 + 0.279 b^2)/4$.
 9.114 (a) $mb^2/7$. (b) $m(7a^2 + 10b^2)/70$.
 9.117 (a) $5ma^2/18$. (b) $3.61 ma^2$.
 9.118 (a) $0.994 ma^2$. (b) $2.33 ma^2$.
 9.119 $m(3a^2 + 4L^2)/12$.
 9.120 $1.329 mh^2$.
 9.121 (a) $0.241 mh^2$. (b) $m(3a^2 + 0.1204 h^2)$.
 9.122 $m(b^2 + h^2)/10$.
 9.124 $ma^2/3$; $a/\sqrt{3}$.
 9.126 $I_x = I_y = ma^2/4$; $I_z = ma^2/2$.
 9.127 $1.160 \times 10^{-6} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$; 0.341 in .
 9.128 $837 \times 10^{-9} \text{ kg}\cdot\text{m}^2$; 6.92 mm .
 9.130 $2 mr^2/3$; $0.816r$.
 9.131 (a) 2.30 in . (b) $20.6 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$; 2.27 in .
 9.132 (a) $\pi pl^2 [6a^2t(5a^2/3l^2 + 2a/l + 1) + d^2/l/4]$. (b) 0.1851 .
 9.133 (a) 27.5 mm to the right of A. (b) 32.0 mm .
 9.135 $I_x = 7.11 \times 10^{-3} \text{ kg}\cdot\text{m}^2$; $I_y = 16.96 \times 10^{-3} \text{ kg}\cdot\text{m}^2$; $I_z = 15.27 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.
 9.136 $I_x = 175.5 \times 10^{-3} \text{ kg}\cdot\text{m}^2$; $I_y = 309.10^{-3} \text{ kg}\cdot\text{m}^2$; $I_z = 154.4 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.
 9.138 $I_x = 334 \times 10^{-6} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$; $I_y = I_z = 1.356 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$.
 9.139 $I_x = 344 \times 10^{-6} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$; $I_y = 132.1 \times 10^{-6} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$; $I_z = 453 \times 10^{-6} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$.
 9.141 (a) $13.99 \times 10^{-3} \text{ kg}\cdot\text{m}^2$. (b) $20.6 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.
 (c) $14.30 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.
 9.142 $I_x = 28.3 \times 10^{-3} \text{ kg}\cdot\text{m}^2$; $I_y = 183.8 \times 10^{-3} \text{ kg}\cdot\text{m}^2$; $k_x = 42.9 \text{ mm}$; $k_y = 109.3 \text{ mm}$.
 9.143 $30.5 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$.
 9.145 (a) $26.4 \times 10^{-3} \text{ kg}\cdot\text{m}^2$. (b) $31.2 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.
 (c) $8.58 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.
 9.147 $I_x = 0.0392 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$; $I_y = 0.0363 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$; $I_z = 0.0304 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$.
 9.148 $I_x = 0.323 \text{ kg}\cdot\text{m}^2$; $I_y = I_z = 0.419 \text{ kg}\cdot\text{m}^2$.
 9.149 $I_{xy} = 2.50 \times 10^{-3} \text{ kg}\cdot\text{m}^2$; $I_{yz} = 4.06 \times 10^{-3} \text{ kg}\cdot\text{m}^2$; $I_{zx} = 8.81 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.
 9.150 $I_{xy} = 286 \times 10^{-6} \text{ kg}\cdot\text{m}^2$; $I_{yz} = I_{zx} = 0$.
 9.151 $I_{xy} = -1.726 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$; $I_{yz} = 0.507 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$; $I_{zx} = -2.12 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$.
 9.152 $I_{xy} = -538 \times 10^{-6} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$; $I_{yz} = -171.4 \times 10^{-6} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$; $I_{zx} = 1120 \times 10^{-6} \text{ lb}\cdot\text{ft}\cdot\text{s}^2$.

9.155 $I_{xy} = -8.04 \times 10^{-3} \text{ kg}\cdot\text{m}^2$; $I_{yz} = 12.90 \times 10^{-3} \text{ kg}\cdot\text{m}^2$;
 $I_{zx} = 94.0 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.
9.156 $I_{xy} = 0$; $I_{yz} = 48.3 \times 10^{-6} \text{ kg}\cdot\text{m}^2$;
 $I_{zx} = -4.43 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.

9.157 $I_{xy} = 47.9 \times 10^{-6} \text{ kg}\cdot\text{m}^2$; $I_{yz} = 102.1 \times 10^{-6} \text{ kg}\cdot\text{m}^2$;
 $I_{zx} = 64.1 \times 10^{-6} \text{ kg}\cdot\text{m}^2$.

9.158 $I_{xy} = -m' R_1^3/2$; $I_{yz} = m' R_1^3/2$; $I_{zx} = -m' R_2^3/2$.

9.159 $I_{xy} = wa^3(1 - 5\pi)g$; $I_{yz} = -11\pi wa^3/g$;
 $I_{zx} = 4wa^3(1 + 2\pi)/g$.

9.160 $I_{xy} = -11wa^3/g$; $I_{yz} = wa^3(\pi + 6)/2g$; $I_{zx} = -wa^3/4g$.

9.162 (a) $mac/20$. (b) $I_{xy} = mab/20$; $I_{yz} = mbc/20$.

9.165 $18.17 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.

9.166 $11.81 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.

9.167 $5Wa^2/18g$.

9.168 $4.41 \gamma ta^4/g$.

9.169 $281 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.

9.170 $0.354 \text{ kg}\cdot\text{m}^2$.

9.173 (a) $1/\sqrt{3}$. (b) $\sqrt{7/12}$.

9.174 (a) $b/a = 2$; $c/a = 2$. (b) $b/a = 1$; $c/a = 0.5$.

9.175 (a) 2. (b) $\sqrt{2/3}$.

9.179 (a) $K_1 = 0.363ma^2$; $K_2 = 1.583ma^2$; $K_3 = 1.720ma^2$.

(b) $(\theta_x)_1 = (\theta_x)_1 = 49.7^\circ$; $(\theta_y)_1 = 113.7^\circ$; $(\theta_z)_1 = 45.0^\circ$

$(\theta_x)_2 = 90.0^\circ$; $(\theta_y)_2 = 135.0^\circ$; $(\theta_z)_2 = 73.5^\circ$; $(\theta_x)_3 = 23.7^\circ$.

9.180 (a) $K_1 = 14.30 \times 10^{-3} \text{ kg}\cdot\text{m}^2$; $K_2 = 13.96 \times 10^{-3} \text{ kg}\cdot\text{m}^2$;
 $K_3 = 20.6 \times 10^{-3} \text{ kg}\cdot\text{m}^2$.

(b) $(\theta_x)_1 = (\theta_y)_1 = 90.0^\circ$; $(\theta_z)_1 = 0^\circ$; $(\theta_x)_2 = 3.42^\circ$;

$(\theta_y)_2 = 86.6^\circ$.

$(\theta_z)_2 = 90.0^\circ$; $(\theta_x)_3 = 93.4^\circ$; $(\theta_y)_3 = 3.43^\circ$; $(\theta_z)_3 = 90.0^\circ$

9.182 (a) $K_1 = 0.1639Wa^2/g$; $K_2 = 1.054Wa^2/g$; $K_3 = 1.115Wa^2/g$.

(b) $(\theta_x)_1 = 36.7^\circ$; $(\theta_y)_1 = 71.6^\circ$; $(\theta_z)_1 = 59.5^\circ$; $(\theta_x)_2 = 74.9^\circ$;

$(\theta_y)_2 = 54.5^\circ$; $(\theta_z)_2 = 140.5^\circ$; $(\theta_x)_3 = 57.5^\circ$; $(\theta_y)_3 = 138.8^\circ$;

$(\theta_z)_3 = 112.4^\circ$

9.183 (a) $K_1 = 2.26\gamma ta^4/g$; $K_2 = 17.27\gamma ta^4/g$; $K_3 = 19.08\gamma ta^4/g$.

(b) $(\theta_x)_1 = 85.0^\circ$; $(\theta_y)_1 = 36.8^\circ$; $(\theta_z)_1 = 53.7^\circ$; $(\theta_x)_2 = 81.7^\circ$;

$(\theta_y)_2 = 54.7^\circ$; $(\theta_z)_2 = 143.4^\circ$; $(\theta_x)_3 = 9.70^\circ$; $(\theta_y)_3 = 99.0^\circ$;

$(\theta_z)_3 = 86.3^\circ$.

9.185 $I_x = 16ah^3/105$; $I_y = ha^3/5$.

9.186 $\pi a^3 b/8$; $a/2$.

9.188 $\bar{I}_x = 1.874 \times 10^6 \text{ mm}^4$; $\bar{I}_y = 5.82 \times 10^6 \text{ mm}^4$.

9.189 (a) $3.13 \times 10^6 \text{ mm}^4$. (b) $2.41 \times 10^6 \text{ mm}^4$.

9.191 -2.81 in^4 .

9.193 (a) $ma^2/3$. (b) $3ma^2/2$.

9.195 $I_x = 0.877 \text{ kg}\cdot\text{m}^2$; $I_y = 1.982 \text{ kg}\cdot\text{m}^2$;

$I_z = 1.652 \text{ kg}\cdot\text{m}^2$.

9.196 $0.0442 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$.

CHAPTER 10

10.1 65.0 N ↓.

10.2 132.0 lb →.

10.3 $39.0 \text{ N}\cdot\text{m}$ ↓.

10.4 $1320 \text{ lb}\cdot\text{in.}$ ↗.

10.5 (a) 60.0 N C, 8.00 mm ↓. (b) 300 N C, 40.0 mm ↓.

10.6 (a) 120.0 N C, 16.00 mm ↓. (b) 300 N C, 40.0 mm ↓.

10.9 $Q = 3 P \tan \theta$.

10.10 $Q = P[(l/a)] \cos^3 \theta - 1]$.

10.12 $Q = 2 P \sin \theta / \cos (\theta/2)$.

10.14 $Q = (3P/2) \tan \theta$.

10.15 $M = Pl/2 \tan \theta$.

10.16 $M = Pl(\sin \theta + \cos \theta)$.

10.17 $M = \frac{1}{2} Wl \tan \alpha \sin \theta$.

10.18 (a) $M = Pl \sin 2\theta$. (b) $M = 3Pl \cos \theta$. (c) $M = Pl \sin \theta$.

10.19 $85.2 \text{ lb}\cdot\text{ft}$ ↓.

10.20 22.8 lb ↗ 70.0° .

10.23 38.7° .

10.24 68.0° .

10.27 36.4° .

10.28 67.1° .

10.30 25.0° .

10.31 39.7° and 69.0° .

10.32 390 mm .

10.33 330 mm .

10.35 38.7° .

10.36 52.4° .

10.37 22.6° .

10.38 51.1° .

10.39 59.0° .

10.40 78.7° , 324° , 379° .

10.43 12.03 kN ↘.

10.44 20.4° .

10.45 2370 lb ↖.

10.46 2550 lb ↖.

10.48 $300 \text{ N}\cdot\text{m}$, $81.8 \text{ N}\cdot\text{m}$.

10.49 $\eta = 1/(1 + \mu \cot \alpha)$.

10.50 $\eta = \tan \theta / \tan (\theta + \phi_s)$.

10.52 37.6 N , 31.6 N .

10.53 $\mathbf{A} = 250 \text{ N}$ ↑; $\mathbf{M}_A = 450 \text{ N}\cdot\text{m}$ ↗.

10.54 1050 N ↑.

10.57 0.833 in. ↓.

10.58 0.625 in. →.

10.60 25.0° .

10.61 39.7° and 69.0° .

10.62 390 mm .

10.69 $\theta = -45.0^\circ$, unstable; $\theta = 135.0^\circ$, stable.

10.70 $\theta = -63.4^\circ$, unstable; $\theta = 116.6^\circ$, stable.

10.71 $\theta = 90.0^\circ$ and $\theta = 270^\circ$, unstable; $\theta = 22.0^\circ$ and

$\theta = 158.0^\circ$, stable.

10.72 $\theta = 0$ and $\theta = 180.0^\circ$, unstable; $\theta = 75.5^\circ$ and

$\theta = 284^\circ$, stable.

10.73 59.0° , stable.

10.74 78.7° , stable; 324° , unstable; 379° , stable.

10.77 357 mm .

10.78 252 mm .

10.80 9.39° and 90.0° , stable; 34.2° , unstable.

10.81 17.11° , stable; 72.9° , unstable.

10.83 49.1° .

10.86 16.88 m .

10.87 54.8° .

10.88 37.4° .

10.89 $P < kl/2$.

10.91 $k > 6.94 \text{ lb/in.}$

10.92 15.00 in.

10.93 $P < 2kL/9$.

10.94 $P < kL/18$.

10.96 $P < 160.0 \text{ N}$.

10.98 $P < 764 \text{ N}$.

10.100 (a) $P < 10.00 \text{ lb}$. (b) $P < 20.0 \text{ lb}$.

10.101 60.0 lb ↓.

10.102 $600 \text{ lb}\cdot\text{in.}$ ↓.

10.103 500 N ↑.

10.105 $M = 7Pa \cos \theta$

10.107 19.40° .

10.108 7.13 in.

10.110 $\theta = 0$, unstable; $\theta = 137.8^\circ$, stable.

10.112 (a) 22.0° . (b) 30.6° .

CHAPTER 11

- 11.1** 97.5 ft, 49.5 ft/s, 17 ft/s².
11.2 1.000s, 15.00 ft, -6.00 ft/s²; 2.00 s, 14.00 ft, 6.00 ft/s².
11.3 (a) 102.9 mm, -35.6 mm/s, -11.40 mm/s².
 (b) -36.1 mm/s, 72.1 mm/s².
11.4 (a) 0 mm, 960 mm/s →, 9220 mm/s² or 9.22 m/s² ←.
 (b) 14.16 mm ←, 87.9 mm/s →, 3110 mm/s² or 3.11 m/s² →.
11.5 0.667 s, 0.259m, -8.56 m/s.
11.7 (a) 0.586 s and 3.414 s. (b) 0 m. (c) 3.656 m.
11.9 (a) 77.5 ft/s. (b) 7.75 s.
11.10 1.427 ft/s, 0.363 ft.
11.11 $x(t) = t^4/108 + 10t + 24$ m.
 $v(t) = t^3/27 + 10$ m/s.
11.12 (a) 6.00 m/s⁴. (b) $a = 6t^2$, $v = 2t^3 - 8$, $x = t^4/2 - 8t + 8$.
11.15 800 m/s² ↑.
11.16 (a) -2.43×10^6 ft/s². (b) 1.366×10^{-3} s.
11.17 (a) 5.89 ft/s. (b) 1.772 ft.
11.18 167.1 mm/s² ↑, 15.19 m/s² ↑.
11.21 (a) 2.52 m²/s², 4.70 m/s.
11.22 (a) 10.00 ft. (b) 1.833 ft/s, 0.440 ft/s².
11.23 (a) 42.0 ft. (b) 12.86 ft/s.
11.24 (a) 29.3 m/s. (b) 0.947 s.
11.25 (a) 4.76 mm/s. (b) 0.171 s.
11.26 1.995 m/s².
11.27 (a) -0.0525 m/s². (b) 6.17 s.
11.28 (a) 7.15 mi. (b) -275×10^{-6} ft/s². (c) 49.9 min.
11.31 (a) $2.36 v_0 T$, $\pi v_0/T$. (b) $0.363 v_0$.
11.32 $r + \frac{d_{\max}}{2} \cos \theta, -v_{\max} \sin \theta, -\frac{d_{\max}}{2} \dot{\theta} \sin \theta - \frac{2v_{\max}^2}{d_{\max}} \cos \theta$.
11.33 (a) 2.0 m/s². (b) 60.0 m/s.
11.34 (a) -0.417 m/s². (b) 18.00 km/h.
11.35 (a) 6.0 s. (b) 180.0 ft.
11.36 (a) 252 ft/s. (b) 1076 ft.
11.39 11.60 s, 50.4 m.
11.40 (a) 1.563 m/s². (b) 3.13 m/s².
11.41 (a) -3.20 ft/s² and 3.72 ft/s². (b) 3.41 s before A reaches the exchange zone.
11.42 (a) 15.05 s, 734 ft from the initial point of A.
 (b) A: 42.5 mi/h. B: 23.7 mi/h.
11.43 (a) $\mathbf{a}_A = 0.767$ ft/s² ←, $\mathbf{a}_B = 0.834$ ft/s² →.
 (b) 20.7 s. (c) 51.8 mi/h.
11.44 (a) 1.330 s. (b) 4.68 m below the man.
11.47 (a) 8.00 m/s ↑. (b) 4.00 m/s ↑. (c) 12.00 m/s ↑.
 (d) 8.00 m/s ↑.
11.48 (a) $\mathbf{a}_E = 2.40$ ft/s² ↑, $\mathbf{a}_C = 4.80$ ft/s² ↓.
 (b) 12.00 ft/s ↑.
11.49 (a) 0.125 m/s ↑. (b) 0.5154 m/s ∠ 14°.
11.50 (a) 18 ft/s² ←, 6 ft/s² ↑. (b) 9 ft/s ←, 2.25 ft ←.
11.51 (a) 200 mm/s →. (b) 600 mm/s →.
 (c) 200 mm/s ←. (d) 400 mm/s →.
11.52 (a) $\mathbf{a}_A = 13.33$ mm/s² ←, $\mathbf{a}_B = 20.0$ mm/s² ←.
 (b) 13.33 mm/s² →. (c) 70.0 mm/s →. 440 mm →.
11.55 (a) 2.5 s. (b) 7.5 in. ↓.
11.56 (a) 1.000 s. (b) 3.00 in. ↓.
11.57 (a) $\mathbf{a}_A = 345$ mm/s² ↓, $\mathbf{a}_B = 240$ mm/s² ↑.
 (b) $(\mathbf{v}_A)_0 = 43.3$ mm/s ↑, $(\mathbf{v}_C)_0 = 130.0$ mm/s →.
 (c) 728 mm →.
11.58 (a) 10.00 mm/s →. (b) $\mathbf{a}_A = 2.00$ mm/s² ↑,
 $\mathbf{a}_C = 6.00$ mm/s² →. (c) 175.0 mm ↑.
11.61 88 ft.
11.62 (b) 5.83 s.
11.63 (a) 10 s to 26 s, $a = -5.00$ m/s²;
 41 s to 46 s, $a = 3.00$ m/s²; otherwise $a = 0$.
 (b) 1383 m. (c) 9.00 s, 49.5 s.
11.64 (a) Same as Prob. 11.63. (b) 420 m. (c) 10.69 s, 40.0 s.
11.65 (a) 162 ft. (b) 18 s and 30 s.
11.66 (a) 44.8 s. (b) 103.3 m/s².
11.69 (a) 0.600 s. (b) 0.200 m/s, 2.84 m.
11.70 (a) 60.0 m/s, 1194 m. (b) 59.3 m/s.
11.71 (a) A: 52.2 s, B: 52.0 s. (b) 1.879 m.
11.72 9.39 s.
11.73 8.54 s, 58.3 mi/h.
11.74 77.5 ft.
11.75 5.67 s.
11.78 (a) 18.00 s. (b) 178.8 m. (c) 34.7 km/h.
11.79 (a) 5.01 min. (b) 19.18 mi/h.
11.80 (a) 2.00 s. (b) 1.200 ft/s, 0.600 ft/s.
11.83 (a) 2.96 s. (b) 224 ft.
11.84 (a) 163.0 in/s². (b) 114.3 in/s².
11.85 (a) 15.49 s. (b) 4.65 m/s. (c) 2.90 m/s, 8.50 m.
11.89 (a) 6.28 m/s ∠ 37.2°. (b) 7.49 m.
11.90 (a) 67.1 mm/s ∠ 63.4°, 256 mm/s² ∠ 69.4°.
 (b) 8.29 mm/s ∠ 36.2°, 336 mm/s² ∠ 86.6°.
11.91 (a) $(-12.57 \text{ in/s})\mathbf{i}$, $(-39.5 \text{ in/s}^2)\mathbf{j}$. (b) $y = x^2/8 - 1$.
11.92 (a) max: 15.00 ft/s, min: 5.00 ft/s
 (b) min: $t = 2\pi N$ s, $x = 20\pi N$ ft, $y = 5$ ft, $v_x = 5$ ft/s,
 $v_y = 0$, $\theta = 0$.
 max: $t = (2N + 1)\pi$ s, $x = 20\pi(N + 1)$ ft, $y = 15$ ft,
 $v_x = 15$ ft/s, $v_y = 0$, $\theta = 0$.
11.95 $\sqrt{R^2(1 + w_n^2 t^2) + c^2}$, $Rw_n \sqrt{4 + w_n^2 t^2}$.
11.97 1140 ft.
11.98 (a) 2.94 s. (b) 84.9 m. (c) 10.62 m.
11.99 (a) 115.3 km/h ≤ v_0 ≤ 148.0 km/h.
 (b) $h = 0.788$ m, $\alpha = 6.66^\circ$; $h = 1.068$ m, $\alpha = 4.05^\circ$.
11.100 15.38 ft/s < v_0 < 35.0 ft/s.
11.102 (a) Meets max. height requirement. (b) 0.937 m.
11.103 (a) Ball clears the net. (b) 7.01 m from the net.
11.105 22.9 ft/s.
11.106 16.20 m/s < v_0 < 21.0 m/s.
11.107 (a) 29.8 ft/s. (b) 29.6 ft/s.
11.108 37.7 m/s < v_0 < 44.3 m/s.
11.111 (a) 10.38°. (b) 9.74°.
11.112 (a) 4.17°. (b) 285 m. (c) 15.89 s.
11.113 (a) 14.66°. (b) 0.1074 s.
11.114 (a) 4.98 m. (b) 23.8°.
11.117 17.80 ft/s ∠ 50.9°.
11.118 $\mathbf{v}_A = 125$ mm/s ↑, $\mathbf{v}_B = 75$ mm/s ↓, $\mathbf{v}_C = 175$ mm/s ↓.
11.119 (a) 91.0 ft/s ∠ 47.0°. (b) 364 ft ∠ 47.0°. (c) 293 ft.
11.120 3.20 km/h ∠ 17.8°.
11.123 (a) 4 ft/s ↑. (b) 6 ft/s² ↓.
11.124 (a) 8.53 in/s ∠ 54.1°. (b) 6.40 in/s ∠ 54.1°.
11.125 (a) 0.979 m. (b) 12.55 m/s ∠ 86.5°.
11.126 (a) 0.835 mm/s² ∠ 75°. (b) 8.35 mm/s ∠ 75°.
11.127 (a) 5.18 ft/s ∠ 15°. (b) 1.232 ft/s ∠ 15°.
11.128 10.54 ft/s ∠ 81.3°.
11.129 5.96 m/s ∠ 82.8°.
11.131 15.79 km/h ∠ 26.0°.
11.133 500 m.
11.134 97.6 km/h.
11.135 12.13 m/s.
11.136 (a) 0.407 ft/s². (b) 0.0333 ft/s². (c) 0.00593 ft/s².
11.137 8.56 s.

- 11.138** (a) 10.20 mm/s². (b) 25.2 s.
11.139 (a) 178.9 m. (b) 1.118 m/s².
11.141 (a) 189.5 km/h $\approx$ 54.0°. (b) 21.8 m/s² $\approx$ 5.3°.
11.143 (a) 1.047i - 33.726j m/s². (b) -47.55i - 8.64j m/s.
11.144 1467.9 m.
11.145 (a) 281 m. (b) 209 m.
11.146 (a) 27.6 m. (b) 34.0 m.
11.147 (a) 0.634 m. (b) 9.07 m.
11.149 (a) 14.48 m/s. (b) 21.3 m.
11.151 $(R^2 + c^2)/2w_n R$.
11.152 2.50 ft.
11.153 149.8 Gm.
11.154 1425 Gm.
11.155 16 200 mi/h.
11.156 7740 mi/h.
11.159 1.606 h.
11.161 (a) $(1.624 \text{ in/s})\mathbf{e}_r - (15.56 \text{ in/s})\mathbf{e}_\theta$
 (b) $(-49.9 \text{ in/s}^2)\mathbf{e}_r + (-9.74 \text{ in/s}^2)\mathbf{e}_\theta$
 (c) $(-3.25 \text{ in/s}^2)\mathbf{e}_r$.
11.162 (a) $3\pi b\mathbf{e}_\theta$ and $-4\pi^2 b\mathbf{e}_r$. 14.48. (b) $\theta = 2N\pi$, $N = 0, 1, 2, \dots$.
11.163 13.280 m/s $\approx$ 27.08°, 0.2437 m/s² $\approx$ 30.00°.
11.164 (b) 1.787 m/s².
11.165 (a) $\mathbf{v} = b k \mathbf{e}_\theta$, $\mathbf{a} = -(bk^2/2)\mathbf{e}_r$.
 (b) $\mathbf{v} = 2bk\mathbf{e}_r + 2bk\mathbf{e}_\theta$, $\mathbf{a} = 2bk^2\mathbf{e}_r + 4bk^2\mathbf{e}_\theta$.
11.166 (a) $a = 4b\theta^2$. (b) directed toward point A.
11.169 $\dot{r} = 370 \text{ ft/s}$, $\ddot{r} = 57.9 \text{ ft/s}^2$, $\dot{\theta} = -0.0924 \text{ rad/s}$,
 $\ddot{\theta} = 0.0315 \text{ rad/s}^2$.
11.170 (a) $\dot{r} = -dw/2$, $\dot{\theta} = w/2$. (b) $\ddot{r} = -\sqrt{3} dw^2/4$, $\ddot{\theta} = 0$.
11.171 185.7 km/h.
11.172 61.8 mi/h, 49.7°.
11.175 $be^{i\theta^2}(\theta^2 + 4)^{\frac{1}{2}}\omega^2$.
11.176 $\frac{b}{\theta^4}(36 + 4\theta^2 + \theta^4)^{\frac{1}{2}}\omega^2$.
11.177 $v = 2\pi\sqrt{A^2 + n^2B^2\cos^2 2\pi nt}$,
 $a = 4\pi^2\sqrt{A^2 + n^4B^2\sin^2 2\pi nt}$
11.179 (a) $v = \sqrt{A^2 + B^2}$, $a = \sqrt{(1 + 16\pi^2)A^2 + B^2}$.
 (b) $v = 2\pi A$, $a = 4\pi^2 A$.
11.180 $\tan^{-1}[R(2 + w_n^2)/c\sqrt{4 + w_n^2}]$.
11.181 (a) $\theta_x = 90^\circ$, $\theta_y = 123.7^\circ$, $\theta_z = 33.7^\circ$.
 (b) $\theta_x = 103.4^\circ$, $\theta_y = 134.3^\circ$, $\theta_z = 47.4^\circ$.
11.182 (a) 1.00 s and 4.00 s. (b) 1.500 m, 24.5 m.
11.183 (a) 9.6 s. (b) 543.0 m.
11.185 (a) 111.4 km/h $\approx$ 10.50°. (b) 2.96 km.
11.187 (a) $\mathbf{a}_B = 2.00 \text{ in/s}^2 \uparrow$, $\mathbf{a}_C = 3.00 \text{ in/s}^2 \downarrow$. (b) 0.667 s.
 (c) 0.667 in. $\uparrow$.
11.188 (a) 38.1 m/s, 20.4 m. (b) 41.1 m/s, 29.6 m.
11.189 (a) 3.21 ft/s² $\approx$ 22.4°. (b) 6.43 ft/s² $\approx$ 22.4°.
11.190 $1.097\mathbf{e}_r + 19.71\mathbf{e}_n \text{ m/s}^2$.
11.191 (a) 23.4 ft/s. (b) 103.2 ft.

CHAPTER 12

- 12.1** (a) 844 lb. (b) 26.2 slugs.
12.2 (a) 0°: 4.987 lb, 45°: 5.000 lb, 90°: 5.013 lb.
 (b) 5.000 lb. (c) 0.1554 lb·s²/ft.
12.3 $2.84 \times 10^6 \text{ kg}\cdot\text{m/s}$.
12.5 0.242 mi.
12.6 (a) 1449 ft. (b) 10.0 s.
12.7 (a) 18.84 s. (b) 36.14 m.
12.8 (a) 110.5 km/h. (b) 85.6 km/h. (c) 69.9 km/h.
12.9 (a) 40.1 m. (b) 47.0 m.
12.10 (a) 2.22 s. (b) 3.32 m.
12.11 51.0 m.
12.12 (a) 234 m. (b) 3.33 kN (tension).
12.15 (a) (1): 10.73 ft/s² $\downarrow$, (2): 16.10 ft/s² $\downarrow$, (3): 0.749 ft/s² $\downarrow$.
 (b) (1): 14.65 ft/s $\downarrow$, (2): 17.94 ft/s $\downarrow$, (3): 3.87 ft/s $\downarrow$.
 (c) (1): 1.864 s, (2): 1.242 s, (3): 26.7 s.
12.16 $\mathbf{a}_A = 0.997 \text{ ft/s}^2 \nearrow 15^\circ$, $\mathbf{a}_B = 1.619 \text{ ft/s}^2 \nearrow 15^\circ$.
12.17 (a) 765 lb. (b) 1016 lb.
12.18 (a) 0.986 m/s² $\leq 25^\circ$. (b) 51.7 N.
12.19 (a) 1.794 m/s² $\leq 25^\circ$. (b) 58.2 N.
12.20 (a) 16.19 kN. (b) 2.45 m/s².
12.23 $\mathbf{a}_1 = 19.53 \text{ m/s}^2 \nearrow 65^\circ$, $\mathbf{a}_2 = 4.24 \text{ m/s}^2 \searrow 65^\circ$.
12.24 1.598 km.
12.25 (a) 335 m. (b) 73.6 mm/s $\downarrow$.
12.27 $\sqrt{k/m}(\sqrt{l^2 + x_0^2} - l)$.
12.28 (a) 10.00 N. (b) 103.1 N.
12.29 (a) 8.94 ft/s² $\leftarrow$, 18.06 lb.
 (b) 12.38 ft/s² $\leftarrow$, 15.38 lb. (c) Same as (b).
12.30 20.26 kg.
12.31 (a) 2.43 lb. (b) $\mathbf{a}_A = 3.14 \text{ ft/s}^2 \rightarrow$,
 $\mathbf{a}_B = 0.881 \text{ m/s}^2 \rightarrow$, $\mathbf{a}_C = 5.41 \text{ m/s}^2 \downarrow$.
12.34 0.0740 m/s² $\nearrow 20^\circ$, 137.2 N.
12.35 (a) 5.94 m/s² $\approx 75.6^\circ$. (b) 3.74 m/s $\approx 20^\circ$.
12.36 (a) 49.9°. (b) 6.85 N.
12.37 (a) 80.4 N. (b) 2.30 m/s.
12.38 (a) 22.55 s. (b) 6.379°.
12.39 3.47 m/s.
12.40 $3.01 \text{ m/s} \leq v \leq 3.85 \text{ m/s}$.
12.42 $9.00 \text{ ft/s} < v_C < 12.31 \text{ ft/s}$.
12.43 $2.42 \text{ ft/s} < v < 13.85 \text{ ft/s}$.
12.44 (a) 122.2 lb. (b) 145.6 lb.
12.45 (a) 668 ft. (b) 120.0 lb $\uparrow$.
12.46 434 N.
12.47 (a) 4.63 m/s². (b) 1.962 m/s². (c) 0.1842 m.s².
12.48 77.23 rpm.
12.49 (a) 2.91 N. (b) 13.09°.
12.50 1126 N $\leq 25.6^\circ$.
12.51 (a) 12.19 m/s. (b) 2290 N.
12.53 (a) 0.1858 W. (b) 10.28°.
12.55 7.67 m/s.
12.56 (a) 12.00 m/s. (b) $2.05 \times 10^{-3} \text{ N}$.
12.57 0.236.
12.58 3.71 m.
12.61 0.400.
12.62 (a) 0.1834. (b) left: 10.39°, right 169.6°.
12.63 (a) 2.98 ft/s. (b) left: 19.29°, right 160.7°.
12.64 0°, 180°, and 69.6°.
12.65 (a) no sliding, 0.611 N $\nearrow 75^\circ$. (b) sliding, 0.957 N $\searrow 40^\circ$.
12.66 (a) 289.1 lb.
12.67 -2.17 lb and 64.9 lb.
12.68 2.00 s.
12.69 (a) 7.47 N $\nearrow 45^\circ$. (b) 6.94 m/s² $\approx 45^\circ$.
12.71 (a) 126.6 N. (b) 5.48 m/s² $\rightarrow$. (c) 4.75 m/s² $\downarrow$.
12.72 (a) 142.7 N. (b) 6.18 m/s² $\rightarrow$. (c) 4.10 m/s² $\downarrow$.
12.74 $v_r = v_0 \sin 2\theta / \sqrt{\cos 2\theta}$, $v_\theta = v_0 \sqrt{\cos 2\theta}$.
12.77 (a) 0. (b) $8m v_0^2/r_0$.
12.78 $413 \times 10^{21} \text{ lb}\cdot\text{s}^2/\text{ft}$.
12.79 $383 \times 10^3 \text{ km}$, $238 \times 10^3 \text{ mi}$.
12.80 (a) 35 800 km, 22 200 mi. (b) 3.07 km/s, $10.09 \times 10^3 \text{ ft/s}$.
12.81 (b) 24.8 m/s².
12.82 (a) $1.998 \times 10^{30} \text{ kg}$. (b) 276 m/s².

- ## CHAPTER 13

- 13.41** 41.8°.
13.44 2.30 m/s.
13.45 (a) 27.4°. (b) 3.81 ft.
13.46 (a) 57.2 kW. (b) 269 kW.
13.47 (a) 2.75 kW. (b) 3.35 kW.
13.48 14.80 kN.
13.51 (a) 14.95 kW. (b) 45.4 kW.
13.52 (a) 17.75 kW. (b) 46.7 kW.
13.54 (a) 8.00 hp. (b) 7.91 hp.
13.55 (a) $k_1 k_2 / (k_1 + k_2)$. (b) $k_1 + k_2$.
13.57 (a) 5.12 m/s. (b) 4.20 m/s.
13.58 49.0 ft/s.
13.59 23.1 ft/s.
13.62 (a) 533 lb/ft. (b) 37.0 ft.
13.64 (a) 2.48 m/s ←. (b) 1.732 m/s ↑.
13.65 (a) 2.92 m/s. (b) $(-33.9 \text{ N})\mathbf{i} + (33.3 \text{ N})\mathbf{j}$.
13.66 (a) 43.5°. (b) 8.02 ft/s ↓.
13.68 0.269 m.
13.69 0.1744 m.
13.70 731 N.
13.71 (max) 5520 N at D ; (min) 731 N just above B .
13.72 14.34 ft/s, 13.77 lb ↑.
13.74 Loop 1: (a) 25.1 ft/s. (b) 1.500 lb ←.
 Loop 2: (a) 24.1 ft/s. (b) 1.000 lb.
13.76 Loop 1: (a) $\sqrt{5gr}$. (b) 3 W →.
 Loop 2: (a) $\sqrt{4gr}$. (b) 2 W →.
13.77 0.488 m.
13.78 $3/5I$.
13.80 $V = -\ln xyz$.
13.81 (a) $(k - 1)a^2/2$, not conservative. (b) 0, conservative.
13.82 (a) $P_x = x/R$, $P_y = y/R$, $P_z = z/R$, where
 $R = (x^2 + y^2 + z^2)^{1/2}$.
 (b) $U_{OABD} = -\Delta V_{OD} = a\sqrt{3}$.
13.85 (a) 62.5 MJ/kg. (b) 11.18 km/s.
13.86 (a) 9.56 km/s. (b) 2.39 km/s.
13.87 (a) 50.1×10^9 ft·lb. (b) 115.9×10^9 ft·lb.
13.88 (a) 1.918×10^6 ft·lb/lb. (b) 10.51×10^6 ft·lb/lb.
13.89 25.1 Mm/h.
13.90 6.48 km/s.
13.93 $v_r = \pm 3.87 \text{ m/s}$, $v_\theta = 1.000 \text{ m/s}$.
13.94 (a) 0.720 m. (b) 0.834 m/s.
13.95 3.77 in, $(28.04 \text{ ft/s})\mathbf{e}_r + (7.96 \text{ ft/s})\mathbf{e}_r$.
13.96 (a) 14.36 ft/s. (b) 1.225 ft.
13.97 (a) 4.14 ft/s. (b) 16.58 ft/s.
13.100 $27.6 \times 10^3 \text{ km/h}$.
13.101 (a) 7960 ft/s. (b) 4820 ft/s.
13.102 (a) 16 800 ft/s. (b) 32 700 ft/s.
13.103 14.20 km/s.
13.106 (a) 7.35 km/s. (b) 45.0°.
13.107 68.9°.
13.108 $r_{\max} = r_0(1 + \sin \alpha)$, $r_{\min} = (1 - \sin \alpha)r_0$.
13.109 3450 m/s.
13.110 (a) $11.32 \times 10^3 \text{ ft/s}$. (b) $13.68 \times 10^3 \text{ ft/s}$.
13.111 $30.9 \times 10^3 \text{ ft/s}$, 58.9°.
13.115 (b) $v_{\text{esc}}\sqrt{\alpha/(1 + \alpha)} < v_0 < v_{\text{esc}}\sqrt{(1 + \alpha)/(2 + \alpha)}$.
13.119 4 min 19 s.
13.120 (a) 3.64 s. (b) 27.3 s.
13.121 17.86 lb.
13.123 6.26 s.
13.124 (a) 2280 lb. (b) 3.00 s.
13.125 0.278.
13.126 (a) 18.16 s. (b) 1.94 km.

- 13.129** (a) 14.78 s. (b) 693 lb (tension).
13.130 (a) 29.6 s. (b) 2500 lb (tension).
13.131 (a) 5.28 s. (b) 17.05 kN (compression).
13.132 (a) 0.549 s. (b) 56.8 N.
13.134 (a) 3730 lb. (b) 7450 lb.
13.136 223 MPa.
13.138 15.36 mi/h.
13.139 76.9 lb.
13.140 1.449 kips.
13.141 6.21 W.
13.142 2.68 kN.
13.145 (a) 1.67 mi/h $\leftarrow$. (b) 0.190 s.
13.146 (a) car A. (b) 115.2 km/h.
13.147 65.0 kN.
13.148 (a) 9.32 ft·lb, 0.932 lb·s.
(b) 7.99 ft·lb, 0.799 lb·s.
13.149 497 ft/s.
13.150 (a) 2.80 ft/s $\leftarrow$. (b) 0.229 ft/s $\leftarrow$.
13.151 (a) 1.694 m/s $\downarrow$. (b) 0.1619 J.
13.152 (a) 778.9 m/s. (b) 4.65 J. (c) 19.74 N.
13.155 (a) $v_A = 0.594$ m/s $\leftarrow$, $v_B = 1.156$ m/s $\rightarrow$. (b) 2.99 J.
13.156 $(1 - e^2)mv^2$.
13.157 $0.728 \leq e \leq 0.762$.
13.158 (a) 3.00 lb. (b) $2.00 \text{ lb} \leq W_B \leq 6.00 \text{ lb}$.
13.161 (a) $v_0(1 - e)/2$ and $v_0(1+e)/2$. (b) $v_0(1 - e)^2/4$ and $v_0(1 + e)^2/4$.
(c) $v_0(1 + e)^{n-1}/2^{n-1}$. (d) $0.698v_0$.
13.163 0.294 m/s $\leftarrow$.
13.164 $\mathbf{v}'_A = 0.711 v_0 \angle 39.3^\circ$, $\mathbf{v}'_B = 0.636 v_0 \angle 45^\circ$.
13.165 (a) $0.848v_0 \angle 27.0^\circ$. (b) $0.456v_0 \angle 57.6^\circ$.
13.166 $\mathbf{v}'_A = 6.37 \text{ m/s} \angle 77.2^\circ$, $\mathbf{v}'_B = 1.802 \text{ m/s} \angle 40^\circ$.
13.167 $\mathbf{v}'_A = 1.322 \text{ m/s} \angle 70.9^\circ$, $\mathbf{v}'_B = 3.85 \text{ m/s} \angle 27.0^\circ$.
13.168 (a) 70.2° . (b) 0.322 m/s.
13.169 0.837.
13.172 13.09 m/s $\angle 26.6^\circ$.
13.174 (a) 20.6 mi/h. (b) 0.203.
13.175 (a) 0.294 m. (b) 54.4 mm.
13.176 (a) 0.324. (b) 14.30 ft/s.
13.177 (a) 2.90 m/s. (b) 100.5 J.
13.179 (a) 8.89 mm. (b) 3758 N.
13.180 (a) 0.588. (b) 148.7 kN/m.
13.182 (a) $\mathbf{v}'_A = 0$, $\mathbf{v}'_B = 0$.
(b) $\mathbf{v}'_A = 1.201 \text{ m/s} \leftarrow$, $\mathbf{v}'_B = 0.400 \text{ m/s} \rightarrow$.
13.183 45.5 mm.
13.184 (a) 26.65 ft/s $\angle 30^\circ$.
(b) 31.93 ft/s $\angle 39.0^\circ$.
13.185 3.47 in.
13.186 (a) 0.923. (b) 1.278 m.
13.188 (a) $\mathbf{v}'_A = 2.36 \text{ ft/s} \angle 83.8^\circ$, $\mathbf{v}'_B = 3.23 \text{ ft/s} \rightarrow$. (b) 1.97 in.
13.190 102.6 mi/h.
13.191 1.688 ft·lb.
13.194 0.283.
13.195 (a) 13.31 N $\rightarrow$. (b) 4.49 N $\downarrow$. (c) 13.31 N $\leftarrow$.
13.197 (a) 217 mm. (b) 69.1 mm.
13.198 (a) $\mathbf{v}'_A = \mathbf{v}'_B = \mathbf{v}'_C = 1.368 \text{ m/s}$. (b) 0.668 m. (c) 1.049 m.
13.200 0.107 m.

CHAPTER 14

- 14.1** (a) 4.46 m/s $\leftarrow$. (b) 0.409 m/s $\leftarrow$.
14.2 10.67 km/h $\leftarrow$, 4.27 km/h $\leftarrow$, and 4.27 km/h $\leftarrow$.
14.3 (a) 4.25 ft/s $\rightarrow$. (b) 4.25 ft/s $\rightarrow$.
14.4 (a) 0.800 oz. (b) 900 ft/s $\rightarrow$.
14.7 (a) 3.79 km/h $\rightarrow$, 2.77 km/h $\rightarrow$.
(b) 5.54 km/h $\rightarrow$, 2.77 km/h $\rightarrow$.
(c) 5.54 km/h $\rightarrow$, 3.60 km/h $\rightarrow$.
14.8 $v_A = 1.013 \text{ m/s} \leftarrow$, $v_B = 0.338 \text{ m/s} \leftarrow$, $v_C = 0.150 \text{ m/s} \leftarrow$.
14.9 $-(600 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{i} - (1070.0 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{j} + (370.0 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{k}$.
14.10 (a) $(22.78 \text{ m})\mathbf{i} + (15.00 \text{ m})\mathbf{j} + (11.67 \text{ m})\mathbf{k}$.
(b) $(38.0 \text{ kg} \cdot \text{m/s})\mathbf{i} + (32.0 \text{ kg} \cdot \text{m/s})\mathbf{j} + (40.0 \text{ kg} \cdot \text{m/s})\mathbf{k}$.
(c) $-(826.67 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{i} - (602.22 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{j} + (211.11 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{k}$.
14.11 (a) $\mathbf{v}_A = (4.00 \text{ ft/s})\mathbf{j}$, $\mathbf{v}_B = (1.000 \text{ ft/s})\mathbf{i}$, $\mathbf{v}_C = (3.00 \text{ ft/s})\mathbf{k}$.
(b) $(1.20 \text{ ft} \cdot \text{lb} \cdot \text{s})\mathbf{i} + (0.60 \text{ ft} \cdot \text{lb} \cdot \text{s})\mathbf{j} - (2.40 \text{ ft} \cdot \text{lb} \cdot \text{s})\mathbf{k}$.
14.12 (a) $\mathbf{v}_A = (10.00 \text{ ft/s})\mathbf{j}$, $\mathbf{v}_B = (5.00 \text{ ft/s})\mathbf{i}$, $\mathbf{v}_C = (10.00 \text{ ft/s})\mathbf{k}$.
(b) $(6.00 \text{ ft} \cdot \text{lb} \cdot \text{s})\mathbf{i} + (3.00 \text{ ft} \cdot \text{lb} \cdot \text{s})\mathbf{j} - (6.00 \text{ ft} \cdot \text{lb} \cdot \text{s})\mathbf{k}$.
14.15 $(114.4 \text{ m})\mathbf{i} - (76.1 \text{ m})\mathbf{j} + (8.75 \text{ m})\mathbf{k}$.
14.16 $(1180 \text{ m})\mathbf{i} + (140 \text{ m})\mathbf{j} + (155 \text{ m})\mathbf{k}$.
14.19 $x = 45.2 \text{ ft}$, $y = 54.5 \text{ ft}$.
14.20 (a) 2.00 s. (b) 92.8 mi/h.
14.21 $(81.5 \text{ ft})\mathbf{i} + (351 \text{ ft})\mathbf{k}$.
14.22 (a) 8.00 ft/s $\rightarrow$. (b) 36.6° , $v_C = 10.39 \text{ ft/s}$, $v_D = 8.72 \text{ ft/s}$.
14.24 $v_A = 431 \text{ m/s}$, $v_B = 395 \text{ m/s}$, $v_C = 528 \text{ m/s}$.
14.25 $v_A = 646 \text{ m/s}$, $v_B = 789 \text{ m/s}$, $v_C = 176 \text{ m/s}$.
14.26 $v_A = 919 \text{ m/s}$, $v_B = 717 \text{ m/s}$, $v_C = 619 \text{ m/s}$.
14.31 friction: 2.97 J, first impact: 3007 J, second impact: 24.3 J.
14.32 (a) 23.6 ft·lb. (b) 2.85 ft·lb.
14.33 (woman) 382 ft·lb, (man) 447 ft·lb.
14.35 (b) $E_A = 180.0 \text{ kJ}$, $E_B = 320 \text{ kJ}$.
14.37 (a) $\mathbf{v}_B = \frac{m_A v_0}{m_A + m_B} \rightarrow$. (b) $h = \frac{m_A}{m_A + m_B} \frac{v_0^2}{2g}$.
14.38 $\mathbf{v}_A = 4.11 \text{ m/s} \angle 46.9^\circ$, $\mathbf{v}_B = 17.39 \text{ m/s} \angle 16.7^\circ$.
14.39 (a) $\mathbf{v}_{B/A} = 11.59 \text{ ft/s} \angle 30^\circ$. (b) $\mathbf{v}_A = 3.76 \text{ ft/s} \rightarrow$.
14.40 $\mathbf{v}_A = 3.11 \text{ ft/s} \leftarrow$, $\mathbf{v}_B = 4.66 \text{ ft/s} \rightarrow$.
14.41 $v_A = 7.50 \text{ ft/s}$, $v_B = 6.50 \text{ ft/s}$, $v_C = 11.25 \text{ ft/s}$.
14.42 $v_A = 10.61 \text{ ft/s}$, $v_B = 9.19 \text{ ft/s}$, $v_C = 5.30 \text{ ft/s}$.
14.45 $v_A = 0.218 \text{ m/s} \angle 53.1^\circ$ and $v_B = 1.813 \text{ m/s} \angle 43.8^\circ$.
14.46 $(200 \text{ ft/s})\mathbf{i} + (172 \text{ ft/s})\mathbf{j} + (1560 \text{ ft/s})\mathbf{k}$.
14.47 (a) $v_C = 11.00 \text{ ft/s}$, $v_D = 5.50 \text{ ft/s}$. (b) 0.786.
14.48 $x = 181.7 \text{ mm}$, $y = 0$, $z = 139.4 \text{ mm}$.
14.51 (a) $\mathbf{v}_B = 2.40 \text{ m/s} \angle 53.1^\circ$, $\mathbf{v}_C = 2.56 \text{ m/s} \rightarrow$, (b) $c = 1.059 \text{ m}$.
14.52 (a) $\mathbf{v}_A = 2.40 \text{ m/s} \downarrow$, $\mathbf{v}_B = 3.00 \text{ m/s} \angle 53.1^\circ$, (b) $a = 1.864 \text{ m}$.
14.55 (a) $\mathbf{v}_A = 2.25 \text{ ft} \uparrow$, $\mathbf{v}_B = 2.25 \text{ ft/s} \downarrow$, $\mathbf{v}_C = 3.90 \text{ ft/s} \rightarrow$. (b) 11.1 in.
14.56 (a) 2.00 ft/s $\rightarrow$. (b) 0.760 ft. (c) 5.29 rad/s $\downarrow$.
14.57 1086.5 N.
14.58 $\rho A_2 v_2^2 - \rho A_1 v_1^2 \cos \theta$.
14.59 drag = 26.3 lb $\rightarrow$, lift = 12.74 lb $\uparrow$.
14.60 drag = 34.6 lb $\rightarrow$, lift = 16.76 lb $\uparrow$.
14.61 (a) 14.8 kN. (b) 27.7 kN.
14.62 90.6 N $\leftarrow$.
14.64 $D_x = 329 \text{ N}$, $D_y = 0$, $C_x = -203 \text{ N}$, $C_y = 271 \text{ N}$.
14.66 (a) $\theta = 35.4^\circ$. (b) 187.3 N $\angle 53.8^\circ$.
14.67 (a) 26.0 m/s. (b) 230 N $\angle 48.4^\circ$.
14.68 $C_x = 90.0 \text{ N}$, $C_y = 2360 \text{ N}$, $D_x = 0$, $D_y = 2900 \text{ N}$.
14.69 100 kg/s.
14.70 7580 lb.
14.71 33.6 kN $\leftarrow$.
14.72 7180 lb.
14.74 (a) 9690 lb, 3.38 ft. (b) 6960 lb, 9.43 ft.
14.76 (a) $3.03 \text{ m/s}^2 \angle 18^\circ$. (b) 922 km/h.
14.77 (a) 30.6 m/s. (b) $96.1 \text{ m}^3/\text{s}$. (c) 55 100 N·m/s.
14.78 (a) 3.23 MW. (b) 0.464.
14.79 213 m.
14.80 (a) 15 450 hp. (b) 28 060 hp. (c) 0.551.
14.83 (a) $m_0 e^{qL/m_0 v_0}$. (b) $v_0 e^{-qL/m_0 v_0}$.
14.86 (a) $m(v^2 + gy)/l$. (b) $\mathbf{R} = mg(1 - y/l)\mathbf{j}$.

- 14.87** (a) $mg y/l$. (b) $m[g(l - y) + v^2]/l \uparrow$.
14.88 $\sqrt{gh} \tan h(\sqrt{gh} t/L)$.
14.89 10.10 ft/s.
14.90 4.75 ft/s.
14.91 533 kg/s.
14.94 (a) 90.0 m/s². (b) 35.9×10^3 km/h.
14.95 7930 m/s.
14.96 (a) 1800 m/s. (b) 9240 m/s.
14.99 87.2 mi.
14.100 (a) 92.8 ft/s² $\uparrow$. (b) 780 ft/s² $\uparrow$. (c) 119.3 mi. (d) 14660 mi/h.
14.101 186.8 km/h.
14.102 (a) 31.2 km. (b) 197.5 km.
14.106 (a) 1.595 m/s. (b) 0.370 m.
14.107 (a) 5.20 km/h $\rightarrow$. (b) 4.00 km/h $\rightarrow$.
14.108 (a) 6.05 ft/s. (b) 6.81 ft/s.
14.110 $\mathbf{v}_A = 15.38$ ft/s $\rightarrow$, $\mathbf{v}_B = 5.13$ ft/s $\leftarrow$.
14.112 $\mathbf{A}_x = 55.5$ lb $\rightarrow$, $\mathbf{A}_y = 20.2$ lb $\downarrow$, $\mathbf{m}_A = 41.4$ lb-ft $\downarrow$.
14.114 $\mathbf{D} = 2.29$ kN $\uparrow$, $\mathbf{C} = 1.712$ kN $\uparrow$.
14.115 414 rpm.
14.116 Case 1: (a) 0.333 g $\downarrow$. (b) $0.817\sqrt{gl}$.
 Case 2: (a) $gy/l \downarrow$. (b) $\sqrt{gl}$.

CHAPTER 15

- 15.1** (a) 29.6 rad/s. (b) 32.2 rev.
15.2 (a) 0.50 rad, -4.71 rad/s, -34.50 rad/s². (b) 0, -1.934 rad/s, 36.46 rad/s².
15.3 (a) 0.253 rad, -0.927 rad/s, -36.55 rad/s². (b) 0, 0, 0.
15.4 (a) -3.01 rad/s². (b) 13 800 rev.
15.5 (a) 150 rev. (b) 2100 rev.
15.6 (a) 0.855 rad/s. (b) 3.71° .
15.9 (a) 9.55 rev. (b) ∞ . (c) 7.82 s.
15.10 $-(0.450 \text{ m/s})\mathbf{i} - (1.200 \text{ m/s})\mathbf{j} + (1.500 \text{ m/s})\mathbf{k}$, $(12.60 \text{ m/s}^2)\mathbf{i} + (7.65 \text{ m/s}^2)\mathbf{j} + (9.90 \text{ m/s}^2)\mathbf{k}$.
15.11 $(0.750 \text{ m/s})\mathbf{i} + (1.500 \text{ m/s})\mathbf{k}$, $(12.75 \text{ m/s}^2)\mathbf{i} + (11.25 \text{ m/s}^2)\mathbf{j} + (3.00 \text{ m/s}^2)\mathbf{k}$.
15.12 $-(37.4 \text{ in/s})\mathbf{i} + (12.00 \text{ in/s})\mathbf{j} - (15.60 \text{ in/s})\mathbf{k}$, $-(126.1 \text{ in/s}^2)\mathbf{i} - (74.3 \text{ in/s}^2)\mathbf{j} + (246 \text{ in/s}^2)\mathbf{k}$.
15.13 $-(18.72 \text{ in/s})\mathbf{i} + (6.00 \text{ in/s})\mathbf{j} - (7.80 \text{ in/s})\mathbf{k}$, $-(3.46 \text{ in/s}^2)\mathbf{i} - (27.6 \text{ in/s}^2)\mathbf{j} + (73.1 \text{ in/s}^2)\mathbf{k}$.
15.16 66 700 mi/h, 19.47×10^{-3} ft/s².
15.17 (a) 1525 ft/s, 0.1112 ft/s². (b) 1168 ft/s, 0.0852 ft/s². (c) 0, 0.
15.18 (a) 2.50 rad/s $\uparrow$, 1.500 rad/s² $\downarrow$. (b) 771 mm/s² $\searrow 76.5^\circ$.
15.19 12.00 rad/s² $\uparrow$ or 12.00 rad/s² $\downarrow$.
15.22 left: 3.49 s; middle: 6.98 s; right: 13.96 s.
15.23 (a) 0.500 ft/s $\rightarrow$, 1.500 ft/s² $\leftarrow$. (b) 4.24 ft/s² $\searrow 45^\circ$.
15.24 (a) 300 rpm $\uparrow$, 100 rpm $\downarrow$. (b) $\mathbf{a}_B = 1974$ in/s² $\leftarrow$, $\mathbf{a}_C = 658$ in/s² $\rightarrow$.
15.25 (a) A: 15.00 rad/s $\uparrow$; B: 7.50 rad/s $\downarrow$. (b) A: 75.0 ft/s² $\uparrow$; B: 37.5 ft/s² $\downarrow$.
15.26 (a) C: 120 rpm; B: 275 rpm. (b) A: 23.7 m/s² $\uparrow$; B: 19.90 m/s² $\downarrow$.
15.27 (a) 10.00 rad/s. (b) A: 7.50 m/s²; B: 3.00 m/s² $\downarrow$. (c) 4.00 m/s² $\downarrow$.
15.28 (a) 0.400 rad/s² $\downarrow$. (b) 1.528 rev.
15.29 (a) 3.00 rad/s² $\downarrow$. (b) 4.00 s.
15.30 (a) 1.975 rad/s² $\uparrow$. (b) 6.91 rad/s $\uparrow$.
15.31 (a) 15.28 rev. (b) 10.14 s.
15.32 (a) 15.52 s. (b) $\omega_A = 445$ rpm $\uparrow$, $\omega_B = 371$ rpm $\downarrow$.
15.33 (a) $\alpha_A = 3.40$ rad/s² $\downarrow$, $\alpha_B = 1.963$ rad/s² $\downarrow$. (b) 9.23 s.
15.36 $b\omega_0^2/2\pi \rightarrow$.
15.37 $bv^2/2\pi r^3 \downarrow$.
15.38 $\mathbf{v}_B = 140.8$ ft/s $\rightarrow$, $\mathbf{v}_C = 0$, $\mathbf{v}_0 = 136.0$ ft/s $\searrow 15^\circ$, $\mathbf{v}_E = 99.6$ ft/s $\searrow 45^\circ$.
15.39 (a) 0.378 rad/s $\downarrow$. (b) 6.42 in/s $\uparrow$.
15.40 (a) 0.231 rad/s $\downarrow$. (b) $-(1.00 \text{ m/s})\mathbf{i} - (0.577 \text{ m/s})\mathbf{j}$.
15.41 (a) 3.00 rad/s $\downarrow$. (b) 1.30 m/s $\searrow 67.4^\circ$.
15.44 (a) 10.00 rad/s $\uparrow$. (b) $-(7.40 \text{ m/s})\mathbf{i} - (1.00 \text{ m/s})\mathbf{j}$.
15.45 (a) $-(1.40 \text{ m/s})\mathbf{i} - (1.00 \text{ m/s})\mathbf{j}$. (b) $x = 100.0$ mm, $y = -140.0$ mm.
15.47 (a) 0.583 rad/s $\downarrow$. (b) 1.537 ft/s $\searrow 77.48^\circ$.
15.48 (a) $\omega_B = \omega_C = \omega_D = \frac{1}{2}\omega_A \uparrow$. (b) $\omega_S = 0.25 \omega_A \downarrow$.
15.49 (a) $\omega_B = \omega_C = \omega_D = 150$ rpm $\downarrow$. (b) $\omega_S = 195$ rpm $\downarrow$.
15.50 (a) 48.0 rad/s $\downarrow$. (b) 3.39 m/s $\searrow 45^\circ$.
15.51 (a) 5.65 m/s $\uparrow$. (b) 9000 rpm, (c) 1500.
15.53 (a) 200 rad/s $\uparrow$. (b) 24.0 rad/s $\downarrow$.
15.55 (a) $(6.00 \text{ rad/s})\mathbf{k}$ or 6.00 rad/s $\uparrow$. (b) $(360 \text{ mm/s})\mathbf{i} - (672 \text{ mm/s})\mathbf{j}$ or 762 mm/s $\searrow 61.8^\circ$.
15.56 (a) 540 mm/s $\rightarrow$. (b) 457 mm/s $\searrow 61.8^\circ$.
15.57 (a) 4.38 rad/s $\downarrow$, 12.25 in/s $\uparrow$. (b) 0, 42.0 in/s $\downarrow$. (c) 4.38 rad/s $\uparrow$, 12.25 in/s $\downarrow$.
15.58 (a) 22.9° and 192.6° . (b) 5.60 rad/s $\downarrow$ and 5.60 rad/s $\uparrow$.
15.61 (a) $\mathbf{v}_P = 0$, $\omega_{BD} = 39.3$ rad/s $\uparrow$. (b) $\mathbf{v}_P = 6.28$ m/s $\downarrow$, $\omega_{BD} = 0$.
15.62 $\mathbf{v}_P = 6.52$ m/s $\downarrow$, $\omega_{BD} = 20.8$ rad/s $\uparrow$.
15.63 (a) 12.00 rad/s $\uparrow$. (b) 3.90 m/s $\searrow 67.4^\circ$.
15.64 $\omega_{DE} = 2.55$ rad/s $\downarrow$, $\omega_{BD} = 0.955$ rad/s $\uparrow$.
15.65 $\omega_{BD} = 4.00$ rad/s $\uparrow$, $\omega_{EB} = 0.600$ rad/s $\uparrow$.
15.68 (a) 3.33 rad/s $\uparrow$. (b) 2.00 m/s $\searrow 56.3^\circ$.
15.69 (a) 1.500 m. (b) 5.00 m/s $\downarrow$.
15.70 14.76 in/s $\rightarrow$.
15.71 (a) 338 mm/s $\leftarrow$, 0. (b) 710 mm/s $\leftarrow$, 2.37 rad/s $\downarrow$.
15.72 $(1 - r_A/r_C)\omega_{ABC}$.
15.74 (a) 1.714 in. below A. (b) 75.0 ft/s $\rightarrow$. (c) 53.2 ft/s $\searrow 41.2^\circ$.
15.75 $x = 0$, $z = 9.34$ ft.
15.76 (a) 3.00 rad/s $\uparrow$. (b) 300 mm/s $\leftarrow$. (c) 180.0 mm/s (wound).
15.77 (a) 3.00 rad/s $\downarrow$. (b) 180 mm/s $\rightarrow$. (c) 300 mm/s (unwound).
15.78 (a) 50 mm to the right of the axle. (b) $\mathbf{v}_B = 750$ mm/s $\downarrow$, $\mathbf{v}_D = 1.950$ m/s $\uparrow$.
15.79 (a) 25 mm to the right of O. (b) 420 mm/s $\uparrow$.
15.80 (a) A: 300 mm to the left of A. C: 600 mm to the left of C. (b) $\omega_A = 4.00$ rad/s $\downarrow$, $\omega_C = 2.00$ rad/s $\uparrow$.
15.82 (a) 0.467 rad/s $\uparrow$. (b) 3.49 ft/s $\searrow 59.2^\circ$.
15.83 (a) 3.08 rad/s $\uparrow$. (b) 83.3 m/s $\searrow 73.9^\circ$.
15.86 (a) 0.122 rad/s $\uparrow$. (b) 22.76 mm/s $\searrow 15^\circ$.
15.87 (a) 0.133 rad/s $\uparrow$. (b) 18.22 mm/s $\searrow 15^\circ$.
15.88 (a) $(v_A/l) \sin \beta / \cos(\beta - \theta)$. (b) $v_A \cos \theta / \cos(\beta - \theta)$.
15.89 (a) 6.72 ft/s $\searrow 45^\circ$. (b) 2.75 rad/s $\downarrow$. (c) 6.57 ft/s $\searrow 21.2^\circ$.
15.90 (a) 0.900 rad/s $\downarrow$. (b) 411 mm/s $\searrow 20.5^\circ$.
15.91 (a) 1.00 rad/s $\downarrow$. (b) 1.04 m/s $\rightarrow$.
15.94 (a) 1.58 rad/s $\downarrow$. (b) 28.0 in/s $\searrow 78.3^\circ$.
15.95 (a) $\omega_{AB} = 1.200$ rad/s $\downarrow$, $\omega_{DE} = 0.450$ rad/s $\downarrow$. (b) 5.25 in/s $\leftarrow$.
15.96 (a) 5.00 rad/s $\uparrow$. (b) 3.00 m/s $\downarrow$.
15.97 (a) 2.49 rad/s $\uparrow$. (b) 3.73 rad/s $\downarrow$. (c) 0.835 m/s $\searrow 53.6^\circ$.
15.98 (a) $\omega_{AB} = 1.177$ rad/s $\downarrow$, $\omega_{DE} = 2.50$ rad/s $\downarrow$. (b) 29.4 in/s $\leftarrow$.
15.99 Space centroid: quarter circle, $r = 15$ in, centered at O. Body centroid: semicircle, $r = 7.5$ in., centered midway between A and B.
15.100 Space centroid: lower rack. Body centroid: circumference of gear.

- 15.102** $\omega_{BD} = 0.955 \text{ rad/s} \downarrow$, $\omega_{DE} = 2.55 \text{ rad/s} \uparrow$.
15.103 $\omega_{BD} = 4.000 \text{ rad/s} \uparrow$, $\omega_{EB} = 0.600 \text{ rad/s} \uparrow$.
15.105 (a) $0.50 \text{ rad/s}^2 \downarrow$. (b) $\mathbf{a}_A = 3.25 \text{ m/s}^2 \uparrow$, $\mathbf{a}_E = 0.75 \text{ m/s}^2 \uparrow$.
15.106 (a) $0.20 \text{ m/s}^2 \downarrow$. (b) $2.20 \text{ m/s}^2 \uparrow$.
15.107 (a) $0.900 \text{ m/s}^2 \rightarrow$. (b) $1.800 \text{ m/s}^2 \leftarrow$.
15.108 (a) 0.600 m from A. (b) 0.200 m from A.
15.109 (a) $51.3 \text{ in/s}^2 \downarrow$. (b) $184.9 \text{ in/s}^2 \angle 16.1^\circ$.
15.110 (a) $1.039 \text{ rad/s}^2 \downarrow$. (b) $(2.60 \text{ ft/s}^2)\mathbf{i} + (4.50 \text{ ft/s}^2)\mathbf{j}$ or $5.20 \text{ ft/s}^2 \angle 60^\circ$.
15.111 (a) $1430 \text{ m/s}^2 \downarrow$. (b) $1430 \text{ m/s}^2 \uparrow$, (c) $1430 \text{ m/s}^2 \searrow 60^\circ$.
15.112 (a) $13.35 \text{ in/s}^2 \searrow 61.0^\circ$. (b) $12.62 \text{ in/s}^2 \angle 64.0^\circ$.
15.113 $\mathbf{a}_A = 56.6 \text{ in/s}^2 \searrow 58.0^\circ$, $\mathbf{a}_B = 80.0 \text{ in/s}^2 \uparrow$, $\mathbf{a}_C = 172.2 \text{ in/s}^2 \searrow 25.8^\circ$.
15.114 $\mathbf{a}_A = 48.0 \text{ in/s}^2 \uparrow$, $\mathbf{a}_B = 85.4 \text{ in/s}^2 \searrow 69.4^\circ$, $\mathbf{a}_C = 82.8 \text{ in/s}^2 \searrow 65.0^\circ$.
15.115 (a) $2.00 \text{ rad/s}^2 \downarrow$. (b) $0.224 \text{ m/s}^2 \searrow 63.4^\circ$.
15.118 (a) 92.5 in/s^2 . (b) 278 in/s^2 .
15.120 $148.3 \text{ m/s}^2 \downarrow$.
15.121 $296 \text{ m/s}^2 \uparrow$.
15.122 $\mathbf{a}_D = 1558 \text{ m/s}^2 \searrow 45^\circ$, $\mathbf{a}_E = 337 \text{ m/s}^2 \angle 45^\circ$.
15.124 (a) $242 \text{ in/s}^2 \leftarrow$. (b) $403 \text{ in/s}^2 \searrow 72.5^\circ$.
15.125 $694 \text{ in/s}^2 \leftarrow$.
15.127 $2.10 \text{ m/s}^2 \angle 47.1^\circ$.
15.128 (a) $1.47 \text{ rad/s}^2 \uparrow$. (b) $1.575 \text{ m/s}^2 \searrow 47.1^\circ$.
15.129 (a) $228 \text{ rad/s}^2 \uparrow$. (b) $92.0 \text{ rad/s}^2 \downarrow$.
15.130 (a) $138.1 \text{ ft/s}^2 \searrow 78.6^\circ$. (b) $203 \text{ ft/s}^2 \angle 19.5^\circ$.
15.132 (a) $4.18 \text{ rad/s}^2 \downarrow$. (b) $2.43 \text{ rad/s}^2 \downarrow$.
15.133 (a) $8.15 \text{ rad/s}^2 \uparrow$. (b) $0.896 \text{ rad/s}^2 \downarrow$.
15.134 (a) $3.70 \text{ rad/s}^2 \downarrow$. (b) $3.70 \text{ rad/s}^2 \downarrow$.
15.136 $\mathbf{v}_D = 1.382 \text{ m/s} \downarrow$, $\mathbf{a}_D = 0.695 \text{ m/s}^2 \downarrow$.
15.138 $\mathbf{v}_B = b\omega \cos \theta$, $\mathbf{a}_B = b\alpha \cos \theta - b\omega^2 \sin \theta$.
15.139 $\mathbf{v}_B \sin \beta/l \cos \theta$.
15.140 $(v_B \sin \beta/l)^2 \sin(\theta/\cos^3 \theta)$.
15.141 $v_x = v[1 - \cos(vt/r)]$, $v_y = v \sin(vt/r)$.
15.142 $\omega = bv_A(b^2 + x_A^2)^{-1/2}$, $\alpha = 2bx_A v_A/(b^2 + x_A^2)^{3/2}$.
15.143 $v_{B_x} = v_A - lb^2 v_A/(b^2 + x_A^2)^{3/2} \rightarrow$, $(v_{B_y})_x = lb x_A v_A/(b^2 + x_A^2)^{3/2} \uparrow$.
15.144 $\omega_{BD} = b\omega(b + l \cos \theta)/(l^2 + b^2 + 2bl \cos \theta) \downarrow$, $\mathbf{v}_E = bl\omega \sin \theta/(l^2 + b^2 + 2bl \cos \theta) \searrow$, $\tan^{-1}[(b \sin \theta/(l + b \cos \theta))]$.
15.145 $bl\omega^2(l^2 - b^2) \sin \theta/(l^2 + b^2 + 2bl \cos \theta) \uparrow$.
15.147 $\omega = v_0 \sin^2 \theta/r \cos \theta \uparrow$, $\alpha = (v_0/r)^2 (1 + \cos^2 \theta) \tan^3 \theta \uparrow$.
15.148 $(v_p)_x = r\omega \left[\cos \frac{r\omega t}{R-r} - \cos \omega t \right]$, $(v_p)_y = r\omega \left[\sin \frac{r\omega t}{R-r} + \sin \omega t \right]$.
15.149 Path is the y axis. $\mathbf{v} = (R\omega \sin \omega t)\mathbf{j}$, $\mathbf{a} = (R\omega^2 \cos \omega t)\mathbf{j}$.
15.150 $2.40 \text{ m/s} \searrow 73.9^\circ$.
15.151 $2.87 \text{ m/s} \searrow 44.8^\circ$.
15.152 (a) $1.815 \text{ rad/s} \downarrow$. (b) $16.42 \text{ in/s} \searrow 20^\circ$.
15.153 (a) $5.16 \text{ rad/s} \downarrow$. (b) $1.399 \text{ m/s} \searrow 60^\circ$.
15.154 (a) $3.81 \text{ rad/s} \downarrow$, $6.53 \text{ m/s} \angle 16.26^\circ$. (b) $3.00 \text{ rad/s} \downarrow$, $4.00 \text{ m/s} \rightarrow$.
15.155 (a) $11.25 \text{ rad/s} \uparrow$. (b) $75.0 \text{ in/s} \rightarrow$.
15.160 (a) $1.78 \times 10^{-3} \text{ m/s}^2$ west. (b) $1.36 \times 10^{-3} \text{ m/s}^2$ west. (c) $1.36 \times 10^{-3} \text{ m/s}^2$ west.
15.161 (a) $54 \text{ rad/s}^2 \downarrow$. (b) $33.9 \text{ ft/s}^2 \angle 45^\circ$.
15.162 0.0234 m/s^2 west.
15.164 (a) $0.520 \text{ m/s} \searrow 82.6^\circ$. (b) $50.0 \text{ mm/s}^2 \searrow 9.8^\circ$.
15.165 (a) $0.520 \text{ m/s} \searrow 37.4^\circ$. (b) $50.0 \text{ mm/s}^2 \searrow 69.8^\circ$.
15.166 (a) $1006 \text{ mm/s} \angle 72.6^\circ$. (b) $1811 \text{ mm/s}^2 \angle 32.0^\circ$.
15.167 (a) $1018 \text{ mm/s} \searrow 70.5^\circ$. (b) $1537 \text{ mm/s}^2 \searrow 2.4^\circ$.
15.168 (1) $303 \text{ mm/s}^2 \rightarrow$; (2) $168.5 \text{ mm/s}^2 \searrow 57.7^\circ$.
15.169 (3) $483 \text{ mm/s}^2 \leftarrow$; (4) $168.5 \text{ mm/s}^2 \searrow 57.7^\circ$.
15.170 $0.750 \text{ m/s} \angle 71.3^\circ$, $2.13 \text{ m/s}^2 \searrow 61.9^\circ$.
15.171 $2.79 \text{ rad/s} \downarrow$, $2.13 \text{ rad/s}^2 \downarrow$.
15.174 (a) $0.436 \text{ rad/s} \uparrow$. (b) $0.271 \text{ rad/s}^2 \uparrow$.
15.175 (a) $0.354 \text{ rad/s} \uparrow$. (b) $0.125 \text{ rad/s}^2 \uparrow$.
15.176 $7.86 \text{ rad/s} \uparrow$, $81.1 \text{ rad/s}^2 \uparrow$.
15.177 $3.81 \text{ rad/s} \downarrow$, $81.4 \text{ rad/s}^2 \downarrow$.
15.178 $1.526 \text{ rad/s} \downarrow$, $57.6 \text{ rad/s}^2 \downarrow$.
15.181 (a) $3.61 \text{ rad/s} \uparrow$. (b) $86.6 \text{ in/s} \angle 30^\circ$. (c) $563 \text{ in/s}^2 \searrow 46.1^\circ$.
15.182 (a) $3.61 \text{ rad/s} \downarrow$. (b) $86.6 \text{ in/s} \searrow 30^\circ$. (c) $563 \text{ in/s}^2 \searrow 46.1^\circ$.
15.183 $51.5 \text{ m/s}^2 \searrow 44.4^\circ$.
15.184 (a) $(33.0 \text{ rad/s})\mathbf{i} - (44.0 \text{ rad/s})\mathbf{k}$. (b) $(4.80 \text{ m/s})\mathbf{i} + (3.60 \text{ m/s})\mathbf{k}$.
15.185 (a) $(44.0 \text{ rad/s})\mathbf{i} - (33.0 \text{ rad/s})\mathbf{k}$. (b) $(3.60 \text{ m/s})\mathbf{i} + (4.80 \text{ m/s})\mathbf{k}$.
15.186 (a) $(1.5 \text{ rad/s})\mathbf{i} - (3.5 \text{ rad/s})\mathbf{j} - (3.0 \text{ rad/s})\mathbf{k}$. (b) $(640 \text{ mm/s})\mathbf{i} - (360 \text{ mm/s})\mathbf{j} + (740 \text{ mm/s})\mathbf{k}$.
15.187 (a) $(0.60 \text{ rad/s})\mathbf{i} - (2.00 \text{ rad/s})\mathbf{j} + (0.75 \text{ rad/s})\mathbf{k}$. (b) $(20.0 \text{ in/s})\mathbf{i} + (15.0 \text{ in/s})\mathbf{j} + (24.0 \text{ in/s})\mathbf{k}$.
15.188 $(118.4 \text{ rad/s}^2)\mathbf{i}$.
15.189 $(230 \text{ rad/s}^2)\mathbf{i} - (2.5 \text{ rad/s}^2)\mathbf{k}$.
15.190 (a) $(6.28 \text{ rad/s}^2)\mathbf{i}$. (b) $(8.38 \text{ rad/s}^2)\mathbf{k}$.
15.193 (a) $-(0.600 \text{ m/s})\mathbf{i} + (0.750 \text{ m/s})\mathbf{j} - (0.600 \text{ m/s})\mathbf{k}$. (b) $-(6.15 \text{ m/s}^2)\mathbf{i} - (3.00 \text{ m/s}^2)\mathbf{j}$.
15.195 (a) $-(20.0 \text{ rad/s}^2)\mathbf{j}$. (b) $-(4.00 \text{ ft/s}^2)\mathbf{i} + (10.00 \text{ ft/s}^2)\mathbf{k}$. (c) $-(10.25 \text{ ft/s}^2)\mathbf{j}$.
15.196 $-(3.46 \text{ ft/s}^2)\mathbf{i} - (5.13 \text{ ft/s}^2)\mathbf{j} + (8.66 \text{ ft/s}^2)\mathbf{k}$.
15.197 (a) $\omega_1 / \sin \beta$. (b) $\omega_1 / \tan \beta$. (c) $\omega_1^2 / \tan \beta$.
15.198 (a) $(0.0375 \text{ rad/s}^2)\mathbf{i}$. (b) $-(0.1434 \text{ m/s})\mathbf{i} + (0.204 \text{ m/s})\mathbf{j} - (0.1228 \text{ m/s})\mathbf{k}$. (c) $-(0.696 \text{ m/s}^2)\mathbf{i} - (0.0358 \text{ m/s}^2)\mathbf{j} + (0.0430 \text{ m/s}^2)\mathbf{k}$.
15.199 (a) $(28.4 \text{ rad/s})\mathbf{i} + (5.24 \text{ rad/s})\mathbf{j}$. (b) $(25.8 \text{ rad/s})\mathbf{i}$.
15.200 (a) $(135.1 \text{ rad/s}^2)\mathbf{k}$. (b) $(5.77 \text{ m/s}^2)\mathbf{i} - (232 \text{ m/s}^2)\mathbf{j}$.
15.203 $-(33.3 \text{ in/s})\mathbf{j}$.
15.204 $(15.0 \text{ in/s})\mathbf{j}$.
15.205 $-(34.5 \text{ mm/s})\mathbf{i}$.
15.206 $-(30.0 \text{ in/s})\mathbf{j}$.
15.207 $(45.7 \text{ in/s})\mathbf{j}$.
15.210 $(\omega_2/\cos 25^\circ)(-\sin 25^\circ\mathbf{i} + \cos 25^\circ\mathbf{k})$.
15.211 $(\omega_1 \cos 25^\circ)(-\sin 25^\circ\mathbf{i} + \cos 25^\circ\mathbf{k})$.
15.212 (a) $(1.463 \text{ rad/s})\mathbf{i} + (0.1052 \text{ rad/s})\mathbf{j} + (0.0841 \text{ rad/s})\mathbf{k}$. (b) $-(1.725 \text{ in/s})\mathbf{i}$.
15.213 (a) $-(4.15 \text{ rad/s})\mathbf{i} + (0.615 \text{ rad/s})\mathbf{j} - (2.77 \text{ rad/s})\mathbf{k}$. (b) $(0.30 \text{ m/s})\mathbf{k}$.
15.216 $-(45.0 \text{ in/s}^2)\mathbf{j}$.
15.217 $(205 \text{ in/s}^2)\mathbf{j}$.
15.218 $-(9.51 \text{ mm/s}^2)\mathbf{j}$.
15.219 $-(8.76 \text{ mm/s}^2)\mathbf{j}$.
15.220 (a) $(-3.00 \text{ ft/s})\mathbf{i} + (6.00 \text{ ft/s})\mathbf{j} - (20.94 \text{ ft/s})\mathbf{k}$. (b) $(6.28 \text{ rad/s}^2)\mathbf{i}$. (c) $(-62.87 \text{ ft/s}^2)\mathbf{i} - (9.00 \text{ ft/s}^2)\mathbf{j} + (12.57 \text{ ft/s}^2)\mathbf{k}$.
15.221 (a) $-(24.94 \text{ ft/s})\mathbf{k}$. (b) $(1.00 \text{ rad/s}^2)\mathbf{j} + (8.38 \text{ rad/s}^2)\mathbf{k}$. (c) $-(60.62 \text{ ft/s}^2)\mathbf{i} - (16.00 \text{ ft/s}^2)\mathbf{j} - (10.00 \text{ ft/s}^2)\mathbf{k}$.
15.222 (a) $-(1.215 \text{ m/s})\mathbf{i} + (1.620 \text{ m/s})\mathbf{k}$. (b) $-(30.4 \text{ m/s}^2)\mathbf{j}$.
15.223 (a) $-(1.215 \text{ m/s})\mathbf{i} - (1.080 \text{ m/s})\mathbf{j} + (1.620 \text{ m/s})\mathbf{k}$. (b) $(19.44 \text{ m/s}^2)\mathbf{i} - (30.4 \text{ m/s}^2)\mathbf{j} - (12.96 \text{ m/s}^2)\mathbf{k}$.
15.224 (a) $(1.200 \text{ m/s})\mathbf{i} + (0.500 \text{ m/s})\mathbf{j} - (1.200 \text{ m/s})\mathbf{k}$. (b) $-(7.20 \text{ m/s}^2)\mathbf{i} - (14.40 \text{ m/s}^2)\mathbf{k}$.
15.227 (a) $(0.750 \text{ m/s})\mathbf{i} + (1.299 \text{ m/s})\mathbf{j} - (1.732 \text{ m/s})\mathbf{k}$. (b) $(27.1 \text{ m/s}^2)\mathbf{i} + (5.63 \text{ m/s}^2)\mathbf{j} - (15.00 \text{ m/s}^2)\mathbf{k}$.
15.228 (a) $(129.9 \text{ mm/s})\mathbf{i} + (75.0 \text{ mm/s})\mathbf{j} + (86.6 \text{ mm/s})\mathbf{k}$. (b) $(45.0 \text{ mm/s}^2)\mathbf{i} - (112.6 \text{ mm/s}^2)\mathbf{j} + (60.0 \text{ mm/s}^2)\mathbf{k}$.

- 15.230** $\mathbf{v}_C = -(45.0 \text{ in/s})\mathbf{i} + (36.6 \text{ in/s})\mathbf{j} - (31.2 \text{ in/s})\mathbf{k}$,
 $\mathbf{a}_C = -(303 \text{ in/s}^2)\mathbf{i} - (384 \text{ in/s}^2)\mathbf{j} + (208 \text{ in/s}^2)\mathbf{k}$.
- 15.231** (a) $\omega_1 + (R/r)(\omega_1 - \omega_2)\mathbf{k}$. (b) $\omega_1(\omega_1 - \omega_2)(R/r)\mathbf{j}$.
- 15.232** $-(41.6 \text{ in/s}^2)\mathbf{i} - (61.5 \text{ in/s}^2)\mathbf{j} + (103.9 \text{ in/s}^2)\mathbf{k}$.
- 15.233** (a) $(0.0375 \text{ rad/s}^2)\mathbf{i}$.
 (b) $-(0.143 \text{ m/s})\mathbf{i} + (0.205 \text{ m/s})\mathbf{j} - (0.123 \text{ m/s})\mathbf{k}$.
 (c) $-(0.0696 \text{ m/s}^2)\mathbf{i} - (0.0358 \text{ m/s}^2)\mathbf{j} + (0.0430 \text{ m/s}^2)\mathbf{k}$.
- 15.234** $\mathbf{v}_A = -(1.39 \text{ m/s})\mathbf{i} + (0.80 \text{ m/s})\mathbf{j} - (1.20 \text{ m/s})\mathbf{k}$,
 $\mathbf{a}_A = -(20.8 \text{ m/s}^2)\mathbf{i} - (11.09 \text{ m/s}^2)\mathbf{j} + (33.3 \text{ m/s}^2)\mathbf{k}$.
- 15.235** $\mathbf{v}_A = -(1.39 \text{ m/s})\mathbf{i} + (0.80 \text{ m/s})\mathbf{j} - (1.20 \text{ m/s})\mathbf{k}$,
 $\mathbf{a}_A = -(22.5 \text{ m/s}^2)\mathbf{i} - (10.09 \text{ m/s}^2)\mathbf{j} + (34.9 \text{ m/s}^2)\mathbf{k}$.
- 15.236** (a) $-(1.37 \text{ ft/s})\mathbf{i} + (3.76 \text{ ft/s})\mathbf{j} + (1.88 \text{ ft/s})\mathbf{k}$.
 (b) $(1.22 \text{ ft/s}^2)\mathbf{i} - (0.342 \text{ ft/s}^2)\mathbf{j} - (0.410 \text{ ft/s}^2)\mathbf{k}$.
- 15.239** (a) $(4.33 \text{ ft/s})\mathbf{i} - (6.18 \text{ ft/s})\mathbf{j} + (5.30 \text{ ft/s})\mathbf{k}$.
 (b) $(2.65 \text{ ft/s}^2)\mathbf{i} - (2.64 \text{ ft/s}^2)\mathbf{j} - (3.25 \text{ ft/s}^2)\mathbf{k}$.
- 15.240** (a) $(27.2 \text{ in/s}^2)\mathbf{i} - (6.75 \text{ in/s}^2)\mathbf{j}$.
 (b) $(12.80 \text{ in/s}^2)\mathbf{i} - (7.68 \text{ in/s}^2)\mathbf{k}$.
- 15.241** (a) $-(1.600 \text{ in/s}^2)\mathbf{i} + (6.75 \text{ in/s}^2)\mathbf{j}$.
 (b) $(12.80 \text{ in/s}^2)\mathbf{i} + (7.68 \text{ in/s}^2)\mathbf{k}$.
- 15.242** $-(5.04 \text{ m/s})\mathbf{i} - (1.200 \text{ m/s})\mathbf{k}$.
 $-(9.60 \text{ m/s}^2)\mathbf{i} - (25.9 \text{ m/s}^2)\mathbf{j} + (57.6 \text{ m/s}^2)\mathbf{k}$.
- 15.243** $-(0.720 \text{ m/s})\mathbf{i} - (1.200 \text{ m/s})\mathbf{k}$,
 $-(9.60 \text{ m/s}^2)\mathbf{i} + (25.9 \text{ m/s}^2)\mathbf{j} - (11.52 \text{ m/s}^2)\mathbf{k}$.
- 15.244** (a) $r\omega_2^2 \sin 30^\circ \mathbf{j} - (r\omega_2^2 \cos 30^\circ + 2r\omega_1\omega_2)\mathbf{k}$.
 (b) $-r(\omega_1^2 + \omega_2^2 + 2\omega_1\omega_2 \cos 30^\circ)\mathbf{i} + r\omega_1^2 \cos 30^\circ \mathbf{k}$.
 (c) $-r\omega_2^2 \sin 30^\circ \mathbf{j} + r(2\omega_1^2 \cos 30^\circ + \omega_2^2 \cos 30^\circ + 2\omega_1\omega_2)\mathbf{k}$.
- 15.245** (a) $(0.610 \text{ m/s})\mathbf{k}$, $-(0.880 \text{ m/s}^2)\mathbf{i} + (1.170 \text{ m/s}^2)\mathbf{j}$.
 (b) $(5.20 \text{ m/s})\mathbf{i} - (0.390 \text{ m/s})\mathbf{j} - (1.000 \text{ m/s})\mathbf{k}$,
 $-(4.00 \text{ m/s}^2)\mathbf{i} - (3.25 \text{ m/s}^2)\mathbf{k}$.
- 15.248** $(36.0 \text{ ft/s})\mathbf{i} - (64.0 \text{ ft/s})\mathbf{j}$.
- 15.249** (a) $5.00 \text{ ft/s}^2 \rightarrow$. (b) $5.63 \text{ in} \leftarrow$.
- 15.252** $\alpha_{BD} = 306 \text{ rad/s}^2 \uparrow$, $\alpha_{DE} = 737 \text{ rad/s}^2 \uparrow$.
- 15.253** (a) $1080 \text{ rad/s}^2 \downarrow$. (b) $460 \text{ ft/s}^2 \searrow 64.9^\circ$.
- 15.255** $49.4 \text{ m/s}^2 \searrow 26.0^\circ$.
- 15.256** (a) $(0.450 \text{ m/s})\mathbf{k}$, $(4.05 \text{ m/s}^2)\mathbf{i}$. (b) $-(1.350 \text{ m/s})\mathbf{k}$, $-(6.75 \text{ m/s}^2)\mathbf{i}$.
- 15.258** $(40.0 \text{ in/s})\mathbf{k}$.
- 15.259** $(9.00 \text{ in/s})\mathbf{i} - (7.80 \text{ in/s})\mathbf{j} + (7.20 \text{ in/s})\mathbf{k}$,
 $(9.00 \text{ in/s}^2)\mathbf{i} - (22.1 \text{ in/s}^2)\mathbf{j} - (5.76 \text{ in/s}^2)\mathbf{k}$.
- 15.28** 107.6 rev.
- 15.29** 74.5 s.
- 15.30** $20.4 \text{ rad/s}^2 \downarrow$.
- 15.31** $32.7 \text{ rad/s}^2 \uparrow$.
- 15.33** (a) $5.66 \text{ ft/s}^2 \downarrow$. (b) $7.52 \text{ ft/s} \downarrow$.
- 15.34** (1): (a) $8.00 \text{ rad/s}^2 \uparrow$. (b) $14.61 \text{ rad/s} \uparrow$.
 (2): (a) $6.74 \text{ rad/s}^2 \uparrow$. (b) $13.41 \text{ rad/s} \uparrow$.
 (3): (a) $4.24 \text{ rad/s}^2 \uparrow$. (b) $10.64 \text{ rad/s} \uparrow$.
 (4): (a) $5.83 \text{ rad/s}^2 \uparrow$. (b) $8.82 \text{ rad/s} \uparrow$.
- 15.36** (a) $6.06 \text{ rad/s}^2 \downarrow$. (b) $11.28 \text{ N} \nearrow$.
- 15.39** (a) No slipping on A; slipping on B.
 (b) $\alpha_A = 61.8 \text{ rad/s}^2 \uparrow$; $\alpha_B = 9.66 \text{ rad/s}^2 \downarrow$.
- 15.40** (a) No slipping at either cylinder.
 (b) $\alpha_A = 15.46 \text{ rad/s}^2 \uparrow$, $\alpha_B = 7.73 \text{ rad/s}^2 \downarrow$.
- 15.41** (a) $\alpha_A = 12.50 \text{ rad/s}^2 \uparrow$, $\alpha_B = 33.3 \text{ rad/s}^2 \uparrow$.
 (b) $\omega_A = 240 \text{ rpm} \downarrow$, $\omega_B = 320 \text{ rpm} \uparrow$.
- 15.42** (a) $\alpha_A = 12.50 \text{ rad/s}^2 \uparrow$, $\alpha_B = 33.3 \text{ rpm} \uparrow$.
 (b) $\omega_A = 90.0 \text{ rpm} \uparrow$, $\omega_B = 120.0 \text{ rpm} \downarrow$.
- 15.43** (a) $\alpha_A = 9.16 \text{ rad/s}^2 \uparrow$, $\alpha_B = 38.2 \text{ rad/s}^2 \uparrow$.
 (b) $\mathbf{C} = 54.9 \text{ N} \uparrow$, $\mathbf{M}_C = 2.64 \text{ N}\cdot\text{m} \uparrow$.
- 15.44** (b) $\omega_0/(1 + m_B/m_A) \downarrow$.
- 15.48** (a) $18.40 \text{ ft/s}^2 \rightarrow$. (b) $9.20 \text{ ft/s}^2 \leftarrow$. (c) $z = 24.0 \text{ in}$.
- 15.49** (a) 12.0 in. from end A. (b) $9.20 \text{ ft/s}^2 \rightarrow$.
- 15.50** (a) $2.50 \text{ m/s}^2 \rightarrow$. (b) 0.
- 15.51** (a) $3.75 \text{ m/s}^2 \rightarrow$. (b) $1.25 \text{ m/s}^2 \leftarrow$.
- 15.52** (a) 0, $-1.374 \text{ rad/s}^2 \mathbf{j}$. (b) $-(0.515 \text{ ft/s}^2)\mathbf{i}$, $-1.030 \text{ rad/s}^2 \mathbf{j}$.
- 15.55** $\mathbf{a}_A = 2.71 \text{ m/s}^2 \uparrow$ and $\mathbf{a}_B = 1.496 \text{ m/s}^2 \uparrow$.
- 15.56** 170.9 mm.
- 15.57** (a) $53.1 \text{ rad/s}^2 \downarrow$. (b) $\mathbf{a} = 39.3 \text{ ft/s}^2 \downarrow$.
- 15.58** (a) $0.741 \text{ rad/s}^2 \uparrow$. (b) 0.857 m/s^2 .
- 15.59** (a) 2800 N. (b) $15.11 \text{ rad/s}^2 \downarrow$.
- 15.60** $T_A = 359 \text{ lb}$, $T_B = 312 \text{ lb}$.
- 15.63** (a) $\frac{3g}{2L} \downarrow$. (b) $\frac{g}{4} \uparrow$. (c) $\frac{5g}{4} \downarrow$.
- 15.64** (a) $\frac{2g}{L} \downarrow$. (b) $\frac{g}{3} \uparrow$. (c) $\frac{5g}{3} \downarrow$.
- 15.65** (a) $\frac{3g}{L} \downarrow$. (b) $1.323 \text{ g} \searrow 49.1^\circ$. (c) $2.18 \text{ g} \searrow 66.6^\circ$.
- 15.66** (a) $0.25 \text{ g} \uparrow$. (b) $5 \text{ g/4} \downarrow$.
- 15.67** (a) 0. (b) $g \downarrow$.
- 15.69** (a) $5v_0/2r \uparrow$. (b) v_0/μ_{kg} . (c) $v_0^2/2\mu_{kg}$.
- 15.70** (a) $v_0/r \uparrow$. (b) v_0/μ_{kg} . (c) $v_0^2/2\mu_{kg}$.
- 15.71** (a) 1.597 s. (b) 9.86 ft/s. (c) 19.85 ft.
- 15.72** (a) 1.863 s. (b) 9.00 ft/s. (c) 22.4 ft.
- 15.76** (a) $107.1 \text{ rad/s}^2 \downarrow$. (b) 21.4 N $\leftarrow$, 39.2 N $\uparrow$.
- 15.77** (a) 150 mm. (b) $125 \text{ rad/s}^2 \downarrow$.
- 15.78** (a) $12.08 \text{ rad/s}^2 \downarrow$. (b) 0.750 lb $\leftarrow$, 4.00 lb $\uparrow$.
- 15.79** (a) $8.05 \text{ rad/s}^2 \downarrow$. (b) 24.0 in.
- 15.80** (a) 1522.9 N. (b) 1341.8 N.
- 15.83** 13.64 kN $\rightarrow$.
- 15.84** (a) 1.5 g $\downarrow$. (b) 0.25 mg $\uparrow$.
- 15.85** (a) 9g/7. (b) 4mg/7 $\uparrow$.
- 15.86** (a) 0.6727 ft-lb. (b) 1999.2 lb.
- 15.87** (a) 43.6 rad/s^2 . (b) 21.0 N $\leftarrow$, 54.6 N $\uparrow$.
- 15.88** (a) $3.72 \text{ rad/s}^2 \downarrow$. (b) 1.462 lb.
- 15.94** $r^2 g \sin \beta / (r^2 + \bar{k}^2)$.
- 15.95** (a) 2.27 m (7.46 ft). (b) 0.649 m (2.13 ft).
- 15.98** (a) $17.78 \text{ rad/s}^2 \downarrow$, 2.13 m/s $\rightarrow$. (b) 0.122.
- 15.99** (a) $26.7 \text{ rad/s}^2 \downarrow$, 3.20 m/s $\rightarrow$. (b) 0.0136.
- 16.102** (a) no sliding. (b) $15.46 \text{ rad/s}^2 \downarrow$, 10.30 ft/s^2 .
- 16.103** (a) no sliding. (b) $23.2 \text{ rad/s}^2 \downarrow$, 15.46 ft/s^2 .
- 16.104** (a) slides. (b) $4.29 \text{ rad/s}^2 \uparrow$, $9.66 \text{ ft/s}^2 \rightarrow$.

CHAPTER 16

- 16.1** (a) $R_A = 60.31 \text{ lb} \searrow 84.2^\circ$ and $N_B = 28.5 \text{ lb} \leftarrow$.
 (b) $\mu = 0.1023$.
- 16.2** (a) $18.59 \text{ ft/s}^2 \rightarrow$. (b) 0.577.
- 16.3** (a) 25.8 ft/s^2 . (b) 12.27 ft/s^2 . (c) 13.32 ft/s^2 .
- 16.4** (a) 3.20 m/s^2 . (b) $\mathbf{A} = 3.82 \text{ N} \uparrow$, $\mathbf{B} = 20.7 \text{ N} \uparrow$.
- 16.5** (a) 4.09 m/s^2 . (b) 42.5 N.
- 16.6** (a) 5270 N $\uparrow$. (b) 4120 N.
- 16.9** (a) $5.00 \text{ m/s}^2 \rightarrow$. (b) $0.311 \text{ m} \leq h \leq 1.489 \text{ m}$.
- 16.10** (a) $2.55 \text{ m/s}^2 \rightarrow$. (b) $h \leq 1.047 \text{ m}$.
- 16.11** 195.9 kg.
- 16.12** 229 N.
- 16.14** (a) $4.91 \text{ m/s}^2 \searrow 30^\circ$. (b) $F_A = 0$, $F_B = 68.0 \text{ N}$ compression.
- 16.15** (a) 173.2 N $\rightarrow$. (b) 15.02 rad/s. (c) $86.6 \text{ rad/s}^2 \uparrow$.
- 16.18** $B_y = 16.48 \text{ lb}$ and $D_y = 17.62 \text{ lb}$.
- 16.19** (a) $30.6 \text{ ft/s}^2 \searrow 84.1^\circ$. (b) $\mathbf{A} = 0.505 \text{ lb} \searrow 30^\circ$,
 $\mathbf{B} = 1.285 \text{ lb} \searrow 30^\circ$.
- 16.20** Block: $17.01 \text{ ft/s}^2 \searrow 58.5^\circ$; platform: $31.3 \text{ ft/s}^2 \searrow 30^\circ$.
- 16.25** 125.7 N-m.
- 16.26** 9480 rev.
- 16.27** 93.5 rev.

16.105 (a) slides. (b) $12.88 \text{ rad/s}^2 \nearrow$, $3.22 \text{ ft/s}^2 \leftarrow$.

16.107 (a) $6.63 \text{ ft/s}^2 \rightarrow$. (b) $3.79 \text{ ft/s}^2 \rightarrow$. (c) $0.355 \text{ ft} \rightarrow$.

16.108 (a) $72.4 \text{ rad/s}^2 \nearrow$. (b) $7.24 \text{ m/s}^2 \downarrow$.

16.109 (a) $2.64 \text{ m/s}^2 \leftarrow$. (b) $11.87 \text{ N} \leftarrow$.

16.111 (a) 0.298 . (b) $0.536 \text{ g} \rightarrow$.

16.112 (a) 0.322 . (b) $0.566 \text{ g} \rightarrow$.

16.113 $8.26 \text{ N} \leftarrow$.

16.114 (a) $0.125 \text{ g/r} \downarrow$. (b) $0.125 \text{ g} \rightarrow$, $0.125 \text{ g} \downarrow$.

16.115 $m_B g \sin \theta / [2r \{m_h + m_B (1 + \cos \theta)\}]$.

16.116 $3.43 \text{ lb} \nearrow 70.5^\circ$, $0.1550 \text{ ft}\cdot\text{lb} \downarrow$.

16.117 (a) $\frac{g}{L[\frac{1}{3} + \sin^2 \theta]}$. (b) $\frac{mg}{1 + 3 \sin^2 \theta} \uparrow$.

16.118 (a) $27.6 \text{ rad/s}^2 \downarrow$. (b) $5.714 \text{ lb} \uparrow$.

16.119 (a) $0.510 \text{ rad/s}^2 \downarrow$. (b) $\mathbf{F}_A = 31.80 \text{ lb} \nearrow 78.7^\circ$ and $\mathbf{F}_B = 13.79 \text{ lb} \searrow 11.3^\circ$.

16.120 $mg \sin \theta / (1 + 3 \sin \theta)$.

16.121 (a) $6.26 \text{ rad/s}^2 \downarrow$. (b) $13.22 \text{ N} \leftarrow$.

16.124 $6.40 \text{ N} \leftarrow$.

16.125 $7.10 \text{ lb} \rightarrow$.

16.126 $5.51 \text{ lb} \rightarrow$.

16.127 $67.62 \text{ N} \nearrow 56.0^\circ$.

16.128 $75.13 \text{ N} \uparrow$.

16.129 $25.9 \text{ N} \searrow 60^\circ$.

16.131 (a) $37.8 \text{ ft/s}^2 \searrow 26.1^\circ$. (b) $48.4 \text{ lb} \uparrow$.

16.134 (a) $4.36 \text{ rad/s}^2 \nearrow$. (b) $31.36 \text{ lb} \uparrow$.

16.135 (a) $36.3 \text{ N}\cdot\text{m} \nearrow$. (b) $231 \text{ N} \leftarrow$, $524 \text{ N} \uparrow$.

16.136 (a) $82.3 \text{ N}\cdot\text{m} \nearrow$. (b) $147.2 \text{ N} \leftarrow$, $479 \text{ N} \uparrow$.

16.137 $\mathbf{B} = 805 \text{ N} \leftarrow$, $\mathbf{D} = 426 \text{ N} \rightarrow$.

16.138 $\mathbf{B} = 525 \text{ N} \nearrow 38.1^\circ$, $\mathbf{D} = 322 \text{ N} \searrow 15.7^\circ$.

16.139 (a) $24.8 \text{ rad/s}^2 \downarrow$. (b) $29.5 \text{ lb} \uparrow$.

16.140 (a) $19.3 \text{ ft}\cdot\text{lb} \downarrow$. (b) $81.9 \text{ lb} \uparrow$.

16.143 (a) $\alpha_A = \frac{2}{5} \frac{g}{r} \nearrow$ and $\alpha_B = \frac{2}{5} \frac{g}{r} \downarrow$. (b) $\frac{1}{5} mg$. (c) $\frac{4}{5} g \downarrow$.

16.146 (a) $50.2 \text{ N} \nearrow 60.3^\circ$. (b) 0.273 .

16.148 (a) $17.03 \text{ ft/s}^2 \searrow 20^\circ$. (b) $42.7 \text{ rad/s}^2 \nearrow$.

16.151 $M_{\max} = 10.39 \text{ lb}\cdot\text{in.}$ located 20.8 in. below A.

16.153 20.6 ft .

16.154 17.34 ft .

16.156 (a) $2\mu g / (1 + 3\mu)$. (b) 1.000 g .

16.157 (a) $0.513 \text{ g/L} \downarrow$. (b) $0.912 \text{ mg} \uparrow$. (c) $0.241 \text{ mg} \rightarrow$.

16.158 (a) $1.519 \text{ g/L} \downarrow$. (b) $0.260 \text{ g} \downarrow$. (c) $0.740 \text{ mg} \uparrow$.

16.160 (1): (a) $1.200 \text{ g/c} \downarrow$. (b) $0.671 \text{ g} \nearrow 63.4^\circ$.

(2): (a) $24 \text{ g/17c} \downarrow$. (b) $12 \text{ g/17} \downarrow$.

(3): (a) $2.40 \text{ g/c} \downarrow$. (b) $0.500 \text{ g} \downarrow$.

16.162 (a) $51.2 \text{ rad/s}^2 \downarrow$. (b) $21.0 \text{ N} \uparrow$.

16.163 (a) $59.8 \text{ rad/s}^2 \downarrow$. (b) $20.4 \text{ N} \uparrow$.

17.13 $\omega_A = \frac{2n}{n^2 + 1} \sqrt{\frac{\pi M_0}{I_0}}$.

17.16 (a) $\sqrt{3g/L}$, $2.50 \text{ W} \uparrow$. (b) $5.67 \text{ rad/s} \downarrow$, $4.50 \text{ lb} \uparrow$.

17.17 (a) 0.289 l . (b) $1.861 \sqrt{g/l}$, $2.00 \text{ mg} \uparrow$.

17.18 $11.52 \text{ rad/s} \nearrow$.

17.19 16.23 ft/s .

17.20 (a) $3.94 \text{ rad/s} \downarrow$, $271 \text{ lb} \searrow 5.25^\circ$.

(b) $5.58 \text{ rad/s} \downarrow$, $701 \text{ lb} \uparrow$.

17.23 7.09 rad/s .

17.24 (a) -0.250 rpm . (b) 0.249 rpm .

17.25 $\sqrt{4gs/3}$.

17.26 $\sqrt{gs}$.

17.27 (a) 5.18 ft/s . (b) 0.042 .

17.29 (a) 5.00 rad/s . (b) $24.9 \text{ N} \uparrow$.

17.30 (a) $1.142 \sqrt{\frac{g}{r}} \downarrow$. (b) $1.553 \text{ mg} \uparrow$.

17.31 (a) $[10g(R - r)(1 - \cos \beta)/7]^{1/2}$.

(b) $mg(17 - 10 \cos \beta)/7$.

17.32 (a) 2.06 ft . (b) 4.00 lb .

17.33 (a) $7.43 \text{ ft/s} \downarrow$. (b) 4.00 lb .

17.35 (a) $11.57 \text{ rad/s} \downarrow$. (b) $27.8 \text{ rad/s} \nearrow$.

17.36 $\mathbf{v}_A = 0.775 \sqrt{gl} \leftarrow$, $\mathbf{v}_B = 0.775 \sqrt{gl} \nearrow 60^\circ$.

17.37 $1.170 \text{ rad/s} \downarrow$, $5.07 \text{ m/s} \leftarrow$.

17.38 $[3g(\cos \theta_0 - \cos \theta_2)/L]^{1/2} \downarrow$.

17.39 $3.71 \text{ rad/s} \nearrow$, $7.74 \text{ ft/s} \uparrow$.

17.40 $15.54 \text{ ft/s} \rightarrow$.

17.42 $2.69 \text{ m/s} \downarrow$.

17.43 $84.7 \text{ rpm} \downarrow$.

17.44 $110.8 \text{ rpm} \downarrow$.

17.45 $3.25 \text{ m/s} \downarrow$.

17.46 $4.43 \text{ m/s} \downarrow$.

17.47 $0.770 \text{ m/s} \leftarrow$.

17.48 (a) 44.3 hp . (b) 118.1 hp .

17.49 (a) $39.8 \text{ N}\cdot\text{m}$. (b) $95.5 \text{ N}\cdot\text{m}$. (c) $229 \text{ N}\cdot\text{m}$.

17.50 1146 rpm .

17.51 10.87 lb .

17.52 179.1 mm .

17.53 $0.335 \text{ lb}\cdot\text{in.}$

17.54 3.87 rad/s .

17.55 $24.6 \text{ ft}\cdot\text{lb}$.

17.58 3.88 s .

17.59 $(1 + \mu_k^2) r \omega_0 / [2\mu_k(1 + \mu_k)g]$.

17.62 $\omega_0 / (1 + m_A/m_B)$.

17.63 (a) $\omega_A = 686 \text{ rpm} \nearrow$, $\omega_B = 514 \text{ rpm} \downarrow$. (b) $4.18 \text{ lb}\cdot\text{s} \uparrow$.

17.64 (a) 5.15 lb . (b) 2.01 lb .

17.65 $\mathbf{X} = m\mathbf{v}$, $d = \bar{k}^2 \omega / \bar{v}$.

17.69 2.79 ft .

17.70 (a) $r^2 g t \sin \beta / (r^2 + \bar{k}^2) \searrow \beta$.

(b) $u_s \geq \bar{k}^2 \tan \beta / (r^2 + \bar{k}^2)$.

17.71 (a) $2.55 \text{ m/s} \uparrow$. (b) 10.53 N .

17.72 (a) $8.05 \text{ ft/s} \rightarrow$. (b) $2.68 \text{ ft/s} \rightarrow$.

17.74 (a) $8.41 \text{ m/s} \downarrow$. (b) 16.82 N .

17.75 (a) 0.557 s . (b) 16.82 N .

17.77 (a) $2.50 \bar{v}_0 / r$. (b) $\bar{v}_0 / \mu_k g$.

17.78 (a) 2.50 s . (b) 16.95 ft/s .

17.79 $\frac{5}{6} \omega_0$.

17.80 10.19 rpm .

17.81 A and B: $159.1 \text{ rpm} \downarrow$; platform $20.9 \text{ rpm} \nearrow$.

17.82 18.07 rad/s .

17.83 (a) 2.54 rad/s . (b) 1.902 J .

CHAPTER 17

17.1 $12.77 \text{ N}\cdot\text{m}$.

17.2 8690 rev .

17.3 9.60 in.

17.4 0.841 .

17.5 (a) $19.20 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$. (b) 11.46 rev .

17.6 (a) 293 rpm . (b) 15.92 rev .

17.7 19.77 rev .

17.10 $109.4 \text{ lb} \rightarrow$.

17.11 (a) 6.35 rev . (b) 7.14 N .

17.12 (a) 2.54 rev . (b) 17.86 N .

- 17.86** 37.2 rpm.
17.87 $\omega_{BC} = 36.6 \text{ rpm} \downarrow$ and $\omega_A = 16.87 \text{ rpm} \uparrow$.
17.88 2.51 m/s.
17.89 18.83 rad/s, 0.0508 kg·m².
17.90 (a) 31.1 rad/s. (b) 18.13 ft/s.
17.91 (a) 15.00 rad/s. (b) 20.5 ft/s.
17.94 1.542 m/s.
17.95 2.01 ft/s $\leftarrow$.
17.96 0.400 r .
17.97 (a) 24.4 rad/s $\downarrow$. (b) 1545 lb $\rightarrow$.
17.98 (a) 10.00 in. (b) 22.6 rad/s $\downarrow$.
17.101 (a) 2.16 m/s $\rightarrow$. (b) 4.87 kN $\nearrow 66.9^\circ$.
17.102 (a) 158.0 mm. (b) 1.992 m/s $\rightarrow$.
17.103 (a) 0.90 $v_0/L \downarrow$. (b) 0.10 $v_0 \rightarrow$.
17.104 2.40 rad/s $\downarrow$.
17.105 1.667 in.
17.106 $\omega = \frac{\sqrt{2gh}(c + \cos \theta)}{r(1 + c)^{\frac{3}{2}}} \downarrow$ and $v = \frac{\sqrt{2gh}(c + \cos \theta)}{(1 + c)^{\frac{3}{2}}} \rightarrow$.
17.107 $\frac{\pi}{3}L$.
17.108 (a) $mv_0/M \rightarrow$. (b) $mv_0/MR \uparrow$.
17.109 (a) 1.500 R . (b) 1.000 R .
17.112 $\omega = 2.4 \frac{v_0}{L} \uparrow$ and $\bar{v} = 0.721v_0 \nearrow 56.3^\circ$.
17.115 2.38 m/s.
17.116 4.867 rad/s $\uparrow$.
17.117 (a) $0.437\sqrt{g/L}$. (b) 5.12° .
17.118 (a) 0.250 $\omega_0 \downarrow$. (b) 0.9375. (c) 1.50° .
17.119 48.7° .
17.120 1887 ft/s.
17.121 725 mm.
17.122 447 mm.
17.123 $0.606\sqrt{gL} \rightarrow$.
17.124 $0.866\sqrt{gL} \rightarrow$.
17.127 (a) 3.00 rad/s $\uparrow$. (b) 0.938 m/s $\uparrow$.
17.128 (a) 2.60 rad/s $\downarrow$. (b) 1.635 m/s $\searrow 53.4^\circ$.
17.131 $1.250 v_0/r$.
17.132 (a) $\mathbf{v}_A = 0$, $\omega_A = v_1/r \downarrow$, $\mathbf{v}_B = v_1 \rightarrow$, $\omega_B = 0$.
 (b) $\mathbf{v}'_A = 0.286 v_1 \rightarrow$, $\mathbf{v}'_B = 0.514 v_1 \rightarrow$.
17.133 (a) $\mathbf{v}_A = (v_0 \sin \theta)\mathbf{j}$, $\mathbf{v}_B = (v_0 \cos \theta)\mathbf{i}$, $\omega_A = (v_0/r)(-\sin \theta\mathbf{i} + \cos \theta\mathbf{j})$, $\omega_B = 0$.
 (b) $0.714 v_0 \cos \theta\mathbf{i}$.
17.134 $\omega_{AB} = 2.68 \text{ rad/s} \downarrow$, $\omega_{BC} = 13.39 \text{ rad/s} \downarrow$.
17.135 (a) 106.7 rev. (b) 6.98 s.
17.136 70.1 lb $\downarrow$.
17.137 (a) 18.22 ft/s. (b) 359.7 lb $\uparrow$. (c) 234.2 ft·lbs.
17.139 (a) 53.1° . (b) $1.095\sqrt{gL} \searrow 53.1^\circ$.
17.140 $\mathbf{A} = 100.1 \text{ N} \uparrow$, $\mathbf{B} = 43.9 \text{ N} \rightarrow$.
17.142 0.778 ω_0 .
17.143 (a) 418 rpm. (b) -20.4 J .
17.145 (a) 68.6 rpm. (b) 2.82 J.
18.9 $(1.843 \text{ lb}\cdot\text{ft}\cdot\text{s})\mathbf{i} - (0.455 \text{ lb}\cdot\text{ft}\cdot\text{s})\mathbf{j} + (1.118 \text{ lb}\cdot\text{ft}\cdot\text{s})\mathbf{k}$.
18.10 $-(2.03 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{i} + (4.16 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{j} + (0.675.03 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{k}$.
18.11 $0.500 \text{ m}^2\omega_1\mathbf{i} - m(L^2 + 0.250 r^2)(r\omega_1/L)\mathbf{j}$.
18.12 (a) 0.485 rad/s. (b) 0.01531 rad/s.
18.15 (a) $(5.65 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{i} - (1.885 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{j} + (12.57 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{k}$.
 (b) 25.4° .
18.16 (a) $(5.65 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{i} - (1.885 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{j} + (12.57 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{k}$.
 (b) 154.6° .
18.17 (a) $(1.078 \text{ lb}\cdot\text{s}\cdot\text{ft})\mathbf{i} - (0.647 \text{ lb}\cdot\text{s}\cdot\text{ft})\mathbf{k}$. (b) 31.0° .
18.18 (a) $(1.078 \text{ lb}\cdot\text{s}\cdot\text{ft})\mathbf{i} - (0.647 \text{ lb}\cdot\text{s}\cdot\text{ft})\mathbf{k}$.
 (b) $(1.078 \text{ lb}\cdot\text{s}\cdot\text{ft})\mathbf{i} - (0.647 \text{ lb}\cdot\text{s}\cdot\text{ft})\mathbf{k}$.
18.21 93.6 kg.
18.22 2.57 s.
18.25 (a) 0. (b) $(3F\Delta t/ma)(\mathbf{i} - 4\mathbf{k})$.
18.26 (a) $-(F\Delta t/m)\mathbf{i}$. (b) $(3 F\Delta t/8ma)(\mathbf{j} + 4\mathbf{k})$.
18.27 (a) $-(0.300 \text{ m/s})\mathbf{i}$. (b) $-(0.962 \text{ rad/s})\mathbf{i} - (0.577 \text{ rad/s})\mathbf{j}$.
18.28 (a) $(0.300 \text{ m/s})\mathbf{j}$.
 (b) $-(3.46 \text{ rad/s})\mathbf{i} + (1.923 \text{ rad/s})\mathbf{j} - (0.857 \text{ rad/s})\mathbf{k}$.
18.31 (a) $0.1250 \omega_0(-\mathbf{i} + \mathbf{j})$. (b) $0.0884 a\omega_0\mathbf{k}$.
18.32 (a) $0.1031 ma\omega_0\mathbf{k}$. (b) $-0.01473 ma\omega_0\mathbf{k}$.
18.33 $(0.0248 \text{ rad/s})\mathbf{i} - (0.277 \text{ rad/s})\mathbf{j} - (0.360 \text{ rad/s})\mathbf{k}$.
18.34 (a) -0.726 rad/s .
 (b) $-(2160 \text{ ft/s})\mathbf{i} - (4860 \text{ ft/s})\mathbf{j} + (860 \text{ ft/s})\mathbf{k}$.
18.35 (a) $t_A = 0.129 \text{ s}$, $t_B = 1.086 \text{ s}$. (b) $-(50.6 \text{ mm/s})\mathbf{j}$.
18.36 (a) 0.941 s. (b) $(0.0169 \text{ rad/s})\mathbf{j}$. (c) $-(39.2 \text{ mm/s})\mathbf{j}$.
18.39 $0.1250 mr^2(\omega_2^2 + 2\omega_1^2)$.
18.40 0.349 ft·lb.
18.41 0.978 ft·lb.
18.42 12.67 ft·lb.
18.43 15.47 J.
18.44 $0.1250 ma^2\omega^2$.
18.45 $0.203 ma^2\omega^2$.
18.47 237 J.
18.49 27.0 J.
18.50 46.2 J.
18.51 $0.1000 m\bar{v}_0^2$.
18.53 16.75 ft·lb.
18.54 39.9 ft·lb.
18.55 $0.500 \text{ m}^2\omega_1\omega_2\mathbf{i}$.
18.56 $(0.204 \text{ ft}\cdot\text{lb})\mathbf{k}$.
18.57 $(2.22 \text{ ft}\cdot\text{lb})\mathbf{k}$.
18.58 $(5.30 \text{ lb}\cdot\text{ft})\mathbf{k}$.
18.59 $(3.38 \text{ N}\cdot\text{m})\mathbf{i}$.
18.60 $\frac{1}{4}mr^2\omega^2\sin\beta\cos\beta\mathbf{k}$.
18.64 $\frac{1}{4}mr^2\alpha\sin\beta\cos\beta\mathbf{j} + \frac{1}{4}mr^2\omega^2\sin\beta\cos\beta\mathbf{k}$.
18.65 $\mathbf{C} = 0.1667 mb\omega^2\sin\beta\cos\beta\mathbf{i}$.
 $\mathbf{D} = -0.1667 mb\omega^2\sin\beta\cos\beta\mathbf{i}$.
18.66 $\mathbf{A} = -(4.97 \text{ lb})\mathbf{i}$, $\mathbf{B} = -(1.656 \text{ lb})\mathbf{j}$.
18.67 $\mathbf{A} = -(1.103 \text{ lb})\mathbf{j} - (0.920 \text{ lb})\mathbf{k}$.
 $\mathbf{B} = (1.103 \text{ lb})\mathbf{j} + (0.920 \text{ lb})\mathbf{k}$.
18.68 $\mathbf{A} = (14.4 \text{ N})\mathbf{k}$, $\mathbf{B} = -(14.4 \text{ N})\mathbf{k}$.
18.71 (a) $3M_0/mb^2\cos^2\beta$. (b) $\mathbf{C} = -\mathbf{D} = (M_0\tan\beta/2b)\mathbf{k}$.
18.72 (a) $(14.49 \text{ rad/s}^2)\mathbf{j}$. (b) $\mathbf{A} = -(1.125 \text{ lb})\mathbf{k}$, $\mathbf{B} = -(0.375 \text{ lb})\mathbf{k}$.
18.73 (a) $(0.873 \text{ lb}\cdot\text{ft})\mathbf{i}$. (b) $\mathbf{A} = -\mathbf{B} = -(0.218 \text{ lb})\mathbf{j} + (0.262 \text{ lb})\mathbf{k}$.
18.74 (a) $(2.67 \text{ N}\cdot\text{m})\mathbf{i}$. (b) $\mathbf{A} = -\mathbf{B} = (2.00 \text{ N})\mathbf{j}$.
18.75 (a) $(0.1301 \text{ lb}\cdot\text{ft})\mathbf{i}$. (b) $\mathbf{A} = -\mathbf{B} = -(0.0331 \text{ lb})\mathbf{i} + (0.0331 \text{ lb})\mathbf{j}$.
18.76 $\mathbf{A} = -\mathbf{B} = -(0.449 \text{ lb})\mathbf{j} - (0.383 \text{ lb})\mathbf{k}$.
18.79 $\mathbf{A} = -\mathbf{B} = (1.527 \text{ N})\mathbf{j}$.
18.81 (a) 10.47 N·m. (b) 10.47 N·m.
18.82 24.0 N $\uparrow$.

CHAPTER 18

- 18.1** $0.250 \text{ m}^2\omega_2\mathbf{j} + 0.500 \text{ m}^2\omega_1\mathbf{k}$.
18.2 $-(0.0408 \text{ slug}\cdot\text{ft}^2/\text{s})\mathbf{i} + (0.1398 \text{ slug}\cdot\text{ft}^2/\text{s})\mathbf{j}$.
18.3 $0.247 \text{ slug}\cdot\text{ft}^2/\text{s}$, $\theta_z = 48.6^\circ$, $\theta_y = 41.4^\circ$, $\theta_z = 90^\circ$.
18.5 $(0.1125 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{j} + (0.675 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{k}$.
18.7 $0.432 \text{ m}^2\omega$, 20.2° .
18.8 9.7° .

- 18.84** 1.138°; up.
18.85 10.20 rad/s.
18.86 (a) 27.0°. (b) 8.09 rad/s.
18.87 (a) 7.53 rad/s. (b) 7.00 rad/s.
18.88 4.84 rad/s.
18.90 7.89 rad/s.
18.91 15.24 rad/s.
18.93 $\mathbf{A} = -\mathbf{B} = (0.1906 \text{ lb})\mathbf{k}$.
18.94 7.87 rad/s.
18.95 (a) $\mathbf{C} = -(592 \text{ N})\mathbf{j}$ and $\mathbf{D} = (592 \text{ N})\mathbf{j}$. (b) $\mathbf{C} = \mathbf{D} = 0$.
18.96 35.5 rpm.
18.99 $-(45.0 \text{ N})\mathbf{i}$, $(3.38 \text{ N}\cdot\text{m})\mathbf{i} + (10.13 \text{ N}\cdot\text{m})\mathbf{k}$.
18.100 (a) $\mathbf{A} = (1.786 \text{ kN})\mathbf{i} + (143.5 \text{ kN})\mathbf{j}$,
 $\mathbf{B} = -(1.786 \text{ kN})\mathbf{i} + (150.8 \text{ kN})\mathbf{j}$, (b) $-(35.7 \text{ kN}\cdot\text{m})\mathbf{k}$.
18.101 $\mathbf{C} = -(7.81 \text{ lb})\mathbf{i} + (7.43 \text{ lb})\mathbf{k}$,
 $\mathbf{D} = -(7.81 \text{ lb})\mathbf{i} - (7.43 \text{ lb})\mathbf{k}$.
18.102 $\mathbf{C} = -(12.58 \text{ lb})\mathbf{i} + (9.43 \text{ lb})\mathbf{k}$,
 $\mathbf{D} = -(12.58 \text{ lb})\mathbf{i} - (9.43 \text{ lb})\mathbf{k}$.
18.103 $\mathbf{D} = -(22.0 \text{ N})\mathbf{i} + (26.8 \text{ N})\mathbf{j}$, $\mathbf{E} = -(21.2 \text{ N})\mathbf{i} - (5.20 \text{ N})\mathbf{j}$.
18.104 (a) $(0.392 \text{ N}\cdot\text{m})\mathbf{k}$. (b) $\mathbf{D} = -(21.0 \text{ N})\mathbf{i} + (28.0 \text{ N})\mathbf{j}$,
 $\mathbf{E} = -(21.0 \text{ N})\mathbf{i} - (4.00 \text{ N})\mathbf{j}$.
18.107 2930 rpm.
18.109 45.9 rpm, 533 rpm.
18.111 442 rpm.
18.112 68.1°.
18.113 $\cos^{-1} \left[\frac{2d^2\dot{\psi}}{(d^2 + h^2)\dot{\phi}} \right]$.
18.114 (a) 128.3 rad/s. (b) 2.17 in.
18.115 23.7°.
18.116 (a) 52.7 rad/s. (b) 6.44 rad/s.
18.117 (a) 4.89 rpm. (b) 4.96 rpm, 396 rpm.
18.124 (a) 13.19°. (b) 1242 rpm (retrograde)
18.126 24.8 rev/h.
18.127 (a) 12.85°. (b) 5.78 rev/h. (c) 20.7 rev/h.
18.128 (a) 109.4 rpm, $\gamma_x = 90^\circ$, $\gamma_y = 100.05^\circ$, $\gamma_z = 10.05^\circ$.
(b) $\theta_x = 90^\circ$, $\theta_y = 113.9^\circ$, $\theta_z = 23.9^\circ$.
(c) precession: 47.1 rpm; spin: 64.6 rpm.
18.130 (a) $\theta_x = 90.0^\circ$, $\theta_y = 26.0^\circ$, $\theta_z = 64.0^\circ$.
(b) precession, 0.847 rad/s (retrograde); spin: 0.1593 rad/s.
18.131 (a) $40.0^\circ < \theta < 140.0^\circ$. (b) 5.31 rad/s. (c) 5.58 rad/s.
18.132 (a) 2.00 rad/s. (b) 8.94 rad/s.
18.135 (a) 41.2°. (b) 5.52 rad/s.
18.136 (a) 4.23 rad/s. (b) 12.50 rad/s.
18.139 (a) 47.0°. (b) precession: 15.25 rad/s; spin: 307 rad/s.
18.140 (a) 76.3°. (b) precession: 9.62 rad/s; spin: 294 rad/s. (c) 36.5°.
18.148 $(0.234 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{j} + (1.250 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{k}$.
18.150 (a) 0. (b) $(F\Delta t/ma) (2.50\mathbf{i} - 1.454\mathbf{j} + 2.19\mathbf{k})$.
18.151 4.29 kN·m.
18.153 $\mathbf{D} = -(7.12 \text{ lb})\mathbf{j} + (4.47 \text{ lb})\mathbf{k}$, $\mathbf{E} = -(1.822 \text{ lb})\mathbf{j} + (4.47 \text{ lb})\mathbf{k}$.
18.154 $\mathbf{D} = 0$; $\mathbf{M}_D = (11.23 \text{ N}\cdot\text{m}) \cos^2 \theta \mathbf{i} +$
 $(11.23 \text{ N}\cdot\text{m}) \sin \theta \cos \theta \mathbf{j} - (2.81 \text{ N}\cdot\text{m}) \sin \theta \cos \theta \mathbf{k}$.
18.155 (a) $\theta_x = 52.5^\circ$, $\theta_y = 37.5^\circ$, $\theta_z = 90^\circ$.
(b) 53.8 rev/h. (c) 6.68 rev/h.
18.156 axis: 32.0°, precession: 1.126 rpm, and spin: 0.344 rpm.
18.157 (a) 4.00 rad/s. (b) 5.66 rad/s.
19.3 1.273 in., 150.8 ft/s².
19.4 (a) 0.324 s, 3.08 Hz. (b) 12.91 mm, 4.84 m/s².
19.5 (a) 0.308 s, 3.25 Hz. (b) 1.021 m/s, 20.8 m/s².
19.6 (a) 267 rpm. (b) 5.36 ft/s.
19.7 (a) 11.29°. (b) 1.933 m/s².
19.8 (a) 0.557 Hz. (b) 293 mm/s.
19.11 (a) 0.0352 s. (b) 6.34 ft/s $\uparrow$, 64.4 ft/s² $\downarrow$.
19.12 0.445 ft $\uparrow$, 2.27 ft/s $\downarrow$, 114.7 ft/s² $\downarrow$.
19.13 (a) 1.288°. (b) 0.874 ft/s, 0.760 ft/s².
19.14 (a) 4.91 mm, 5.81 Hz, 0.1791 m/s.
(b) 491 N, (c) 0.1592 m/s $\uparrow$.
19.17 (a) 0.517 s, 1.934 Hz. (b) 0.365 m/s, 4.43 m/s².
19.18 2.87 s.
19.19 $\sqrt{\frac{k}{2m}}$.
19.20 (a) 0.361 s, 2.77 Hz. (b) 0.765 m/s, 13.30 m/s².
19.23 4.
19.24 (a) 6.80 kg. (b) 0.583 s.
19.25 192 lb/ft.
19.26 (a) 21.7 kg. (b) 1011 kg/m³.
19.28 (a) 22.3 MN/m. (b) 266 Hz.
19.30 (a) 858 N/mm. (b) 149.5 rpm.
19.31 (a) 3.56 kg. (b) 43.7 kg.
19.34 16.26°.
19.35 (a) 1.737 s. (b) 1.864 s. (c) 2.05 s.
19.36 28.1 in.
19.37 (a) 0.293 s. (b) 0.215 m/s.
19.38 (a) 1.047 rad/s. (b) 16.42 in.
19.39 (a) 0.227 s. (b) 333 mm/s.
19.41 (a) 0.491 s. (b) 9.60 in/s.
19.42 (a) 0.1957 s. (b) 171.7 ft/s².
19.44 75.5°.
19.45 0.346 Hz.
19.48 (a) 2.79 s. (b) 1.933 m.
19.49 (a) 1.617 s. (b) 1.676 s.
19.50 (a) 227 mm. (b) 1.352 s.
19.51 (a) $6.33\sqrt{\frac{b}{g}}$. (b) $6.67\sqrt{\frac{b}{g}}$.
19.55 (a) 2.21 Hz. (b) 115. 3 N/m.
19.56 $\frac{1}{2\pi} \sqrt{\frac{6k}{5m} + \frac{9g}{10l}}$ Hz.
19.57 $\frac{1}{2\pi} \sqrt{\frac{2k}{3m} + \frac{4g}{3L}}$ Hz.
19.59 (a) 0.426 s. (b) 15.44 ft/s.
19.60 (a) 88.1 mm/s. (b) 85.1 mm/s.
19.62 82.2 mm/s $\uparrow$.
19.63 6.57 kg·m².
19.64 (a) 21.3 kg. (b) 1.836 s.
19.67 0.672 in.
19.68 8.60 ft.
19.69 19.02 mm.
19.70 $\frac{1}{2\pi} \sqrt{\frac{k}{5m}}$ Hz.
19.71 3.18 s.
19.72 $6.28\sqrt{R/g}$.
19.75 $l/\sqrt{12}$.
19.76 75.5°.
19.77 $0.159\sqrt{(2k/3m) + (4g/3L)}$.
19.78 2.10 Hz.
19.79 (a) 0.715 s. (b) 0.293 ft/s.
19.80 0.821 s.

CHAPTER 19

- 19.1** 10 mm, 3.18 Hz.
19.2 4.33 ft/s, 0.551 Hz.

- 19.83** 1.327 s.
19.85 1.834 s.
19.86 2.39 s.
19.87 $2\pi\sqrt{(12r^2 + 2l^2)/3gl}$.
19.89 (a) $\sqrt{(6ka^2 - 3mgl)/(2\pi)}$. (b) $\sqrt{mgl/2k}$.
19.90 2.29 Hz.
19.91 0.911 Hz.
19.92 $0.1312\sqrt{g/r}$.
19.95 $0.276\sqrt{g/l}$.
19.96 $\frac{1}{2\pi}\sqrt{\frac{12k}{7m} + \frac{8g}{7\sqrt{3l}}}$ Hz.
19.97 $1.814l/\sqrt{gr}$.
19.98 0.352 s.
19.99 (a) 37.1 mm. (b) 260 mm.
19.100 (a) 160.0 N/m. (b) 40.0 N/m.
19.101 $\sqrt{\frac{2k}{3m}} < \omega_f < \sqrt{\frac{4k}{3m}}$.
19.102 50.1 lb.
19.105 $\omega_f < 8.46$ rad/s, (no out-of-phase solution).
19.106 3.21 m/s².
19.107 (a) 0.450 rad/s. (b) 2.70 m/s².
19.109 $\omega_f > \sqrt{2g/l}$.
19.110 (a) 1.034 in. (b) $-0.1033 \sin \pi t$ (lb).
19.112 651 rpm.
19.114 22.0 mm.
19.115 0.0999 in.
19.116 $\omega_f < 322$ rpm.
19.117 39.1 kg.
19.119 149.3 mm.
19.121 Force transmissibility: $1/(1 - \omega_f^2/\omega_n^2)$,
 Displacement transmissibility: $1/(1 - \omega_f^2/\omega_n^2)$.
19.122 (a) 4.17%. (b) 84.9 Hz.
19.123 8.04%.
19.125 (a) 1399 rpm. (b) 0.01670 in.
19.132 (a) 6.49 kip · s/ft. (b) 230 kips/ft.
19.133 5.48 N·m·s.
19.134 (a) 6490 lb/ft. (b) 0.1939 s.
19.135 (a) 0.118. (b) 38.4 mm.
19.136 56.9 mm.
19.137 8.82 N.
19.138 106.5 mm/s ↑.
19.139 0.1791 in.
19.140 10.61 lb·s/in.
19.141 ≥ 0.707 .
19.143 (a) 147 kip/ft. (b) 0.0292.
19.144 0.0162 in.
19.145 (a) 0.127. (b) 462 N·s/m.
19.146 0.487.
19.148 (a) 71.8 N. (b) 39.0 N.
19.149 (a) 4.90 in. (b) 30.3 lb.
19.151 (a) $m\ddot{x} + c\dot{x} + kx = (k \sin \omega_f t + c\omega_f \cos \omega_f t)\delta_m$.
 (b) $x = x_m \sin(\omega_f t - \varphi + \psi)$, where
 $x_m = \delta_m \sqrt{k^2 + (c\omega_f)^2} / \sqrt{(k - m\omega_f^2)^2 + (c\omega_f)^2}$,
 $\tan \varphi = c\omega_f/(k - m\omega_f^2)$, $\tan \psi = c\omega_f/k$.
19.153 $R < 2\sqrt{L/C}$.
19.154 (a) E/R . (b) L/R .
19.157 (a) $c(\dot{x}_A - \dot{x}_m) + kx_A = 0$
 $m\ddot{x}_m + c(\dot{x}_m - \dot{x}_A) = P_m \sin \omega_f t$
 (b) $R(\dot{q}_A - \dot{q}_m) + (1/C)q_A = 0$
 $L\ddot{q}_m + R(\dot{q}_m - \dot{q}_A) = E_m \sin \omega_f t$
19.159 0.760 lb·s² · ft, 8.66 in.
19.160 (a) 6.82 lb. (b) 33.4 lb/ft.
19.161 1.785 s.
19.162 (a) 0.0139. (b) 0.0417 lb·s/ft.
19.165 (a) $0.07246\theta + 0.3375\dot{\theta} + 1.25\theta = 0$.
 (b) -19.05×10^{-6} degrees.
19.168 (a) $m\ddot{x} + 2T(2x/l) = 0$. (b) $\pi\sqrt{ml/T}$.
19.169 0.045 in.

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APPENDIX B

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